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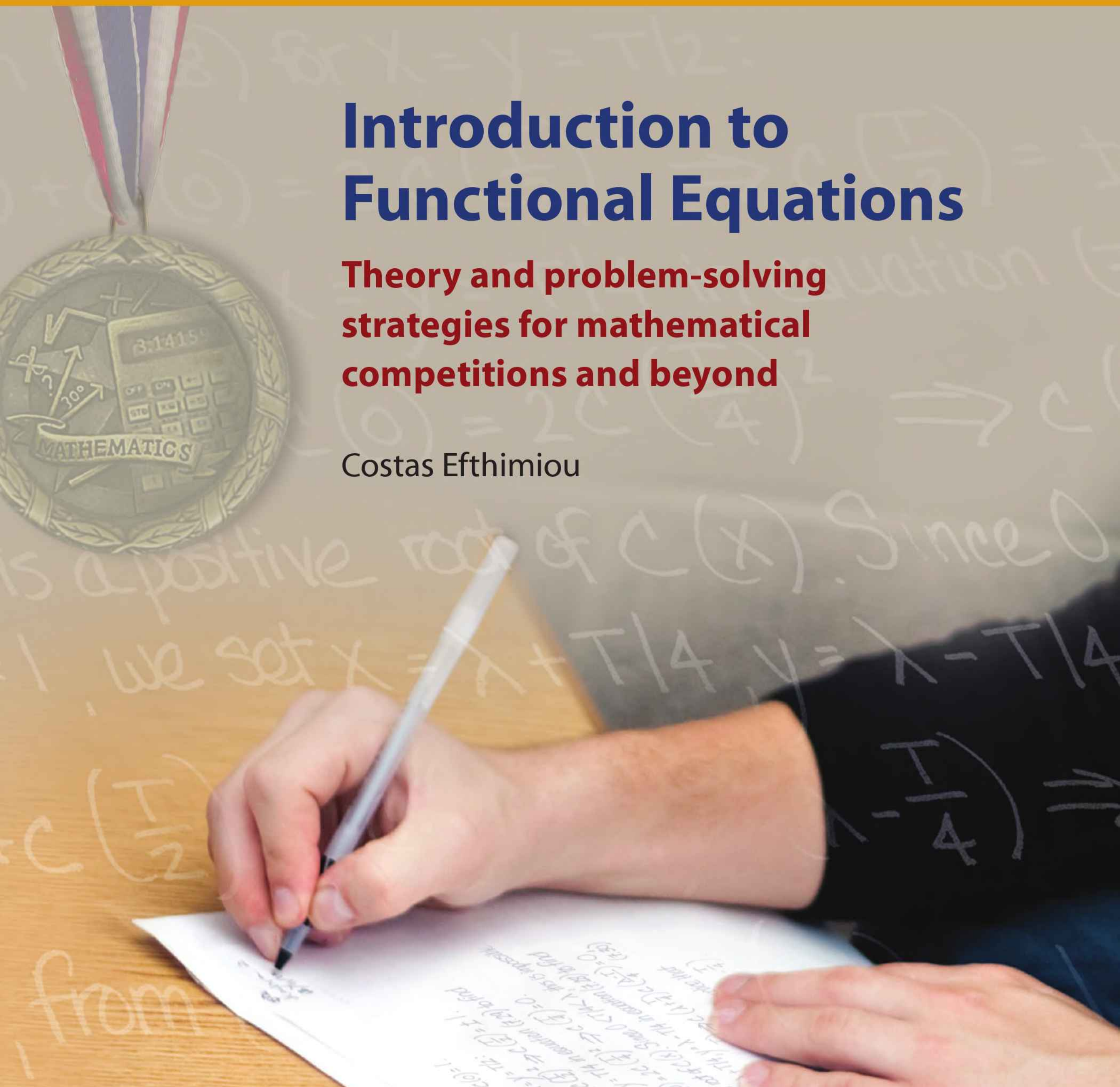

Introduction to Functional Equations

Theory and problem-solving strategies for mathematical competitions and beyond

Costas Efthimiou

MSRI AMS
MATHEMATICAL SCIENCES RESEARCH INSTITUTE
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To My Mother ♡ ♡ ♡ ♡ ♡ ♡

♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡ ♡

who taught me that love is not
measured by how many things
one can offer to a child but
by how many things
a parent sacrifices
for the child

♡

Contents

Introduction

One of the most beautiful mathematical topics I encountered as a student was functional equations: the study of functions which satisfy given equations, such as

$$f(x+y) = f(x) + f(y).$$

Functional equations arise in all areas of mathematics and even more so in science, engineering, and the social sciences. They appear at all levels of mathematics: from the elementary definition of an even function as one that satisfies the functional equation

$$f(x) = f(-x),$$

to the forefront of research. For example, during the last quarter century, the celebrated Yang–Baxter functional equation has been at the heart of many different areas of mathematics and theoretical physics, such as lattice integrable systems, factorized scattering in quantum field theory, braid and knot theory, and quantum groups, to name a few. The Yang–Baxter equation is a system of N^6 functional equations for the $(N^2 \times N^2)$-matrix $\boldsymbol{R}(x)$ whose entries are functions of x:

$$\sum_{\alpha,\beta,\gamma=1}^{N} \boldsymbol{R}_{jk}^{\alpha\beta}(x-y)\, \boldsymbol{R}_{i\alpha}^{\ell\gamma}(x)\, \boldsymbol{R}_{\gamma\beta}^{mn}(y) = \sum_{\alpha,\beta,\gamma=1}^{N} \boldsymbol{R}_{ij}^{\alpha\beta}(y)\, \boldsymbol{R}_{\beta k}^{\gamma n}(x)\, \boldsymbol{R}_{\alpha\gamma}^{\ell m}(x-y),$$

where i, j, k, ℓ, m, n take the values $1, 2, \ldots, N$. Amazingly, although this equation appears to impose more constraints (N^6) than the number of unknowns (N^4), it has a rich set of solutions. The study of the Yang–Baxter equation is well beyond our scope; however, the reader will hopefully come to appreciate the role that functional equations play in mathematics and science over the course of reading this book.

Despite being such a rich subject, this topic does not fit perfectly into any of the conventional areas of mathematics and thus a systematic way of studying functional equations is not found in the traditional curriculum. Several good books on the topic exist, but are either relatively hard to find [40] or quite advanced [8, 9, 28]. While this book was in the making, a nice, simple book [39] appeared, which has considerable overlap with, and is complementary to, this book.

We should point out in advance that there is no universal technique for solving functional equations, as will become obvious once the reader solves the problems throughout this book. At the current time, thorough knowledge of many areas of mathematics, lots of imagination, and, perhaps, a bit of luck are the best methods for solving such equations.

The core of the book is the result of a series of lectures I presented to the Putnam team at University of Central Florida soon after my arrival there. My personal belief is that the training of a math team should be based on a systematic approach for each subject of interest for the mathematical competitions (e.g., functional equations, inequalities, finite and infinite sums, etc.) instead of randomly solving hard problems to get experience. Therefore, this book is an attempt to present the fundamentals of the subject in a pedagogical manner, accessible to students who have a background on the theory of functions up to differentiability. Some parts of the book use additional ideas from calculus, but they may be omitted on a first reading.

In particular, the book should be useful to high school students who participate in math competitions and have an interest in International Math Olympiads (IMO). Many of the problems I have included I solved as a high school student[1] in preparation for the national math exam of my home country, Greece. I still enjoy them as much as I enjoyed them then. In fact, I now appreciate their beauty much more, as I have grown to have a better understanding of the subject and its importance.

College students with an interest in the Putnam Competition should also find the book useful, as standard texts do not provide a thorough coverage of functional equations. Finally, we hope that any person with interest in mathematics will find in this book something to his or her interest.

However, a piece of advice is in order here for all readers: The majority of the problems discussed in this book are related to mathematical competitions which target ingenuity and insight, so they are not easy. Approach them with caution. They should serve as a source of enjoyment, not despair!

And a final comment: The interested reader should, perhaps, read the book simultaneously with Smítal's [40] and Small's [39] books, which are at the same level. I cover more functional equations but Smítal presents beautifully the topic of iterations and functional equations of one variable.[2] Similarly, Small's book [39] is a very enjoyable, well-written book and focuses on the most essential aspects of functional equations. Once the reader is done with these three books, he may read Aczél's and Kuczma's authoritative books

[1] Yes, I still have many of my notes! As a result, some of the problems I present might be taken from national or local competitions without reference. I would appreciate a note from readers who discover such omissions or other typos and mistakes.

[2] One can detect some influence from Smítal's book on my presentation in Chapters 16 and 17. I highly recommended this book to any serious student.

[8, 9, 28], which contain a vast amount of information. Beyond that level, the reader will be ready to immerse himself in the current literature on the subject, and perhaps even conduct his own research in the area.

More on How to Approach This Book

Mathematics has a reputation of being a very stiff subject that only follows a pre-determined pattern: definitions, axioms, theorems, proofs. Any deviations from this pre-established pattern are neither welcome nor wise. In this respect, mathematics appears to be very different from the sciences and even from its closest neighbor, theoretical physics. For mathematics uses **demonstrative reasoning** and the sciences use **plausible reasoning**.[3]

As mentioned previously, the majority of the problems in this book are taken from mathematical competitions. As such, they are precious not only for their originality and beauty, but also for the lessons they teach the students regarding mathematical discovery. Without a doubt, the final solutions of these problems must be presented as demonstrative reasoning. However, to reach this stage, a student must first go through the stage of plausible reasoning. Solving a problem first involves guessing the solution. Proving the solution first involves visualization of the proof. Polishing the proof requires trying these steps over and over. Therefore demonstrative reasoning and plausible reasoning are not two distinct, isolated methods; they are just the two faces of the process of discovering new results in mathematics. So, in agreement with Pólya's motto, the problems in this book should be used to "let us learn proving, but also *let us learn guessing.*"

Unfortunately, due to lack of time and space, I have omitted the plausible reasoning which can be experienced by attending Math Circles, and I have presented only the demonstrative reasoning in what I believe to be a polished way. However, students should be assured that the solutions presented could not have been found without hunches and guesses of some kind. The beginning student should thus be neither discouraged nor disappointed if his reasoning, either at the plausible or at the demonstrative stage, fails to be as good as those of experienced solvers. Paul Erdős, one of the most prolific publishers of papers in mathematical history and an extraordinary problem solver, has said: "Nobody blames a mathematician if the first proof of a new theorem is clumsy." Therefore, solve these problems as many times as necessary to improve your solutions. With every new solution you discover, you gain invaluable knowledge that becomes your new weapon to attack new problems. This is not really a new idea; it has been quite familiar to anyone who has tried problem solving. Here is René Descartes' testimony: "Each problem that

[3]The terms are discussed in G. Pólya's classic two-volume work [32] which teaches students the role of guessing in rigorous mathematics and how to become an effective guesser.

I solved became a rule which served afterwards to solve other problems." Any reader who places considerable amount of effort on the problems will benefit from them, even if he does not completely solve all the problems.

Of course, all this assumes that you see beauty in these problems. I cannot define mathematical beauty, but I know it when I see it. It is really love at first sight. Upon completion of the first draft of the manuscript, I showed it to some good students who had not attended my lectures. Given their high academic performance in the standard curriculum, I expected them to make favorable comments about the choice of problems, but there was one student who declared them to be trick problems! He appeared unimpressed and, I think, completely uninterested in them. I had always assumed, even after many years of teaching, that if a student stands well above the crowd in mathematical ability, he will also see well beyond the crowd. However, this incident made me realize that even some students with mathematical talent, when they have been educated in a dry and boring traditional system, often by unmotivated teachers, have never developed mathematical aesthetics and a deep understanding of how the process of mathematical discovery really happens. As a skillful but blind diamond cutter, looking at the diamond, never sees its sparkle and so cannot appreciate the fine cuts, so a student who has never been shown true mathematics but is otherwise a skillful solver may not appreciate the extraordinary beauty of math problems.

Therefore, it is my obligation as I present this book to students to declare: Trick mathematics is the norm. Straightforward proofs are the exceptions. There are patterns, but these are the products of experience and hard work of the giants before us. To unveil a pattern, you must first produce many trick proofs. The trickier the proof, the greater the hidden beauty. Such is the case, for example, with Sharkovskii's theorem, stated in Chapter 16—a fine jewel of modern mathematics.

Appreciating and solving the problems given in mathematical competitions builds strong intuition, develops mathematical ingenuity, and promotes deep understanding for trick proofs. It leads the way to creating new generations of outstanding mathematicians. Dear reader, please approach this book with this spirit in mind.

Costas Efthimiou
Orlando, FL
July 2010

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First of all, I must thank the students who sat in my lectures at the University of Central Florida in 2001. Without them, I would never had written the initial notes from which this book evolved.

Besides those who inspired the project, the author of any book should thank Donald Knuth, Leslie Lamport and the LaTeX3 Project team for their invaluable services to all authors. With the powerful typesetting language of TeX and its derivative, LaTeX, any author can easily pen his thoughts in what he thinks is the most effective way. The preparation of this book would have otherwise been a painful or impossible task.

The figures were created using Daniel Luecking's MFPIC package, and I also thank him for his help in using the package to stand-alone EPS figures.

Many other people helped further the project in different ways, from conception to completion. Roy Choudhury and Vaggelis Alexopoulos skimmed over a half-finished version and gave me words of approval that made me very happy and reinforced my intent to produce a final copy for publication. Joseph Brennan read the chapter on polynomials and offered suggestions. Li Zhou contributed a few problems; Caleb Wiese typed pages of handwritten notes on difference equations; Robert van Gorder proofread the manuscript and suggested several corrections. Christopher Frye proofread the manuscript carefully near completion and detected many typos. Last, but not at least, I thank Silvio Levy, who corrected many typos, pointed out flaws in several of my proofs, and edited the book to the right specifications for the final printing by AMS.

Of course, I am responsible for any surviving mistakes. I apologize to the reader in advance for any confusion that may arise due to them. And finally, I must confess that I have not always followed the suggestions made by the proofreaders: sometimes from lack of space and time, sometimes from personal eccentricity. However, I can only hope that the value of this book has not been compromised as a result.

Part I

BACKGROUND

Chapter 1

Functions

1.1 Sets

If x is an **element** (or **member**) of a set X, we write $x \in X$. Set membership, by definition, does not involve multiplicity or order: thus the set $\{a, a\}$ is really $\{a\}$ and the sets $\{a, b\}$, $\{b, a\}$ are identical. The set with no elements is called the **empty set** and is denoted by $\varnothing$ or $\{\}$.

The elements of a set are determined by a common property such as that in the following problem.

Problem 1.1 (IMO 1971). *Prove that for every natural number m, there exists a finite set S_m of points in a plane with the following property: For every point A in S_m, there are exactly m points in S_m which are at unit distance from A.*

Solution ([6], December 1971, p. 40). For $m = 1$ we take S_1 to be the end-points of a unit interval. For $m = 2$ we take S_2 to be the vertices of an equilateral triangle.

For $m > 2$ we construct S_m by induction. Given $S_k = \{\mathrm{A}_1, \mathrm{A}_2, \ldots\}$, let S_{k+1} be the set that includes all points of S_k and Σ_k, where $\Sigma_k = \{\mathrm{A}'_1, \mathrm{A}'_2, \ldots\}$ is the set of points obtained by translating the points of S_k by unit vector $\vec{u}$: $\mathrm{A}_i \overset{\vec{u}}{\mapsto} \mathrm{A}'_i$. That is, for each value of the label i, the points A_i, A'_i are at a unit distance. Each point in S_k and Σ_k has exactly k points at unit distance. By construction then, each point in S_{k+1} has at least $k+1$ points at unit distance. To ensure that there will be no more than $k+1$ points at unit distance, the vector $\vec{u}$ must be selected appropriately to avoid any unwanted equilateral triangles. For each point A_i in S_k we draw a circle C_i of radius 1 and center A_i. Let P_i^1, P_i^2, ... be the intersection points of C_i with the other circles. The (finite) set of vectors $\overrightarrow{\mathrm{A}_i\mathrm{P}_i^\alpha}$ for all values of i and α are the vectors to be excluded as candidates for $\vec{u}$. $\square$

If every element of the set A also belongs to the set B, we say that A is a **subset** of B, and we write $A \subset B$. If two sets A, B satisfy $A \subset B$ and $B \subset A$, the two sets are **equal**; we write $A = B$.

The notation $A \subset B$ allows the possibility that $A = B$; in the rare event that we wish to exclude this possibility, we write $A \subsetneqq B$, and say that A is a **proper subset** of B. The notation $B \supset A$ is equivalent to $A \subset B$, and says that B is a **superset** of A.

Let $A, B \subset S$. The **union** of A and B, denoted by $A \cup B$, is defined as the set of $x \in S$ such that $x \in A$ or $x \in B$.

Let $A, B \subset S$. The **intersection** of A and B, denoted by $A \cap B$, is defined as the set of elements $x \in S$ such that $x \in A$ and $x \in B$. Two sets A, B are called **disjoint** if $A \cap B = \varnothing$.

Let $A, B \subset S$. The **difference** $B \setminus A$ is defined as the set of elements $x \in B$ that do not belong to A. In the particular case that $A \subset B$, we call $B \setminus A$ the **complement of** A **in** B, or simply the **complement of** A (if B can be inferred from the context). The complement of A is often denoted by $A^{\complement}$.

The **power set** of S, denoted by $\mathcal{P}(S)$, is the set of all subsets of S:

$$\mathcal{P}(S) := \{A : A \subset S\}.$$

The **Cartesian product** of an n-tuple of sets $(A_1, A_2, \ldots, A_n)$ is defined as the set whose elements consist of all n-tuples whose i-th entry is an element of A_i: in symbols,

$$A_1 \times A_2 \times \cdots \times A_n := \{\, (a_1, a_2, \ldots, a_n) : a_i \in A_i \text{ for } i = 1, 2, \ldots, n \,\}. \tag{1.1}$$

If $A_1 = A_2 = \cdots = A_n =: A$, we usually write

$$\underbrace{A \times A \times \cdots \times A}_{n \text{ times}} =: A^n.$$

Problem 1.2 (Putnam 1961). *Let Ω be a set of n points, where $n > 2$. Let Σ be a nonempty subset of $\mathcal{P}(\Omega)$ that is closed*[1] *with respect to unions, intersections, and complements. If k is the number of members of Σ, what are the possible values of k? Give a proof.*

Solution [21]. Since Σ is nonempty, there is a set A in Σ. Its complement $A^{\complement}$ is also in Σ and so is the intersection $A \cap A^{\complement} = \varnothing$ and the union $A \cup A^{\complement} = \Omega$.

Let $\mathcal{M}$ now be the following subset of Σ:

$$\mathcal{M} = \{M \in \Sigma : \text{there is no } A \in \Sigma,\ A \neq \varnothing, \text{ such that } A \subsetneqq M\}.$$

[1]That is, if A and B are members of Σ, then so are $A \cup B$, $A \cap B$, $A^{\complement} = \Omega \setminus A$, and $B^{\complement} = \Omega \setminus B$. This information was provided in the test.

In other words, $\mathcal{M}$ contains the 'minimal' sets of Σ. Obviously, $\mathcal{M}$ is a finite set.

Given two distinct minimal sets M_1 and M_2, they must be disjoint, since $M_1 \cap M_2 = B \in \Sigma$ and thus $B \subset M_1, M_2$, which, by definition, cannot be true for any nonempty set.

Now let M be the union of all minimal sets:

$$M = \bigcup_{M_i \in \mathcal{M}} M_i.$$

Its complement $M^{\complement}$ can be $\varnothing$ or some nonempty set. If it is nonempty, there is a minimal set M_0 such that $M_0 \subset M^{\complement}$; thus $M_0 \cap M^{\complement} \neq \varnothing$. However, $M_0 \cap M^{\complement} = M_0 \cap (\Omega \setminus M_0 \setminus \dots) = \varnothing$. A contradiction. Therefore $M^{\complement} = \varnothing$ or $M = \Omega$.

Finally, let C be any random set in Σ. Then

$$\begin{aligned} C = C \cap \Omega = C \cap \Big(\bigcup_{M_i \in \mathcal{M}} M_i \Big) \\ = \bigcup_{M_i \in \mathcal{M}} (C \cap M_i). \end{aligned}$$

Each of the intersections $C \cap M_i$ is either the empty set or M_i.

To summarize, we have proved that: Given a partition of Ω in p disjoint sets $M_1, M_2, \dots, M_p$, where $1 \leq p \leq n$, each member of Σ can be written as the union of a subcollection of the sets $M_1, M_2, \dots, M_p$. And since for each set M_i we have two choices (to include it or not to include it), we have 2^p possible subcollections. Therefore, $k = 2^p$, with all values $p = 1, 2, \dots, n$ allowed. $\square$

Topological Concepts

Definition 1.3. Let $\mathcal{O}$ be a subset of the power set $\mathcal{P}(X)$ satisfying these properties:

1. $\varnothing \in \mathcal{O}, \ X \in \mathcal{O}$.
2. The intersection of a finite number of elements of $\mathcal{O}$ also belongs to $\mathcal{O}$.
3. The union of any number of elements of $\mathcal{O}$ also belongs to $\mathcal{O}$.

We say that $\mathcal{O}$ is a **topology** on X. The pair $(X, \mathcal{O})$ is called a **topological space**. We also say that X is a topological space, especially if the topology can be inferred from the context. It is common to refer to the elements of a topological space as **points**. The elements of the topology $\mathcal{O}$ are called the **open sets** of X (with respect to the topology). The complement $X \setminus O$ of an open set O is called a **closed** set.

Definition 1.4. Let $(X, \mathcal{O})$ be a topological space. A set $N \subset X$ is called a **neighborhood** of the point $p \in X$ if p lies in N and there exists an open set $O \in \mathcal{O}$ that contains p and is contained in N: in symbols, $p \in O \subset N$.

Nothing in this definition requires a neighborhood to be open. However, when we refer to a **neighborhood** of a point $a \in \mathbb{R}$, we shall have in mind — unless otherwise specified — an open interval of the form $(a-\delta, a+\delta)$, where $\delta > 0$ is called the **radius** of the neighborhood.

Given a topological space X and a subset S of X, the following definitions cover some of the most interesting points and sets related to S:

1. A point ℓ is called a **limit point** of the set S if every neighborhood of ℓ contains at least one point of S different from ℓ. The set S' of all limit points of S is called the **derived set** of S.
2. A point i is called an **interior point** if there exists a neighborhood of i that is contained in S. The set $\operatorname{Int} S$ of all interior points of S is called the **interior set** of S.
3. A point e is called an **exterior point** of S if there exists a neighborhood of e that is disjoint from S. The set $\operatorname{Ext} S$ of all exterior points of S is called the **exterior set** of S.
4. A point b is called a **boundary point** of S if every neighborhood of b intersects both S and $X \setminus S$. The set ∂S of all boundary points of S is called the **boundary** of S.
5. A point i_0 is called an **isolated point** of S if $i_0 \in S$ and i_0 lies in the exterior of $S \setminus \{i_0\}$. In other words, there is a neighborhood $N(i_0)$ of i_0 that does not contain any other point of S except i_0.
6. The smallest closed set $\bar{S}$ containing S is called the **closure** of S.
7. A subset S of X is called **dense** in X if $\bar{S} = X$.

There are many useful relations among the sets defined above. Let X be a topological space and let A, B, O, C be subsets of X. Then:

1. C is closed if and only if $\overline{C} = C$.
2. C is closed if and only if $C' \subset C$.
3. C is closed if and only if $\partial C \subset C$.
4. O is open if and only if $\operatorname{Int} O = O$.
5. O is open if and only if $\partial O \cap O = \varnothing$.
6. $\overline{A} = A \cup A'$.
7. $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

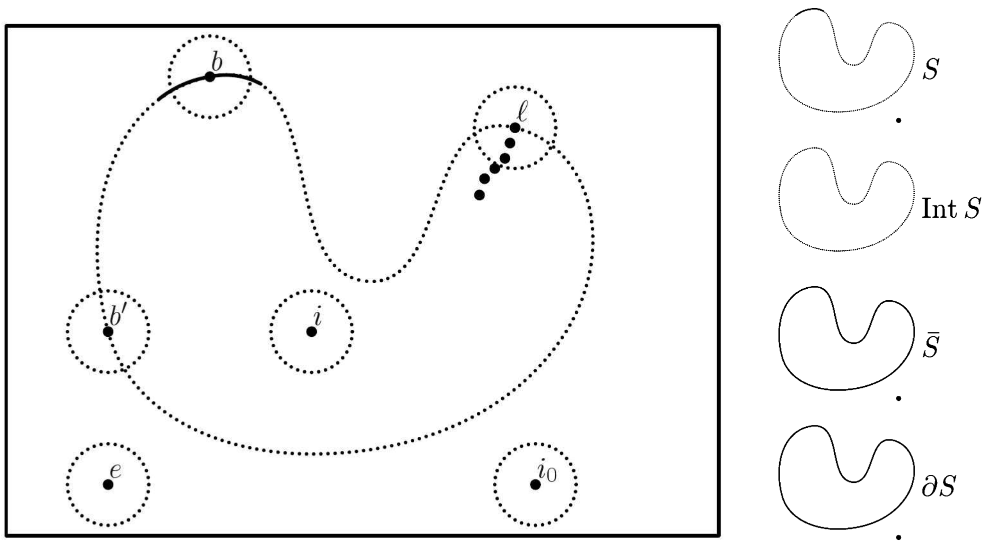


Figure 1.1: Terminology of points and sets in a topological space. The left side focuses on points: i is an interior point of the set S shown; e is exterior to S (and equivalently, interior to the complement of S); b and b' are boundary, the first being in S, the second in its complement. The set S was chosen to contain the isolated point i_0 in addition to the irregularly shaped region: note that $i_0 \in S \cap \overline{S \setminus \{i_0\}}$. A limit point ℓ of S is the limit of a converging sequence of points in S distinct from ℓ.

8. $\overline{\overline{A}} = \overline{A}$.
9. $\overline{A \cap B} \subset \overline{A} \cap \overline{B}$.
10. $\overline{A \setminus B} \subset \overline{A} \setminus \overline{B}$.
11. $\overline{X \setminus A} = \overline{X} - \operatorname{Int} A$.
12. $\overline{A} = X \setminus \operatorname{Ext} A$.
13. $\partial A = X \setminus (\operatorname{Int} A \cup \operatorname{Ext} A)$.
14. $\partial A = \overline{A} \cap \overline{(X \setminus A)}$.
15. $\partial A = \overline{A} \setminus \operatorname{Int} A$.

Example 1.5. An interesting set is the set of rational numbers

$$\mathbb{Q} = \{m/n : m \in \mathbb{Z},\, n \in \mathbb{Z}^*\}$$

as a subset of $\mathbb{R}$. Then $\overline{\mathbb{Q}} = \mathbb{R}$, $\partial\mathbb{Q} = \mathbb{R}$, $\operatorname{Int}\mathbb{Q} = \varnothing$. Since $\operatorname{Int}\mathbb{Q} \neq \mathbb{Q}$, $\mathbb{Q}$ is not open; since $\partial\mathbb{Q} \supset \mathbb{Q}$, $\mathbb{Q}$ is not closed either. □

Problem 1.6 (Putnam 1956). *Suppose that each set X of points in the plane has an associated set $\overline{X}$ of points called its cover. Suppose further that*

$$\overline{X \cup Y} \supset \overline{\overline{X}} \cup \overline{Y} \cup Y, \tag{1.2}$$

where $\cup$ *designates point set sum (or union) and* $\supset$ *denotes set inclusion. Prove*

(a) $\overline{X} \supset X$,

(b) $\overline{\overline{X}} = \overline{X}$,

(c) $X \supset Y$ *implies* $\overline{X} \supset \overline{Y}$.

Prove conversely that (a), (b), and (c) imply (1.2).

Solution. In (1.2) we set $X = Y$ to find

$$\overline{X \cup X} \supset \overline{\overline{X}} \cup \overline{X} \cup X \Rightarrow \overline{X} \supset \overline{\overline{X}} \cup \overline{X} \cup X.$$

From the last relation we conclude that

$$\overline{X} \supset X$$

which is the relation (a) sought and

$$\overline{X} \supset \overline{\overline{X}}.$$

However, relation (a), after replacing X by $\overline{X}$, implies

$$\overline{\overline{X}} \supset \overline{X}.$$

The last two relations give the desired relation (b).

Finally, if $X \supset Y$ then $X \cup Y = X$. Therefore relation (1.2) is equivalent to

$$\overline{X} \supset \overline{\overline{X}} \cup \overline{Y} \cup Y,$$

from which we conclude $\overline{X} \supset \overline{Y}$.

To prove the converse, notice that for any two sets X and Y

$$X \cup Y \supset X \overset{(c)}{\Rightarrow} \overline{X \cup Y} \supset \overline{X} \overset{(b)}{\Rightarrow} \overline{X \cup Y} \supset \overline{\overline{X}}.$$

Similarly

$$X \cup Y \supset Y \overset{(c)}{\Rightarrow} \overline{X \cup Y} \supset \overline{Y}.$$

From the last two relations we see that

$$\overline{X \cup Y} \supset \overline{\overline{X}} \cup \overline{Y}.$$

And since $\overline{Y} \supset Y$, on the right-hand side we can write $\overline{Y} = \overline{Y} \cup Y$, thus obtaining (1.2). □

1.2 Relations

A **relation** on a set $A \neq \varnothing$ is a subset $R \subset A \times A$ of the Cartesian product $A \times A$. Usually, instead of writing $(x, y) \in R$, we write xRy; we say that x is **related** to y.

Definition 1.7. A relation $\sim$ on the set A is called an **equivalence relation** if it satisfies the following axioms for all $a, b, c \in A$:

1. *Reflexivity*: $a \sim a$.
2. *Symmetry*: $a \sim b$ implies $b \sim a$.
3. *Transitivity*: if $a \sim b$ and $b \sim c$ then $a \sim c$.

The expression $x \sim y$ is then read "x is equivalent to y".

Definition 1.8. Given an equivalence relation on A, the **equivalence class** of x is the set

$$[x] \equiv \{\, a \in A : a \sim x \,\}.$$

Exercise 1.9. *Let* $[x]$ *and* $[y]$ *be two equivalence classes. Show that either* $[x] = [y]$ *or* $[x] \cap [y] = \varnothing$.

A **partition** of a set A is a family of pairwise disjoint subsets $\{A_i\}$ of A such that $A = \bigcup_i A_i$. Using the last problem, we see that an equivalence relation in a set A defines a partition of A.

Example 1.10. In the set of integers $\mathbb{Z}$ we fix an integer k and we define an equivalence relation by

$$m \sim n \iff m - n \text{ is divisible by } k.$$

Since any integer n can be uniquely written in the form

$$n = n'\,k + r \quad \text{for } 0 \leq r < k \text{ and } n', k, r \in \mathbb{Z},$$

we conclude that two integers n, m belong to the same equivalence class if and only if they have the same residue r when divided by k. Therefore, this equivalence relation partitions the set of integers into k subsets:

$$A_r = \{k\,n' + r : n' \in \mathbb{Z}\} = [r] \quad \text{for } r = 0, 1, \ldots, k-1.$$

The set of equivalence classes is

$$\mathbb{Z}_k = \{[0], [1], \ldots, [k-1]\}.$$

Often we write $\mathbb{Z}_k = \mathbb{Z}/\!\sim$ to show that $\mathbb{Z}_k$ is the set $\mathbb{Z}$ with the relation $\sim$ "factored out". $\square$

Definition 1.11. A relation $\preccurlyeq$ on the set A is called an **order relation** if it satisfies the following axioms for all $a, b, c \in A$:

1. Reflexivity: $a \preccurlyeq a$.
2. Symmetry: If $a \preccurlyeq b$ and $b \preccurlyeq a$, then $a = b$.
3. Transitivity: If $a \preccurlyeq b$ and $b \preccurlyeq c$, then $a \preccurlyeq c$.

The expression $x \preccurlyeq y$ is then read "x precedes y".

If every element of A is comparable to every other, that is, if for all $a, b \in A$ we have

4. Trichotomy: either $a \preccurlyeq b$ or $b \preccurlyeq a$,

then we say that $\preccurlyeq$ is a **total order**. If there are elements that are not comparable to one another, then we say the order is **partial**.

Example 1.12. Given a set A, let's consider on $\mathcal{P}(A)$ the relation of inclusion. That is, take $\preccurlyeq\, = \,\subset$, in the sense that for any $S, T \in \mathcal{P}(A)$, we have $S \preccurlyeq T$ if and only if $S \subset T$. This is a partial order on $\mathcal{P}(A)$ if A has more than one element, because two singlets $\{a\}$ and $\{b\}$ are not included in each other if $a \neq b$. $\square$

Example 1.13. The set of reals $\mathbb{R}$ equipped with $\leq$ is totally ordered. $\square$

1.3 Functions

The Notion of a Function

Suppose a subset F of $A \times B$ satisfies the following properties:

- $(x, y) \in F$ and $(x, y') \in F$ implies $y = y'$.
- For every $x \in A$, there is y such that $(x, y) \in F$.

We say that the triple $f = (F, A, B)$ is a **function** (or a **map**), and we write $f : A \to B$, pronounced "f maps A into B". We call A the **domain** of f, and B its **codomain**. Finally, the set of ordered pairs F is the graph of f; it is also denoted by Γ_f.

We think of the function f as a correspondence that assigns to an element x in the domain the element y of the codomain such that $(x, y) \in \Gamma_f$. We write also $f(x) = y$. Thus

$$\Gamma_f = \{\, (x, f(x)) : x \in A \,\} \subset A \times B.$$

It follows from the definition that the domain and codomain are integral parts of a function: for two functions to be equal, their domains, codomains, and graphs must be equal. That is, $f : A \to B$ and $g : C \to D$ are equal if and only if $A = C$, $B = D$, and $f(x) = g(x)$ for all $x \in A$.

For $x \in A$ and $y = f(x)$, we call y the **image** of x, and x the **preimage** of y. If $A' \subset A$, the subset of B given by $f(A') = \{f(a) : a \in A'\} \subset B$ is called the **image** of A'. The image $f(A)$ of the domain is also called the image of the function.[2]

Comment. Although the most common objects of our study will be real-valued functions ($B \subset \mathbb{R}$) of a real variable ($A \subset \mathbb{R}$), the definitions given here are more general. The elements of the domain may be vectors, matrices, or any other mathematical object, and likewise for the codomain. □

Example 1.14. The **identity function** $\mathrm{id}_A : A \to A$ maps each element of A to itself: $\mathrm{id}_A(x) = x$. □

Example 1.15. Suppose $c \in B$. The **constant function** with value c is the function $f : A \to B$ that maps the whole domain to c, that is, $f(x) = c$ for all $x \in A$. □

Suppose $f : A \to \mathbb{R}$ is a real-valued function defined on a subset $A \subset \mathbb{R}$. The graph $\Gamma_f = \{\, (x, f(x)) : x \in A \,\} \subset A \times B$ is often of help in visualizing the function's behavior (see Problem 1.17 on the next page). However, for many functions such a pictorial representation may not be possible and we must treat the graph as a set.

Example 1.16. The **Dirichlet function**, defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}, \end{cases}$$

is one for which a sketch of the graph would not be much help: since both $\mathbb{Q}$ and its complement are dense in $\mathbb{R}$, the graph would look like two horizontal lines, no matter how much we magnify it. □

Problem 1.17. *Draw the graph of a function* $f : \mathbb{R} \to \mathbb{R}$ *that satisfies*

(a) $f(x + 1) = f(x) - 2$ *for all* $x \in \mathbb{R}$, *and*

(b) $f(x) = x^2$ *for* $x \in [0, 1)$.

[2] I avoid the term "range", which can mean either the codomain or the image of the domain.

Solution. Using induction it is easy to see that

$$f(x+n) = f(x) - 2n \quad \text{for all } n \in \mathbb{N}.$$

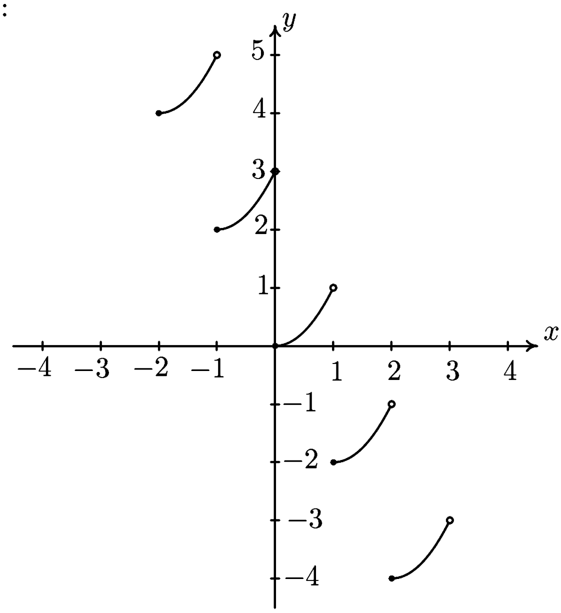
Let $y \in \mathbb{R}$ be an arbitrary real number. We can always write it as $y = x + n$, where n is the integer part of y, also denoted by $\lfloor y \rfloor$, and x is the fractional part of y, denoted by $\{y\}$ and belonging to the interval $[0, 1)$. Then, the previous equation gives

$$f(y) = f(\{y\}) - 2n \quad \text{for } n \le y < n+1.$$

Since the fractional part satisfies the inequality $0 \le \{y\} < 1$, condition (b) yields $f(\{y\}) = \{y\}^2$. However, $\{y\} = y - n$ and we finally have

$$f(y) = (y-n)^2 - 2n \quad \text{for } n \le y < n+1.$$

Here is a sketch of the graph:



□

Properties of Functions

A function $f : A \to B$ is called **surjective** or **onto** if its image and codomain coincide: $f(A) = B$. Equivalently, every element in B has a preimage under f in A. Obviously, if a function $f : A \to B$ is not surjective, we can find a surjective function with the same "content" as f by just replacing the codomain B with $f(A)$.

Exercise 1.18. *Let $f : A \to B$ be a function and let A', B' be subsets of A. Prove:*

$$f(A' \cap B') \subset f(A') \cap f(B');$$
$$f(A' \cup B') = f(A') \cup f(B').$$

□

A function $f : A \to B$ is called **injective** or **one-to-one** if any two distinct elements in A are mapped to distinct elements in B:

$$x_1 \neq x_2 \;\Rightarrow\; f(x_1) \neq f(x_2) \quad \text{for all } x_1, x_2 \in A.$$

Equivalently, f is injective if

$$f(x_1) = f(x_2) \;\Rightarrow\; x_1 = x_2 \quad \text{for all } x_1, x_2 \in A.$$

A function f is called **bijective** if it is both injective and surjective.

The **restriction** of a function $f : A \to B$ to a set $A' \subset A$ is the function $g : A' \to B$ defined by $g(x) = f(x)$ for all $x \in A'$. It is often denoted by $f|_{A'}$. Equivalently, one can say that f is the **extension** of g to the set $A \supset A'$.

Let A, B be sets with total order, both orders being denoted by $\leq$. We say that $f : A \to B$ is **nondecreasing** if $f(x) \leq f(y)$ whenever $x, y \in A$ satisfy $x \leq y$. A nondecreasing function for which, in addition, $f(x) \neq f(y)$ whenever $x \neq y$ (equivalently, $f(x) < f(y)$ whenever $x < y$) is **strictly increasing**, or **order-preserving**. **Nonincreasing** functions and **strictly decreasing** (or **order-reversing**) functions are defined analogously. A (strictly) increasing or decreasing function is also called (strictly) **monotonic**.

A function $f : A \to B$ is said to be **upper bounded** if there exists $M \in B$ such that $f(x) \leq M$ for all $x \in A$. It is called **lower bounded** if there exists $m \in B$ such that $m \leq f(x)$ for all $x \in A$. The function is **bounded** if it is both lower and upper bounded. For a lower bounded function, we call the greatest of its lower bounds the **infimum**; for an upper bounded function, we call its least upper bound the **supremum**.

Operations on Functions

Given two real-valued functions $f : A \to \mathbb{R}$ and $g : A \to \mathbb{R}$ with the same domain, we define their sum $f + g$ (difference $f - g$, product $f \cdot g$, and so on) to be a function h defined on the same domain whose image $h(x)$ of $x \in A$ is the sum (difference, product, and so on) of the images $f(x)$, $g(x)$ of the original functions. (The quotient f/g is defined on the subset of A where g does not vanish.)

Let $f : A \to B$ and $g : B \to C$ be functions. We can define a new function by applying f and g sequentially; this new function $h : A \to C$, called the **composite** (or **composition**) of f and g, is given by $h(x) = g(f(x))$ for all $x \in A$. We also write $h = g \circ f$. Composition of functions satisfies associativity, that is, $(f \circ g) \circ h = f \circ (g \circ h)$, where the domains and codomains are assumed such that all operations are well defined.

For a function f whose domain contains the codomain, the notation $f^n(x)$ will indicate[3] the n-fold composition of f with itself: $f^2 = f \circ f$, and $f^n = f \circ f^{n-1}$ for $n > 2$. We call f^n the n-th **iterate** of f. For numerical functions, this is to be distinguished from the notation for powers of f; apart from the case of trigonometric functions (see footnote), we write $f(x)^2$ or $(f(x))^2$, and not $f^2(x)$, to indicate the square of $f(x)$.

Inverse Function

A bijective function $f : A \to B$ provides a unique association between the elements of the domain and codomain. This allows us to define a closely related function $g : B \to A$, by saying that $x = g(y)$ if and only if $y = f(x)$. This is called the **inverse function** of f and is typically denoted by f^{-1} instead of g. Obviously f^{-1} is also bijective.

Theorem 1.19. *A bijective function f has a unique inverse function f^{-1}.*

Comparing the graphs of f and f^{-1},

$$\begin{aligned}\Gamma_f &= \{(x,y) : y = f(x),\ x \in A\} \quad \text{and}\\ \Gamma_{f^{-1}} &= \{(y,x) : x = f^{-1}(y),\ y \in B\}\\ &= \{(y,x) : y = f(x),\ x \in A\},\end{aligned}$$

we notice that $\Gamma_{f^{-1}}$ is obtained from Γ_f by reversing each ordered pair (x,y) in Γ_f, and conversely. Geometrically, the two graphs are the reflection of one another in the line $y = x$.

Exercise 1.20. *Let $f : A \to B$ and $g : B \to C$ be two bijective functions. Prove that* (a) $(f^{-1})^{-1} = f$; (b) $f^{-1} \circ f = \mathrm{id}_A$; (c) $f \circ f^{-1} = \mathrm{id}_B$; *and* (d) $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. □

Problem 1.21 (IMO 1973). *Let G be the set of functions $f : \mathbb{R} \to \mathbb{R}$ of the form $f(x) = a\,x + b$, where a and b are real numbers and $a \neq 0$. Suppose that G satisfies the following conditions:*

(a) *If $f, g \in G$ then $g \circ f \in G$.*

(b) *If $f \in G$ then $f^{-1} \in G$.*

(c) *For each $f \in G$ there exists a number $x_f \in \mathbb{R}$ such that $f(x_f) = x_f$.*

Prove that there exists a number $k \in \mathbb{R}$ such that $f(k) = k$ for all $f \in G$.

[3] I break this rule only for the trigonometric functions: $\sin^2 x$ stands for $(\sin x)^2$, not $\sin(\sin x)$. This tradition is so strong and widespread that I found it impossible to reverse it.

Solution. If $a = 1$ for a function $f(x) = x + b$, then condition (c) requires that $b = 0$. In this case, all points k of $\mathbb{R}$ satisfy $f(k) = k$. Therefore we need to show the claim for $a \neq 1$.

Let $a, a' \neq 1$ and let

$$f(x) = a\,x + b, \quad g(x) = a'\,x + b',$$

be two functions in G. Then, condition (c) requires that there are two points x_f and x_g (not necessarily distinct) such that

$$f(x_f) = x_f \;\Rightarrow\; x_f = -\frac{b}{a-1},$$

and

$$g(x_g) = x_g \;\Rightarrow\; x_g = -\frac{b'}{a'-1}.$$

According to condition (b), both

$$f^{-1} = \frac{1}{a}\,x - \frac{b}{a} \quad \text{and} \quad g^{-1} = \frac{1}{a'}\,x - \frac{b'}{a'}$$

are in G. Then, according to condition (a),

$$f \circ g(x) = aa'\,x + ab' + b,$$

and

$$f^{-1} \circ g^{-1}(x) = \frac{1}{aa'}\,x - \frac{b' + ba'}{aa'},$$

and

$$f \circ g \circ f^{-1} \circ g^{-1} = x + (ab' + b) - (b' + ba')$$

are also elements of G. Since there is an x_0 for this latter function such that $f \circ g \circ f^{-1} \circ g^{-1}(x_0) = x_0$, we conclude that

$$(ab' + b) - (b' + ba') = 0 \;\Rightarrow\; \frac{b}{1-a} = \frac{b'}{1-a'} \;\Rightarrow\; x_f = x_g. \qquad \square$$

If $B' \subset B$, we denote by $f^{-1}(B')$ the set of preimages of elements of B':

$$f^{-1}(B') = \{x \in A : f(x) = y \text{ for some } y \in B'\}.$$

Exercise 1.22. *Let A' be a subset of A and B', B'' be two subsets of B. Prove:*

$$\begin{aligned} f^{-1}(f(A')) &\supset A'; \\ f(f^{-1}(B')) &\subset B'; \\ f^{-1}(B' \cap B'') &= f^{-1}(B') \cap f^{-1}(B''); \\ f^{-1}(B' \cup B'') &= f^{-1}(B') \cup f^{-1}(B''). \end{aligned} \qquad \square$$

1.4 Limits and Continuity

Limits

Consider a function $f : A \to \mathbb{R}$ defined on a subset A of $\mathbb{R}$, and a point ξ in the closure $\overline{A}$ of A. We say that f has a **limit** $\ell \in \mathbb{R}$ at ξ, and we write $\lim_{x\to\xi} f(x) = \ell$, if the following condition is satisfied: *For any* $\varepsilon > 0$*, there exists* $\delta > 0$ *such that*

$$|f(x) - \ell| < \varepsilon \quad \textit{for all } x \in A \textit{ such that } 0 < |x - \xi| < \delta.$$

In other words, $f(x)$ gets arbitrarily close to ℓ as x approaches ξ. Note that ξ need not be in the domain of definition of f.

In the language of neighborhoods, f has a limit ℓ at ξ if and only if, given any neighborhood J of ℓ in $\mathbb{R}$, there exists a neighborhood I of ξ in $\mathbb{R}$ such that $f(A \cap I \setminus \{\xi\}) \subset J$.

This formulation immediately allows the notion of limit to be extended to functions from A into any topological space B, such as the Cartesian plane: $f : A \to B$ has a **limit** $\ell \in B$ at ξ if, given any neighborhood J of ℓ in B, there exists a neighborhood I of ξ in $\mathbb{R}$ such that $f(A \cap I \setminus \{\xi\}) \subset J$.

We say that f has a **right-hand limit** ℓ at ξ, written $\lim_{x\to\xi^+} f(x) = \ell$, if $f(x)$ gets arbitrarily close to ℓ as x approaches ξ *from the right*, regardless of what happens to the left of ξ. In terms of epsilons and deltas: *We write* $\lim_{x\to\xi^+} f(x) = \ell$ *if, for any* $\varepsilon > 0$*, there exists* $\delta > 0$ *such that*

$$|f(x) - \ell| < \varepsilon \quad \textit{for all } x \in A \textit{ such that } x \in (\xi, \xi+\delta). \tag{1.5}$$

Note that $\lim_{x\to\xi^+} f(x) = \ell$ if and only if $\lim_{x\to\xi} \bar{f}(x) = \ell$, where $\bar{f}$ is the restriction of f to the subset $A \cap [\xi, +\infty)$.

The **left-hand limit**, denoted by $\lim_{x\to\xi^-} f(x)$, is defined similarly, using the condition $x \in (\xi-\delta, \xi)$ in (1.5), or the restriction of f to the subset $A \cap (-\infty, \xi]$.

Exercise 1.23. *Let the domains of* f *and* g *contain some neighborhood of* $\xi \in \mathbb{R}$ *(possibly with* ξ *itself removed). Suppose we have* $\lim_{x\to\xi} f(x) = \ell_1$ *and* $\lim_{x\to\xi} g(x) = \ell_2$*. Prove:*

(a) $\lim_{x\to\xi}(\lambda f(x)) = \lambda\,\ell_1$*, for any* $\lambda \in \mathbb{R}$.

(b) $\lim_{x\to\xi}(f(x) + g(x)) = \ell_1 + \ell_2$.

(c) $\lim_{x\to\xi} f(x)\,g(x) = \ell_1\ell_2$.

(d) $\lim_{x\to\xi} f(x)/g(x) = \ell_1/\ell_2$ *provided that* $\ell_2 \neq 0$ *and that, for some* $\delta > 0$*, we have* $g(x) \neq 0$ *whenever* $0 < |x - \xi| < \delta$. □

Theorem 1.24. *Let $f(x)$ and $g(x)$ exist and satisfy $f(x) \leq g(x)$ for all x in a neighborhood of ξ in $\mathbb{R}$ (possibly with ξ itself removed). If the limits of f, g as $x \to \xi$ exist, then*

$$\lim_{x\to\xi} f(x) \leq \lim_{x\to\xi} g(x).$$

Theorem 1.25 (Squeeze Theorem)**.** *Let $f(x)$, $g(x)$ and $h(x)$ exist and satisfy $f(x) \leq g(x) \leq h(x)$ for all x in a neighborhood of ξ in $\mathbb{R}$ (possibly with ξ itself removed). If $\lim_{x\to\xi} f(x) = \ell$ and $\lim_{x\to\xi} h(x) = \ell$, then*

$$\lim_{x\to\xi} g(x) = \ell.$$

Recall that a **sequence** is a function with domain $\mathbb{N}$, but we use notation such as x_n rather than $x(n)$ for the value of the function at n, and the sequence itself may be written $\{x_n\}_{n\in\mathbb{N}}$ or $\{x_n\}$. More generally, the domain of a sequence can be $\mathbb{Z} \cap [n_0, \infty)$, for some $n_0 \in \mathbb{Z}$.

A sequence $\{x_n\}_{n\in\mathbb{N}}$ of real numbers is said to **converge** to a limit $\ell \in \mathbb{R}$, and we write $\lim_{n\to\infty} x_n = \ell$ or $x_n \to \ell$, if for any $\varepsilon > 0$ there exists n_0 such that $|x_n - \ell| < \varepsilon$ for all $n \geq n_0$.

Results analogous to Theorems 1.24 and 1.25 also apply to sequences.

Continuity

Let A be a subset of $\mathbb{R}$. A function $f : A \to B$ is called **continuous** at $a \in A$ if $\lim_{x\to a} f(x) = f(a)$. It is called **right-continuous** at a if $\lim_{x\to a^+} f(x) = f(a)$, and **left-continuous** at a if $\lim_{x\to a^-} f(x) = f(a)$. We say that f is continuous (or right-continuous, or left-continuous) on an open interval $I \subset \mathbb{R}$ if f has the same property at every $x \in I$.

Using the notion of open sets, we can extend the concept of continuity to functions between two topological spaces. A function $f : X \to Y$ is said to be **continuous** at $a \in X$ if, given any open set J containing $f(a)$, there exists an open set I containing a and such that $f(I) \subset J$. The function f is continuous on X if it is continuous at every point $x \in X$.

Theorem 1.26. *Let B be any topological space and let $A \subset \mathbb{R}$ contain the interval $(\xi-\epsilon, \xi+\epsilon)$ for some $\xi \in \mathbb{R}$ and $\epsilon > 0$.*

(a) *A function $f : A \to B$ is continuous at ξ if and only if*

$$\lim_{n\to\infty} f(x_n) = f(\xi)$$

for every sequence $\{x_n\}$ converging to ξ.

(b) *A function $f : A \to B$ is right-continuous at ξ if and only if*

$$\lim_{n\to\infty} f(x_n) = f(\xi)$$

for every sequence $\{x_n\}$ converging to ξ from the right (this condition means that $\{x_n\}$ converges to ξ and $x_n \geq \xi$ for every n). A similar statement holds for left continuity.

A function that is not continuous at a point ξ of its domain is called **discontinuous** at ξ. A discontinuity can occur for one of the following reasons:

(i) The right-hand and left-hand limits are equal but different from the value of the function (a **removable discontinuity**).

(ii) The right-hand and left-hand limits exist but they have different values (a **jump discontinuity**).

(iii) The right-hand limit or the left-hand limit does not exist (an **essential discontinuity**).

Example 1.27. The function of Problem 1.17 has jump discontinuities at all integer points. □

We also apply the terminology above when ξ is in the closure of the domain, but not in the domain itself. By definition, f cannot be continuous at ξ in this case, but we talk of removable, jump, and essential discontinuities depending on whether the left-hand and right-hand limits exist and agree, or exist and disagree, or don't both exist.

Example 1.28. The function $f(x) = 1/x^2$, defined on $\mathbb{R} \setminus \{0\}$, has an essential discontinuity at $x = 0$: since $f(x)$ is unbounded in any interval $(0, \epsilon)$, there can be no limit of $f(x)$ at 0. □

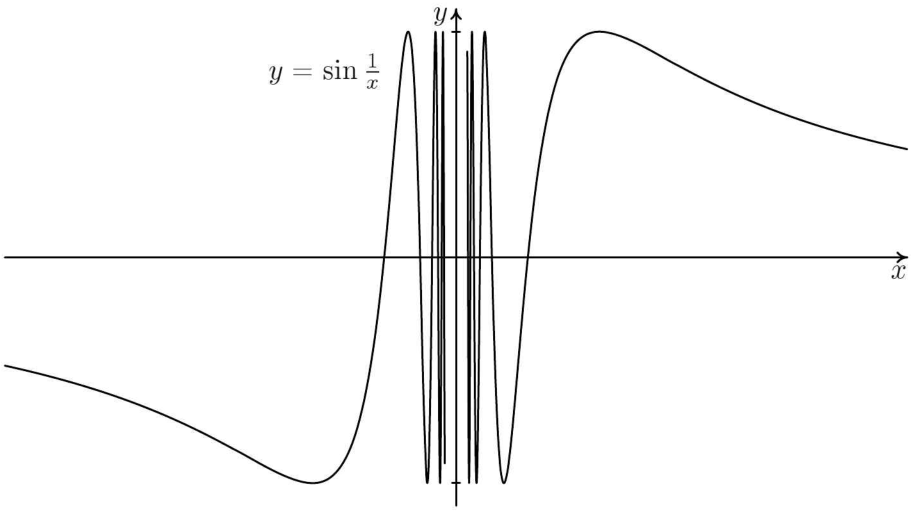
Example 1.29. The same is true of the function

$$f(x) = \sin\frac{1}{x},$$

again defined only for $x \neq 0$. We see from the graph above that $f(x)$ oscillates violently between -1 and 1 as x gets close to 0. To prove formally that f has no right-hand limit at 0, consider the sequences given by

$$a_n = \frac{2}{(4n+1)\pi} \quad \text{and} \quad b_n = \frac{2}{(4n-1)\pi}.$$

They both converge to 0 from the right, but $f(a_n) = \sin\left(2\pi n + \frac{\pi}{2}\right) = 1$ and $f(b_n) = \sin\left(2\pi n - \frac{\pi}{2}\right) = -1$, which would be in violation of Theorem 1.26(b) if there were a right-hand limit. □



Example 1.30. The function $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ given by

$$f(x) = x \sin \frac{1}{x}$$

has a discontinuity at $x = 0$, since this point does not belong to the domain. But the limit as $x \to 0$ exists and equals 0, as follows from Theorem 1.25, for instance. In other words, the right-hand and left-hand limits at 0 are both 0, and we are in case (i) of the previous page: a removable discontinuity.

The adjective "removable" is explained by the fact that, in this case, we can define (or redefine) the function at the point of discontinuity to obtain a continuous function. Indeed, let's extend f to a function $\bar{f}$ on all of $\mathbb{R}$, by setting $\bar{f}(x) = f(x)$ for $x \neq 0$, and $\bar{f}(0) = 0$. Then $\bar{f}$ is continuous at all points: we have removed the discontinuity. (In this situation we usually won't bother giving a different name to the extension: we will just say "set $f(0) = 0$", even though we're defining a new function distinct from f.) $\square$

The previous discussion may have created the impression that functions are mostly continuous and they may have only a small set of points of discontinuity. However, this is not correct. The Dirichlet function of Example 1.16 is discontinuous *everywhere*. The function

$$f(x) = \begin{cases} x & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}, \end{cases}$$

which is perhaps the simplest modification of the Dirichlet function, is continuous only at one point, $x = 0$.

Exercise 1.31. *Let $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ be two continuous functions at $x = \xi$. Prove:*

(a) *$f+g$ is continuous at $x=\xi$.*

(b) *fg is continuous at $x=\xi$.*

(c) *f/g is continuous at $x=\xi$ provided that $g(\xi)\neq 0$.*

(d) *$f\circ g$ is continuous at $x=\xi$.* □

The next theorem, saying essentially that the graph of a continuous function cannot change signs unless it crosses the x-axis, finds many applications, as does its generalization, the Intermediate Value Theorem.

Theorem 1.32 (Bolzano). *If a function f is continuous on the interval $[a,b]$ and $f(a)f(b)<0$, then there exists a point $\xi\in(a,b)$ such that $f(\xi)=0$.*

Theorem 1.33 (Intermediate Value Theorem). *If a function $f(x)$ is continuous on the interval $[a,b]$, it takes on every value between $f(a)$ and $f(b)$.*

See Problem 1.17 for an example of how the conclusion of these theorems can fail to hold when the function is discontinuous.

1.5 Differentiation

First Derivative

A function $f: A\to\mathbb{R}$, where A contains ξ in its interior, is **differentiable** at ξ if the limit

$$\lim_{x\to\xi}\frac{f(x)-f(\xi)}{x-\xi}=\lim_{h\to 0}\frac{f(h+\xi)-f(\xi)}{h}$$

exists. This limit is then called the **derivative** of f at $x=\xi$ and is denoted by $f'(\xi)$ or $(df(x)/dx)|_\xi$.

The function f is called differentiable on an open interval $I\subset\mathbb{R}$ if f is differentiable at every $x\in I$. Geometrically, the derivative $f'(\xi)$ gives the slope of the tangent line at the point $(\xi, f(\xi))$ of Γ_f.

Using right-hand and left-hand limits, we can extend the concept of differentiation to *differentiation from the right* or *from the left.*

Theorem 1.34. *A function that is differentiable at a point ξ is continuous at ξ.*

Differentiability of the function $f(x)$ at x is equivalent to the continuity of the function

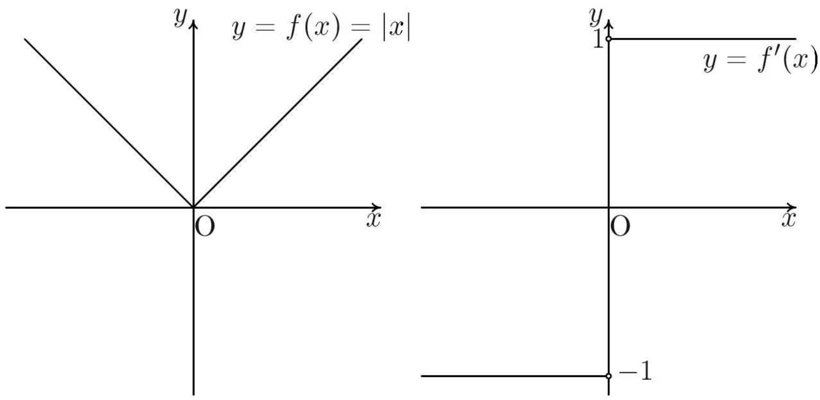
$$F(h)=\frac{f(h+x)-f(x)}{h}$$

at $h=0$, with the variable x playing the role of a parameter specifying the point of interest. Therefore any discontinuity at $h=0$, other than a removable one, signals problems.

Example 1.35. For the function $f(x) = |x|$, we have

$$F(h) = \frac{|x+h| - |x|}{h}.$$

For $x = 0$, this becomes $F(h) = |h|/h = \operatorname{sgn}(h)$, defined for $h \neq 0$. This is a discontinuous function with a jump discontinuity at $x = 0$. Therefore $f(x) = |x|$ does not have a derivative at $x = 0$. The discontinuity in $F(h)$ appears as a sharp bend in the graph of $f(x)$.



Note, however, that $|x|$ is differentiable from the right and from the left. □

Example 1.36. The function

$$f(x) = \begin{cases} x \sin \dfrac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

implies

$$F(h) = \sin \frac{1}{h}$$

when we set $x = 0$. This has an essential discontinuity at $h = 0$. Therefore $f(x)$ has no derivative at $x = 0$. In contrast, if we take

$$f(x) = \begin{cases} x^2 \sin \dfrac{1}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

we obtain

$$F(h) = h \sin \frac{1}{h},$$

which is continuous at $h = 0$; therefore $f'(0)$ exists and equals 0. □

Continuous functions may appear to fail to be differentiable only at a small set of points. This view was held by mathematicians until Weierstrass[4] explicitly showed it to be incorrect. We present Weierstrass's example after some additional results on the derivative.

Exercise 1.37. *Let f and g be differentiable at x and let c be a constant. Prove:*

(a) $(f+g)'(x) = f'(x) + g'(x)$.

(b) $(cf)'(x) = cf'(x)$.

(c) $(fg)'(x) = f(x)g'(x) + f'(x)g(x)$.

(d) $(f/g)'(x) = \dfrac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}$, *for* $g(x) \neq 0$.

(e) $(f(g(x))' = f'(g(x))\, g'(x)$. □

The following three theorems, each generalizing the previous one, are among the most useful results on differentiable functions.

Theorem 1.38 (Rolle). *If f is continuous on $[a,b]$, is differentiable on (a,b) and satisfies $f(a) = f(b) = 0$, there is a $\xi \in (a,b)$ such that $f'(\xi) = 0$.*

That is, the graph of such an f, restricted to (a,b), has a horizontal tangent. One could try to use the zero derivative to show the existence of a local maximum or minimum for f between a and b (see page 28 for the connection). But in fact it's enough for f to be *continuous* to guarantee the existence of an *absolute* maximum or minimum in (a,b); see Theorem 1.47.

Theorem 1.39 (Lagrange's Mean Value Theorem). *If f is continuous on $[a,b]$ and differentiable on (a,b), there is a $\xi \in (a,b)$ such that*

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

That is, the graph of such an f has the property that every secant has at least one parallel tangent (the tangency occurring between the intersections of the secant).

Theorem 1.40 (Cauchy's Mean Value Theorem). *If f, g are continuous on $[a,b]$, differentiable on (a,b), and $g(x) \neq 0$ for any $x \in (a,b)$, there is a $\xi \in (a,b)$ such that*

$$\frac{f'(\xi)}{g'(\xi)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

[4] Bolzano seems to have done so before Weierstrass but his work was seemingly ignored.

Problem 1.41 (Putnam 1939). *Let $f(x)$ be defined for $x \in [a,b]$. Assuming appropriate properties of continuity and differentiability, prove for $x \in (a,b)$ that*

$$\frac{\frac{f(x)-f(a)}{x-a} - \frac{f(b)-f(a)}{b-a}}{x-b} = \frac{1}{2} f''(\xi),$$

where ξ is some number in (a,b).

Solution ([21]). Consider the function

$$F(y) = \begin{vmatrix} f(y) & y^2 & y & 1 \\ f(x) & x^2 & x & 1 \\ f(a) & a^2 & a & 1 \\ f(b) & b^2 & b & 1 \end{vmatrix},$$

where x is considered fixed and $y \in [a,b]$. Then $F(y)$ is continuous on $[a,b]$ and differentiable on (a,b). Moreover, $F(a) = F(x) = 0$ (since two rows of the determinant are equal to each other). Therefore the function $F(y)$ satisfies the conditions of Rolle's theorem on the interval $[a,x]$ and there exists a point $\alpha \in (a,x)$ such that $F'(\alpha) = 0$. Similarly, there exists a point $\beta \in (x,b)$ such that $F'(\beta) = 0$.

But, from the previous results, the function

$$F'(y) = \begin{vmatrix} f'(y) & 2y & 1 & 0 \\ f(x) & x^2 & x & 1 \\ f(a) & a^2 & a & 1 \\ f(b) & b^2 & b & 1 \end{vmatrix}$$

also satisfies the conditions of Rolle's theorem on the interval $[\alpha, \beta]$. Therefore, there exists $\xi \in (\alpha, \beta)$ such that $F''(\xi) = 0$, where

$$F''(y) = \begin{vmatrix} f''(y) & 2 & 0 & 0 \\ f(x) & x^2 & x & 1 \\ f(a) & a^2 & a & 1 \\ f(b) & b^2 & b & 1 \end{vmatrix}.$$

Upon expanding the determinant, we arrive at the result sought. □

Example 1.42 (Weierstrass's nowhere differentiable function). Let

$$f(x) = \sum_{k=0}^{\infty} a^k \cos(b^k \pi x),$$

where $0 < a < 1$ and b is an odd positive integer. I leave the proof of continuity to the reader (who may simply accept it), and show only that f is nowhere differentiable, if ab is suitably large.

Let $F(h) = \big(f(x+h) - f(x)\big)/h$, as usual. Write $F = A_n + R_n$, where

$$A_n(h) = \sum_{k=0}^{n-1} a^k \, \frac{\cos[b^k\pi(x+h)] - \cos(b^k\pi x)}{h}$$

is the sum of the first n terms of the sum defining F, and

$$R_n = \sum_{k=n}^{\infty} a^k \, \frac{\cos[b^k\pi(x+h)] - \cos(b^k\pi x)}{h} \tag{1.6}$$

is the remainder. By the triangle inequality,

$$|A_n(h)| \le \sum_{k=0}^{n-1} a^k \left| \frac{\cos[b^k\pi(x+h)] - \cos(b^k\pi x)}{h} \right|.$$

For each k, we apply Lagrange's mean value theorem to the function $\cos(b^k\pi x)$ on $[x, x+h]$, to find $\xi \in (x, x+h)$ such that

$$\frac{\cos[b^k\pi(x+h)] - \cos(b^k\pi x)}{h} = b^k\pi \, \sin(b^k\pi\xi).$$

It follows that

$$\left| \frac{\cos[b^k\pi(x+h)] - \cos(b^k\pi x)}{h} \right| = b^k\pi \, |\sin(b^k\pi\xi)| \le b^k\pi;$$

thus

$$|A_n(h)| \le \pi \sum_{k=0}^{n-1} a^k b^k = \pi \, \frac{(ab)^n - 1}{ab - 1} < \pi \, \frac{(ab)^n}{ab - 1},$$

if we take $ab > 1$.

Next we turn to the remainder term R_n. Set $b^n x = N_n + f_n$, where N_n is an integer part and f_n a fractional part, the latter defined here in the interval $[-1/2, 1/2]$. If $h = (1 - f_n)/b^n$, then $\frac{2}{3} b^n \le 1/h \le 2b^n$. Also, for any $k \ge n$,

$$\cos[b^k\pi(x+h)] = (-1)^{N_n+1} \quad \text{and} \quad \cos(b^k\pi x) = (-1)^{N_n+1} \cos(b^{k-n}\pi f_n).$$

Therefore, for such h, all summands in (1.6) are of the same sign, and we have

$$\begin{aligned} |R_n(h)| &= \left| \sum_{k=n}^{\infty} a^k \, \frac{1 - \cos(b^{k-n}\pi f_n)}{h} \right| \\ &> a^n \, \frac{1 - \cos(\pi f_n)}{h} \ge \tfrac{2}{3}(ab)^n \, [1 - \cos(\pi f_n)] > \tfrac{2}{3}(ab)^n. \end{aligned}$$

Since $F = A_n + R_n$, we have, for some $h \leq \frac{3}{2} b^{-n}$,

$$|F(h)| \geq |R_n| - |A_n| > (ab)^n \frac{2ab - (2 + 3\pi)}{3(ab - 1)}.$$

If $2ab > 2 + 3\pi$, the right-hand side grows without bound as $n \to \infty$, while h tends to 0. Therefore, F cannot be continuous, implying that f cannot be differentiable. Since x has been left unspecified, the result holds for any x in the domain of f. □

Higher Derivatives

The derivative f' of a function f may or may not be differentiable. If it is, we denote its derivative by f'', and so on. The n-th derivative of f is denoted by $f^{(n)}$, or by $d^n f / dx^n$.

Theorem 1.43 (Leibniz). *If f, g are n-times differentiable, then*

$$\frac{d^n}{dx^n}[f(x)g(x)] = \sum_{k=0}^{n} \binom{n}{k} f^{(k)}(x)\, g^{(n-k)}(x).$$

Theorem 1.44 (Taylor). *If f is $(n+1)$-times differentiable on $[a, x]$, with the first n derivatives continuous, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} (x-a)^k + R_n,$$

where R_n, known as the ***remainder of order*** *n, is equal to*

$$R_n = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1} \quad \textit{for some } \xi \in (a, x).$$

If the derivative $f^{(n)}$ exists for any $n \geq 1$, the function is called **smooth**. For a smooth function, one can construct the **Taylor series**

$$\sum_{n=0}^{\infty} \frac{1}{n!} f^{(n)}(a)\,(x-a)^n,$$

which does not necessarily converge. If it converges to $f(x)$ on some neighborhood I of a with radius r, we say that f is **analytic** at the point a. If $f(x)$ is analytic for all points of I, we say that f is analytic on I.

Problem 1.45 (Putnam 1998). *Let f be a real function on the real line with continuous third derivative. Prove that there exists a point a such that*

$$f(a) \cdot f'(a) \cdot f''(a) \cdot f'''(a) \geq 0.$$

Solution ([27]). If any of the $f(a), f'(a), f''(a), f'''(a)$ is zero, then the statement is true. We thus have to prove it in the case when $f(x)$,$f'(x)$,$f''(x)$,$f'''(x)$ are nonzero. In such a case, by continuity, all $f(x), f'(x), f''(x), f'''(x)$ are strictly positive or strictly negative.

Without loss of generality, we assume that $f'''(x) > 0$ and $f''(x) > 0$—otherwise we make the changes of variable $x \mapsto -x$ and $f \mapsto -f$, respectively, which reverse the signs of these derivatives without changing the sign of the product $f(x)f'(x)f''(x)f'''(x)$.

Since $f'''(x) > 0$, the function f'' is strictly increasing. Then

$$f'(x) - f'(0) = \int_0^x f''(t)dt \geq \int_0^x f''(0)dt = xf''(0) \ ,$$

or $f'(x) \geq f'(0) + xf''(0)$. Similarly, since $f''(x) > 0$, the function f' is strictly increasing. Then

$$f(x) - f(0) = \int_0^x f'(t)dt \geq \int_0^x f'(0)dt = xf'(0) \ ,$$

or $f(x) \geq f(0) + xf'(0)$. We can now choose a large enough number $x = a$ such that $f'(a) > 0$ and $f(a) > 0$. $\square$

Comment. As the authors of [27] observe, a function cannot be positive and concave everywhere nor negative and convex everywhere. Therefore, there is an interval for which

$$f(x)\, f''(x) > 0.$$

Assuming differentiability of f as many times as necessary and, by the above reasoning, substituting f successively by f', f'', $\ldots$, $f^{(n)}$, we have

$$\begin{aligned} f'(x)\, f'''(x) &> 0, \\ f''(x)\, f^{(4)}(x) &> 0, \\ f'''(x)\, f^{(5)}(x) &> 0, \\ &\ldots, \\ f^{(n)}(x)\, f^{(n+2)}(x) &> 0. \end{aligned}$$

Can you now say if there is a point a such that

$$f(a)f'(a)f^{(n)}(a)f^{(n+1)}(a) > 0, \quad \text{for } n \geq 3?$$

Applications of Derivatives

Definition 1.46. Let $f : D \to \mathbb{R}$ be a function, where D is a subset of $\mathbb{R}$, and let $x_0 \in D$. We say that f has

- an **absolute maximum** at x_0 if $f(x) \le f(x_0)$ for all $x \in D$;
- a **local maximum** at x_0 if $f(x) \le f(x_0)$ for all x in some neighborhood of x_0 in D;
- an **absolute minimum** at x_0 if $f(x) \ge f(x_0)$ for all $x \in D$;
- a **local minimum** at x_0 if $f(x) \ge f(x_0)$ for all x in some neighborhood of x_0 in D.

An **extremum** is either a minimum or a maximum (qualified as local or absolute as the case may be).

Theorem 1.47 (Extreme Value Theorem). *If f is continuous on $[a,b]$, then f attains an absolute maximum and an absolute minimum somewhere in $[a,b]$. If, in addition, $f(a) = f(b)$, then f attains an absolute maximum or an absolute minimum somewhere in (a,b).*

The second statement follows easily from the first: either the maximum and the minimum in $[a,b]$ are equal — and then the function is constant and any point in (a,b) attains the maximum — or the maximum (or the minimum) is distinct from $f(a) = f(b)$, and so must be achieved in (a,b).

Problem 1.48 (Bulgaria 2000). *Let*

$$f(x) = \frac{x^2+4x+3}{x^2+7x+14}, \quad g(x) = \frac{x^2-5x+10}{x^2+5x+20}.$$

(a) *Find the maximum value of $f(x)$.*

(b) *Find the maximum value of $g(x)$.*

(c) *Find the maximum value of $g(x)^{f(x)}$.*

Solution. The quadratic polynomials $x^2+7x+14$, $x^2-5x+10$, $x^2+5x+20$ have negative discriminants and therefore they are positive for all values of x. Thus $g(x) > 0$ and the function $g(x)^{f(x)}$ is well defined.

The quadratic polynomial x^2+4x+3 has roots -1 and -3. Therefore $f(x)$ is positive for $x \in (-\infty,-3) \cup (-1,+\infty)$ and negative for $x \in (-3,-1)$.

Finally,

$$g(x) = \frac{x^2-5x+10}{x^2+5x+20} = \frac{(x^2+5x+20)-10(x+1)}{x^2+5x+20} = 1 - 10\,\frac{x+1}{x^2+5x+20}.$$

That is, $g(x) \le 1$ for $x \ge -1$ and $g(x) \ge 1$ for $x \le -1$.

(a) The maximum value of $f(x)$ is 2. Indeed,

$$\begin{aligned} f(x) \le 2 \quad &\Leftrightarrow \quad \frac{x^2+4x+3}{x^2+7x+14} \le 2 \\ &\Leftrightarrow \quad x^2+4x+3 \le 2x^2+14x+28 \\ &\Leftrightarrow \quad 0 \le x^2+10x+25 = (x+5)^2. \end{aligned}$$

The maximum value is attained for $x = -5$.

(b) As in part (a),

$$g(x) \leq 3 \;\Leftrightarrow\; 0 \leq (x+5)^2.$$

The function $g(x)$ attains a maximum value of 3 for $x = -5$.

(c) For $x \in (-\infty, -3]$, since $g(x) > 0$, it follows that $\ln g(x) \leq \ln 3$. Also $f(x) \leq 2$. Then

$$f(x) \ln g(x) \;\leq\; 2\ln 3, \quad \text{so} \quad g(x)^{f(x)} \leq 9.$$

For $x \in [-3, -1]$, we have $g(x) \geq 1$ and $f(x) \leq 0$. Then

$$g(x)^{|f(x)|} \geq 1, \quad \text{so} \quad g(x)^{f(x)} \leq 1.$$

For $x \in [-1, +\infty]$, we have $0 < g(x) \leq 1$ and $f(x) \geq 0$. Then

$$g(x)^{f(x)} \leq 1.$$

So, the maximum value of $g(x)^{f(x)}$ is 9, attained at $x = -5$. The reader should notice that we could have simply said that, since the maxima of f and g both occur at $x = -5$, then the maximum of g^f is equal to $(\max g)^{(\max f)} = 3^2 = 9$. $\square$

This problem demonstrates that quite some work is necessary to find extrema of a function by algebraic methods. In addition, having numbers that always conspire to simplify the calculations is impossible. The study of extrema is systematized by the derivatives of a function. We shall call **critical points** of a function f those points c at which $f'(c) = 0$ or f' does not exist.

Theorem 1.49. *Extreme values of a function can only occur at critical points and endpoints.*

However, a critical point may not necessarily be a point of extreme value. The next two theorems provide sufficient criteria for a critical point to lead to an extremum.

Theorem 1.50 (First Derivative Test). *Let f be differentiable in a neighborhood of a critical point c in the interior of the domain.*

If the derivative f' changes sign as it crosses c, then f has a local extremum at c: a local maximum if $f' > 0$ for $x < c$ and $f' < 0$ for $x > c$, and a local minimum in the opposite case.

Corollary. *Let $f : A \to \mathbb{R}$ be defined on an* ***interval*** *$A \subset \mathbb{R}$ and differentiable in the interior of A. Then:*

(a) *f is strictly increasing if and only if $f' > 0$.*

(b) *f is nondecreasing if and only if $f' \geq 0$.*

(c) *f is nonincreasing if and only if $f' \leq 0$.*

(d) *f is strictly decreasing if and only if $f' < 0$.*

In each case, the inequality is supposed to hold at all points in the domain.

Theorem 1.51 (Second Derivative Test). *Let f be defined and differentiable in a neighborhood of a critical point c. Then f has a local minimum at c if $f''(c) > 0$. It has a local maximum if $f''(c) < 0$.*

Exercise 1.52. *If $f''(x_0) = f'''(x_0) = \cdots = f^{(2n)}(x_0) = 0$, but $f^{(2n+1)}(x_0) \neq 0$, discuss the behavior of f in the neighborhood of x_0. The point x_0 is called a* ***point of inflection.*** □

1.6 Solved Problems

We now present some solved problems on functions, selected for their relevance to later chapters.

Problem 1.53 (Singapore 2002). *Let $f(x)$ be a function which satisfies*

$$f(29 + x) = f(29 - x) \quad \textit{for all } x \in \mathbb{R}.$$

If $f(x)$ has exactly three real roots α, β, γ, determine the value of $\alpha + \beta + \gamma$.

Solution. Consider the function F given by $F(x) = f(29 + x)$. (If we translate the graph of f to the left by 29 units, we obtain the graph of F; alternatively, we can think of translating the coordinate axes to the right).

Clearly, the functional equation satisfied by the function becomes

$$F(x) = F(-x);$$

that is, F is an even function. Therefore its roots must be located symmetrically with respect to the origin. Since there are three roots, one must be 0, and the others must be x_0 and $-x_0$ for some x_0. Returning to the original function f, its roots are obtained from those of F by adding 29, since $0 = F(x)$ means $0 = f(29 + x)$. Thus the roots of f are 29, $x_0 + 29$ and $-x_0 + 29$, and their sum is 87. □

Problem 1.54 ([14], Problem 7). *Let* $f_0(x) = \frac{1}{1-x}$, *and* $f_n(x) = f_0(f_{n-1}(x))$, $n = 1, 2, 3, \ldots$. *Evaluate* $f_{1976}(1976)$.

Solution. We notice that

$$\begin{aligned} f_0(x) &= \frac{1}{1-x}, \\ f_1(x) &= f_0(f_0(x)) = \frac{1-x}{-x}, \\ f_2(x) &= f_0(f_1(x)) = x, \\ f_3(x) &= f_0(f_2(x)) = \frac{1}{1-x} = f_0(x). \end{aligned}$$

From these results we conclude inductively that

$$f_{3k+r}(x) = f_r(x) \quad \text{for } k = 0, 1, 2, \ldots \text{ and } r = 0, 1, 2.$$

Therefore

$$f_{1976}(x) = f_2(x).$$

In particular, $f_{1976}(1976) = f_2(1976) = 1976$. □

Problem 1.55. *Let* $f(x) = x^2 - 2$ *with* $x \in [-2, 2]$. *Show that the equation*

$$f^n(x) = x$$

has 2^n *real roots.*

Solution. Since $x \in [-2, 2]$ we set $x = 2\cos\theta$, with $0 \le \theta \le 2\pi$. Then

$$\begin{aligned} f(\cos\theta) &= 2[2\cos^2\theta - 1] = 2\cos(2\theta), \\ f(f(\cos\theta)) &= [2\cos(2\theta)]^2 - 2 = 2[2\cos^2(4\theta) - 1] = 2\cos(4\theta), \end{aligned}$$

and so on. By induction, we easily verify that

$$f^n(\cos\theta) = 2\cos(2^n\theta).$$

The given equation, in the new notation, becomes

$$2\cos(2^n\theta) = 2\cos\theta,$$

with solutions $2^n\theta = 2k\pi \pm \theta$, $k \in \mathbb{Z}$ or

$$\theta_k^- = k\,\frac{2\pi}{2^n - 1} \quad \text{and} \quad \theta_k^+ = k\,\frac{2\pi}{2^n + 1} \quad \text{for } k \in \mathbb{Z}.$$

The distinct solutions are those for which $0 \leq \theta < 2\pi$. Therefore,

$$\begin{aligned} \theta_k^- &= k\,\frac{2\pi}{2^n - 1} \quad \text{for } k = 0, 1, \ldots, 2^{n-1} - 1, \\ \theta_k^+ &= k\,\frac{2\pi}{2^n + 1} \quad \text{for } k = 1, \ldots, 2^{n-1}. \end{aligned}$$

Counting, these are exactly 2^n in number. □

This problem actually appeared in an International Mathematical Olympiad:

Problem 1.56 (IMO 1976). *Let $P_1(x) = x^2 - 2$ and $P_j(x) = P_1(P_{j-1}(x))$ for $j = 2, 3, \ldots$. Show that for any positive integer n, the roots of the equation $P_n(x) = x$ are real and distinct.*

In this statement the domain is not specified to be $[-2, 2]$. The problem is actually taken from the theory of *orthogonal polynomials.* The functions $T_N(\cos\theta) = \cos(N\theta)$ are known as **Chebyshev polynomials** (of the first kind). When written in terms of the variable $x = \cos\theta$, they are indeed polynomials. The function $f^n(x)$ of the problem is just $2\,T_{2^n}(x)$.

Here is a related problem that was proposed for the IMO but did not make the cut:

Problem 1.57 (IMO 1978 Longlist). *Given the expression*

$$P_n(x) = \frac{1}{2^n}\left[\left(x + \sqrt{x^2 - 1}\right)^n + \left(x - \sqrt{x^2 - 1}\right)^n\right],$$

prove that $P_n(x)$ satisfies the identity

$$P_n(x) - x\,P_{n-1}(x) + \tfrac{1}{4}\,P_{n-2}(x) = 0,$$

and that $P_n(x)$ is a polynomial in x of degree n.

The Chebyshev polynomials $T_n(x)$ satisfy the recursion relation

$$T_n(x) - 2xT_{n-1}(x) + T_{n-2}(x) = 0,$$

and are given by the expression

$$T_n(x) = \tfrac{1}{2}\left[(x + \sqrt{x^2 - 1})^n + (x - \sqrt{x^2 - 1})^n\right].$$

Thus the problem just involves a multiplicative rewriting of the Chebyshev polynomials:

$$P_n(x) = 2^{n-1}\,T_n(x).$$

You can try to solve it anyway without this information.

Problem 1.56, in the way stated, is not immediately related to the Chebyshev polynomials, and it takes some time — even for a more experienced person — to make the connection.[5] However, Problem 1.57 is a standard exercise from college textbooks. Here is another variant:

Problem 1.58 (Sweden 1996). *For all integers $n \geq 1$ the functions p_n are defined for $x \geq 1$ by*

$$p_n(x) = \tfrac{1}{2}\big[\big(x+\sqrt{x^2-1}\big)^n + \big(x-\sqrt{x^2-1}\big)^n\big].$$

Show that $p_n(x) \geq 1$ and that $p_{mn}(x) = p_m(p_n(x))$.

Again, you can try to solve it independently of what is presented here.

The next problem and its solution share several common ideas with the solution of the previous problem.

Problem 1.59 (Turkey 1998). *Let $\{a_n\}$ be the sequence of real numbers defined by $a_1 = t$ and*

$$a_{n+1} = 4a_n\,(1-a_n) \quad \textit{for } n \geq 1.$$

For how many distinct values of t do we have $a_{1998} = 0$?

Solution. We define the function $f(x) = 4x(1-x)$. Then the given sequence becomes[6] $a_1 = t$, $a_2 = f(t)$, $a_3 = f^2(t)$, and so on. Therefore, the problem asks us to find the distinct roots of the equation $f^{1997}(t) = 0$.

The image $f(x)$ of x will be in $[0,1]$ if $x \in [0,1]$. To have any roots, t must be in $[0,1]$. In such a case we can set $t = \sin^2\theta$, with $\theta \in [0,\pi/2]$. Then

$$f(t) = f(\sin^2\theta) = 4\sin^2\theta(1-\sin^2\theta) = (2\sin\theta\,\cos\theta)^2 = \sin^2(2\theta).$$

Inductively,

$$\begin{aligned} f^2(t) &= f(f(t)) = f(\sin^2(2\theta)) = \sin^2(4\theta),\\ \dots &\quad \dots \quad \dots\\ f^n(t) &= \sin^2(2^n\theta). \end{aligned}$$

The roots of $f^n(t) = 0$ are, then, those satisfying $2^n\theta = k\pi$, $k \in \mathbb{Z}$ or, more precisely,

$$\theta = \frac{k\pi}{2^n} \quad \text{for } k = 0,1,2,\dots,2^{n-1}.$$

For $n = 1997$, this gives $2^{1996}+1$ distinct values of t. □

[5] At least that was the situation in 1976. After this, the theory of iterations became increasingly known and popular and the IMO 1976 problem also became another textbook problem. We discuss the topic of iterations in Chapter 16.

[6] For more details on this idea, see Chapter 16.

The function $f(x) = \lambda x(1-x)$ is known as the **logistic function** and plays an important role in the topic of *chaos*. (See Section 16.7.) Sometimes it is also called the **population growth model**. This name is motivated by the following interpretation. Let $n = 1, 2, \ldots$ count the generations of a species and let p_n be the population of the species at the n-th generation. If the population enjoys an unlimited food supply and habitat, it will grow according to the law $p_{n+1} = Ap_n$ (a geometric progression). However, as the population grows, stress develops if resources are limited. As a result, a fraction of the population dies. This "removed" population is given by $-Bp_n^2$. Therefore, the population of each generation follows the law $p_{n+1} = Ap_n - Bp_n^2$. If we define $a_n = \lambda p_n$ and $\lambda = A/B$, then the last equation is written equivalently as $a_{n+1} = \lambda a_n(1-a_n)$.

Here is an extension of Bolzano's theorem to some discontinuous functions:

Problem 1.60 ([1], Problem E1336, November 1958). *For the function $f : [0,1] \to \mathbb{R}$, $f(0) > 0$, $f(1) < 0$ and there exists a continuous function g such that $h = f + g$ is increasing. Prove that there exists a $\xi \in (0,1)$ such that $f(\xi) = 0$.*

Solution ([1], May 1959). Set $E = \{x : f(x) \geq 0\}$. This set is nonempty since $0 \in E$. It is also bounded since at most it can be $[0,1)$. Let $s \leq 1$ be its supremum. Since h is increasing, we have, for any $x \in E$ and $s \geq x$,

$$h(s) \geq h(x) = f(x) + g(x) \geq g(x).$$

By taking the limit $x \to s$ of this inequality, we get $h(s) \geq g(s)$, since g is continuous; this, in turn, implies $f(s) \geq 0$.

Again from the increasing property of h, we have $h(s) \leq h(1)$ with $h(1) = g(1) + f(1) < g(1)$ and $h(s) = g(s) + f(s) \geq g(s)$. Therefore $g(1) > h(1) \geq h(s) \geq g(s)$. Since g is continuous, it takes all values between $g(1)$ and $g(s)$. In particular it takes the value $h(s)$. That is, there exists a $\xi \geq s$ such that $g(\xi) = h(s)$.

Now we observe that

$$h(\xi) \geq h(s) \;\Rightarrow\; h(\xi) \geq g(\xi) \;\Rightarrow\; f(\xi) \geq 0.$$

In other words, $\xi \in E$. However, by the definition of s, t cannot be greater than s; thus $\xi = s$. Consequently, the equation $g(\xi) = h(s)$ gives $f(\xi) = 0$. $\square$

Problem 1.61 (IMO 1968). *Let f be a real-valued function defined for all real numbers x such that, for some positive a, the equation*

$$f(x+a) = \tfrac{1}{2} + \sqrt{f(x) - f(x)^2} \tag{1.7}$$

holds for all x.

(a) *Prove that the function $f(x)$ is periodic.*

(b) *For $a = 1$ give an example of a non-constant function with the required properties.*

Solution. (a) Setting $x = y + a$ in equation (1.7), we have

$$\begin{aligned}
f(y+2a) &= \tfrac{1}{2} + \sqrt{f(y+a) - f(y+a)^2} \\
&= \tfrac{1}{2} + \sqrt{\tfrac{1}{2} + \sqrt{f(y) - f(y)^2} - \left(\tfrac{1}{2} + \sqrt{f(y) - f(y)^2}\right)^2} \\
&= \tfrac{1}{2} + \sqrt{\tfrac{1}{2} + \sqrt{f(y) - f(y)^2} - \tfrac{1}{4} - f(y) + f(y)^2 - \sqrt{f(y) - f(y)^2}} \\
&= \tfrac{1}{2} + \sqrt{\tfrac{1}{4} - f(y) + f(y)^2} \\
&= \tfrac{1}{2} + \sqrt{\left(\tfrac{1}{2} - f(y)\right)^2} \\
&= \tfrac{1}{2} + \left|\tfrac{1}{2} - f(y)\right|.
\end{aligned}$$

From this equation (or the given one) we see that $f(x) \geq \frac{1}{2}$. Therefore, the absolute value on the last line is equal to $f(y) - \frac{1}{2}$ and thus

$$f(y+2a) = f(y).$$

The function $f(x)$ is periodic with period $2a$.

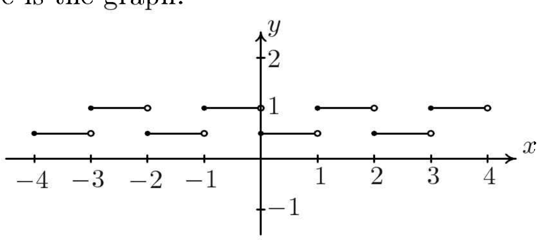
(b) The simplest nonconstant function is one taking two values. So, let

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } x \in [0,1), \\ 1 & \text{if } x \in [1,2), \end{cases}$$

with the rest of the values determined by periodicity:

$$f(x) = \begin{cases} \frac{1}{2} & \text{if } x \in [2n,\, 2n+1), \\ 1 & \text{if } x \in [2n+1,\, 2n+2), \end{cases}$$

for all $n \in \mathbb{Z}$. Here is the graph:

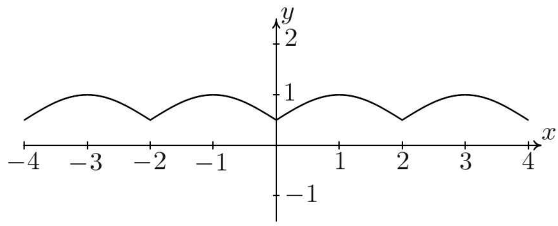


One might dislike this function being discontinuous. If a continuous function is sought, one can use

$$f(x) = \frac{1}{2} + \frac{1}{2}\sin\left(\frac{\pi x}{2}\right) \quad \text{for } x \in [0,2),$$

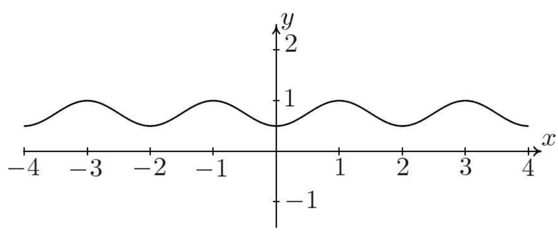
with the rest of the values determined by periodicity. The result can be written nicely in the form

$$f(x) = \frac{1}{2} + \frac{1}{2}\left|\sin\left(\frac{\pi x}{2}\right)\right| \quad \text{for } x \in \mathbb{R}.$$



Still, the function is not differentiable at the points $x = 2n$, $n \in \mathbb{Z}$. We can give another example that is differentiable at these points too:

$$f(x) = \frac{1}{2} + \frac{1}{2}\sin^2\left(\frac{\pi x}{2}\right) \quad \text{for } x \in \mathbb{R}.$$



□

Problem 1.62 ([1], Problem 11233, June/July 2006). *Show that for positive integer n and for $x \neq 0$,*

$$\frac{d^n}{dx^n}\left(x^{n-1}\sin\frac{1}{x}\right) = \frac{(-1)^n}{x^{n+1}}\sin\left(\frac{1}{x} + \frac{n\pi}{2}\right).$$

Solution. We can prove this easily by induction. The identity is true for $n = 1$:

$$\begin{aligned}\frac{d}{dx}\sin\frac{1}{x} &= -\frac{1}{x^2}\cos\frac{1}{x}\\ &= -\frac{1}{x^2}\sin\left(\frac{1}{x} + \frac{\pi}{2}\right).\end{aligned}$$

Let it be true for some $n = k$:

$$\frac{d^k}{dx^k}\left(x^{k-1}\sin\frac{1}{x}\right) = \frac{(-1)^k}{x^{k+1}}\sin\left(\frac{1}{x} + \frac{k\pi}{2}\right).$$

Using this and Theorem 1.43, we find

$$\begin{aligned}
\frac{d^{k+1}}{dx^{k+1}}\left(x^k \sin\frac{1}{x}\right) &= \frac{d^{k+1}}{dx^{k+1}}\left(x\, x^{k-1}\sin\frac{1}{x}\right)\\
&= (k+1)\,\frac{d^k}{dx^k}\left(x^{k-1}\sin\frac{1}{x}\right)+x\,\frac{d^{k+1}}{dx^{k+1}}\left(x^{k-1}\sin\frac{1}{x}\right)\\
&= (k+1)\,\frac{(-1)^k}{x^{k+1}}\sin\left(\frac{1}{x}+\frac{k\pi}{2}\right)+x\,\frac{d}{dx}\frac{(-1)^k}{x^{k+1}}\sin\left(\frac{1}{x}+\frac{k\pi}{2}\right).
\end{aligned}$$

The derivative of the product (second summand on the right-hand side) gives two terms, one of which exactly cancels the remaining term. So, finally

$$\begin{aligned}
\frac{d^{k+1}}{dx^{k+1}}\left(x^k \sin\frac{1}{x}\right) &= \frac{(-1)^{k+1}}{x^{k+2}}\cos\left(\frac{1}{x}+\frac{k\pi}{2}\right)\\
&= \frac{(-1)^{k+1}}{x^{k+2}}\sin\left(\frac{1}{x}+\frac{(k+1)\pi}{2}\right).
\end{aligned}$$

Therefore the identity is true for $n = k+1$ and consequently for all $n \in \mathbb{N}^*$. □

By the same technique, you may generalize the previous result to the following.

Problem 1.63. *If f is an n-times differentiable function, then*

$$\frac{d^n}{dx^n}\left[x^{n-1} f\left(\frac{1}{x}\right)\right] = \frac{(-1)^n}{x^{n+1}} f^{(n)}\left(\frac{1}{x}\right).$$

If this is too straightforward, a more challenging problem would be:

Problem 1.64. *If f is an n-times differentiable function, then find*

$$\frac{d^n}{dx^n}\left[x^m f\left(\frac{1}{x}\right)\right].$$

If you fail, the answer (but not the proof) is found in the January 2008 issue of [1].

Problem 1.65 ([1], Problem E3214, June/July 1987). *Let f be a real function with $n+1$ derivatives on $[a,b]$. Suppose $f^{(k)}(a) = f^{(k)}(b) = 0$ for $k = 0, 1, \ldots, n$. Prove that there is a number $\xi \in (a,b)$ such that $f^{(n+1)}(\xi) = f(\xi)$.*

Following the solution given by R. Brooks ([1], vol. **96**, p. 740), we split the solution into two parts: a simple lemma (the case of $n = 0$) and the proof for a general n.

Lemma 1.66. *Let f be a differentiable function on $[a, b]$. Suppose $f(a) = f(b) = 0$. Prove that there is a number $\xi \in (a, b)$ such that $f'(\xi) = f(\xi)$.*

Proof. Consider the function

$$g(x) = e^{-x} f(x),$$

which satisfies the conditions of Rolle's theorem on $[a, b]$. Then there exists a $\xi \in (a, b)$ such that $g'(\xi) = 0$ or

$$e^{-\xi} \left(f(\xi) - f'(\xi)\right) = 0.$$

From this, it is obvious that $f'(\xi) = f(\xi)$. $\square$

Using the lemma above we can now prove the statement in the general case.

Solution. Consider the function

$$g(x) = \sum_{k=0}^{n} f^{(k)}(x),$$

which satisfies the conditions of the previous lemma. Then there is a $\xi \in (a, b)$ such that $g'(\xi) = g(\xi)$, which is easily seen to give

$$f^{(n+1)}(\xi) = f(\xi). \qquad \square$$

L. M. Levine has shown that the result holds true without requiring the vanishing of the derivatives at $x = b$. (See [1], vol. **96**, p. 740.)

Part II

BASIC EQUATIONS

Chapter 2

A Primer on Functional Relations

2.1 Fundamental Notions

Toward a Definition of Functional Equations

Functional equations are encountered even in elementary mathematics. Some of the most common definitions of properties of functions are given using functional equations. Consider, for example, the definition of an **even function**,

$$f(x) = f(-x) \quad \text{for all } x,$$

the definition of an **odd function**,

$$f(x) = -f(-x) \quad \text{for all } x,$$

and the definition of a **periodic function**,

$$\text{there exists } T \text{ such that} \qquad f(x+T) = f(x) \quad \text{for all } x.$$

However, a satisfactory formal definition for functional equations is elusive. If the only criterion were that a function be involved, just about any equation would qualify: algebraic equations (those involving algebraic functions), differential equations (involving functions and their derivatives), integral equations (involving functions and their integrals), difference equations (involving the evaluation of functions at evenly spaced points), integro-differential equations (involving functions, their derivatives, and their integrals), and so on.

Example 2.1 (Truesdell equation)**.** Truesdell [44] attempted to unify into a single framework the special functions of analysis using the differential-difference equation

$$\frac{\partial F(z,\alpha)}{\partial z} = F(z, \alpha+1).$$

In this book we shall refrain from such advanced topics. □

Although we will meet some functional equations that contain derivatives and integrals, I will adopt the standard convention that excludes differential and integral equations.[1] Some authors, such as Saaty [34] and Hille [52], add difference equations[2] to the list of exclusions. However, I side with those, such as Aczél [8], who regard difference equations as functional equations, since it is a natural fit. Finally, there are certain advanced types of equations, such as operator functional equations, which, for the purposes of this book, I will pretend do not exist.

What finally remains are the so-called *algebroid functional equations*. And yet, a solid general definition is still not easy to give. Trying to leave aside the difficulties so we can move on, we will start with this working definition:

Definition 2.2. An **algebroid functional equation** for the unknown function $f(x)$ is an equation of the form

$$F(x, f(x)) = 0 \quad \text{for all } x \in D, \tag{2.1}$$

where F is a known function of two variables and D is a given set.

Unfortunately, this definition hides some important aspects of functional equations. The function F may contain parameters. Often these parameters appear in the form of variables that also take values in D. Other times the parameters can be written as values of the function f itself. For example,

$$F(x, y, f(x), f(y)) = 0 \quad \text{for all } x, y \in D \tag{2.2}$$

is a functional equation for $f(x)$, with y and $f(y)$ appearing as "parameters". This use of parameters adds infinite flexibility to the class of functional equations, making a rigid, yet widely applicable, formal definition impossible. For our purposes, we shall regard (2.1) as a shorthand notation for all such possibilities.

Warning: This terminology is not standard. Practitioners of the subject talk of functions of one or more variables and functional equations of one or more variables: for example, a *function of one variable* $f(x)$ may satisfy a *functional equation in two variables*, $f(x + y) = f(x) + f(y)$. I have chosen to reserve the term "variable" for the arguments of the unknown function (independent variable), calling "parameters" the copies of the independent and dependent variables that may appear in a given functional equation. What is usually

[1] There is extensive literature on these equations. The reader not already familiar with them might consult [15, 26] for differential equations and [30, 43] for integral equations.

[2] For difference equations, see [19, 22], for example.

called a parameter is for me just a constant. Therefore the functional equation $f(f(x)+y) = f(x)+y+a$ is an equation for the function $f(x)$ of one variable and contains the parameter y and the constant a. The functional equation $f(\vec{x}+\vec{y}) = f(\vec{x})+f(\vec{y})$ is an equation for the function $f(\vec{x}) = f(x_1, x_2, \ldots, x_n)$ of n variables and contains the $n+1$ parameters $\vec{y}, f(\vec{y})$.

My convention is motivated by the fact that I want to consider functional equations such as $f(f(x)) = a$ and $\vec{f}(\vec{f}(\vec{x})) = \vec{a}$ as having the same type: both are equations with no parameters. According to the standard definition, one is an equation in one variable and the other is an equation in many variables.

In general, functional equations that contain parameters are easier to solve than those that do not.[3] □

The set D in (2.1) is the **domain** of the solutions of the functional equation. If no domain is given explicitly, the assumption is that D should be taken as the maximal set of values of x for which the functional equation is truly an identity.

Defining the domain of the solutions of a functional equation is very important: different domains lead to different sets of solutions, not necessarily related in a predictable way.

Example 2.3. Consider the functional equation

$$f(xy) = f(x) + f(y),$$

one which we will study quite thoroughly. If we seek functions $f : \mathbb{R}^* \to \mathbb{R}$, the admissible solutions are $f(x) = 0$ or $f(x) = \ln|x|$. (See Chapter 5.) However, if we seek functions defined on all of $\mathbb{R}$, then $f(x) = 0$ is the unique solution, as we can see by setting $y = 0$ in the functional equation, thus obtaining $f(0) = f(x) + f(0)$. □

A function $f(x)$ that satisfies (2.1) is called a **particular solution** for the functional equation. Solving the equation means finding the set of all possible particular solutions. This set is strongly dependent on the properties imposed on the function $f(x)$: thus, one may search for measurable, invertible, bounded, monotonic, continuous, integrable, or differentiable functions satisfying (2.1), to give a few examples. Each condition on $f(x)$ will result, in general, in a different set of solutions.

Example 2.4. The identity function $f(x) = x$ is a particular solution of the equation $f(x+y) = f(x) + f(y)$ with $x, y \in \mathbb{R}$. The set of all continuous solutions is $\{f(x) = cx : c \in \mathbb{R}\}$. If we allow discontinuous solutions, the set of solutions increases drastically. (See Sections 5.1 and 11.2.) □

[3] In [28], the term **iterative functional equations** is used for equations with no parameters. Although not all such equations contain iterates of functions, the term is motivated by the fact that iterates are essential for the methods and techniques used to obtain the solutions. (See Part V of this book.)

Definition 2.2 may be generalized to a functional equation containing an unknown function of several variables,

$$F(x_1, x_2, \ldots, x_n, f(x_1, x_2, \ldots, x_n)) = 0,$$

to a system of functional equations for several unknown functions $f_1(x)$, $f_2(x)$, $\ldots$, $f_N(x)$,

$$\begin{aligned} F_1(x, f_1(x), f_2(x), \ldots, f_N(x)) &= 0, \\ F_2(x, f_1(x), f_2(x), \ldots, f_N(x)) &= 0, \\ &\cdots\cdots \\ F_m(x, f_1(x), f_2(x), \ldots, f_N(x)) &= 0, \end{aligned}$$

or a combination of the above,

$$\begin{aligned} F_1(x_1, \ldots, x_n, f_1(x_1, \ldots, x_n), \ldots, f_N(x_1, \ldots, x_n)) &= 0, \\ F_2(x_1, \ldots, x_n, f_1(x_1, \ldots, x_n), \ldots, f_N(x_1, \ldots, x_n)) &= 0, \\ &\cdots\cdots \\ F_m(x_1, \ldots, x_n, f_1(x_1, \ldots, x_n), \ldots, f_N(x_1, \ldots, x_n)) &= 0. \end{aligned}$$

Of course, similar expressions with parameters are possible.

Hidden but not Removed

I have claimed that our Definition 2.2 of functional equations excludes differential and integral functional equations according to the commonly used conventions. However, strictly speaking, this statement is not exactly right. Higher concepts in analysis, such as differentiation and integration, are based on simpler concepts of functions and, in particular, the far-reaching idea of limit. Therefore, it is not surprising to discover differential and integral functional equations hidden inside our algebroid functional equations.

Consider, for example, the following situation. For a continuous function of two variables $g(x, y)$ such that $g(x, 0) = 0$, let $f(x)$ be a continuous function of one variable satisfying the algebroid equation

$$f(x + y) - f(x) - y\, f(x) = y\, g(x, y) \ .$$

For $y = 0$, this equation appears to reduce to the trivial identity $0 = 0$. However, if we rewrite it as

$$\frac{f(x + y) - f(x)}{y} - f(x) = g(x, y) \ ,$$

at $y = 0$ it reduces to the differential equation

$$f'(x) = f(x) \ .$$

Such relations are perhaps interesting and worth exploring as an independent topic. For the purposes of this book, I have made no effort to understand or explore it at a deeper level beyond this superficial comment.

Another Cautionary Remark

A comprehensive theory of functional equations, grouping them according to fundamental underlying principles and providing a structured way to solve them, has not yet been achieved. Here is an example of why this has proved so hard:

Example 2.5. The **associativity functional equation**

$$f(x, f(y, z)) = f(f(x, y), z)$$

and the closely related equation

$$f(x, f(y, z)) = f(y, f(x, z))$$

share virtually every property that might be thought to matter: both involve a single function of two variables; both include just one additional parameter; and both have very similar structures. But while the general solution of the associativity equation is

$$f(x, y) = g^{-1}(g(x) + g(y)),$$

where g is an arbitrary function of one variable (try it out!), the second equation is solved by

$$f(x, y) = g^{-1}(h(x) + g(y)),$$

with *two* arbitrary functions! The two answers are *very* different. (For the method of solution, see Sections 6.2.1 and 6.2.3 of [8].) □

Hunches and intuition aside, we do not fully understand what the underlying characteristics are that separate two equations that appear similar. This is what makes it so difficult to devise a finer classification of functional equations and a general theory for how to solve them. At the current time, we must rely on a classification based on empirical data. And such is the organization of this book.

The Inverse Problem

So far we have presented the functional equation as the starting point for which one desires to know the solution. However, the inverse problem is very interesting too: Given a function (or a set of functions), what functional equation (or equations) characterizes it (them)? If known, such a functional equation (or equations) can be used as the basis for the axiomatic definition of the function(s).

Example 2.6. One can define the trigonometric functions $\sin x$ and $\cos x$ (and thus all of trigonometry) as follows.

Definition 2.7. Let the functions $S : \mathbb{R} \to \mathbb{R}$ and $C : \mathbb{R} \to \mathbb{R}$ satisfy the functional equation

$$C(x-y) = C(x)C(y) + S(x)S(y) \quad \text{for all } x, y \in \mathbb{R} \tag{2.3}$$

and the following condition: There exists a positive number λ such that

$$C(0) = S(\lambda) = 1,$$

and

$$C(x) > 0 \text{ and } S(x) > 0 \quad \text{for all } x \in (0, \lambda).$$

We call the solution thus obtained the **analytic sine** and **analytic cosine**.

Of course, one must prove that this definition is meaningful; in other words, that there exists such a pair of functions and it is unique. If we had chosen

$$S(x+y) = S(x)C(y) + C(x)S(y) \quad \text{for all } x, y \in \mathbb{R}$$

in place of the functional equation written above, the lack of uniqueness is easily seen. The functions

$$C(x) = a^x \cos x \quad \text{and} \quad S(x) = a^{x-\lambda} \sin x$$

satisfy all requirements with $\lambda = \pi/2$ and any value of $a \in \mathbb{R}_+^*$. Therefore, an infinite number of solutions are obtained. We study the problem of characterization of the sine and cosine functions in Section 7.1. There, we will prove that the data given above for the analytic sine and cosine admit a unique solution, obviously given by $S(x) = \sin x$, $C(x) = \cos x$.

Leaving aside the issue of existence and uniqueness, one can easily show that the functions $S(x)$, $C(x)$ have all the properties of sine and cosine: $C(\lambda) = S(0) = 0$, $C(x)^2 + S(x)^2 = 1$, $|S(x)| \leq 1$, $|C(x)| \leq 1$, etc. (See Problem 2.23 on page 53.) □

Notice that, in the definition of analytic sine and analytic cosine, besides the functional equation, we used functional inequalities too. This is our next topic of discussion.

Functional Relations

Functional relations are a generalization of functional equations, where the desired function satisfies rules involving relations not limited to equalities. Again, some of the most common definitions of properties of functions are given using functional relations. For example, the definition of a **strictly increasing function**,

$$x < y \;\Rightarrow\; f(x) < f(y),$$

involves a functional inequality, as do the related definitions of nondecreasing, nonincreasing, and strictly decreasing functions. In general:

Definition 2.8. A **functional relation** for the unknown function $f(x)$ is a relation of the form

$$F(x, f(x)) \;\sim\; G(x, f(x)) \quad \text{for all } x \in D, \tag{2.4}$$

where F and G are known functions of two variables, D is a given set, and $\sim$ is some relation (usually an equivalence or order relation) defined on the codomain of F and G.

Problem 2.9 ([3], Problem 3039, May 2006). *Let a, b be fixed nonzero real numbers. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f\Big(x - \frac{b}{a}\Big) + 2x \;\le\; \frac{a}{b}x^2 + \frac{2b}{a} \;\le\; f\Big(x + \frac{b}{a}\Big) - 2x \quad \textit{for all } x \in \mathbb{R}.$$

Solution. Letting $y = x - b/a$, we see that the left inequality becomes

$$f(y) \;\le\; \frac{a}{b}y^2 + \frac{2b}{a}.$$

Similarly, letting $y = x + b/a$, we see that the right inequality becomes

$$f(y) \;\ge\; \frac{a}{b}y^2 + \frac{2b}{a}.$$

Hence

$$f(y) \;=\; \frac{a}{b}y^2 + \frac{2b}{a} \quad \text{for all } y \in \mathbb{R}.$$

□

2.2 Beginning Problems

As a leisurely introduction to solving functional equations, this section offers simple problems that can be understood even by a student who is just beginning to learn functions. As we progress with our subject in the next section and later chapters, we shall study progressively more difficult problems.

Problem 2.10. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$2f(x + y) + 6y^3 = f(x + 2y) + x^3 \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. What possibilities are there for f? Substituting $y = 0$ in the functional equation, we get $2f(x) = f(x) + x^3$, which implies $f(x) = x^3$.

For $f(x) = x^3$, the left-hand side of the functional equation becomes

$$2x^3 + 6x^2y + 6xy^2 + 8y^3,$$

while the right-hand side becomes

$$2x^3 + 6x^2y + 12xy^2 + 8y^3.$$

But $6xy^2$ and $12xy^2$ are different except for very special values of the variables. Therefore the only possible solution, $f(x) = x^3$, does not work. The equation has no solution. □

The previous problem points out that, when solving a functional equation, we must check that the derived functions indeed satisfy the given equation (inverse). However, in the following, we systematically omit this step whenever everything works out fine.

Problem 2.11. *Let $f : \mathbb{R}_+^* \to \mathbb{R}_+$ be such that*

$$f(x^2) = \frac{1}{x} \quad \textit{for all } x \in \mathbb{R}_+^*.$$

Find f.

Solution. In the defining equation we set $y = x^2$. Then

$$f(y) = \frac{1}{\sqrt{y}}. \qquad □$$

Problem 2.12. *Let $f : \mathbb{R}^* \to \mathbb{R}$ be such that*

$$f\left(1 + \frac{1}{x}\right) = x^2 + \frac{1}{x^2} \quad \textit{for all } x \in \mathbb{R}^*.$$

Find f.

Solution. In the defining equation we set $y = 1 + \frac{1}{x}$; that is, $x = \frac{1}{y-1}$. Then

$$f(y) = \frac{1}{(y-1)^2} + (y-1)^2. \qquad □$$

Problem 2.13. *Find a particular solution $f : \mathbb{R}^* \to \mathbb{R}$ of the functional equation*

$$f(x) - f(x+1) = f(x)\,f(x+1) \quad \textit{for all } x \in \mathbb{R} \setminus \{0, 1\}.$$

Solution. The function $f_p(x) = 1/x$ satisfies the equation. Indeed

$$\begin{aligned} f_p(x) - f_p(x+1) &= \frac{1}{x} - \frac{1}{x+1} = \frac{1}{x(x+1)} \\ &= f_p(x)\, f_p(x+1) \quad \text{for } x \neq 0, -1. \end{aligned}$$

□

Problem 2.14. (a) *Show that the function*

$$f_p(x) = a \cos^{-1} x \quad \textit{for } 0 \le x \le 1,$$

where a is a real constant, satisfies the functional equation

$$f\big(xy - \sqrt{(1-x^2)(1-y^2)}\big) = f(x) + f(y).$$

(b) *Find a particular solution $f : [1, \infty) \to \mathbb{R}$ of the functional equation*

$$f\big(xy - \sqrt{(x^2-1)(y^2-1)}\big) = f(x) + f(y).$$

(c) *Find a particular solution $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f\big(x\sqrt{1+y^2} + y\sqrt{1+x^2}\big) = f(x) + f(y).$$

(d) *Find a particular solution $f : [-1, 1] \to \mathbb{R}$ of the functional equation*

$$f\left(\frac{x+y}{1-xy}\right) = f(x) + f(y).$$

(e) *Find a particular solution $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f\left(\frac{x+y}{1+xy}\right) = f(x) + f(y).$$

Solution. (a) Let $X = \cos^{-1} x$ and $Y = \cos^{-1} y$. Then $x = \cos X$ and $y = \cos Y$ and

$$f(x) + f(y) = a(X+Y).$$

On the other hand, we have

$$\begin{aligned} xy - \sqrt{(1-x^2)(1-y^2)} &= \cos X \cos Y - \sqrt{\sin^2 X \sin^2 Y} \\ &= \cos X \cos Y - \sin X \sin Y \\ &= \cos(X+Y), \end{aligned}$$

where the second equality is true because $X, Y \in [0, \pi/2]$, so the sines are nonnegative. Therefore

$$f\big(xy - \sqrt{(1-x^2)(1-y^2)}\big) = a \cos^{-1}\big(\cos(X+Y)\big) = a\,(X+Y),$$

from which the relation sought follows immediately.

(b)–(e) Similarly, we see that these functional equations are satisfied by the functions $a \cosh^{-1} x$, $a \sinh^{-1} x$, $a \tan^{-1} x$, and $a \tanh^{-1} x$, respectively. □

Problem 2.15. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x+y) = r\,y + f(x) \quad \textit{for all } x \in \mathbb{R} \textit{ and } y \in \mathbb{R}.$$

Solution. Substituting $x = 0$ in the defining functional equation, we find

$$f(y) = r\,y + c,$$

where $c = f(0)$. □

Even a weaker version of the defining condition, involving only $y \in \mathbb{R}_+^*$, still leads to the same set of functions:

Problem 2.16. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x+y) = r\,y + f(x) \quad \textit{for all } x \in \mathbb{R} \textit{ and } y \in \mathbb{R}_+^*.$$

Solution. Substituting $x = 0$ in the defining functional equation, we find

$$f(y) = r\,y + c \quad \text{for } y > 0,$$

where $c = f(0)$. We then substitute $x = -y$, with $y > 0$, to find

$$f(0) = r\,y + f(-y) \;\Rightarrow\; f(-y) = -r\,y + c.$$

Therefore

$$f(x) = r\,x + c \quad \text{for all } x.$$ □

Problem 2.17. *Let $\varepsilon > 0$ be fixed. Suppose $f : \mathbb{R} \to \mathbb{R}$ is periodic of period T for every $T \in (0, \varepsilon]$:*

$$f(x+T) = f(x) \quad \textit{for all } x \in \mathbb{R} \textit{ and } T \in (0, \varepsilon].$$

Show that f is constant.

Solution. Fixing x, we see immediately from the functional equation that f is constant in the interval $[x, x+\varepsilon]$. Since $\mathbb{R}$ is the union of intervals of the form $[n\varepsilon, (n+1)\varepsilon]$ for $n \in \mathbb{Z}$, it follows that f is constant everywhere. Formally, for $x \in (n\varepsilon, (n+1)\varepsilon]$, with (say) $n < 0$, we apply induction to obtain that

$$f(x) = f(x+\varepsilon) = f(x+2\varepsilon) = \cdots = f(x + (-n)\varepsilon),$$

which equals $f(0)$ since $x - n\varepsilon \in (0, \varepsilon]$. The case where $n > 1$ is treated analogously. □

Problem 2.18. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(xy) = y^r f(x) \quad \textit{for all } x, y \in \mathbb{R},$$

where the constant r is assumed such that y^r is well defined for all real y.

Solution. Setting $x = 1$ in the defining equation we find $f(y) = c\,y^r$, with $c = f(1)$. □

Problem 2.19. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(xy) = y^r f(x) \quad \textit{for all } x \in \mathbb{R} \textit{ and } y \in \mathbb{R}_+^*.$$

Solution. Setting $x = 0$ in the defining equation we find

$$f(0)\,(1 - y^r) = 0.$$

If $r \neq 0$, since this equation must be true for all positive y, we must have $f(0) = 0$. If $r = 0$, then the value $f(0)$ is undetermined.

Then we substitute $x = 1$, $x = -1$ in the defining equation to find

$$\begin{aligned} f(y) &= a\,y^r \quad \text{for } y > 0, \\ f(-y) &= b\,y^r \quad \text{for } y > 0, \end{aligned}$$

where we have set $a = f(1)$ and $b = f(-1)$. From this we see that for $r \neq 0$

$$f(x) = \begin{cases} a\,x^r, & \text{if } x > 0, \\ 0, & \text{if } x = 0, \\ b\,|x|^r, & \text{if } x < 0, \end{cases}$$

while for $r = 0$

$$f(x) = \begin{cases} a, & \text{if } x > 0, \\ c, & \text{if } x = 0, \\ b, & \text{if } x < 0. \end{cases}$$

□

Problem 2.20. *Find functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(\lambda x, \lambda y) = \lambda^r f(x, y) \quad \textit{for all } \lambda, x, y,$$

where r is a fixed number such that λ^r is well defined for all λ.

Solution. Setting $\lambda = 1/x$, $x \neq 0$, we find

$$f(x, y) = x^r f\left(1, \frac{y}{x}\right).$$

Thus, if $g(y) = f(1, y)$ is any real-valued function, we obtain a solution,

$$f(x, y) = x^r g\left(\frac{y}{x}\right),$$

of the given equation. □

Comment. It is straightforward to generalize the above problem to one in n variables: The equation

$$f(\lambda x_1, \lambda x_2, \ldots, \lambda x_n) = \lambda^r f(x_1, x_2, \ldots, x_n) \quad \text{for all } \lambda, x_1, x_2, \ldots, x_n \tag{2.5}$$

is solved by

$$f(x_1, x_2, \ldots, x_n) = x_1^r g\left(\frac{x_2}{x_1}, \ldots, \frac{x_n}{x_1}\right),$$

where g is any function in $n - 1$ variables. □

Comment. The functional equation (2.5) is known as the **Euler equation** and a function satisfying it is called **homogeneous of degree** r. □

Problem 2.21. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

(a) *$f(\lambda x) = f(x)$ for all $x \in \mathbb{R}$, where λ is a real constant distinct from ± 1, and*

(b) *f is continuous at $x = 0$.*

Solution. We study three cases.

(i) If $\lambda = 0$, it is immediate that $f(x) = f(0)$.

(ii) If $|\lambda| < 1$, repeated applications of the defining functional equation give

$$f(x) = f(\lambda x) = f(\lambda^2 x) = \cdots = f(\lambda^n x)$$

for any natural number n. Then, by the continuity of f at $x = 0$,

$$f(0) = f(\lim_{n\to\infty} \lambda^n x) = \lim_{n\to\infty} f(\lambda^n x) = \lim_{n\to\infty} f(x) = f(x).$$

(iii) If $|\lambda| > 1$, we define $\mu = 1/\lambda$ and in the defining equation we substitute $x = \mu y$: $f(y) = f(\mu y)$. Then

$$f(y) = f(\mu y) = f(\mu^2 y) = \cdots = f(\mu^n y)$$

for any natural number n. Using the continuity of f at $x = 0$, we conclude again that

$$f(x) = f(0).$$

Thus, in all cases, the only possible functions are the constants. □

Question. *Let $h(x)$ be any continuous periodic function with period T. Define*

$$f(x) = h(a + T \log_\lambda x),$$

where $a \in \mathbb{R}$ and $\lambda \in \mathbb{R}_+^* \setminus \{1\}$. Show that

$$f(\lambda x) = f(x) \quad \text{for all } x \in \mathbb{R}_+^*.$$

However, f is not constant. How can this be, in view of Problem 2.21?

Answer. Although f satisfies the functional equation of the problem, it does not satisfy the second condition: continuity at $x = 0$. This condition was essential in proving that the function is constant. If we eliminate this condition, then nonconstant solutions of the functional equation are possible, as demonstrated explicitly by the given function.

Problem 2.22. *Find the function $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x - y) = f(x)\, f(y) \quad \text{for all } x, y.$$

Solution. The identically vanishing function $f(x) \equiv 0$ is a solution. To search for others, assume $x_0 \in \mathbb{R}$ is such that $f(x_0) \neq 0$. Setting $x = x_0$ and $y = 0$ in the defining solution, we find $f(x_0) = f(x_0)\, f(0)$, and therefore $f(0) = 1$. Setting $y = x$, we now find $f(0) = f(x)^2$ or $f(x) \equiv \pm 1$. The solution $f(x) \equiv -1$ is inconsistent with the defining relation. Hence $f(x) \equiv 1$. □

Problem 2.23. *Consider the analytic sine and analytic cosine function* (see page 46). *Prove:*

(a) $S(0) = C(\lambda) = 0$.

(b) $C(x)^2 + S(x)^2 = 1$.

(c) *The two functions $C(x)$, $S(x)$ are bounded. In particular, $|C(x)| \leq 1$ and $|S(x)| \leq 1$.*

(d) $C(\lambda - x) = S(x)$ *and* $S(\lambda - x) = C(x)$.

(e) *The functions $C(x)$, $S(x)$ are periodic.*

(f) $S(x + y) = S(x)C(y) + S(y)C(x)$.

(g) $C(-x) = C(x)$ *and* $S(-x) = -S(x)$.

(h) *For the functions $C(x)$, $S(x)$, the trigonometric identities for $C(x \pm y)$ and $S(x \pm y)$ and their corollaries are valid.*

(i) *The functions $C(x)$, $S(x)$ are continuous in $\mathbb{R}$.*

Solution. (a) In the defining equation (2.3), we set $x = y = 0$ to find

$$C(0) = C(0)^2 + S(0)^2,$$

which, when combined with $C(0) = 1$, gives $S(0) = 0$. Similarly, if, in the defining equation (2.3), we set $x = y = \lambda$, we find $C(\lambda) = 0$.

(b) This identity is now immediate if, in the defining equation (2.3), we set $x = y$.

(c) If we write the previous result in the form $|C(x)|^2+|S(x)|^2 = 1$, we conclude that $|C(x)|^2 \leq 1$, $|S(x)|^2 \leq 1$, or equivalently $|C(x)| \leq 1$, $|S(x)| \leq 1$.

(d) In the defining equation (2.3), we set $x = \lambda$ to find $C(\lambda - y) = S(y)$. Now set $y = \lambda - x$ to find $C(x) = S(\lambda - x)$ too.

(e) In the result (d), place $x - \lambda$ in place of x:

$$\begin{aligned} C(2\lambda - x) &= S(x - \lambda) = -S(\lambda - x) = -C(x), \\ S(2\lambda - x) &= C(x - \lambda) = +C(\lambda - x) = +S(x). \end{aligned}$$

Substituting once more $-x - 2\lambda$ for x, we arrive at the result sought:

$$\begin{aligned} C(4\lambda + x) &= -C(-x - 2\lambda) = -C(2\lambda + x) = C(-x) = C(x), \\ S(4\lambda + x) &= S(-x - 2\lambda) = -S(2\lambda + x) = -S(-x) = S(x). \end{aligned}$$

(f) Using the result (d), we can write

$$\begin{aligned} S(x + y) &= C(\lambda - x - y) = C((\lambda - x) - y) \\ &= C(\lambda - x)C(y) + S(\lambda - x)S(y) \\ &= S(x)C(y) + C(x)S(y). \end{aligned}$$

(g) If, in the defining equation (2.3), we set $x = 0$, we immediately see that the analytic cosine is even: $C(-y) = C(y)$. Now, in the equation proved in part (f), we set $y = -x$, obtaining

$$S(0) = S(x)C(-x) + S(-x)C(x),$$

and therefore

$$0 = C(x)\big(S(x) + S(-x)\big). \tag{2.6}$$

If $C(x) \neq 0$, then $S(x) = -S(-x)$. This is the case, for example, for any $x \in (0, \lambda)$. For x' satisfying $C(x') = 0$, no information is obtained from (2.6). However, in this case, part (b) implies $S(x') \pm 1$. Take an arbitrary $y \in (0, \lambda)$. Then

$$C(x' + y) = C(x' - (-y)) = C(x')C(y) + S(x')S(-y) = \pm S(y).$$

This result implies that $C(x' + y) \neq 0$ and therefore $S(x' + y) = -S(-x' - y)$. By part (e), this equation is equivalently written $S(x')C(y) = -S(-x')C(y)$, or $S(x') = -S(-x')$.

(h) This part is a straightforward application of the formulæ found so far. For example, setting $x = -y$ in the defining equation (2.3), we find

$$C(2x) = C(x)^2 - S(x)^2 = 2C(x)^2 - 1 = 1 - 2S(x)^2.$$

This can also be written as

$$C(x) = \pm\sqrt{\frac{1 + C(2x)}{2}},$$

with the plus sign chosen if $x \in (0, \lambda)$. One can continue similarly.

One can show with these results that $C(x)$ is strictly decreasing in the interval $[0, 2\lambda]$ and strictly increasing in the interval $[2\lambda, 4\lambda]$. $S(x)$ is strictly increasing in the interval $[-\lambda, \lambda]$ and strictly decreasing in the interval $[\lambda, 3\lambda]$. The reader may want to fill in the details.

(i) Since $0 \leq C(x) \leq 1$ for any $x \in (0, \lambda)$, the limit $\lim_{x\to 0^+} C(x)$ exists and lies in the interval $[0, 1]$. Let's indicate it by ℓ. Then from the equation $C(2x) = 2C(x)^2 - 1$ we find that ℓ satisfies the equation $\ell = 2\ell^2 - 1$. The positive root of this equation is $\ell = 1$. Since $C(x)$ is even, the root from the other side exists and is also 1. Therefore

$$\lim_{x\to 0} C(x) = 1 = C(0);$$

that is, $C(x)$ is continuous at $x = 0$.

Using the equation $C(2x) = 1 - 2S(x)^2$, we also see that $S(x)$ is continuous at $x = 0$ too:

$$\lim_{x\to 0} S(x) = 0 = S(0).$$

Now using the equation

$$C(x + h) = C(x)C(h) - S(x)S(h),$$

and taking the limit $h \to 0$, we find

$$\lim_{h\to 0} C(x + h) = C(x) \lim_{h\to 0} C(h) - S(x) \lim_{h\to 0} S(h),$$

or

$$\lim_{h\to 0} C(x + h) = C(x).$$

The function $C(x)$, and thus also $S(x)$, are continuous at all points. □

Chapter 3

Equations for Arithmetic Functions

3.1 The Notion of Difference Equations

In this chapter we will sometimes depart from the notation a_n for sequences, using the notation $a(n)$ employed for functions in general. Another term for a sequence, common in number theory, is **arithmetic function**.

Consider an arithmetic function (sequence) with domain $\mathbb{N}$. A **difference functional equation** is an equation that relates a number of terms of the sequence

$$f(a_{n+1}, a_n, a_{n-1}, \ldots, n) = 0,$$

where f is some given function and n takes values, usually, from some integer n_0 on. If the difference equation can be solved for a_{n+1}, say

$$a_{n+1} = g(a_n, a_{n-1}, \ldots, n), \quad n \geq 0,$$

we speak of a **recursion relation**.

Example 3.1. The equation

$$a_{n+3}^3 + a_{n+2} \ln a_{n+3} + a_{n+1} \tan a_{n+3} - \frac{a_n}{a_{n+2}} = 0, \quad n \geq 0,$$

is a difference equation that relates any term a_{n+3} of the sequence to the three preceding terms. It is not easy, however, to solve for a_{n+3}.

The equations

$$a_{n+1} = a_n + \omega, \tag{3.1}$$

$$a_{n+1} = \lambda\, a_n, \tag{3.2}$$

$$a_{n+1} = a_n + a_{n-1}, \tag{3.3}$$

are some of the best known recursion relations. Equation (3.1) defines an **arithmetic progression** with **step** ω; we have

$$a_n = a_0 + n\omega \quad \text{for } n \geq 0.$$

Equation (3.2) defines a **geometric progression** with **ratio** λ:

$$a_n = \lambda^n a_0 \quad \text{for } n \geq 0.$$

Equation (3.3) defines a sequence that depends on the first two terms a_0 and a_1. If $a_0 = a_1 = 1$, we get the **Fibonacci sequence**, whose members are the famous **Fibonacci numbers**; the first few of them are

$$1, \quad 1, \quad 2, \quad 3, \quad 5, \quad 8, \quad 13, \quad \ldots.$$

Similarly, if $a_0 = 1$ and $a_1 = 3$, we get the related **Lucas sequence**, which starts with

$$1, \quad 3, \quad 4, \quad 7, \quad 11, \quad 18, \quad 29, \quad \ldots$$

and whose members are called **Lucas numbers**. □

Exercise 3.2. *Prove:*

(a) *If* $\{a_n\}$ *is an arithmetic progression, then*

$$\sum_{k=1}^{n} a_k = \frac{a_1 + a_n}{2}\, n.$$

(b) *If* $\{a_n\}$ *is a geometric progression with ratio* λ*, then*

$$\sum_{k=1}^{n} a_k = \frac{a_n\lambda - a_1}{\lambda - 1}.$$

If $|\lambda| < 1$*, then*

$$\sum_{k=1}^{\infty} a_k = \frac{a_1}{1-\lambda}.$$ □

A Formal Definition

We can make the definition of a difference equation a bit more formal by the introduction of two **operators** — that is, functions whose domain is a set of functions. We will usually denote operators by capital letters with a hat (see examples below).

We start by considering certain important operators that act on the set of real-valued arithmetic functions (sequences).[1]

The **(left) shift operator** $\hat{T}$, applied to an arithmetic function a, gives the function $\hat{T}a$ defined by

$$\hat{T}a(n) = a(n+1).$$

Note that, unless the domain of a is all of $\mathbb{Z}$, the domain of $\hat{T}a$ does not coincide with the domain of a; it's translated, or shifted, one unit to the left. For $a : \mathbb{N} \to \mathbb{R}$, for instance, the shift operator $\hat{T}a$ has domain $\{-1\} \cup \mathbb{N}$.

Applying the operator $\hat{T}$ successively m times, we obtain the n-th power of $\hat{T}$, formally defined by $\hat{T}^m = \hat{T}\,\hat{T}^{m-1}$, where $m > 1$. We can easily see that $\hat{T}^m$ is a left shift by m units:

$$\hat{T}^m a(n) = a(n+m).$$

The **difference operator** $\hat{\Delta}$ is defined by

$$\hat{\Delta}a(n) = a(n+1) - a(n) \tag{3.4}$$

for any arithmetic function a. This equality may be rewritten as

$$\hat{\Delta}a(n) = \hat{T}a(n) - a(n) = \big((\hat{T} - \hat{I})a\big)(n), \tag{3.5}$$

where we have introduced the **identity operator** $\hat{I}$ defined by

$$\hat{I}a(n) = a(n)$$

for any arithmetic function a. It follows from (3.5) that the difference operator has the elegant (if more abstract) expression

$$\hat{\Delta} = \hat{T} - \hat{I}$$

in terms of the shift and identity operators. Equivalently,

$$\hat{T} = \hat{\Delta} + \hat{I}.$$

We can apply the difference operator many times: $\hat{\Delta}^m = \hat{\Delta}\,\hat{\Delta}^{m-1}$, where $m > 1$. For $m = 2$, we have

$$\begin{aligned}\hat{\Delta}^2 a(n) &= \hat{\Delta}\big(\hat{\Delta}a(n)\big) = \hat{\Delta}\big(a(n+1) - a(n)\big)\\ &= \big(a(n+2) - a(n+1)\big) - \big(a(n+1) - a(n)\big)\\ &= a(n+2) - 2a(n+1) + a(n).\end{aligned}$$

[1] We say real-valued for concreteness; integer- or complex- or vector-valued sequences will work just as well.

This chain of equalities can be expressed equivalently by writing

$$\hat{\Delta}^2 a(n) == (\hat{T}-\hat{I})^2 a(n) = (\hat{T}^2 - 2\hat{T} + \hat{I})a(n),$$

where we have used the equalities

$$\hat{I}\hat{S} = \hat{S} = \hat{S}\hat{I},$$

valid for any operator $\hat{S}$. Inductively we can show that

$$\hat{\Delta}^m = (\hat{T}-\hat{I})^m = \sum_{k=0}^{m} \binom{m}{k} (-1)^{m-k} \hat{T}^k.$$

From the equality $\hat{T} = \hat{\Delta} + \hat{I}$ of the previous page we also get, inductively,

$$\hat{T}^m = (\hat{\Delta}+\hat{I})^m = \sum_{k=0}^{m} \binom{m}{k} \hat{\Delta}^k.$$

The last equation implies that any term $a(n+m)$ of the sequence $\{a(n)\}$ can be written in terms of the m differences $\hat{\Delta}^k a(n)$, $k = 1, 2, \ldots, m$. We have thus arrived at an alternative definition of a difference equation:

Definition 3.3. An equation relating the value of the function $a(n)$ with one or more of its differences $\hat{\Delta}^k a(n)$, $k = 1, 2, \ldots,$ is called a **difference equation**.

Example 3.4. Some examples of difference equations are

$$\begin{aligned}
\hat{\Delta}a(n) + 5a(n) &= 0,\\
\hat{\Delta}^2 a(n) + \hat{\Delta}a(n) + a(n) &= n^2+1,\\
\left(\hat{\Delta}a(n)\right)^2 + a(n)^2 &= 1,\\
a(n)\,\hat{\Delta}^3 a(n) &= \cos(3n).
\end{aligned}$$

If the action of a difference operator is spelled out and we perform the obvious simplifications, these equations become (reverting to sequence notation)

$$\begin{aligned}
a_{n+1} + 4a_n &= 0,\\
a_{n+2} - a_{n+1} - a_n &= n^2+1,\\
a_{n+1}^2 - 2a_{n+1}\,a_n + 2a_n^2 &= 1,\\
a_n\,a_{n+3} - 3a_n\,a_{n+2} + 3a_n\,a_{n+1} - a_n^2 &= \cos(3n).
\end{aligned}$$ □

3.2 Multiplicative Functions

Definition 3.5. An arithmetic function f defined on $\mathbb{N}$ is called **multiplicative** if f is not identically zero and

$$f(m\,n) = f(m)\,f(n) \quad \text{for all } \textit{relatively prime } m, n \in \mathbb{N}^*. \tag{3.6}$$

It is called **completely multiplicative** if

$$f(m\,n) = f(m)\,f(n) \quad \text{for all } n, m. \tag{3.7}$$

Example 3.6. The **Möbius function** $\mu(n)$ is defined as follows: If $n = 1$, then $\mu(1) = 1$. If $n \neq 1$, let $p_1, p_2, \ldots, p_k$ be the prime factors of n, $n = p_1^{a_1} p_2^{a_2} \ldots p_k^{a_k}$. Then

$$\mu(n) = \begin{cases} (-1)^k & \text{if } a_1 = a_2 = \cdots = a_k = 1, \\ 0 & \text{otherwise.} \end{cases}$$

It is easy to check that the Möbius function is multiplicative but not completely multiplicative. □

The proofs of the next four theorems are left to the reader as easy exercises.

Theorem 3.7. *If f, g are multiplicative functions, the product fg and quotient f/g (defined on a suitable domain) are also multiplicative functions. The same holds if "multiplicative" is replaced by "completely multiplicative".*

Theorem 3.8. *If f is multiplicative, then $f(1) = 1$.*

Theorem 3.9. *Let f be an arithmetic function such that $f(1) = 1$. Then f is multiplicative if and only if*

$$f(p_1^{a_1} p_2^{a_2} \ldots p_k^{a_k}) = f(p_1^{a_1})\,f(p_2^{a_2})\,\ldots\,f(p_k^{a_k}),$$

for all distinct prime numbers p_i and all integers $a_i \geq 1$.

Theorem 3.10. *A multiplicative function f is completely multiplicative if and only if*

$$f(p^a) = f(p)^a,$$

for all prime numbers p and integers $a \geq 1$.

3.3 Linear Difference Equations

A difference equation of the form

$$a_{n+k} + p_1(n)\, a_{n+k-1} + \cdots + p_k(n)\, a_n = g(n), \tag{3.8}$$

where the $p_i(n)$, for $i = 1, 2, \ldots, k$, are fixed functions, is called a **linear difference equation of order** k. It is called **homogeneous** if, in addition, $g(n) \equiv 0$, and **inhomogeneous** otherwise. In physics, the function g is related to an input that tries to impose (force) a particular behavior on the system. For this reason, $g(n)$ is sometimes called the **forcing term**. The term *linear* means that if two sequences $\{b_n\}$ and $\{c_n\}$ are solutions of the homogeneous equation, so are the sequences $\{b_n + c_n\}$ and $\{\lambda b_n\}$, where λ is a constant. This linearity property can be easily verified for difference equations of the form (3.8) with $g \equiv 0$.

In order to solve a k-th order difference equation, initial data must be given. The initial data are usually the first k terms:

$$a_0, a_1, \ldots, a_{k-1} = \text{given}. \tag{3.9}$$

The equation (3.8) together with the initial data (3.9) constitutes an **initial value problem**.

Theorem 3.11. *The initial value problem* (3.8) *and* (3.9) *has a unique solution.*

Proof. Given (3.9), we can find a_k from (3.8), then a_{k+1}; then we can find a_{k+2}, and so on. Therefore, by induction, each a_n is determined uniquely. □

Obviously, this theorem not only guarantees that a solution exists, but it also determines how it can be constructed. However, such an inductive approach is very inefficient, requiring a lot of work and time in many cases. Therefore, a more concise approach is desirable; such an approach is presented in the remainder of the section.

Theory of Solutions

In this subsection we present general statements about the solutions of linear difference equations. In the next subsection we then describe how to find the solutions of such difference equations building on the results of this subsection.

Definition 3.12. Let $f_1, f_2, \ldots, f_r$ be arithmetic functions defined on $\mathbb{N}$ and consider $n_0 \in \mathbb{N}$. We say that $f_1, f_2, \ldots, f_r$ are **linearly dependent** for $n \geq n_0$ if there are constants $\alpha_1, \alpha_2, \ldots, \alpha_r$, *not all of which are zero*, satisfying

$$\alpha_1 f_1(n) + \alpha_2 f_2(n) + \cdots \alpha_r f_r(n) = 0 \quad \text{for all } n \geq n_0. \tag{3.10}$$

(Note that, if we choose $n_1 \geq n_0$, linear dependence for $n \geq n_0$ implies linear dependence for $n \geq n_1$ as well.)

If $f_1, f_2, \ldots, f_r$ are not linearly dependent for $n \geq 0$, we say they are **linearly independent**.

Note. To show that $f_1, f_2, \ldots, f_r$ are linearly independent, we don't need to consider *all* integers n in (3.10). Suppose we have found *some* set S of integers such that

$$\alpha_1 f_1(n) + \alpha_2 f_2(n) + \cdots \alpha_r f_r(n) = 0 \quad \text{for } n \in S \tag{3.11}$$

implies $\alpha_1 = \cdots = \alpha_r = 0$. Then $f_1, f_2, \ldots, f_r$ are linearly independent. Indeed, the validity of (3.10) (with $n_0 = 0$) implies that of (3.11), which implies $\alpha_1 = \cdots = \alpha_r = 0$, which, by Definition 3.12, means the functions cannot be linearly dependent for $n \geq 0$.

Exercise 3.13. *Show that the arithmetic functions 3^n, $n3^n$, $n^2 3^n$ are linearly independent.*

Solution. Let $\alpha_1, \alpha_2, \alpha_3$ be such that

$$\alpha_1 3^n + \alpha_2 n 3^n + \alpha_3 n^2 3^n = 0.$$

Let's form the system (3.11) for the easiest values of n, namely 0, 1 and 2:

$$\begin{aligned} \alpha_1 \qquad\qquad\qquad &= 0,\\ 3\alpha_1 + \ \ 3\alpha_2 + \ \ 3\alpha_3 &= 0,\\ 9\alpha_1 + 18\alpha_2 + 27\alpha_3 &= 0. \end{aligned}$$

The unique solution is $\alpha_1 = \alpha_2 = \alpha_3 = 0$. By the note, the three functions are linearly independent. □

Given solutions $x_1, x_2, \ldots, x_r$ of (3.8), there is a straightforward method to check for linear independence, as explained below.

Definition 3.14. The **Casoratian** of the solutions $x_1, x_2, \ldots, x_r$ is the arithmetic function C defined by

$$C(n) = \det \begin{bmatrix} x_1(n) & x_2(n) & \cdots & x_r(n) \\ x_1(n+1) & x_2(n+1) & \cdots & x_r(n+1) \\ \vdots & \vdots & & \vdots \\ x_1(n+r-1) & x_2(n+r-1) & \cdots & x_r(n+r-1) \end{bmatrix}.$$

Theorem 3.15. *A k-tuple of solutions $x_1, \ldots, x_k$ of an order-k homogeneous difference equation*

$$x(n+k) + p_1(n)\, x(n+k-1) + \cdots + p_k(n)\, x(n) = 0$$

is linearly independent if and only if $C(0) \neq 0$.

Proof. Taking $S = \{0, 1, \ldots, k-1\}$ in the note preceding Exercise 3.13, we see that linear independence will be proved if the equations

$$\begin{array}{llll} \alpha_1 x_1(0) & + \alpha_2 x_2(0) & + \cdots + \alpha_k x_k(0) & = 0, \\ \alpha_1 x_1(1) & + \alpha_2 x_2(1) & + \cdots + \alpha_k x_k(1) & = 0, \\ \alpha_1 x_1(2) & + \alpha_2 x_2(2) & + \cdots + \alpha_k x_k(2) & = 0, \\ \quad\cdots & \quad\cdots & \qquad\quad\cdots & \\ \alpha_1 x_1(k-1) & + \alpha_2 x_2(k-1) & + \cdots + \alpha_k x_k(k-1) & = 0 \end{array} \tag{3.12}$$

are satisfied only when $\alpha_1 = \cdots = \alpha_k = 0$. We can view this as a system of homogeneous equations in $\alpha_1, \ldots, \alpha_k$. Then linear algebra tells us that $\alpha_1 = \cdots = \alpha_k = 0$ is the unique solution if and only if the determinant of the matrix of coefficients is nonzero. But this determinant is

$$\begin{vmatrix} x_1(0) & x_2(0) & \cdots & x_k(0) \\ x_1(1) & x_2(1) & \cdots & x_k(1) \\ \vdots & \vdots & & \vdots \\ x_1(k-1) & x_2(k-1) & \cdots & x_k(k-1) \end{vmatrix} = C(0).$$

Thus $C(0) \neq 0$ implies linear independence.

Conversely, suppose that $C(0) = 0$ and let $(\alpha_1, \ldots, \alpha_k) \neq (0, \ldots, 0)$ be such that (3.12) holds. We claim that

$$\alpha_1 x_1(n) + \alpha_2 x_2(n) + \cdots + \alpha_k x_k(n) = 0 \tag{3.13}$$

for all $n \geq 0$; that is, $x_1, \ldots, x_k$ are linearly dependent for $n \geq 0$.

We already know that (3.13) holds when $n = 0, \ldots, k-1$. Next take $n = k$. The functions x_i, being solutions of our homogeneous difference equation, each satisfy

$$x_i(k) + p_1(0)\, x_i(k-1) + \cdots + p_k(0)\, x_i(0) = 0.$$

Adding together $p_k(0)$ times the first equation (3.12), $p_{k-1}(0)$ times the second equation, and so on, up to $p_1(0)$ times the last equation, we obtain (3.13) for $n = k$.

Inductively, we use the fact that

$$x_i(m+k) + p_1(m)\, x_i(m+k-1) + \cdots + p_k(m)\, x_i(m) = 0$$

to show that (3.13) holds for $n = m+k$ if it holds for $n = m, \ldots, n = m+k-1$. Therefore it holds for all n, and we are done. $\square$

The attentive reader may have remarked that the same argument (starting at n_0 instead of 0) proves a slightly more general result:

Theorem 3.16. *A k-tuple of solutions $x_1, \ldots, x_k$ of an order-k homogeneous difference equation is linearly dependent for $n \geq n_0$ if and only if $C(n_0) \neq 0$.*

Example 3.17. The equation

$$a_{n+3} - 7a_{n+1} + 6a_n = 0$$

has the solutions

$$x_1(n) = 1, \quad x_2(n) = (-3)^n, \quad x_3(n) = 2^n.$$

The Casoratian for these solutions is

$$C(n) = \det \begin{bmatrix} 1 & (-3)^n & 2^n \\ 1 & (-3)^{n+1} & 2^{n+1} \\ 1 & (-3)^{n+2} & 2^{n+2} \end{bmatrix} = -20 \cdot 2^n \cdot (-3)^n,$$

so these solutions are linearly independent. □

Theorem 3.18 (Abel). *Let*

$$x(n+k) + p_1(n)\, x(n+k-1) + \cdots + p_k(n)\, x(n) = 0$$

be a homogeneous difference equation. For any k-tuple of solutions and any $n_0 \in \mathbb{N}$, the Casoratian satisfies

$$C(n_0 + 1) = (-1)^k p_k(n_0) C(n_0).$$

By induction, this gives, for any $n \geq n_0$,

$$C(n) = (-1)^{k(n-n_0)} \Big(\prod_{i=n_0}^{n-1} p_k(i) \Big) C(n_0).$$ □

Corollary. *Suppose that $p_k(n) \neq 0$ for all $n \geq n_0$. Then $C(n) \neq 0$ if and only if $C(n_0) \neq 0$.*

Example 3.19. The equation

$$x(n+2) - \frac{3n-2}{n-1}\, x(n+1) + \frac{2n}{n-1}\, x(n) = 0$$

has solutions $x_1(n) = n$ and $x_2(n) = 2^n$. We have

$$C(n) = \det \begin{bmatrix} n & 2^n \\ n+1 & 2^{n+1} \end{bmatrix}.$$

Therefore

$$C(0) = \det \begin{bmatrix} 0 & 1 \\ 1 & 2 \end{bmatrix} = -1 \neq 0,$$

and the two solutions are linearly independent. □

Definition 3.20. A k-tuple of solutions $x_1, \ldots, x_k$ of an order-k homogeneous difference equation is said to form a **fundamental set** of solutions if there is no n_0 such that $x_1, \ldots, x_k$ are linearly dependent for $n \geq n_0$.

It follows from Theorem 3.16 that the solutions $x_1, \ldots, x_k$ form a fundamental set if and only if $C(n) \neq 0$ for all $n \geq 0$.

Theorem 3.21. *The homogeneous difference equation*

$$x(n+k) + p_1(n)\, x(n+k-1) + \cdots + p_k(n)\, x(n) \;=\; 0$$

has a fundamental set of solutions if $p_k(n) \neq 0$ for all $n \geq 0$.

Proof. Let x_i be the solution of the equation with initial conditions

$$x(0) = 0, \quad \ldots, \quad x(i) = 1, \quad \ldots, \quad x(k-1) = 0$$

(see page 62 for which such a solution exists and is unique). Clearly, the matrix defining the Casoratian $C(0)$ for the k-tuple of solutions $x_0, \ldots, x_{k-1}$ is the identity, so $C(0) = 1$. By Abel's theorem it follows that $C(n) \neq 0$ for all n, so the solutions form a fundamental set. □

Now we can take advantage of the linearity of linear difference equations to prove the following:

Lemma 3.22. *If x_1, x_2 are solutions to a homogeneous linear difference equation, then $x_1 + x_2$ and $\lambda\, x_1$ are also solutions for any constant λ. More generally, any linear combination of solutions is also a solution.*

Theorem 3.23. *Let $x_1, x_2, \ldots, x_k$ be a fundamental set of solutions of the homogeneous equation. Then the general solution for the equation is given by*

$$x(n) \;=\; \sum_{i=1}^{k} \beta_i\, x_i(n), \tag{3.14}$$

where the β_i are constants.

Proof. Let x be a solution; we must find coefficients β_i making (3.14) true. Clearly we must have

$$\begin{bmatrix} x_1(0) & x_2(0) & \cdots & x_k(0) \\ x_1(1) & x_2(1) & \cdots & x_k(1) \\ \vdots & \vdots & & \vdots \\ x_1(k-1) & x_2(k-1) & \cdots & x_k(k-1) \end{bmatrix} \begin{bmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{bmatrix} = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(k-1) \end{bmatrix}.$$

The $k \times k$ matrix on the left has determinant $C(0)$, which is nonzero because we're dealing with a fundamental set of solutions. There remains to show that (3.14) is satisfied for $n \geq k$. This is done inductively, using essentially the same argument as in the proof of Theorem 3.15, except that (3.13) is replaced by (3.14). □

Solving Difference Equations

We now know that a fundamental set of solutions to a linear difference equation determines all solutions. We must still learn how to obtain such a fundamental set.

Homogeneous Equations with Constant Coefficients

Take first the homogeneous linear difference equation *with constant coefficients* given by

$$a(n+k) + p_1\, a(n+k-1) + p_2\, a(n+k-2) + \cdots + p_k\, a(n) \;=\; 0. \tag{3.15}$$

("Constant coefficients" means the p_i do not depend on n.) We assume that $p_k \neq 0$. This guarantees that the equation really is of k-th order.

To find solutions, try out the function $a_n = \lambda^n$. Plugging this into (3.15) and simplifying, we get

$$\lambda^k + p_1\lambda^{k-1} + p_2\lambda^{k-2} + \cdots + p_k \;=\; 0. \tag{3.16}$$

This is known as the **characteristic equation**. Solving this algebraic equation of degree k gives us k roots, called the **characteristic roots**. Since $p_k \neq 0$, they are all nonzero — which was assumed in deriving (3.16). Two cases must be considered.

Case 1. All roots $\lambda_1, \lambda_2, \ldots, \lambda_k$ are distinct. In this case the k functions x_i defined by $x_i(n) = \lambda_i^n$ are solutions (why?); for conciseness we denote this k-tuple of functions by $\{\lambda_1^n, \lambda_2^n, \ldots, \lambda_k^n\}$. In fact it forms a fundamental set, since their Casoratian

$$C(0) \;=\; \det \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_k \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_k^2 \\ \vdots & \vdots & & \vdots \\ \lambda_1^{k-1} & \lambda_2^{k-1} & \cdots & \lambda_k^{k-1} \end{bmatrix} \tag{3.17}$$

can be shown to equal

$$C(0) = \prod_{1 \le i < j \le k} (\lambda_i - \lambda_j) \neq 0.$$

(The determinant in (3.17) is called a **Vandermonde determinant**.) The general solution to the difference equation is then

$$x(n) \;=\; \sum_{i=1}^{k} \beta_i\, \lambda_i^n. \qquad \square$$

Example 3.24. Find the solution to the initial value problem

$$a_{n+2} = a_{n+1} + a_n$$

with initial conditions $a_1 = 1,\ a_2 = 1$.

Solution. The characteristic equation for this difference equation is

$$\lambda^2 - \lambda - 1 = 0,$$

with solutions[2]

$$\lambda_\pm = \frac{1 \pm \sqrt{5}}{2}.$$

The corresponding fundamental set of solutions is thus given by

$$\Big(\frac{1+\sqrt{5}}{2}\Big)^n \quad \text{and} \quad \Big(\frac{1-\sqrt{5}}{2}\Big)^n,$$

giving the general solution

$$a_n = \beta_1\Big(\frac{1+\sqrt{5}}{2}\Big)^n + \beta_2\Big(\frac{1-\sqrt{5}}{2}\Big)^n.$$

Using the initial conditions we get

$$\beta_1 = \frac{1}{\sqrt{5}}, \quad \beta_2 = -\frac{1}{\sqrt{5}}.$$

So the solution is

$$a_n = \frac{1}{\sqrt{5}}\Big[\Big(\frac{1+\sqrt{5}}{2}\Big)^n - \Big(\frac{1-\sqrt{5}}{2}\Big)^n\Big].$$

This is the Fibonacci sequence. If we change the initial conditions to $a_1 = 1$, $a_2 = 3$ we get the Lucas sequence:

$$L_n = \Big(\frac{1+\sqrt{5}}{2}\Big)^n + \Big(\frac{1-\sqrt{5}}{2}\Big)^n.$$

□

Case 2. The roots $\lambda_1, \lambda_2, \ldots, \lambda_r$, with $r < k$, have multiplicities $m_1, m_2, \ldots,$ m_r, adding up to k. A fundamental set of solutions to (3.15) is given by the union of the sets

$$\{\lambda_i^n, n\lambda_i^n, n^2\lambda_i^n, \ldots, n^{m_i-1}\lambda_i^n\},$$

one for each root λ_i. Using the Casoratian, one can show that these are indeed linearly independent. □

[2]The number $\lambda_+ = \frac{1}{2}(1+\sqrt{5})$ is known as the **golden ratio** and it is usually denoted by ϕ. (Notice that $\lambda_- = -1/\phi$.) Since antiquity, this number has played an important role in science, geometry, architecture, art, music, aesthetics, and philosophy. There are several good books which introduce the reader to this amazing number [25, 29, 45].

Example 3.25. Find the solution to the initial value problem

$$a_{n+3} - 7a_{n+2} + 16a_{n+1} - 12a_n = 0,$$

with the initial conditions $a_0 = 0$, $a_1 = 1$, $a_2 = 1$.

Solution. The characteristic equation for this difference equation is

$$\lambda^3 - 7\lambda^2 + 16\lambda - 12 = 0.$$

The fundamental set of solutions corresponding to the above equation is

$$\{2^n, n2^n, 3^n\}$$

and the general solution is

$$a_n = \alpha_0 2^n + \alpha_1 n 2^n + \alpha_2 3^n.$$

Using the initial conditions we get

$$\alpha_0 = 3 \quad \text{for } \alpha_1 = 2, \quad \alpha_2 = -3. \qquad \square$$

One may argue that, in the previous discussion, we pulled the solution out of a hat. We thus present a modified proof that is more explicit. We do so in the case of a second-order difference equation. The generalization for a k-th order equation is straightforward but lengthier.

Given the difference equation

$$a_{n+2} + p_1\, a_{n+1} + p_2\, a_n = 0 \quad \text{for } n = 0, 1, 2, \ldots,$$

let λ_1, λ_2 be the roots (which may be equal) of the characteristic equation

$$\lambda^2 + p_1\, \lambda + p_2 = 0.$$

We can then write the difference equation as

$$(\hat{T} - \lambda_1 \hat{I})(\hat{T} - \lambda_2 \hat{I})a_n = 0,$$

where $\hat{T}$ is the shift operator and $\hat{I}$ the identity operator. We now define a new sequence

$$b_n = (\hat{T} - \lambda_2 \hat{I})a_n \quad \text{for } n = 0, 1, 2, \ldots, \tag{3.18}$$

so that

$$(\hat{T} - \lambda_1 \hat{I})b_n = 0.$$

This last equation is very simple: it's the recursion relation of a geometric progression

$$b_{n+1} = \lambda_1\, b_n$$

with solution

$$b_n = b_0 \lambda_1^n.$$

Returning to the defining equation (3.18) of b_n, we see that a_n is a solution to

$$a_{n+1} - \lambda_2 a_n = b_0 \lambda_1^n.$$

This equation is solved in Problem 3.31 of page 73. The final solution is

$$a_n = \begin{cases} \lambda_2^n a_0 + b_0 \dfrac{\lambda_1^n - \lambda_2^n}{\lambda_1 - \lambda_2} & \text{if } \lambda_1 \neq \lambda_2, \\ \lambda_2^n a_0 + b_0 n\lambda_2^{n-1} & \text{if } \lambda_1 = \lambda_2. \end{cases}$$

By redefining the constants, we can write these solutions as

$$a_n = \begin{cases} \beta_1 \lambda_1^n + \beta_2 \lambda_2^n & \text{if } \lambda_1 \neq \lambda_2, \\ \beta_1 \lambda_2^n + \beta_2 n\lambda_2^n & \text{if } \lambda_1 = \lambda_2. \end{cases}$$

Linear Inhomogeneous Equations

Theorem 3.26. *The general solution to the inhomogeneous linear difference equation*

$$a(n+k) + p_1 a(n+k-1) + \cdots + p_k a(n) = g(n),$$

where $g(n) \not\equiv 0$, may be written as

$$a(n) = x_p(n) + \sum_{i=1}^{k} \beta_i x_i(n),$$

where $\{x_1, x_2, \ldots, x_k\}$ is a fundamental set of solutions to the corresponding homogeneous equation and x_p is a particular solution of the inhomogeneous equation.

Proof. It is straightforward to verify that $a(n)$ is a solution of the inhomogeneous equation.

Inversely, let $x_p(n)$ be a solution of the inhomogeneous equation. Then

$$x_p(n+k) + p_1 x_p(n+k-1) + \cdots + p_k x_p(n) = g(n).$$

Subtracting this from the given difference equation, we see that $a(n) - x_p(n)$ is a solution of the homogeneous equation; therefore

$$a(n) - x_p(n) = \sum_{i=1}^{k} \beta_i x_i(n). \qquad \square$$

Therefore, the solution of inhomogeneous equations is essentially making an educated guess as to what a solution x_p may be. Several methods have been developed to find such a solution. The most useful seems to be the **method of undetermined coefficients**. This method works well when $g(n)$ is of the form a^n, $\sin(bn)$, $\cos(bn)$ or n^ℓ (or some combination of them). Table 3.1 gives the form of x_p for each case of $g(n)$. Once a guess has been made, it must be plugged in and the respective coefficients must be solved for.

Forcing term $g(n)$	**Particular solution x_p**
a^n	$c\,a^n$
n^ℓ	$c_0 + c_1 n + \cdots + c_\ell n^\ell$
$a^n n^\ell$	$a^n\,(c_0 + c_1 n + \cdots + c_\ell n^\ell)$
$\sin(bn)$, $\cos(bn)$	$c_1 \sin(bn) + c_2 \cos(bn)$
$a^n \sin(bn)$, $a^n \cos(bn)$	$a^n\,[c_1 \sin(bn) + c_2 \cos(bn)]$
$a^n n^\ell \sin(bn)$, $a^n n^\ell \cos(bn)$	$a^n\big[(c_0 + c_1 n + \cdots + c_\ell n^\ell)\sin(bn) + (d_0 + d_1 n + \cdots + d_\ell n^\ell)\cos(bn)\big]$

Table 3.1: Most common forcing terms and corresponding particular solutions.

Example 3.27. Find the general solution to the difference equation

$$a_{n+2} + a_{n+1} - 12a_n = n\,2^n.$$

Solution. Using the technique of the previous section, the solution to the homogeneous equation is

$$\beta_1\,3^n + \beta_2\,(-4)^n.$$

Because the forcing term is of the form $b^n n^\ell$, with $\ell = 1$, we guess that $x_p(n)$ has the form $c_1 2^n + c_2 n 2^n$. Plugging this into the equation and solving for the constants, we get

$$x_p(n) = -\tfrac{5}{18}\,2^n - \tfrac{1}{6}\,n2^n.$$

Thus the general solution is

$$a_n = \beta_1 3^n + \beta_2(-4)^n - \tfrac{5}{18}\,2^n - \tfrac{1}{6}\,n2^n. \qquad \square$$

3.4 Solved Problems

Problem 3.28 (UK 1996; Estonia 1997; Greece 1999). *A function defined on the positive integers satisfies*

(a) $f(1) = 1996$,

(b) $f(1) + \cdots + f(n) = n^2 f(n), \quad n > 1$.

Find $f(1996)$.

Solution. In condition (b), substituting $n-1$ in place of n we find

$$f(1) + \cdots + f(n-1) \; = \; (n-1)^2 f(n-1).$$

Then subtracting this equation from the given one, we find a first-order difference equation

$$f(n) \; = \; n^2 f(n) - (n-1)^2 f(n-1),$$

which implies

$$f(n) \; = \; \frac{n-1}{n+1}\, f(n-1).$$

By iteration

$$f(n) \; = \; \frac{n-1}{n+1}\,\frac{n-2}{n}\,\frac{n-3}{n-1} \;\cdots\; \frac{1}{3}\, f(1) \; = \; \frac{2}{n(n+1)}\, f(1).$$

Therefore

$$f(1996) \; = \; \frac{2}{1996 \cdot 1997}\, 1996 \; = \; \frac{2}{1997}.$$

□

Problem 3.29. *The function* F *is defined in the set of natural numbers* $\mathbb{N}^*$ *and satisfies the conditions*

(a) $F(n) = F(n-1) + a^n$,

(b) $F(1) = 1$.

Find $F(n)$.

Solution. In condition (a), in place of n we substitute $2, 3, \ldots, n-1, n$:

$$\begin{aligned} F(2) &= F(1) + a^2 \\ F(3) &= F(2) + a^3 \\ &\cdots \\ F(n-1) &= F(n-2) + a^{n-1} \\ F(n) &= F(n-1) + a^n \end{aligned}$$

Adding these equations we find

$$F(n) \; = \; F(1) + \sum_{k=2}^{n} a^k.$$

If $a \neq 1$, then

$$F(n) = 1 + \frac{a^2(a^{n-1} - 1)}{a - 1}.$$

If $a = 1$, then

$$F(n) = n. \qquad \square$$

Problem 3.30. *A sequence of real numbers a_n satisfying the condition*

$$a_n = \lambda\, a_{n-1} + \omega \quad \text{for } n = 0, 1, 2, \ldots,$$

with $\lambda \neq 0, 1$ and $\omega \neq 0$, is given. Find the expression of a_n in terms of a_0.

Comment. This sequence is known as a **mixed progression** with ratio λ and step ω.

Solution. In the given condition, replace n in turn by 1, 2, 3, …, $n-1$, n:

$$\begin{aligned} a_1 &= \lambda\, a_0 + \omega \\ a_2 &= \lambda\, a_1 + \omega \\ a_3 &= \lambda\, a_2 + \omega \\ &\cdots \\ a_{n-1} &= \lambda\, a_{n-2} + \omega \\ a_n &= \lambda\, a_{n-1} + \omega \end{aligned}$$

Multiply these equations by λ^{n-1}, λ^{n-2}, …, λ, 1 respectively and add the resulting equations to find

$$a_n = \lambda^n\, a_0 + \omega \sum_{k=0}^{n-1} \lambda^k,$$

or

$$a_n = \lambda^n\, a_0 + \omega\, \frac{\lambda^n - 1}{\lambda - 1}. \qquad \square$$

Problem 3.31. *The sequence of real numbers a_n satisfying the condition*

$$a_n = \lambda\, a_{n-1} + \omega\, \mu^{n-1} \quad \text{for } n = 0, 1, 2, \ldots,$$

with $\lambda \neq 0$, generalizes the mixed progression of the previous problem. Find the expression of a_n in terms of a_0.

Solution. In the given condition, replace n in turn by 1, 2, 3, ..., $n-1$, n:

$$\begin{aligned}
a_1 &= \lambda\, a_0 + \omega \\
a_2 &= \lambda\, a_1 + \omega\, \mu \\
a_3 &= \lambda\, a_2 + \omega\, \mu^2 \\
&\cdots \\
a_{n-1} &= \lambda\, a_{n-2} + \omega\, \mu^{n-2} \\
a_n &= \lambda\, a_{n-1} + \omega\, \mu^{n-1}
\end{aligned}$$

Multiply these equations by λ^{n-1}, λ^{n-2}, λ^{n-1}, ..., λ, 1 respectively and add the resulting equations to find:

$$a_n = \lambda^n\, a_0 + \frac{\omega}{\lambda}\mu^n \sum_{k=1}^{n} \left(\frac{\lambda}{\mu}\right)^k .$$

If $\mu = \lambda$,

$$a_n = \lambda^n\, a_0 + \omega\, n \lambda^{n-1},$$

whereas if $\mu \neq \lambda$,

$$a_n = \lambda^n\, a_0 + \omega\, \frac{\lambda^n - \mu^n}{\lambda - \mu}. \qquad \square$$

Problem 3.32. *A sequence of real numbers a_n satisfying the condition*

$$a_n = \lambda\, x^{a_{n-1}} + \omega,$$

where $x > 0$ and $\lambda \neq 0$, is given. Find the expression of a_n in terms of a_0.

Solution. The given recursion relation may be written in the form

$$a_n - \omega = \lambda x^{\omega}\, x^{a_{n-1}-\omega}.$$

To simplify the notation we introduce

$$\beta_n = a_n - \omega \quad \text{and} \quad \mu = \lambda x^{\omega}.$$

Then

$$\beta_n = \mu\, x^{\beta_{n-1}}.$$

If $\mu = 1$ by iteration we can easily find

$$\beta_n = x^{x^{\cdot^{\cdot^{\cdot^{x^{\beta_0}}}}}},$$

where x appears n times. If $\mu \neq 1$, then we write

$$\frac{\beta_n}{\mu} = (x^\mu)^{\beta_{n-1}/\mu},$$

and we define

$$\gamma_n = \frac{\beta_n}{\mu} \quad \text{and} \quad y = x^\mu.$$

Therefore

$$\gamma_n = y^{\gamma_{n-1}}$$

with solution

$$\gamma_n = y^{y^{\cdot^{\cdot^{\cdot^{y^{\gamma_0}}}}}},$$

where y appears n times. This can be rewritten for the original sequence a_n:

$$a_n = \omega + \lambda x^\omega \, y^{y^{\cdot^{\cdot^{\cdot^{y^{\frac{a_0-\omega}{\lambda x^\omega}}}}}}} \qquad \text{for } y = x^{\lambda x^\omega}.$$

□

Problem 3.33 (IMO 1976). *A sequence $\{u_n\}$ is defined by*

$$u_0 = 2, \quad u_1 = \tfrac{5}{2}, \quad u_{n+1} = u_n\,(u_{n-1}^2 - 2) - u_1 \quad \textit{for } n \geq 1, 2, \ldots.$$

Prove that for positive integers n,

$$\lfloor u_n \rfloor = 2^{\lfloor 2^n - (-1)^n \rfloor / 3},$$

where $\lfloor x \rfloor$ denotes the integer part of x.

Solution. Setting $n = 1, 2$ in the given recursion relation we find $u_2 = \frac{5}{2}$ and $u_3 = \frac{65}{8}$. Then we notice that

$$\begin{aligned} u_0 &= 2 = 2^0 + 2^{-0}, \\ u_1 &= \tfrac{5}{2} = 2^1 + 2^{-1}, \\ u_2 &= \tfrac{5}{2} = 2^1 + 2^{-1}, \\ u_3 &= \tfrac{65}{8} = 2^3 + 2^{-3}. \end{aligned}$$

We thus suspect that u_n can be written in the form

$$u_n = 2^{a_n} + 2^{-a_n},$$

for some sequence $\{a_n\}$. Even more, it appears that the first term will be an integer and the second term fractional, implying that the integer part of u_n is solely based on its first term. We thus come to believe that[3]

$$a_n = \frac{2^n - (-1)^n}{3},$$

[3] Notice that 3 always divides $2^n - (-1)^n$. You may try to show this formally if it is not evident or known to you.

which we will prove by induction.

The statement is true for the first terms, as we have seen. Let it be true for all terms up to some index k. Then if

$$u_{k+1} = 2^{a_{k+1}} + 2^{-a_{k+1}} \quad \text{for } a_{k+1} = \frac{2^{k+1} - (-1)^{k+1}}{3},$$

we will show that u_{k+1} satisfies the recursion relation of the problem. First we see that

$$\begin{aligned} u_k\,(u_{k-1}^2 - 2) &= \left(2^{a_k} + 2^{-a_k}\right)\,\left[\left(2^{a_{k-1}} + 2^{-a_{k-1}}\right)^2 - 2\right] \\ &= \left(2^{a_k} + 2^{-a_k}\right)\,\left(2^{2a_{k-1}} + 2^{-2a_{k-1}}\right) \\ &= 2^{a_k + 2a_{k-1}} + 2^{-(a_k + 2a_{k-1})} + 2^{a_k - 2a_{k-1}} + 2^{-(a_k - 2a_{k-1})}. \end{aligned}$$

Then

$$\begin{aligned} a_k + 2a_{k-1} &= \frac{2^k - (-1)^k}{3} + 2\,\frac{2^{k-1} - (-1)^{k-1}}{3} \\ &= \frac{2^k + (-1)^{k+1}}{3} + \frac{2^k - 2\,(-1)^{k+1}}{3} = \frac{2^{k+1} - (-1)^{k+1}}{3} = a_{k+1}, \end{aligned}$$

and

$$\begin{aligned} a_k - 2a_{k-1} &= \frac{2^k - (-1)^k}{3} - 2\,\frac{2^{k-1} - (-1)^{k-1}}{3} \\ &= \frac{2^k + (-1)^{k+1}}{3} - \frac{2^k - 2\,(-1)^{k+1}}{3} = (-1)^{k+1}. \end{aligned}$$

Therefore

$$u_k\,(u_{k-1}^2 - 2) = 2^{a_{k+1}} + 2^{-a_{k+1}} + 2^{(-1)^{k+1}} + 2^{-(-1)^{k+1}}.$$

Notice that the last two terms are 2^1 or 2^{-1} for all k, and their sum is u_1. We thus conclude that

$$u_k\,(u_{k-1}^2 - 2) = u_{k+1} + u_1,$$

and this achieves the proof. $\square$

Chapter 4

Equations Reducing to Algebraic Systems

In this chapter we study functional equations and systems of functional equations that can be solved by a reduction to a set of algebraic equations.

4.1 Solved Problems

Problem 4.1. *Find an even function $f:\mathbb{R}\to\mathbb{R}$ and an odd function $g:\mathbb{R}\to\mathbb{R}$ such that $2^x = f(x)+g(x)$ for all $x\in\mathbb{R}$.*

Solution. Replacing x by $-x$ in the given equation, we get $2^{-x} = f(x)-g(x)$. Solving the system of this equation and the original one, we obtain

$$\begin{aligned} f(x) &= \frac{2^x+2^{-x}}{2} = \cosh(x\ln 2),\\ g(x) &= \frac{2^x-2^{-x}}{2} = \sinh(x\ln 2). \end{aligned}$$ □

Problem 4.2. *Find $f:\mathbb{R}^*\to\mathbb{R}$ such that*

$$f(x)+2f\left(\frac{1}{x}\right) = x \quad \textit{for all } x\in\mathbb{R}^*.$$

Solution. In the defining equation we set $y = 1/x$:

$$f\Big(\frac{1}{y}\Big)+2f(y) = \frac{1}{y}.$$

This and the given equation give the system of equations

$$\begin{aligned} f(x)+2f\Big(\frac{1}{x}\Big) &= x,\\ f\Big(\frac{1}{x}\Big)+2f(x) &= \frac{1}{x}. \end{aligned}$$

This can be solved easily to give

$$f(x) = \frac{2-x^2}{3x}. \qquad \square$$

Problem 4.3. *Find* $f : \mathbb{R} \setminus \{0,1\} \to \mathbb{R}$ *such that*

$$f\Big(\frac{x}{x-1}\Big) - 2\,f\Big(\frac{x-1}{x}\Big) = 0 \quad \textit{for all } x \in \mathbb{R} \setminus \{0,1\}.$$

Solution. In the defining equation we set $y = x/(x-1)$. Then

$$f(y) - 2\,f\Big(\frac{1}{y}\Big) = 0.$$

Now we set $t = 1/y$:

$$f\Big(\frac{1}{t}\Big) - 2\,f(t) = 0.$$

The last two equations give

$$f(t) \equiv 0. \qquad \square$$

Problem 4.4. *Find* $f : \mathbb{R} \setminus \{0,1\} \to \mathbb{R}$ *such that*

$$f(x) + f\Big(\frac{1}{1-x}\Big) = x \quad \textit{for all } x \in \mathbb{R} \setminus \{0,1\}. \tag{4.1}$$

Solution. In the defining equation (4.1) we set $y = 1/(1-x)$, which is equivalent to $x = (y-1)/y$. Then

$$f\Big(\frac{y-1}{y}\Big) + f(y) = \frac{y-1}{y}.$$

From this equation and the given one (4.1), we find

$$f\Big(\frac{1}{1-x}\Big) - f\Big(\frac{x-1}{x}\Big) = x - \frac{x-1}{x}. \tag{4.2}$$

In (4.2) we again set $y = 1/(1-x)$:

$$f(y) - f\Big(\frac{1}{1-y}\Big) = \frac{y-1}{y} - \frac{1}{y-1}. \tag{4.3}$$

Equations (4.1) and (4.3) can easily be solved for $f(x)$ to give

$$f(x) = \frac{x^3 - x + 1}{2x^2 - 2x}. \qquad \square$$

Problem 4.5 (Putnam 1959). *Find all $f : \mathbb{C} \to \mathbb{C}$ such that*

$$f(z) + z\, f(1 - z) = 1 + z \quad \textit{for all } z \in \mathbb{C}. \tag{4.4}$$

Solution. In the defining equation (4.4) we set $w = 1 - z$, or equivalently $z = 1 - w$. Then

$$f(1 - w) + (1 - w)\, f(w) = 2 - w. \tag{4.5}$$

From (4.4) and (4.5) we find easily

$$(1 - z + z^2)\, f(z) = (1 - z + z^2). \tag{4.6}$$

Let $\rho_\pm = \frac{1}{2}(1 \pm i\sqrt{3})$ be the roots of the polynomial $1 - z + z^2$. We can solve (4.6) provided $1 - z + z^2 \neq 0$:

$$f(z) = 1 \quad \text{for all } z \in \mathbb{C} \setminus \{\rho_+, \rho_-\}.$$

For $z = \rho_\pm$, equation (4.6) is an identity. However, the defining relation (4.4) gives

$$\begin{aligned} f(\rho_+) + \rho_+\, f(1 - \rho_+) &= 1 + \rho_+, \\ f(\rho_-) + \rho_-\, f(1 - \rho_-) &= 1 + \rho_-. \end{aligned}$$

Only one of these two equations is independent. This can be seen by (for example) multiplying the first by ρ_- and observing that $\rho_+ + \rho_- = 1$ and $\rho_+\rho_- = 1$; we recover the second equation. So, the value of f at one of the $\rho_\pm$ is arbitrary. Let's say

$$f(\rho_-) = z_0.$$

Then we find

$$f(\rho_+) = 1 + \rho_+\, (1 - z_0). \qquad \square$$

Problem 4.6. *Let $f : \mathbb{R} \to \mathbb{R}$ be such that*

$$f(x + y) + f(x - y) = 2\, f(x)\, \cos y \quad \textit{for all } x, y \in \mathbb{R}. \tag{4.7}$$

Find all functions f.

Solution. Setting $x = 0$ and $y = \theta$ in the defining relation, we find

$$f(\theta) + f(-\theta) \;=\; 2a\,\cos\theta,$$

where $a = f(0)$.

Setting $x = \theta + \pi/2$ and $y = \pi/2$ in the defining relation, we find

$$f(\theta+\pi) + f(\theta) \;=\; 0.$$

Finally, setting $x = \pi/2$ and $y = \theta + \pi/2$ in the defining relation, we find

$$f(\theta+\pi) + f(-\theta) \;=\; -2b\,\sin\theta,$$

where $b = f(\pi/2)$.

We have thus derived three equations for the three unknowns $f(\theta)$, $f(-\theta)$, $f(\theta+\pi)$. The solution is easily found to be

$$f(\theta) \;=\; a\,\cos\theta + b\,\sin\theta. \qquad \square$$

Question. *Let* $f : \mathbb{R} \to \mathbb{R}$ *be such that*

$$f(x+y) + f(x-y) \;=\; 2\,f(x)\,\cosh y \quad \textit{for all } x, y \in \mathbb{R}. \tag{4.8}$$

Why doesn't the method above work for finding f*?*

Answer. With the substitution used in the solution of (4.7), we were able to find a system of three equations for three unknowns $f(\theta)$, $f(-\theta)$, $f(\theta+\pi)$. In particular, the substitution $y = \pi/2$ in (4.7) causes the right-hand side to vanish. This side would contain $f(\theta+\pi/2)$, a fourth unknown, otherwise. However, the modified equation contains the hyperbolic cosine, which does not vanish for any real value of its argument.

Solution of (4.8). To solve (4.8) one can extend the domain of the definition of f from $\mathbb{R}$ to $\mathbb{C}$ by *analytic continuation* and define $f : \mathbb{C} \to \mathbb{C}$ such that

$$f(z+w) + f(z-w) \;=\; 2\,f(z)\,\cosh w \quad \text{for all } z, w \in \mathbb{C}.$$

Setting $(z,w) = (0, it)$, $(z,w) = (i(t+\pi/2), i\pi/2)$, $(z,w) = (i\pi/2, i(t+\pi/2))$ successively, we find

$$\begin{aligned} f(it) + f(-it) &= 2a\,\cos t,\\ f(i(t+\pi)) + f(-it) &= 2a\,\cos t,\\ f(i(t+\pi)) + f(-it) &= -2b\,\sin t, \end{aligned}$$

where $a = f(0)$, $b = f(i\pi/2)$.

We have thus, again, derived three equations for the three unknowns $f(it)$, $f(-it)$, $f(i(t+\pi))$. The solution is easily found to be

$$f(it) = a\cos t + b\sin t.$$

Returning to real values $\theta = it$:

$$f(\theta) = a\cosh\theta + ib\sinh\theta.$$

The function f will be real-valued if a is real and b is imaginary. $\square$

Problem 4.7 ([1], Problem 11030, August/September 2005)**.** *Show that for $d < -1$ there are exactly two real-valued functions f such that*

$$f(x+y) - f(x)f(y) = d\sin x\sin y \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. Letting $x = 0 = y$ we get $f(0)(1 - f(0)) = 0$. Therefore, $f(0)$ can be 0 or 1. If $f(0) = 0$, then by setting $y = 0$ we get $f(x) = 0$ for all $x \in \mathbb{R}$. But $f(x) \equiv 0$ is clearly not a solution; thus $f(0) = 1$.

Replacing (x, y) by (t, t), $(t, -t)$, $(2t, -t)$ sequentially, we obtain the system

$$\begin{aligned} f(2t) &= (f(t))^2 + d\sin^2 t, \\ 1 &= f(t)f(-t) - d\sin^2 t, \\ f(t) &= f(2t)f(-t) - 2d\sin^2 t\cos t. \end{aligned}$$

Eliminating $f(-t)$ and $f(2t)$, we get

$$d\sin^2 t\left(f(t)^2 - 2\cos t\, f(t) + (1 + d\sin^2 t)\right) = 0.$$

If $t \neq n\pi$, with n an integer, then $f(t) = \cos t \pm \sqrt{-d-1}\sin t$ by the quadratic formula.

It remains only to find the values $f(n\pi)$. Setting $t = \pi/2$ in the first equation of the system we get

$$f(\pi) = f(\pi/2)^2 + d = -d - 1 + d = -1.$$

Setting $t = \pi$ in the second equation of the system, we get $f(-\pi) = -1$. Finally, inductively from the given functional equation, we can easily obtain that $f(2k\pi) = 1$ and $f((2k+1)\pi) = -1$ for $k \in \mathbb{Z}$. $\square$

Problem 4.8. *Find $f, g : \mathbb{R} \setminus \{1\} \to \mathbb{R}$ such that*

$$f(2x+1) + 2g(2x+1) = 2x, \tag{4.9}$$

$$f\left(\frac{x}{x-1}\right) + g\left(\frac{x}{x-1}\right) = x. \tag{4.10}$$

Solution. In (4.9) we set $t = 2x + 1$:

$$f(t) + 2\,g(t) \;=\; t - 1.$$

In (4.10) we set $t = x/(x-1)$:

$$f(t) + g(t) \;=\; \frac{t}{t-1}.$$

The last two equations give

$$f(x) \;=\; \frac{x^2 - 4x + 1}{-x + 1}, \quad g(x) \;=\; \frac{x^2 - 3x + 1}{x - 1}. \qquad \square$$

4.2 Group Theory in Functional Equations

Group theory is an essential tool for many branches of mathematics. In this section we shall present the basic ideas and then how they can be used in the solution of some functional equations. To motivate what will be presented in this section, let's solve again Problem 4.4.

Problem 4.9. *Find* $f : \mathbb{R} \setminus \{0, 1\} \to \mathbb{R}$ *such that*

$$f(x) + f\Big(\frac{1}{1-x}\Big) \;=\; x \quad \textit{for all } x \in \mathbb{R} \setminus \{0, 1\}.$$

Solution. If we let $g(x) = 1/(1-x)$, then $g^2(x) = (g \circ g)(x) = (x-1)/x$, and $g^3(x) = x$. The defining equation (4.1), together with the substitutions of x by $g(x)$ and $g^2(x)$, yield the system

$$\begin{aligned} f(x) + f(g(x)) &= x, \\ f(g(x)) + f(g^2(x)) &= g(x), \\ f(g^2(x)) + f(x) &= g^2(x). \end{aligned}$$

Solving the system, we get

$$f(x) \;=\; \frac{x - g(x) + g^2(x)}{2} \;=\; \frac{x^3 - x + 1}{2x^2 - 2x}. \qquad \square$$

So what's important about this solution? Consider the set $G = \{g, g^2, g^3\}$. When this set is equipped with the operation of function composition, some interesting properties emerge. Composition of any three functions satisfies the associative property, $f \circ (h \circ k) = (f \circ h) \circ k$. The function g^3 is the identity element — that is, when composed with any function f, it does not change the function ($g^3 \circ f = f \circ g^3 = f$). Finally, for any function f, there is another function — the inverse function f^{-1} defined in Chapter 1 — such that their composition is the identity function: for $f = g$ we have $f^{-1} = g^2$; for $f = g^2$ we have $f^{-1} = g$; and $f = g^3$, being the identity, is its own inverse. To describe the structure that has been revealed in G, we say that G equipped with the operation of composition is a *group*.

All of this should strike a familiar chord. Recall the set of integers $\mathbb{Z}$. As a set it is not a very interesting object, since there is no relation among its elements. However, when this set is equipped with the operation of addition, interesting properties (as above) emerge. Addition of integers satisfies the associative property

$$m + (n + l) = (m + n) + l;$$

the element 0, when added to any integer n, does not change the integer: $0+n = n+0 = n$; and, for any integer n, there is another integer (traditionally denoted by $-n$) such that their sum is 0: $n+(-n) = (-n)+n = 0$. $\mathbb{Z}$ equipped with the operation of addition is a group. Moreover, for any two integers m and n, $m + n = n + m$. We say that the operation of addition is commutative and that $\mathbb{Z}$ is an *abelian group*.

We will now present some general ideas about groups and use them to solve functional equations. Simple group-theoretic ideas will also be met in some of the later chapters.

Elementary Group Theory

Group theory is an important subject in mathematics due to its many implications and applications. Some of its most important applications are in physics, where it is used to describe symmetries. In this section we show how one may use group theory as a tool to aid us in finding solutions to certain functional equations.

Definition 4.10. Let G be a set and let $*$ be a **binary operation** on G that is **closed**: in other words, a map $* : G \times G \to G$, the image of (x, y) being denoted by $x * y$.

We say that G, endowed with the operation $*$,[1] is a **group** if the following axioms are satisfied:

1. *Associativity*: $x * (y * z) = (x * y) * z$ for all $x, y, z \in G$;
2. *Neutral element*: there exists an $e \in G$ such that $e * x = x * e = x$ for all $x \in G$.
3. *Symmetric elements*: for each $x \in G$, there exists a $x' \in G$ such that $x' * x = x * x' = e$.

[1] This means G is considered together with the operation $*$. The operation is part of the group structure.

If $*$ satisfies in addition the property

4. *Commutativity*: $x * y = y * x$, for $x, y \in G$,

the group is called **abelian** or **commutative**.

The rational numbers apart from 0 provide another example of an abelian group, if we take for $*$ the operation of multiplication. (This is discussed in more detail in Example 4.12 below.) In this case it is traditional to denote the operation by $\times$, or simple juxtaposition.

Another example of a group is the set of bijective functions from a set S to itself, with the operation of composition. The set G discussed right after Problem 4.9 is a **subgroup** of the group of bijections of $\mathbb{R} \setminus \{0, 1\}$ with the operation of composition.

In general, we tend to think of some group operations as being "like addition" and others as being "like multiplication". For an additive operation we call the neutral element *zero* and symmetric elements with respect to zero are called *opposites*; the result of the operation on (x, x) is written as $2x$, and so on. For a multiplicative operation we call the neutral element *unit* and symmetric elements with respect to unity are called *inverses*; the result of the operation on (x, x) is written as x^2, and so on.

Noncommutative group operations are conventionally thought of as multiplicative. (The converse is not always true: the group $G = \{g, g^2, g^3\}$ is written multiplicatively, and is commutative; likewise the product of nonzero rational numbers.)

Example 4.11 (The single element group). The group consisting of the identity element alone is a group. Since we have $G = \{e\}$ and $e * e = e$, it is closed under $*$ and e is its own inverse, while associativity and the neutral element axioms are satisfied trivially. Thus G is a group. □

Example 4.12 (The multiplicative rational numbers). The set of integers together with multiplication cannot be a group since the inverse of $a \in \mathbb{Z}$ is $1/a$, which is not an integer. However, the entire set of all rational numbers, not including zero, of the form a/b, where $a, b \in \mathbb{Z}$ with multiplication, do form a group. It is clear that closure is satisfied since

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

for $a, b, c, d \in \mathbb{Z}$. Since multiplication is associative for integers it must still be associative for rationals. The identity axiom is clearly satisfied by 1. Lastly, the inverse of $a/b \in \mathbb{Q}$ is obviously b/a. We can also see that since multiplication is commutative for integers it is also commutative for the rationals. Thus the rationals together with multiplication form an abelian group. □

Example 4.13 (Matrices). Readers familiar with matrices will recognize that the set of matrices of a given shape, say $n \times m$, forms a group under addition. Matrix multiplication of square matrices is also an important operation, but (like multiplication of rational numbers) it fails to be a group operation unless we throw out elements that don't have an inverse. In matrix theory we say a matrix is **singular** if its determinant is zero. If A and B are nonsingular matrices ($\det A \neq 0$ and $\det B \neq 0$), then the product AB is also nonsingular, since $\det AB = \det A \det B$. Matrix multiplication is associative by definition (though not commutative), and the identity matrix I, of determinant 1, works as the neutral element. Finally, a matrix is invertible if and only if it is nonsingular. Thus the set of nonsingular square matrices of a given size is a group under matrix multiplication. □

Example 4.14 (Cyclic groups). A special type of group that is of interest to functional equations is the cyclic groups. Cyclic groups are groups entirely made of repeated products of a single element; i.e., there exists an element $h \in G$ such that each element can be expressed as h^i, where $i \in \mathbb{N}$. We can see that if there are n elements in the group, then h^n must be the identity. So the set G is $G = \{h, h^2, h^3, \dots, h^{n-1}, h^n = e\}$. We often use $\mathbb{Z}_n$ to denote a cyclic group with n elements. □

Definition 4.15. We say that the set $A = \{a_1 \, a_2, \dots, a_n\}$ **generates** the group G with operation $*$ if every element g of G can be written in the form $g = a_1^{m_1} * a_2^{m_2} * \cdots * a_n^{m_n}$, where $m_1, m_2, \dots, m_n$ are integers. For example, the complex number $i = \sqrt{-1}$ generates the cyclic group $\{i, -1, -i, 1\}$.

Definition 4.16. The **order** of an element $g \in G$, if it exists, is the smallest natural number n such that $g^n = e$. The **order** of a group G, if it is finite, is the number of the elements in G.

Each element of a finite group G has a finite order. If G is a cyclic group with n elements, its order is n (by definition) and the order of any element of G divides n.

Definition 4.17. A **subgroup** H of a group $(G, *)$ is a subset of G which is also a group under the operation $*$.

Every group G has at least two trivial subgroups: the entire group G and the single element group $\{e\}$.

Theorem 4.18 (Lagrange). *For a finite group G, the order of a subgroup H divides the order of the group.*

The ratio of the two orders is the **index** of H in G; it is denoted by $[G : H]$.

Group tables are a way to represent a finite group and its structure. The rows and columns are labeled by elements in the group: say the i-th row corresponds to x_i and the j-th column corresponds to x_j. The (i, j)-th entry of the table is then $x_i x_j$. A group is abelian if its table is symmetric about the diagonal, that, is, if for each pair i, j the (i, j)-th entry of the table is equal to the (j, i)-th entry. A simple example is the group $G = \{-1, 1\}$ under ordinary multiplication:

$\cdot$	-1	1
-1	1	-1
1	-1	1

Before we finish this section, let's look at the Putnam 1961 Problem 1.2 and the IMO 1973 Problem 1.21 again and make some connections with group theory.

In Problem 1.2, we equip the set $P(\Omega)$ with the binary operation

$$A \ominus B = (A \setminus B) \cup (B \setminus A),$$

known as the **symmetric difference** of the sets A and B. It can also be written as

$$A \ominus B = (A \cup B) \setminus (A \cap B) = (A \cap B^{\complement}) \cup (A^{\complement} \cap B)$$

It is easy to show that the operation $\ominus$ is associative, it has a neutral element (the $\varnothing$) and every set has a symmetric element (itself). That is, $(P(\Omega), \ominus)$ is a group. Then, the conditions for forming Σ require that it is a subgroup when equipped with the same operation. By Lagrange's theorem, the order k of Σ must divide the order 2^n of $P(\Omega)$. Therefore k must be a power[2] of 2. Incidentally, notice that, since $A \ominus A = \varnothing$ for all A, every element in $P(\Omega)$ has order 2.

The set G defined in Problem 1.21, equipped with the operation of composition of functions, is easily seen to be a group. The element $f \circ g \circ f^{-1} \circ g^{-1}$ that was used in the solution of the problem is called the **commutator** of f and g and it is often denoted by $[f, g]$. If a group is abelian, then all elements commute and all commutators reduce to the neutral element. Therefore, commutators measure the deviation of a group from being abelian. We can make this concept more precise as follows:

We take all possible commutators of a group and form a set. In general, this set is not a subgroup of G. However, like any subset of G, it generates a subgroup of G. The subgroup of G generated by all commutators is denoted by $[G, G]$ and is called the **commutator subgroup** or **derived subgroup**.

[2]Note that the proof presented for the allowed values of k does not necessary imply that k actually takes these values. To do this a constructive proof is necessary (such as the one presented in Problem 1.2).

Theorem 4.19. *A group is abelian if and only if its commutator subgroup is the single element group* $\{e\}$.

Condition (c) of Problem 1.21 required that G be abelian. This is obvious from the commutator $[f, g]$, which is the identity function, or it can be seen explicitly by a simple calculation:

$$\begin{aligned} f \circ g(x) &= A\,x + B, \\ g \circ f(x) &= A\,x + C, \end{aligned}$$

where $A = aa'$, $B = ab' + b$, $C = a'b + b'$. Condition (c) imposes $B = C$. (Recall the solution of Problem 1.21.)

Using Group Theory to Reduce Functional Equations to Algebraic Systems

If we have a set $G = \{g_1, \ldots, g_n\}$ of functions from a subset of $\mathbb{C}$ to $\mathbb{C}$ which becomes a group when it is endowed with the operation of composition of functions, then we can ask what are the solutions $f(z)$ of the functional equation

$$a_1(z)\,f(g_1(z)) + a_2(z)\,f(g_2(z)) + \cdots + a_n(z)\,f(g_n(z)) = b(z),$$

where $a_1(z), \ldots, a_n(z)$, and $b(z)$ are some given functions.

In such a case the solution to the functional equation can be found easily as follows. Substituting z in the defining equation by $g_1(z), \ldots, g_n(z)$, we obtain a system of n equations with the n unknowns being $f(g_1(z)), \ldots, f(g_n(z))$. We can then solve the system for $f(z)$ at all the values of z where the coefficient matrix is nonsingular. At the points where the coefficient matrix is singular, additional work is necessary. This latter comment is better understood by looking again at the Putnam 1959 problem.

To avoid cumbersome notation, let's take G to be a cyclic group of order n. Then

$$G = \{e, g, g^2, \ldots, g^{n-1}\}$$

with $g^n = e$. Then the algebraic system found after the substitution of z by $g(z), g^2(z), \ldots, g^{n-1}(z)$ is

$$\begin{aligned} a_1(g(z))\,f(g(z)) + a_2(g(z))\,f(g^2(z)) + \cdots + a_n(g(z))\,f(z) &= b(g(z)), \\ a_1(g^2(z))\,f(g^2(z)) + a_2(g^2(z))\,f(g^3(z)) + \cdots + a_n(g^2(z))\,f(g(z)) &= b(g^2(z)), \\ \cdots \quad \cdots \quad & \cdots \\ a_1(g^{n-1}(z))f(g^{n-1}(z)) + a_2(g^{n-1}(z))\,f(z) + \cdots + a_n(g^{n-1}(z))f(g^{n-2}(z)) &= b(g^{n-1}(z)), \\ a_1(z)\,f(z) + a_2(z)\,f(g(z)) + \cdots + a_n(z)\,f(g^{n-1}(z)) &= b(z), \end{aligned}$$

or, after rearranging the terms,

$$\begin{aligned} a_n(g(z))\, f(z) + a_1(g(z))\, f(g(z)) + \cdots + a_{n-1}(g(z))\, f(g^{n-1}(z)) &= b(g(z)), \\ a_{n-1}(g^2(z))\, f(z) + a_n(g^2(z))\, f(g(z)) + \cdots + a_{n-2}(g^2(z))\, f(g^{n-1}(z)) &= b(g^2(z)), \\ \cdots \quad \cdots \quad & \cdots \\ a_2(g^{n-1}(z))\, f(z) + a_3(g^{n-1}(z))\, f(g(z)) + \cdots + a_1(g^{n-1}(z))\, f(g^{n-1}(z)) &= b(g^{n-1}(z)), \\ a_1(z)\, f(z) + a_2(z)\, f(g(z)) + \cdots + a_n(z)\, f(g^{n-1}(z)) &= b(z). \end{aligned}$$

We can then solve the system for $f(z)$ at all values of z where the matrix

$$A(z) = \begin{bmatrix} a_n(g(z)) & a_1(g(z)) & \dots & a_{n-1}(g(z)) \\ a_{n-1}(g^2(z)) & a_n(g^2(z)) & \dots & a_{n-2}(g^2(z)) \\ \vdots & \vdots & \ddots & \vdots \\ a_2(g^{n-1}(z)) & a_3(g^{n-1}(z)) & \dots & a_n(g^{n-1}(z)) \\ a_1(z) & a_2(z) & \dots & a_n(z) \end{bmatrix}$$

satisfies $\det A \neq 0$. Then

$$f(z) = \frac{\det B(z)}{\det A(z)},$$

where

$$B(z) = \begin{bmatrix} b(g(z)) & a_1(g(z)) & \dots & a_{n-1}(g(z)) \\ b(g^2(z)) & a_n(g^2(z)) & \dots & a_{n-2}(g^2(z)) \\ \vdots & \vdots & \vdots & \vdots \\ b(g^{n-1}(z)) & a_3(g^{n-1}(z)) & \dots & a_n(g^{n-1}(z)) \\ b(z) & a_2(z) & \dots & a_n(z) \end{bmatrix}.$$

In general, the algebraic and computational work involved in solving such systems can be very demanding for larger n and nonconstant coefficients.

Having understood the ideas presented in this section, the reader may want to return to some of the functional equations of the previous section and rethink them in the new language. Below, we illustrate the method with an additional example in which the underlying group has order 6 but is not cyclic.

Problem 4.20. *Find all functions $f : \mathbb{C} \setminus \{0,1\} \to \mathbb{C}$ such that*

$$f(z) + 2f\Big(\frac{1}{z}\Big) + 3f\Big(\frac{z}{z-1}\Big) = z.$$

Solution. Let

$$h(z) = \frac{1}{z}, \quad g(z) = \frac{1}{1-z}.$$

Then

$$h^2(z) = z, \quad g^2(z) = \frac{z-1}{z}, \quad g^3(z) = z.$$

Also

$$hg(z) = 1 - z, \quad gh(z) = \frac{z}{z-1},$$

where we have shortened $h \circ g$ and $g \circ h$ into hg and gh, respectively.

The reader is encouraged to verify that $\{\mathrm{id}, h, g, g^2, hg, gh\}$ is a group under composition and to construct the group table.

Substituting z by $h(z)$, $g(z)$, $g^2(z)$, $hg(z)$, and $gh(z)$ in the defining equation, we get

$$\begin{aligned}
f\Big(\frac{1}{z}\Big) + 2f(z) + 3f\Big(\frac{1}{1-z}\Big) &= \frac{1}{z},\\
f\Big(\frac{1}{1-z}\Big) + 2f(1-z) + 3f\Big(\frac{1}{z}\Big) &= \frac{1}{1-z},\\
f\Big(\frac{z-1}{z}\Big) + 2f(\frac{z}{z-1}) + 3f(1-z) &= \frac{z-1}{z},\\
f(1-z) + 2f\Big(\frac{1}{1-z}\Big) + 3f\Big(\frac{z-1}{z}\Big) &= 1-z,\\
f\Big(\frac{z}{z-1}\Big) + 2f\Big(\frac{z-1}{z}\Big) + 3f(z) &= \frac{z}{z-1}.
\end{aligned}$$

These, together with the given equation, form a system of six equations. Solving the system, we obtain

$$f(z) = \frac{1}{24}\Big(2z + \frac{2}{z} + \frac{6}{z-1} + 3\Big) = \frac{2z^3 + z^2 + 5z - 2}{24z(z-1)}. \qquad \square$$

Chapter 5

Cauchy's Equations

In this chapter we study four functional equations that are solved by the linear, power, exponential, and logarithmic functions. These equations were studied by Augustin Louis Cauchy, and since then they have formed the cornerstone of the theory. The derivation of the solutions of these functional equations serves as a vehicle to introduce the reader to the central ideas in functional equations. It is imperative that these simple results and the methodology behind them should be memorized (this can be done with minimal effort), as they appear often in functional problems.

5.1 The First Cauchy Equation

Problem 5.1. *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the functional relation*

$$f(x+y) = f(x) + f(y) \quad \text{for all } x, y \in \mathbb{R}. \tag{5.1}$$

Comment. Equation (5.1) is known as the **first** (or **linear**) **Cauchy functional equation**.

Solution. Setting $x = y = 0$ in the defining equation, we see that $f(0) = 2f(0)$; hence $f(0) = 0$. Then, setting $x = -y$, we obtain $f(0) = f(x) + f(-x)$, which implies

$$f(-x) = -f(x). \tag{5.2}$$

Using induction, we now check that for any natural number $n > 0$,

$$f(x_1 + x_2 + \cdots + x_n) = f(x_1) + f(x_2) + \cdots + f(x_n). \tag{5.3}$$

For $n = 2$, this is true by definition. For the induction step, assume that (5.3) is true for $n = n_0$:

$$f(x_1 + x_2 + \cdots + x_{n_0}) = f(x_1) + f(x_2) + \cdots + f(x_{n_0}).$$

To complete the induction, it suffices to check that (5.3) is true for $n = n_0 + 1$:

$$\begin{aligned} f(x_1 + x_2 + \cdots + x_{n_0} + x_{n_0+1}) &= f((x_1 + x_2 + \cdots + x_{n_0}) + x_{n_0+1}) \\ &= f(x_1 + x_2 + \cdots + x_{n_0}) + f(x_{n_0+1}) \\ &= f(x_1) + f(x_2) + \cdots + f(x_{n_0}) + f(x_{n_0+1}). \end{aligned}$$

Having proved (5.3), we now use it with $x_i = x$ for all i, obtaining

$$f(nx) = n\, f(x). \tag{5.4}$$

Next, if $x = \dfrac{m}{n}\, z$, with $m \in \mathbb{N}$ and $n \in \mathbb{N}^*$, we get

$$f(mz) = n\, f\left(\frac{m}{n} z\right),$$

which, by (5.4), yields

$$\frac{m}{n}\, f(z) = f\left(\frac{m}{n} z\right). \tag{5.5}$$

From this and (5.2), we conclude that

$$f\left(-\frac{m}{n} z\right) = -\frac{m}{n}\, f(z). \tag{5.6}$$

So far we have proved that

$$f(qz) = q\, f(z) \quad \text{for all } z \in \mathbb{R} \text{ and } q \in \mathbb{Q}. \tag{5.7}$$

In particular, if we set $c = f(1)$, we find that

$$f(q) = c\, q \quad \text{for all } q \in \mathbb{Q}.$$

Now take $r \in \mathbb{R}$. There is a sequence of rational numbers (q_n) such that

$$\lim_{n \to +\infty} q_n = r.$$

For the terms of this sequence we have $f(q_n) = c\, q_n$; since f is continuous,

$$f(r) = f\big(\lim_{n \to +\infty} q_n\big) = \lim_{n \to +\infty} f(q_n) = \lim_{n \to +\infty} c\, q_n = c\, r.$$

Thus the solutions to the problem must be of the form $f(x) = c\, x$, for fixed $c \in \mathbb{R}$. It is trivial to see that any $c \in \mathbb{R}$ works. □

Exercise 5.2. *In the first Cauchy equation, restrict the domain to $\mathbb{R}_+$; that is, consider continuous functions $f : \mathbb{R}_+ \to \mathbb{R}$ that satisfy the functional relation*

$$f(x + y) = f(x) + f(y) \quad \textit{for all } x, y \in \mathbb{R}_+.$$

Prove that any such function is still of the form $f(x) = c\, x$; that is, no new solutions are gained by restricting the domain.

5.2 The Second Cauchy Equation

Problem 5.3. *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the functional relation*

$$f(x+y) = f(x)\,f(y) \quad \textit{for all } x, y \in \mathbb{R}. \tag{5.8}$$

Comment. Equation (5.8) is known as the **second** (or **exponential**) **Cauchy functional equation**.

Clearly, $f(x) \equiv 0$ gives a solution, but it's an uninteresting one. So from now on we assume f is not identically zero. We then show that f is *nowhere* zero. Indeed, if $f(x_0) = 0$ for some x_0, then

$$f(x) = f(x - x_0 + x_0) = f(x - x_0)\,f(x_0) = 0 \quad \text{for all } x \in \mathbb{R},$$

and f vanishes identically.

Solution 1. We next show that f is everywhere positive. For any $x \in \mathbb{R}$, the defining relation gives $f(x) = f(x/2)^2$, and so $f(x) \geq 0$; but then $f(x) > 0$.

Now, we already know *one* function that satisfies (5.8): the exponential function $x \mapsto e^x$. This suggests that things may simplify if we express f as an exponential; that is, we should try introducing $g(x) = \ln f(x)$, which is equivalent to $f(x) = e^{g(x)}$. Note that g is well defined since $f(x) > 0$ for all x.

The defining equation (5.8) translates in terms of g as follows:

$$g(x+y) = \ln f(x+y) = \ln(f(x)f(y)) = \ln f(x) + \ln f(y) = g(x) + g(y);$$

that is,

$$g(x+y) = g(x) + g(y) \quad \text{for all } x, y \in \mathbb{R}.$$

We already know that the continuous solutions of this equation are of the form $g(x) = cx$, with $c = g(1)$ any real number. From this, the function f is found to be

$$f(x) = a^x,$$

where we have set $a = e^c = f(1)$; here a can be any positive number. $\square$

In this solution we used as a stepping stone our results for the linear Cauchy functional equation (5.1). However, one might like a proof that is independent of the results of the previous section; we do this next.

Solution 2. Setting $x = y = 0$ in the defining equation, we get $f(0) = f(0)^2$, whose solutions are $f(0) = 0$ and $f(0) = 1$. Under our assumption that f does not vanish identically, therefore, we have $f(0) = 1$.

Then, setting $x = -y$ in the defining equation, we get $f(0) = f(x)\,f(-x)$, which implies

$$f(-x) = f(x)^{-1}. \tag{5.9}$$

Also, it follows from (5.8) by induction that, for any natural number $n > 0$,

$$f(x_1 + x_2 + \cdots + x_n) = f(x_1)\,f(x_2)\,\ldots\,f(x_n). \tag{5.10}$$

This is checked in the same way as (5.3).

Setting $x_i = x$ for all i in (5.10), we obtain

$$f(nx) = f(x)^n. \tag{5.11}$$

Next, if $x = \dfrac{m}{n}\,z$, with $m \in \mathbb{N}$ and $n \in \mathbb{N}^*$, we get

$$f(my) = f\left(\frac{my}{n}\right)^n,$$

which, by (5.11), yields $f(y)^m = f\left(\dfrac{m}{n}y\right)^n$; that is to say

$$f\left(\frac{m}{n}y\right) = f(y)^{m/n}. \tag{5.12}$$

From this and (5.9), we conclude that

$$f\left(-\frac{m}{n}y\right) = f(y)^{-m/n}. \tag{5.13}$$

So far we have proved that

$$f(qz) = f(z)^q \quad \text{for all } z \in \mathbb{R} \text{ and } q \in \mathbb{Q}.$$

In particular, if we set $a = f(1)$, we find that

$$f(q) = a^q \quad \text{for all } q \in \mathbb{Q}.$$

Exactly as in the case of the first Cauchy equation, the continuity of f and the continuity of the function $q \mapsto a^q$ imply that $f(r) = a^r$ for all $r \in \mathbb{R}$. $\square$

5.3 The Third Cauchy Equation

Problem 5.4. *Find all continuous functions $f : \mathbb{R}_+^* \to \mathbb{R}$ that satisfy the functional relation*

$$f(xy) = f(x) + f(y) \quad \textit{for all } x, y \in \mathbb{R}_+^*. \tag{5.14}$$

Comment. Equation (5.14) is known as the **third** (or **logarithmic**) **Cauchy functional equation**.

Note that here the domain is restricted to the positive reals. If we replace $\mathbb{R}_+^*$ by $\mathbb{R}$ in the problem, the only solution is $f(x) \equiv 0$. That solution is of course valid in the restricted domain as well, but again we declare it uninteresting. (See Example 2.3.)

Solution 1. Since the independent variable x takes values in $\mathbb{R}_+^*$, the change of variable $w = \ln x$ leads to a variable taking values in $\mathbb{R}$. Then the function $g : \mathbb{R} \to \mathbb{R}$ given by $g(w) = f(e^w)$, or equivalently $f(x) = g(\ln x)$, satisfies the relation

$$g(\ln x) + g(\ln y) = g\left(\ln(xy)\right) = g(\ln x + \ln y) \quad \text{for all } x, y \in \mathbb{R}_+^*.$$

Since the logarithmic function is surjective onto $\mathbb{R}$, we can write the preceding condition as

$$g(w_1) + g(w_2) = g(w_1 + w_2) \quad \text{for all } w_1, w_2 \in \mathbb{R}.$$

We know from Section 5.1 that the solutions to this functional relation are of the form $g(w) = c\,w$, with $c = g(0) = f(1)$. From this we can work back to the function f:

$$f(x) = f(e^w) = g(w) = c\,w = c \ln x = \frac{\ln x}{1/c} = \frac{\ln x}{\ln e^{1/c}},$$

where, in the last two equalities, we have excluded the case $c = 0$ (corresponding to f being identically zero). The reason for these last manipulations becomes apparent when we introduce $\gamma = e^{1/c} = e^{1/f(1)}$, which leads to

$$f(x) = \frac{\ln x}{\ln \gamma} = \log_\gamma x,$$

the general form of the solution. You should check that γ can be any positive number apart from 1. $\square$

This solution depends on the results of Section 5.1 for the linear Cauchy functional equation (5.1). At the danger of being repetitive, we also write down a solution that does not use those results directly (along the lines of Solution 2 in Section 5.2).

Solution 2. Setting $x = y = 1$ in the defining equation (5.14), we see that

$$f(1) = 2f(1) \;\Rightarrow\; f(1) = 0\ .$$

Also, if $y = 1/x$,

$$f(1) = f(x) + f(x^{-1}) \Rightarrow f(x^{-1}) = -f(x) . \tag{5.15}$$

We now notice that, for any natural number $n > 0$,

$$f(x_1 x_2 \cdots x_n) = f(x_1) + f(x_2) + \cdots + f(x_n) . \tag{5.16}$$

This is verified using the method of induction. For $n = 2$, equation (5.16) is true by definition. Assuming that it is true for some n_0,

$$f(x_1 x_2 \cdots x_{n_0}) = f(x_1) + f(x_2) + \cdots + f(x_{n_0}),$$

we see that it is true for $n_0 + 1$:

$$\begin{aligned} f(x_1 x_2 \cdots x_{n_0} x_{n_0+1}) &= f\left((x_1 x_2 \cdots x_{n_0}) x_{n_0+1}\right) \\ &= f(x_1 x_2 \cdots x_{n_0})\, f(x_{n_0+1}) \\ &= f(x_1)\, f(x_2) \cdots f(x_{n_0})\, f(x_{n_0+1}) . \end{aligned}$$

Setting $x_i = x$, for all i in (5.16),

$$f(x^n) = n\, f(x) . \tag{5.17}$$

If, moreover, $x = y^{m/n}$, $m \in \mathbb{N}$, $n \in \mathbb{N}^*$,

$$f(y^m) = n\, f\left(y^{\frac{m}{n}}\right) \overset{(5.17)}{\Rightarrow} m\, f(y) = n\, f\left(y^{\frac{m}{n}}\right) \Rightarrow f\left(y^{\frac{m}{n}}\right) = \frac{m}{n}\, f(y) . \tag{5.18}$$

From this equation and (5.15), we conclude that

$$f\left(y^{-\frac{m}{n}}\right) = -\frac{m}{n}\, f(y) . \tag{5.19}$$

In other words, so far we have proved that

$$f(y^q) = q\, f(y) , \quad \forall y \in \mathbb{R} , \quad q \in \mathbb{Q} .$$

Now, let $r \in \mathbb{R}$. There is a sequence of rationals $\{q_n\}$ such that

$$\lim_{n\to\infty} q_n = r .$$

For the terms of the sequence $\{q_n\}$, the function f gives

$$f(y^{q_n}) = q_n\, f(y) ,$$

and since it is continuous,

$$f(y^r) = \lim_{n\to\infty} f(y^{q_n}) = \lim_{n\to\infty} q_n\, f(y) = r\, f(y) .$$

Finally, let's set $y = a = \text{const}$ and $x = a^r$. Then we find the function f in the final form:

$$f(x) = f(a) \log_a x = \log_b x ,$$

where we defined a constant b such that $f(a) = 1/\log_a b$. $\square$

Exercise 5.5. *Solve the third Cauchy functional equation with domain $\mathbb{R}^*$; that is, find all continuous functions $f : \mathbb{R}^* \to \mathbb{R}$ that satisfy the functional relation*

$$f(xy) = f(x) + f(y) \quad \textit{for all } x, y \in \mathbb{R}^*.$$

(Recall that, by definition, the continuity requirement in this case boils down to continuity in each of the intervals $(-\infty, 0)$ and $(0, \infty)$.)

5.4 The Fourth Cauchy Equation

Problem 5.6. *Find all continuous functions $f : \mathbb{R}_+ \to \mathbb{R}$ that satisfy the functional relation*

$$f(xy) = f(x)\, f(y) \quad \textit{for all } x, y \in \mathbb{R}_+. \tag{5.20}$$

Comment. Equation (5.20) is known as the **fourth** (or **power**) **Cauchy functional equation**.

Solution 1. Again, $f(x) \equiv 0$ gives an uninteresting solution, so suppose f is not identically zero. We first show that f is nonzero except perhaps at the origin. Indeed, suppose $f(x_0) = 0$ for some $x_0 > 0$. Then

$$f(x) = f\Big(\frac{x}{x_0}\, x_0\Big) = f\Big(\frac{x}{x_0}\Big) f(x_0) = 0 \quad \text{for all } x \in \mathbb{R}_+,$$

and f vanishes identically.

Further, f must be nonnegative everywhere and positive away from 0. Indeed, for any $x \in \mathbb{R}_+$, we can write $f(x) = f(\sqrt{x})^2$.

Once more we introduce $g(x) = \ln f(x)$, but this time only for $x > 0$, since $f(0)$ may vanish. The defining relation (5.20) can be rewritten in terms of g by taking the logarithm of both sides, yielding

$$g(x\, y) = g(x) + g(y) \quad \text{for all } x, y \in \mathbb{R}_+^*.$$

We know from the previous section that the solutions of this equation are $g(x) \equiv 0$ and $g(x) = \log_\gamma x = \ln x / \ln \gamma$, for $\gamma \neq 1$. In the first case we have $f(x) \equiv 1$; in the second we have

$$f(x) = e^{g(x)} = x^c, \tag{5.21}$$

where $c = 1/\ln \gamma \neq 0$. In fact, the function $f(x) \equiv 1$ on $\mathbb{R}_+^*$ is also of the form $f(x) = x^c$, with $c = 0$, so we settle on (5.21) as the general form of our solution on $\mathbb{R}_+^*$.

There remains to extend f to all of $\mathbb{R}_+$. By continuity, we should pick $f(0) = 0$ if $c > 0$, and $f(0) = 1$ if $c = 0$. Clearly these extensions satisfy

(5.20). If $c < 0$, the function $f(x) = x^c$ cannot be extended continuously to the origin, so such values of c do not yield solutions of the fourth Cauchy problem as stated (they do on the domain $\mathbb{R}_+^*$). □

Once more we write down a careful alternative solution independent of the results in Section 5.1, along the lines of Solution 2 in Section 5.2.

Solution 2. One first proves that $f(x) > 0$ (as above). Setting $x = y = 1$ in the defining equation, we see that

$$f(1) = f(1)^2 \Rightarrow f(1)\,(f(1) - 1) = 0 \Rightarrow f(1) = 0,\ 1\ ,$$

and therefore we must have $f(1) = 1$.

Also, if $y = 1/x$ for $x \neq 0$,

$$f(1) = f(x)\,f(x^{-1}) \Rightarrow f(x^{-1}) = f(x)^{-1}\ . \tag{5.22}$$

We now notice that for any natural number $n > 0$

$$f(x_1 x_2 \cdots x_n) = f(x_1)\,f(x_2)\,\cdots\,f(x_n)\ . \tag{5.23}$$

This is verified using the method of induction. For $n = 2$, equation (5.23) is true by definition. Assuming, that it is true for some n_0,

$$f(x_1 x_2 \cdots x_{n_0}) = f(x_1)\,f(x_2)\,\cdots\,f(x_{n_0})\ ,$$

we see that it is true for $n_0 + 1$:

$$\begin{aligned} f(x_1 x_2 \cdots x_{n_0} x_{n_0+1}) &= f\left((x_1 x_2 \cdots x_{n_0}) x_{n_0+1}\right) \\ &= f(x_1 x_2 \cdots x_{n_0})\,f(x_{n_0+1}) \\ &= f(x_1)\,f(x_2)\,\cdots\,f(x_{n_0})\,f(x_{n_0+1})\ . \end{aligned}$$

Setting $x_i = x,\ \forall i$ in (5.23),

$$f(x^n) = f(x)^n\ . \tag{5.24}$$

If, moreover, $x = y^{m/n}$, $m \in \mathbb{N}$, $n \in \mathbb{N}^*$,

$$f(y^m) = f\left(y^{\frac{m}{n}}\right)^n \overset{(5.24)}{\Rightarrow} f(y)^m = f\left(y^{\frac{m}{n}}\right)^n \Rightarrow f\left(y^{\frac{m}{n}}\right) = f(y)^{m/n}\ . \tag{5.25}$$

From this equation and (5.22), we conclude that

$$f\left(y^{-\frac{m}{n}}\right) = f(y)^{-m/n}\ . \tag{5.26}$$

In other words, so far we have proved that

$$f(y^q) = f(y)^q \ , \quad \forall y \in \mathbb{R}_+^* \ , \quad q \in \mathbb{Q} \ .$$

Now, let $r \in \mathbb{R}$. There is a sequence of rationals $\{q_n\}$ such that

$$\lim_{n\to\infty} q_n = r \ .$$

For the terms of the sequence $\{q_n\}$, the function f gives

$$f(y^{q_n}) = f(y)^{q_n} \ ,$$

and since it is continuous,

$$f(y^r) = \lim_{n\to\infty} f(y^{q_n}) = \lim_{n\to\infty} f(y)^{q_n} = f(y)^r \ .$$

Finally, let's set $y = a = \text{const}$ and $x = a^r$. Then we find the function f in the form:

$$f(x) = f(a)^{\log_a x} \ .$$

If $f(a) = 1$, then $f(x) = 1$. If $f(a) \neq 1$, we define a constant c such that $f(a) = a^c$. Then

$$f(x) = x^c \ , \quad c \neq 0 \ . \qquad \square$$

Question. *Define the sign function* $\mathrm{sgn}(x)$ *by*

$$\mathrm{sgn}(x) = \begin{cases} -1 & \text{if } x < 0, \\ 0 & \text{if } x = 0, \\ +1 & \text{if } x > 0. \end{cases}$$

Show that it satisfies $\mathrm{sgn}(x\,y) = \mathrm{sgn}(x)\ \mathrm{sgn}(y)$*; in other words, it is a solution of the functional equation* (5.20)*. However, it is not a power function. Explain.*

Answer. The sign function is not continuous: it has a jump discontinuity at $x = 0$. When we presented the definition of functional equations in Section 2.1, we mentioned that the set of solutions strongly depends on the conditions imposed on the functions. When continuity is assumed, the Cauchy functional equations have a restricted set of solutions, exactly as we have seen above. If this condition is relaxed, then the set of solutions greatly enlarges. Discontinuous solutions of the Cauchy equations will be studied in Chapter 11.

Exercise 5.7. *Solve the fourth Cauchy functional equation with domain* $\mathbb{R}$*; that is, find all continuous functions* $f : \mathbb{R} \to \mathbb{R}$ *that satisfy the functional relation*

$$f(xy) = f(x)f(y) \quad \textit{for all } x, y \in \mathbb{R}.$$

5.5 Solved Problems

Often, functional equations can be solved by a reduction to one of Cauchy's functional equations. In this section, we present several examples.

Problem 5.8 ([46]). *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional relation*

$$f(x+y) = A^y f(x) + A^x f(y) \quad \textit{for all } x, y \in \mathbb{R},$$

where A is a positive constant.

Solution. Assume f is a continuous solution, and define a continuous function g by $g(x) = A^{-x} f(x)$, or equivalently $f(x) = A^x g(x)$. To reexpress the defining functional equation in terms of g, we observe that

$$g(x+y) = A^{-(x+y)} f(x+y) = A^{-x} f(x) + A^{-y} f(y) = g(x) + g(y).$$

We have thus reduced the problem to the first Cauchy functional equation, whose solutions are of the form $g(x) = c\,x$, where c is any constant. Therefore $f(x) = c\,x\,A^x$. □

Problem 5.9. *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \quad \textit{for all } x, y \in \mathbb{R}. \tag{5.27}$$

Comment. This is known as the **Jensen functional equation**. It can also be written in the equivalent form

$$f(x+y) + f(x-y) = 2\,f(x) \quad \text{for all } x, y \in \mathbb{R}.$$ □

Solution. For $y = 0$, equation (5.27) gives

$$f\left(\frac{x}{2}\right) = \frac{f(x) + f(0)}{2} = \frac{f(x) + b}{2},$$

where $b = f(0)$. In this relation we substitute $y + z$ for x to obtain

$$f\left(\frac{y+z}{2}\right) = \frac{f(y+z) + b}{2}.$$

The left-hand side can be rewritten from the defining relation (5.27) and thus

$$\frac{f(y) + f(z)}{2} = \frac{f(y+z) + b}{2},$$

or

$$f(y+z) = f(y) + f(z) - b.$$

Now consider the continuous function g defined by $g(x) = f(x) - b$. In terms of it, the functional equation becomes

$$g(x+y) = g(x) + g(y),$$

whose solutions are of the form $g(x) = c\,x$, where c is any constant. Therefore $f(x) = c\,x + b$, where c and b are arbitrary. □

Problem 5.10. *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f\left(\sqrt{\frac{x^2+y^2}{2}}\right) = \sqrt{\frac{f(x)^2+f(y)^2}{2}} \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. From the defining equation we see that $f(x) \geq 0$ for all $x \in \mathbb{R}_+$, since the square root of a number is a nonnegative number. Define $F(x) = f(\sqrt{x})^2$ for $x \geq 0$. Then the defining equation becomes

$$F\left(\frac{u+v}{2}\right) = \frac{F(u)+F(v)}{2},$$

where $u = x^2$, $v = y^2$. Therefore,

$$F(u) = c\,u + b.$$

The function $F(u)$ will be positive for all values of its argument u if both b and c are nonnegative constants. Therefore,

$$f(x) = \sqrt{c\,x^2 + b}, \quad b, c \geq 0,$$

for $x \geq 0$.

To find the values of f at negative values, we observe that the defining equation is unchanged under the substitution $x \mapsto -x$. This leads to $f(x)^2 = f(-x)^2$; hence $f(-x) = \pm f(x)$. Since f is continuous, its graph for negative values consists of a series of odd and even parts (relative to the branch of positive x) with transitions happening at the roots of f. And since for each nonpositive root we must have a nonnegative root, we conclude that the only possible root is $x = 0$ or $f(x) \equiv 0$. From this discussion we arrive at the following solutions of the given functional equation:

$$\begin{aligned} f(x) &= 0, \\ f(x) &= \sqrt{c\,x^2 + b}, \quad c \geq 0,\ b > 0, \\ f(x) &= a\,|x|, \quad a > 0, \\ f(x) &= a\,x, \qquad a > 0. \end{aligned}$$

□

Problem 5.11. *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) = \alpha^{xy} f(x) f(y) \quad \text{for all } x, y \in \mathbb{R},$$

where α is a positive constant.

Solution. Define the continuous function $g(x) = \alpha^{-x^2/2} f(x)$. Then the functional equation takes the form

$$g(x+y) = g(x) g(y),$$

which has the solutions $g(x) = 0$ or $g(x) = b^x$, where b is a positive constant. Therefore $f(x) = 0$ or $f(x) = b^x \alpha^{x^2/2}$. ☐

Problem 5.12. *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) = f(x) + f(y) + f(x) f(y) \quad \text{for all } x, y \in \mathbb{R}.$$

Solution. Define the continuous function $g(x) = f(x) + 1$. Then the functional equation takes the form

$$g(x+y) = g(x) g(y),$$

which has the solutions $g(x) = 0$ or $g(x) = b^x$, where b a positive constant. Therefore $f(x) = -1$ or $f(x) = b^x - 1$. ☐

Problem 5.13 (Putnam 1947). *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(\sqrt{x^2 + y^2}) = f(x) f(y) \quad \text{for all } x, y \in \mathbb{R}.$$

Solution. If f vanishes identically, it satisfies the equation. In the opposite case, there exists $x_0 \in \mathbb{R}$ such that $f(x_0) \neq 0$; then

$$f(\sqrt{x_0^2 + y^2}) = f(x_0) f(y) = f(x_0) f(-y),$$

or $f(y) = f(-y)$, for all $y \in \mathbb{R}$. This shows that f is an even function. It is enough to search for the form of $f(x)$ when $x \geq 0$.

Define the continuous function $g(x) = f(\sqrt{x})$, $x \geq 0$. Then the functional equation takes the Cauchy form

$$g(u+v) = g(u) g(v) \quad \text{for all } u, v \in \mathbb{R}_+$$

(via the substitutions $u = x^2$ and $v = y^2$). This has the nonvanishing solutions $g(u) = a^u$, where a is any positive constant. Therefore $f(x) = a^{x^2}$. ☐

Problem 5.14. *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(\sqrt[n]{x^n + y^n}) = f(x) + f(y) \quad \textit{for all } x, y \in \mathbb{R}, \tag{5.28}$$

where n is a given positive natural number.

Solution. Here, the reduction to a Cauchy equation is achieved by "factoring" the function f as a composition of functions, one of which is $x \mapsto x^n$. Let $g : \mathbb{R} \to \mathbb{R}$ be defined by $g(x^n) = f(x)$. Then (5.28) is equivalent to

$$g(x^n + y^n) = g(x^n) + g(y^n) \quad \text{for all } x, y \in \mathbb{R}. \tag{5.29}$$

Take first the case of n odd, where $x \mapsto x^n$ is one-to-one and onto. Then, since x^n and y^n can take any real value, 5.29 is equivalent to

$$g(x + y) = g(x) + g(y) \quad \text{for all } x, y \in \mathbb{R},$$

whose solutions, as we know, are of the form $g(x) = c\,x$ for fixed $c \in \mathbb{R}$. Translating back to f, we get $f(x) = g(x^n) = c\,x^n$.

If n is even we use a similar reasoning, but we must consider that the equation $g(x^n) = f(x)$ only defines g for nonnegative values of the argument. Because x^n and y^n range over $[0, +\infty)$ only, (5.29) is equivalent to

$$g(x + y) = g(x) + g(y) \quad \text{for all } x, y \in \mathbb{R}_+.$$

Fortunately this is still enough to guarantee that $g(x) = c\,x$ for fixed $c \in \mathbb{R}$ (see Exercise 5.2). Therefore the solutions in the case of n even are again the functions f of the form $f(x) = g(x^n) = c\,x^n$. □

Problem 5.15 (Romania 1997). *Find all continuous solutions $f : \mathbb{R} \to [0, +\infty)$ such that*

$$f(x^2 + y^2) = f(x^2 - y^2) + f(2xy).$$

Solution. The key here is to observe that the three arguments are related by the equation

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2.$$

Thus, if we set $\alpha = x^2 - y^2$ and $\beta = 2xy$, the defining equation is equivalent to

$$f(\sqrt{\alpha^2 + \beta^2}) = f(\alpha) + f(\beta) \quad \text{for } \alpha, \beta \in \mathbb{R}.$$

(Why are all values of α and β achieved?) From the previous problem we know that the general solution is $f(x) = a\,x^2$, where a is a real constant. □

Problem 5.16. *Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) = x^2y + xy^2 - 2xy + f(x) + f(y) \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. Define the continuous function $g(x) = f(x) - x^3/3 + x^2$. Then the functional equation takes the Cauchy form

$$g(x+y) = g(x) + g(y).$$

This equation has the general solution $g(x) = c\,x$, where a is any real constant. Therefore $f(x) = c\,x + x^3/3 - x^2$. □

Problem 5.17. *Show that the functional equation*

$$f\left(\frac{x+y}{2}\right)^2 = f(x)\,f(y) \tag{5.30}$$

is equivalent to the functional equation

$$f(x)^2 = f(x+y)\,f(x-y). \tag{5.31}$$

Then find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of (5.30).

Comment. Equation (5.31) is known as the **Lobachevsky functional equation**.

Solution. Setting $x = a + b$ and $y = a - b$ in (5.30), we find (5.31).

Setting $y = 0$ in (5.30), we find

$$f\left(\frac{x}{2}\right)^2 = f(x)\,f(0).$$

If $f(0) = 0$, then $f(x/2)^2 = 0$, for all $x \in \mathbb{R}$. This implies that $f(x) = 0$ for all $x \in \mathbb{R}$. If $f(0) \neq 0$, we replace x with $x + y$; hence

$$f\left(\frac{x+y}{2}\right)^2 = f(x+y)\,f(0),$$

or, after using (5.30),

$$f(x)\,f(y) = f(x+y)\,f(0).$$

Set $b = f(0)$ and define the continuous function $g(x) = f(x)/b$. Then the functional equation takes the Cauchy form

$$g(x+y) = g(x)\,g(y).$$

This equation has the solutions $g(x) = 0$ or $g(x) = a^x$, for $a \in \mathbb{R}_+^*$. Working our way back to f we conclude that the general form of the solution is $f(x) = b\,a^x$, with $a \in \mathbb{R}_+^*$ and $b \in \mathbb{R}$. □

Here is an alternative formulation of the previous problem:

Exercise 5.18. *Find all continuous functions that map three successive terms of any arithmetic progression to three successive terms of a geometric progression.*

Can you see the relation?

Problem 5.19 ([46]). *Find all continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) = f(x) + f(y) + a\,(1-b^x)\,(1-b^y), \tag{5.32}$$

where a, b are real constants and $b > 0$.

Solution. Set

$$g(x) = f(x) - a\,(b^x - 1). \tag{5.33}$$

Then the function $g(x)$ satisfies

$$g(x+y) = g(x) + g(y),$$

with solution $g(x) = c\,x$ with c a real constant. Therefore

$$f(x) = a\,(b^x - 1) + c\,x.$$

□

Problem 5.20. *Let $a, b \in \mathbb{R}$. Find the continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y+a) + b = f(x) + f(y) \quad \textit{for all } x, y \in \mathbb{R}. \tag{5.34}$$

Solution. Define the continuous function $g(x) = f(x-a) - b$. Then (5.34) takes the form

$$g(x+y+2a) = g(x+a) + g(y+a),$$

or

$$g(u+v) = g(u) + g(v),$$

where $u = x + a$ and $v = y + a$. By (5.1), $g(u) = c\,u$, where c is a constant. Therefore $f(x) = c(x+a) + b$. □

Problem 5.21 (Romania TST 2006). *Let $r, s \in \mathbb{Q}$. Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that for all $x, y \in \mathbb{Q}$ we have*

$$f\,(x + f(y)) = f(x+r) + y + s. \tag{5.35}$$

Solution. It is easy to verify that $f(x) = \pm(x+s)+r$ are solutions. We show that they are the only solutions.

Adding $z-r$ to both sides of (5.35) and finding the value of f at the points thus obtained, we get

$$f\big(z-r+f\left(x+f(y)\right)\big) = f\big(z+y+s-r+f(x+r)\big). \tag{5.36}$$

Applying (5.35) to each side of (5.36), we obtain

$$f(z)+x+f(y)+s = f(z+y+s)+x+r+s,$$

or

$$f(y)+f(z) = f(y+z+s)+r,$$

which is (5.34). Therefore $f(x) = c(x+s)+r$, where c is a constant. Substituting $f(x) = c(x+s)+r$ into (5.35), we find

$$c\big(x+c(y+s)+r+s\big)+r = c(x+r+s)+r+y+s,$$

or

$$(c^2-1)(y+s) = 0.$$

Hence $c = \pm 1$ and $f(x) = \pm(x+s)+r$. □

As an easy extension of the previous problem, the reader should try:

Exercise 5.22. *Let $a, b \in \mathbb{R}$. Show that $f(x) = \pm(x+a)+b$ are the only continuous solutions to the functional equation*

$$f\left(x+f(y)\right) = f(x+a)+y+b \quad \textit{for all } x, y \in \mathbb{R}.$$

Chapter 6

Cauchy's $\mathbb{NQR}$ Method

In Chapter 5 we encountered many functional equations of the general form

$$f(x+y) = F\left(f(x), f(y)\right); \tag{6.1}$$

for instance, in (5.1), the function $F(a,b)$ is $F(a,b) = a+b$, and in (5.8) it is $F(a,b) = ab$. One can explore functional equations with different functions $F(a,b)$. For example,

$$f(x+y) = f(x) + f(y) + f(x)f(y) \quad \text{and} \quad f(x+y) = \frac{f(x)+f(y)}{1-f(x)\,f(y)}$$

are functional equations of the form (6.1) with, respectively,

$$F(a,b) = a+b+ab \quad \text{and} \quad F(a,b) = \frac{a+b}{1-ab}.$$

One can generalize equation (6.1) further to

$$f(x+y) = F\left(f(x), f(y), f(x-y); x, y\right).$$

Examples include

$$\begin{aligned}
f(x+y) &= k^{xy}\, f(x)f(y), \\
f(x+y)f(x-y) &= f(x)^2, \\
f(x+y)f(x-y) &= f(x)^2\, f(y)^2, \\
f(x+y) + f(x-y) &= 2\, f(x), \\
f(x+y) - f(x-y) &= 2\, f(y), \\
f(x+y) + f(x-y) &= 2\, f(x) + 2\, f(y), \\
f(x+y) + f(x-y) &= 2\, f(x) \cos y,
\end{aligned}$$

with, respectively,

$$
\begin{aligned}
F(a,b,c;x,y) &= k^{xy}\, a\, b, \\
F(a,b,c;x,y) &= a^2/c, \\
F(a,b,c;x,y) &= a^2 b^2/c, \\
F(a,b,c;x,y) &= 2\,a - c, \\
F(a,b,c;x,y) &= 2\,b + c, \\
F(a,b,c;x,y) &= 2\,a + 2\,b - c, \\
F(a,b,c;x,y) &= 2a\,\cos y - c.
\end{aligned}
$$

To include the third and fourth Cauchy functional relations (5.14) and (5.20), we must generalize the model further to allow arbitrary functions of x and y in the left-hand side, rather than only the sum $x + y$:

$$f(G(x,y)) = F\big(f(x), f(y), f(x-y); x, y\big). \tag{6.2}$$

Examples include

$$f(x\,y) = f(x) + f(y) + f(x)f(y)$$

with $G(x,y) = x\,y$ and $F(a,b,c;x,y) = a + b + ab$, and

$$
\begin{aligned}
f(\sqrt[n]{x^n + y^n}) &= f(x) + f(y), \\
f(\sqrt[n]{x^n + y^n}) &= f(x)\, f(y),
\end{aligned}
$$

with $G(x,y) = \sqrt[n]{x^n + y^n}$ and $F(a,b,c;x,y)$ given by $a + b$ or $a\,b$, respectively.

For such functional relations, it is often possible to reconstruct the function in $\mathbb{N}$, then in $\mathbb{Q}$, and finally in $\mathbb{R}$, leading to a successful solution. This method was demonstrated in the solution of Problem 5.1 and in the second solution of Problems 5.3, 5.4, and 5.6. I call it the $\mathbb{NQR}$ **method** (the name is new to the literature).

6.1 The $\mathbb{NQR}$ Method

We now present the sequence of steps to obtain the continuous solutions of the equation

$$f(x+y) = F\left(f(x), f(y), f(x-y); x, y\right). \tag{6.3}$$

In the more general case of (6.2), we follow the same steps; however, the operation of summation in the left-hand side has to be changed according to the nature of the function G.

First of all, we set $x = y = 0$ to find an equation for $f(0)$:

$$f(0) = F\left(f(0), f(0), f(0); 0, 0\right).$$

The value of f at $x = 0$ is then found by solving this equation.

Then, in (6.3), we replace x successively by $2x, 3x, \ldots, (n-1)x$ and $y = x$:

$$\begin{aligned}
f(2x) &= F\left(f(x), f(x), f(0); x, x\right) =: F_2\left(f(x), x\right), \\
f(3x) &= F\left(f(2x), f(x), f(x); 2x, x\right) =: F_3\left(f(x), x\right), \\
f(4x) &= F\left(f(3x), f(x), f(2x); 3x, x\right) =: F_4\left(f(x), x\right), \\
&\ \ \vdots \\
f(nx) &= F\left(f\left((n-1)x\right), f(x), f\left((n-2)x\right); (n-1)x, x\right) =: F_n\left(f(x), x\right).
\end{aligned}$$

In the last equation, we substitute $x = my/n$:

$$f(my) = F_n\left(f\left(\frac{m}{n}y\right), \frac{m}{n}y\right).$$

The left-hand side, however, may be rewritten using the previous result:

$$F_m\left(f(y), y\right) = F_n\left(f\left(\frac{m}{n}y\right), \frac{m}{n}y\right).$$

We assume that this expression can be solved for $f\left(\dfrac{m}{n}y\right)$ to give

$$f\left(\frac{m}{n}y\right) = \mathcal{F}\left(f(y), y; \frac{m}{n}\right),$$

for some function $\mathcal{F}(a, y; x)$. For $y = 1$,

$$f(q) = \mathcal{F}\left(f(1), 1; q\right) \quad \text{for all } q \in \mathbb{Q}_+.$$

Let x be any nonnegative real number. There is a sequence of nonnegative rational numbers $\{q_n\}$ converging to x. Assuming that f and $\mathcal{F}$ are continuous, the last expression gives

$$f(x) = \mathcal{F}\left(f(1), 1; x\right) \quad \text{for all } x \in \mathbb{R}_+.$$

To find the function $f(x)$ for $x < 0$, we relate it to $f(-x)$ using the defining equation (6.3). In particular, setting $y = -x$ in (6.3), we find:

$$\begin{aligned}
f(0) &= F\left(f(x), f(-x), f(2x); x, -x\right) \\
&= F\left(f(x), \mathcal{F}\left(f(1), 1; -x\right), F_2\left(f(x), x\right); x, -x\right).
\end{aligned}$$

This should be solved for $f(x)$ to find the final result:

$$f(x) = \mathcal{G}(x) \quad \text{for } x < 0.$$

6.2 Solved Problems

Problem 6.1. *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the functional relation*

$$f(x+y) = a^{xy} f(x) f(y) \quad \textit{for all } x, y \in \mathbb{R},$$

where a is a positive real constant.

Solution. The function $f \equiv 0$ is clearly a solution. Other solutions must be nonzero everywhere, by the argument given before Solution 1 to the exponential Cauchy equation (page 93). Also as in that problem, the use of $x = y = 0$ in the defining equation leads to $f(0) = 0$ or $f(0) = 1$; hence $f(0) = 1$ for nontrivial solutions. For such solutions, in fact, f is everywhere positive; if not, the intermediate value theorem would imply that f vanishes somewhere.

Setting $y = x, 2x, \ldots, (n-1)x$ in the defining equation, we find

$$\begin{aligned}
f(2x) = f(x+x) &= a^{x^2} f(x)^2, \\
f(3x) = f(x+2x) &= a^{(1+2)x^2} f(x)^3, \\
f(4x) = f(x+3x) &= a^{(1+2+3)x^2} f(x)^4, \\
&\vdots \\
f(nx) = f\left(x+(n-1)x\right) &= a^{(1+2+3+\cdots+(n-1))x^2} f(x)^n,
\end{aligned}$$

or, from Exercise 3.2(a),

$$f(nx) = a^{\frac{n(n-1)}{2}x^2} f(x)^n. \tag{6.4}$$

Next, let $x = my/n$, with $m, n \in \mathbb{N}^*$. Substituting this value in (6.4) yields

$$f(nx) = a^{\frac{(n-1)n}{2}\left(\frac{m}{n}y\right)^2} f\left(\frac{m}{n}y\right)^n.$$

We can also apply (6.4) with (m, y) in place of (n, x), obtaining

$$f(my) = a^{\frac{m(m-1)}{2}y^2} f(y)^m.$$

But these two expressions are equal since $nx = my$:

$$a^{\frac{(n-1)n}{2}\left(\frac{m}{n}y\right)^2} f\left(\frac{m}{n}y\right)^n = a^{\frac{m(m-1)}{2}y^2} f(y)^m.$$

This allows us to solve for $f\left(\frac{m}{n}y\right)$ in terms of $f(y)$:

$$f\left(\frac{m}{n}y\right) = a^{\frac{1}{2}\frac{m}{n}\left(\frac{m}{n}-1\right)y^2} f(y)^{\frac{m}{n}}.$$

Specializing to $y = 1$ and defining $c = f(1) > 0$, we get

$$f(x) = a^{x(x-1)/2}c^x \tag{6.5}$$

for all $x \in \mathbb{Q}_+^*$.

Now take $r \in \mathbb{R}_+$ and a sequence of rationals $\{q_n\}$ converging to r as $n \to \infty$. Since both f and the map $x \mapsto a^{x(x-1)/2}c^x$ are continuous, we can write

$$f(r) = f\big(\lim_{n\to\infty} q_n\big) = \lim_{n\to\infty} f(q_n) = \lim_{n\to\infty} a^{q_n(q_n-1)/2}c^{q_n} = a^{r(r-1)/2}c^r,$$

thus extending[1] the validity of (6.5) to all $x \in \mathbb{R}_+$.

To deal with negative x, we use the fact that $f(0) = 1$ and set $y = -x$ in the defining equation, with $x > 0$. We obtain $1 = a^{-x^2} f(x)\, f(-x)$, which is to say

$$f(-x) = a^{x^2} f(x)^{-1} = a^{-x(-x-1)/2}\, c^{-x}.$$

Hence, the functions solving the problem are those of the form

$$f(x) = a^{x(x-1)/2}\, c^x \quad \text{for all } x \in \mathbb{R}$$

(for fixed $c > 0$), together with the solution $f(x) \equiv 0$. □

Problem 6.2. *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the functional relation*

$$f(x+y)\, f(x-y) = f(x)^2 f(y)^2 \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. Setting $x = y = 0$ in the defining equation, we get $f(0)^4 = f(0)^2$, which has the roots $f(0) = 0$ and $f(0) = \pm 1$. If $f(0) = 0$ the function vanishes identically, because

$$f(2x)\, f(0) = f(x)^4 \quad \text{for all } x, \tag{6.6}$$

as can be seen by setting $x = y$ in the defining equation.

Next we consider the case $f(0) = 1$. Replacing x by $x, 2x, 2^2x, \ldots,$ in (6.6), we find

$$\begin{aligned} f(2x) &= f(x)^4 = f(x)^{2^2}, \\ f(4x) &= f(2x)^4 = f(x)^{4^2}, \\ f(8x) &= f(4x)^4 = f(x)^{8^2}, \end{aligned}$$

[1] From now on we will use the shorthand "by continuity" to signify this process of extending the validity of a functional form from $\mathbb{Q}_+^*$ to $\mathbb{R}_+$ or from $\mathbb{Q}$ to $\mathbb{R}$.

and so on. This suggests that

$$f(nx) = f(x)^{n^2} \tag{6.7}$$

for all $n \in \mathbb{N}^*$, and indeed, we will prove this by induction. For $n = 1$, (6.7) is true. Let's assume it is true for $n = 1, 2, \ldots, m$. To show that it is also true for $n = m + 1$, we take $x = my$ in the defining relation, which yields

$$f\big((m+1)y\big)\, f\big((m-1)y\big) = f(my)^2\, f(y)^2.$$

Solving for $f\big((m+1)y\big)$ and applying (6.7) with $n = m$ and $n = m-1$, we find

$$f\big((m+1)y\big) = f(y)^{-(m-1)^2} f(y)^{2m^2} f(y)^2 = f(y)^{(m+1)^2},$$

which completes the induction step and proves (6.7) for all $n \in \mathbb{N}^*$.

We next apply equation (6.7) in two ways: once for $x = my/n$ with $m, n \in \mathbb{N}^*$ and once for n substituted by m:

$$\begin{aligned} f(nx) &= f(x)^{n^2} = f\Big(\frac{m}{n}y\Big)^{n^2}, \\ f(my) &= f(y)^{m^2}. \end{aligned}$$

From the equality of these two expressions we get

$$f\Big(\frac{m}{n}y\Big) = f(y)^{m^2/n^2}.$$

Setting $y = 1$ and defining $c = f(1) > 0$, we then get $f(x) = c^{x^2}$ for $x \in \mathbb{Q}_+^*$. By continuity, this equality extends to all $x \in \mathbb{R}_+$ (see footnote on the previous page).

To extend f to negative arguments, we go back to the defining equation and set $x = 0$, obtaining $f(y)\, f(-y) = f(y)^2 \cdot 1^2$; that is, $f(-y) = f(y)$. Therefore,

$$f(x) = c^{x^2} \quad \text{for all } x \in \mathbb{R}.$$

This takes care of the case where $f(0) = 1$. Finally, consider the last case, where $f(0) = -1$. Here it is easy to see that the function $-f$ satisfies the same defining equation as f, and takes the value 1 at 0. It follows from the case already proved that $-f(x) = c^{x^2}$, for $c > 0$ fixed; that is, $f(x) = -c^{x^2}$.

Putting everything together, we find that the solutions are of the form $f(x) = c^{x^2}$ or $f(x) = -c^{x^2}$, with $c > 0$, or else $f(x) \equiv 0$. □

Problem 6.3. *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the functional relation*

$$f(x+y) = f(x) + f(y) + f(x)\, f(y) \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. Setting $x = y = z/2$ in the defining equation, we see that

$$f(z) = 2f\left(\frac{z}{2}\right) + f\left(\frac{z}{2}\right)^2 = \left(1 + f\left(\frac{z}{2}\right)\right)^2 - 1,$$

so $f(x) \geq -1$ for all $x \in \mathbb{R}$. In particular, $f(1) \geq -1$. If $f(1) = -1$, then

$$f(x) = f(x - 1 + 1) = f(x-1) + f(1) + f(x-1)\,f(1) = -1 \quad \text{for all } x \in \mathbb{R},$$

that is, $f \equiv -1$.

We now search for other solutions, so $f(1) > -1$. Setting $y = x, 2x, 3x, \ldots,$ in the defining relation, we find

$$\begin{aligned}
&f(2x) = 2f(x) + f(x)^2 = (f(x)+1)^2 - 1,\\
&f(3x) = f(2x) + f(x) + f(2x)\,f(x) = f(x)^3 + 3f(x)^3 + 3f(x) = (f(x)+1)^3 - 1,
\end{aligned}$$

and so on, which suggests that

$$f(nx) = (f(x)+1)^n - 1 \quad \text{for } n \in \mathbb{N}^*. \tag{6.8}$$

We can prove this by induction. For $n = 1$, it is true. Let's assume that it is true for $n = m$:

$$f(mx) = (f(x)+1)^m - 1. \tag{6.9}$$

Set $y = mx$ in the defining relation and apply (6.9) to obtain

$$\begin{aligned}
f((m+1)x) &= f(x) + f(mx) + f(x)\,f(mx)\\
&= f(x) + (f(x)+1)^m - 1 + f(x)\,(f(x)+1)^m - f(x)\\
&= (f(x)+1)^{m+1} - 1.
\end{aligned}$$

Thus (6.8) is true for $n = m + 1$, completing the induction.

Next we take $m, n \in \mathbb{N}^*$ and substitute $x = my/n$ into (6.8), obtaining

$$f(my) = \left(f\left(\frac{m}{n}y\right) + 1\right)^n - 1.$$

But $f(my)$ can be expressed in another way using (6.8):

$$f(my) = (f(y)+1)^m - 1.$$

From the equality of these two expressions we can solve for $f\left(\frac{m}{n}y\right)$ in terms of $f(y)$:

$$f\left(\frac{m}{n}y\right) = (f(y)+1)^{m/n} - 1.$$

In particular, setting $y = 1$ and defining $c = 1 + f(1)$, we conclude that

$$f(x) = c^x - 1 \quad \text{for all } x \in \mathbb{Q}_+^*.$$

(Note that $c > 0$ since $f(1) > -1$ for nonconstant solutions.) Again, the solution extends by continuity to $x \in \mathbb{R}_+$; in particular, $f(0) = 0$.

To deal with negative arguments, we set $y = -x$, with $x > 0$, in the defining equation. We obtain

$$0 = f(x) + f(-x) + f(x)\, f(-x),$$

which implies

$$f(-x) \;=\; \frac{-f(x)}{1 + f(x)} \;=\; c^{-x} - 1.$$

Therefore, the functions solving the problem are those of the form

$$f(x) \;=\; c^x - 1 \quad \text{for all } x \in \mathbb{R}$$

(for fixed $c > 0$), together with the solution $f(x) \equiv -1$. □

The next problem is of the form $f(x + y) = F\,(f(x), f(y), f(xy))$, where $F(a, b, c) = a + b - ab + c$. It does not quite fit into the NQR method. But a combination of an algebraic system and the NQR idea still proves successful.

Problem 6.4 (India TST 2003). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that for all $x, y \in \mathbb{R}$,*

$$f(x + y) + f(x)\, f(y) \;=\; f(x) + f(y) + f(xy). \tag{6.10}$$

Solution. Setting $x = 0 = y$ in (6.10) we get $f(0)^2 = 2\, f(0)$; therefore $f(0) = 0$ or $f(0) = 2$.

If $f(0) = 2$, then by letting $y = 0$ in (6.10) we see that $f(x) = 2$ for all $x \in \mathbb{R}$, i.e., $f \equiv 2$.

Suppose instead that $f(0) = 0$. Let $a = f(1)$. Setting $x = 1$ and $y = -1$ we get $a\, f(-1) \;=\; a + 2\, f(-1)$; so $a \neq 2$ and

$$f(-1) \;=\; \frac{a}{a - 2}.$$

Next we must correlate values of f at opposite arguments. Replacing (x, y) in (6.10) by $(x{-}1, 1)$, $(-x{+}1, -1)$, and $(-x, 1)$, we get

$$f(x) + (a - 2)f(x - 1) \;=\; a, \tag{6.11}$$

$$f(-x) + \left(\frac{2}{a{-}2}\right) f(-x + 1) - f(x - 1) \;=\; \frac{a}{a{-}2}, \tag{6.12}$$

$$f(-x + 1) + (a - 2)f(-x) \;=\; a. \tag{6.13}$$

Eliminating $f(x - 1)$ and $f(-x + 1)$ in (6.11)–(6.13), we get

$$f(x) \;=\; (a - 2)f(-x). \tag{6.14}$$

Replacing x by $-x$ in (6.14) we get

$$f(-x) = (a-2)f(x). \tag{6.15}$$

Combining (6.14) and (6.15) we get $f(x) = (a-2)^2 f(x)$; hence f must vanish identically unless $a-2 = \pm 1$, that is, $a \in \{1,3\}$.

Consider the case $a = 3$. Then (6.11) gives $f(x) = 3 - f(x-1)$. Hence

$$f(2) = 3 - f(1) = 0, \quad f\left(\tfrac{5}{2}\right) = 3 - f\left(\tfrac{3}{2}\right) = f\left(\tfrac{1}{2}\right).$$

On the other hand, replacing (x, y) in (6.10) by $(2, \frac{1}{2})$, we see that

$$f\left(\tfrac{5}{2}\right) = f\left(\tfrac{1}{2}\right) + f(1) = f\left(\tfrac{1}{2}\right) + 3,$$

a contradiction.

Finally, consider the case $a = 1$. Then (6.11) gives $f(x) = f(x-1) + 1$. By induction,

$$f(x+n) = f(x) + n \quad \text{for all } x \in \mathbb{R} \text{ and } n \in \mathbb{Z}.$$

Replacing (x, y) in (6.10) by (x, n), we obtain

$$n\, f(x) = f(nx) \quad \text{for all } x \in \mathbb{R} \text{ and } n \in \mathbb{Z}.$$

Hence if $r = m/n \in \mathbb{Q}$ and $x \in \mathbb{R}$, we have

$$m\, f(x) = f(mx) = f(n \cdot rx) = n\, f(rx),$$

i.e., $f(rx) = r\, f(x)$. Now replacing y in (6.10) by r we get

$$f(x+r) = f(x) + r \quad \text{for all } x \in \mathbb{R} \text{ and } r \in \mathbb{Q}.$$

Moreover, letting $x = y$ in (6.10) we get $f(x)^2 = f(x^2)$. Thus $f(x) \geq 0$ if $x \geq 0$. Since $f(-x) = -f(x)$, it follows that $f(x) \leq 0$ if $x \leq 0$. Now for any $x \in \mathbb{R}$, let $\{r_n\}$ and $\{s_n\}$ be two sequences of rational numbers increasing and decreasing to x, respectively. Then

$$r_n \;\leq\; f(x - r_n) + r_n = f(x) = f(x - s_n) + s_n \;\leq\; s_n,$$

which forces $f(x) = x$ by the squeeze theorem.

Summarizing: the solutions of 6.10 are $f(x) \equiv 0$, $f(x) \equiv 2$, $f(x) = x$. □

Chapter 7

Equations for Trigonometric Functions

7.1 Characterization of the Sine and Cosine

The reader may know several definitions of the basic trigonometric functions:

- Using the sides of a triangle ABC right at A (see Figure 7.1, left), the sine and cosine of an angle are defined by

$$\sin \hat{B} = \frac{b}{a}, \quad \cos \hat{B} = \frac{c}{a}.$$

- Using the trigonometric circle (see Figure 7.1, right), the sine and cosine of an angle θ are defined by

$$\sin \theta = \overline{OB}, \quad \cos \theta = \overline{OA}.$$

- Using differential equations, the sine and cosine can be defined as the two linearly independent functions that solve the equation

$$y''(x) + y(x) = 0.$$

In Section 2.1, we defined the analytic sine and analytic cosine as solutions of the equation

$$C(x-y) = C(x)\,C(y) + S(x)\,S(y) \quad \text{for all } x, y \in \mathbb{R} \tag{7.1}$$

(subject to certain additional properties), expecting S and C to be the usual sine and cosine functions. Indeed, in Problem 2.23 we proved that, if such solution functions exist, they satisfy the same relations that our well-known sine and cosine functions do.

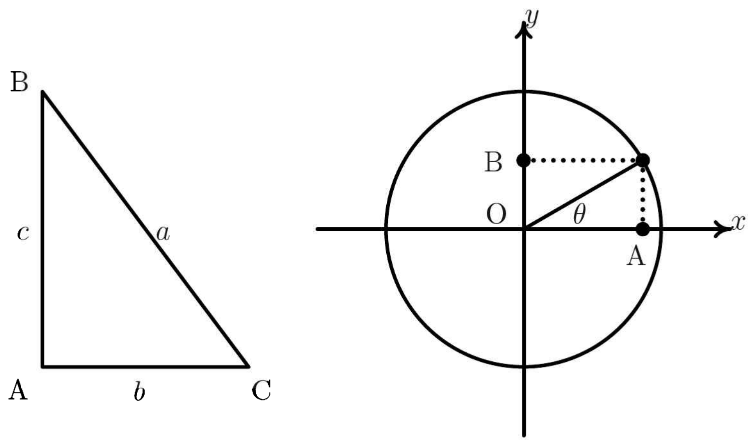


Figure 7.1: A right triangle (left) and the trigonometric circle (right) are often used to define the trigonometric functions.

In this section we return to the functional equation above and, following H. E. Vaughan [65], we establish conditions under which this equation uniquely characterizes the sine and cosine.

Setting $y = x$ in (7.1), we get

$$C(0) = C(x)^2 + S(x)^2. \tag{7.2}$$

In particular, for $x = 0$ we get $C(0) = C(0)^2 + S(0)^2$, hence

$$C(0)\,(1 - C(0)) = S(0)^2. \tag{7.3}$$

Since the right-hand side is positive, the quadratic expression on the left must be positive and thus

$$0 \leq C(0) \leq 1. \tag{7.4}$$

Theorem 7.1. *If $C(0) = 0$, equation (7.1) has a unique solution and it is trivial:*

$$C(x) \equiv 0, \quad S(x) \equiv 0.$$

Proof. If $C(0) = 0$, equation (7.2) can be satisfied only if $C(x)^2 \equiv 0$ and $S(x)^2 \equiv 0$, which implies $C(x) \equiv 0$ and $S(x) \equiv 0$. ☐

Setting $y = 0$ in (7.1), we find

$$C(x)\,(1 - C(0)) = S(x)\,S(0). \tag{7.5}$$

Let's square this expression and use the previous results as follows:

$$
\begin{aligned}
& C(x)^2\,(1-C(0))^2 \;=\; S(x)^2\,S(0)^2 \\
\overset{(7.3)}{\Rightarrow}\; & C(x)^2\,(1-C(0))^2 \;=\; S(x)^2\,C(0)(1-C(0)) \\
\Rightarrow\; & (1-C(0))\,\left[(1-C(0))\,C(x)^2 - C(0)\,S(x)^2\right] \;=\; 0 \\
\overset{(7.2)}{\Rightarrow}\; & (1-C(0))\,\left[(1-C(0))\,C(x)^2 - C(0)(C(0)-C(x)^2)\right] \;=\; 0,
\end{aligned}
$$

or

$$(1-C(0))\,\left[C(x)^2 - C(0)^2\right] \;=\; 0. \tag{7.6}$$

We have now prepared ourselves for our second theorem:

Theorem 7.2. *If* $0 < C(0) < 1$, *equation* (7.1) *has the two solutions*

$$C(x) \;=\; c, \quad S(x) \;=\; b$$

and

$$C(x) \;=\; c, \quad S(x) \;=\; -b,$$

where $0 < c < 1$ *and* $b = \sqrt{c - c^2}$.

Proof. If $0 < C(0) < 1$, equation (7.6) requires that $C(x) = \pm C(0)$. Consider the set

$$G \;=\; \{x : C(x) = C(0) \text{ or } C(x) = -C(0)\}.$$

It includes all points x in the domain of $C(x)$; that is, $G = \mathbb{R}$. Endowed with the operation of addition, G is a group. Now consider the subset

$$H \;=\; \{x : C(x) = C(0)\}.$$

H is either the whole set G (if for all points $x \in \mathbb{R}$, $C(x) = C(0)$) or a proper subset of G (if there are points in $\mathbb{R}$ for which $C(x) = -C(0)$). Let's assume that the second statement is true. According to a theorem of algebra, H will be a subgroup of G if, given $x, y \in H$, also $x + (-y) \in H$. If $x, y \in H$, then $C(x) = C(y) = C(0)$. Then, from equation (7.1) we see that $C(x-y) = C(0)^2 + S(0)^2$ or, using (7.3), $C(x-y) = C(0)$. So, $x - y \in H$ and, indeed, H is a subgroup. Using H, we can define an equivalence relation in G: two elements $a, b \in G$ are equivalent, $a \sim b$, if $a - b \in H$. This partitions G into two sets, H and $H' = \{x : C(x) = -C(0)\}$. Formally, one says that the index of H in G is 2. However, it is known in algebra that $\mathbb{R}$ has no such subgroup. H must thus be the whole G and

$$C(x) \;=\; C(0) \quad \text{for all } x \in \mathbb{R}.$$

Let $c = C(0)$. Then, from (7.2), we find that $S(x) = \pm b$, where $b = \sqrt{c - c^2}$. $\square$

We are thus left with the case

$$C(0) = 1. \tag{7.7}$$

Theorem 7.3. *When $C(0) = 1$, the functions C and S satisfy all identities obtained by replacing* cos *with C and* sin *with S in the usual trigonometric identities involving sine and cosine.*

Proof. The condition $C(0) = 1$ implies that

$$C(x)^2 + S(x)^2 = 1 \tag{7.8}$$

and

$$S(0) = 0. \tag{7.9}$$

Then setting $x = 0$ in the defining equation (7.1), we find that $C(x)$ is an even function:

$$C(y) = C(-y). \tag{7.10}$$

Setting $y = -x$ in (7.1), we find

$$C(2x) = C(x)^2 + S(x)\,S(-x). \tag{7.11}$$

Setting $y = 2x$ in (7.1), we get

$$\begin{aligned} C(x) &= C(x)\,C(2x) + S(x)\,S(2x) \\ &= C(x)\,\big(C(x)^2 + S(x)\,S(-x)\big) + S(x)\,S(2x) \\ &= C(x)^3 + C(x)\,S(x)\,S(-x) + S(x)\,S(2x), \end{aligned}$$

where the middle equality comes from (7.11). We can then solve for $S(x)\,S(2x)$:

$$\begin{aligned} S(x)\,S(2x) &= C(x)\,\big[(1 - C(x)^2) - S(x)\,S(-x)\big] \\ &= C(x)\,\big[S(x)^2 - S(x)\,S(-x)\big] \\ &= C(x)\,S(x)\,[S(x) - S(-x)]\,, \end{aligned}$$

where we have used (7.8). If $S(x) \neq 0$, we conclude from this that

$$S(2x) = C(x)\,[S(x) - S(-x)]\,. \tag{7.12}$$

We will show that this equation is also true when $S(x) = 0$ and thus true for any x. From equation (7.8), we see that

$$S(x)^2 = S(-x)^2.$$

If $S(x) = 0$, then $S(-x) = 0$ too. Also $C(x)^2 = 1$ (from (7.8)) and consequently $C(2x) = 1$ (from (7.11)) and $S(2x) = 0$ (from (7.8)). Therefore, when $S(x) = 0$, equation (7.12) is the trivial identity $0 = 0$.

From the last equation, we can see that for any $x/2$ we have

$$S(x/2) = S(-x/2) \quad \text{or} \quad S(x/2) = -S(-x/2).$$

If $S(x/2) = S(-x/2)$, then $S(x) = 0$ (from (7.12)) and also $S(-x) = 0$. If $S(x/2) = -S(-x/2)$ we get, with the help of (7.12),

$$S(x) = 2\,C(x/2)\,S(x/2). \tag{7.13}$$

Substituting $-x$ for x in this equation we find

$$S(-x) = -2\,C(x/2)\,S(x/2),$$

which establishes that $S(x) = -S(-x)$ if $S(x) \neq 0$. Taking into account all of the previous discussion, we conclude that the function $S(x)$ is odd:

$$S(x) = -S(-x), \tag{7.14}$$

and that (7.13) is also true for all x.

Then, setting $-y$ for y in (7.1) we immediately find

$$C(x+y) = C(x)\,C(y) - S(x)\,S(y). \tag{7.15}$$

Setting $x + y$ for y in (7.1) and then applying (7.15), we obtain

$$\begin{aligned} C(-y) &= C(x)\,C(x+y) + S(x)\,S(x+y) \\ &= C(x)\,[C(x)\,C(y) - S(x)\,S(y)] + S(x)\,S(x+y). \end{aligned}$$

Solving for $S(x)\,S(x+y)$ we get

$$\begin{aligned} S(x)\,S(x+y) &= C(y)(1 - C(x)^2) + C(x)\,S(x)\,S(y) \\ &= C(y)\,S(x)^2 + C(x)\,S(x)\,S(y) \\ &= S(x)\,[S(x)\,C(y) + S(y)\,C(x)]\,, \end{aligned}$$

and, if $S(x) \neq 0$,

$$S(x+y) = S(x)\,C(y) + S(y)\,C(x). \tag{7.16}$$

Obviously this equation is also true if $S(y) \neq 0$ (by renaming the variables). If $S(x) = S(y) = 0$ it is also true, because $C(x), C(y) = \pm 1$ implies $C(x+y) = \pm 1$ by (7.15), hence $S(x+y) = 0$.

All trigonometric identities involving sine and cosine can be derived from the trigonometric counterparts of (7.1), (7.8), (7.10), (7.14), and (7.16). Therefore the same manipulations lead to corresponding formulas involving the functions $S(x)$ and $C(x)$. For instance, to get the formula for $C(x-y)$, one replaces y by $-y$ in (7.16) and uses the fact that S is odd, obtaining

$$S(x-y) = S(x)\,C(y) - S(y)\,C(x). \tag{7.17}$$

Continuing like this,

$$\begin{aligned}
C(2x) &= C(x)^2 - S(x)^2 = 2C(x)^2 - 1 = 1 - 2S(x)^2 \ , \\
S(2x) &= 2S(x)C(x) \ , \\
S(3x) &= S(x)\left[4C(x)^2 - 1\right] = S(x)\left[3 - 4S(x)^2\right] \ , \\
C(x) - C(y) &= 2S\left(\frac{x+y}{2}\right)S\left(\frac{y-x}{2}\right) \ , \\
C(x) + C(y) &= 2C\left(\frac{x+y}{2}\right)C\left(\frac{x-y}{2}\right) \ , \\
S(x) - S(y) &= 2S\left(\frac{x-y}{2}\right)C\left(\frac{x+y}{2}\right) \ , \\
S(x) + S(y) &= 2C\left(\frac{x-y}{2}\right)S\left(\frac{x+y}{2}\right) \ ,
\end{aligned}$$

and so on. □

Although we have shown that $S(x)$ and $C(x)$ satisfy all usual relations of trigonometry, we have not yet shown that they are actually the familiar sine and cosine functions. In fact, equation (7.1) can have solutions that are discontinuous at every point. (See Chapter 11 to fully understand this statement. In fact, you may want to read the remainder of this section only after you have read Chapter 11 and, perhaps, Chapter 12.)

Theorem 7.4. *If* $\lim_{x\to 0+} C(x)$ *exists, both* C *and* S *are continuous at* 0.

Proof. Set $\ell = \lim_{x\to 0+} C(x)$. The equality $C(2x) = 2C(x)^2 - 1$ implies that $\ell = 2\ell^2 - 1$; that is, either $\ell = 1$ or $\ell = -\frac{1}{2}$.

Since $|S(x)| \le 1$, the limit of $S(x)$ as $x \to 0^+$ cannot diverge. Then, if $\ell = -\frac{1}{2}$, from the relation $S(3x) = S(x)[4C(x)^2 - 1]$, as $x \to 0^+$, the right-hand side converges to 0 and thus $\lim_{x\to 0^+} S(x) = 0$. However, from $C(x)^2 + S(x)^2 = 1$, if $\ell = -\frac{1}{2}$, we see that $\lim_{x\to 0^+} S(x)^2 = \frac{3}{4}$, which contradicts the previous result. Therefore, $\ell = 1$.

Since $C(x)$ is even, the limit from the other side exists and is the same: $\lim_{x\to 0} C(x) = 1$. Similarly, since $S(x)$ is odd, $\lim_{x\to 0} S(x) = 0$. It follows, since $C(0) = 1$ and $S(0) = 0$, that C and S are continuous at 0. □

Notice that one could have assumed that $\lim_{x\to 0^+} S(x)$ exists to reach the same conclusions. Using the above result, we can now show that

Theorem 7.5. *Both functions* S *and* C *are continuous at all points.*

Proof. Let y be fixed. Showing that C is continuous at y means proving that $\lim_{x\to y} C(x) = C(y)$; this in turn is equivalent to $C(x) - C(y)$ having limit 0 as $x \to y$. We will use the identity

$$C(x) - C(y) = -2S\left(\frac{x+y}{2}\right) S\left(\frac{x-y}{2}\right). \tag{7.18}$$

(See the last paragraph in the proof of Theorem 7.3.) The second factor on the right-hand side of (7.18) goes to 0 as $x \to y$ since S is continuous at 0. The first factor is bounded; therefore, $C(x) - C(y)$ has limit 0 as desired.

The continuity of S at y likewise follows from the continuity of S at 0 and the formula

$$S(x) - S(y) = 2S\left(\frac{x-y}{2}\right) C\left(\frac{x+y}{2}\right). \qquad \square$$

Theorem 7.6. *$S(x)$ is continuous if and only if there exists $\lambda > 0$ such that $S(x)$ does not change sign in the interval $(0, \lambda)$.*

Proof. (Direct proposition) Let $S(x)$ be continuous. Imagine that we cannot find an interval of the form $(0, \lambda)$ where $S(x)$ maintains its sign. Then, by the intermediate value theorem, for each possible interval there must be a zero of $S(x)$. We consider the set $G = \{x : S(x) = 0\}$ of zeros. If x and y are roots, then equation (7.17) implies that $x - y$ is also a root. Therefore, G endowed with the operation of addition is a group dense in $\mathbb{R}$. For any $x \in \mathbb{R}$ we can find a sequence $\{x_n\}$ in G such that $x_n \to x$. By continuity, $S(x) = S(\lim_{n\to\infty} x_n) = \lim_{n\to\infty} S(x_n) = \lim_{n\to\infty} 0 = 0$. That is, $S(x)$ is identically zero. However, this cannot be true since we are searching for non-trivial solutions. Therefore, there must be an interval $(0, \lambda)$ in which $S(x)$ maintains its sign.

(Inverse proposition) Let $S(x)$ maintain its sign in the interval $(0, \lambda)$. Consider two points x, y in this interval, with $y < x$. Then $(x-y)/2$ and $(x+y)/2$ lie in $(0, \lambda)$ and the right side of equation (7.18) is negative, implying that $C(x) < C(y)$. Thus C is a strictly decreasing function in $(0, \lambda)$. The limit $\lim_{x\to 0^+} C(x)$ therefore exists, and S is continuous by the last two theorems. $\square$

Theorem 7.7. *If $C(x)$ is continuous, then $S(x)$ is differentiable and*

$$S'(x) = S'(0)\, C(x). \tag{7.19}$$

Proof. Since $C(x)$ is continuous, $S(x)$ is continuous too, and there is an interval $(0, \lambda)$ such that $S(x)$ maintains its sign. Take $x \in (0, \lambda)$ and set $\varepsilon = x/n$, partitioning the interval $(0, x)$ into n smaller intervals. Then

$$\begin{aligned} S\left(\frac{\varepsilon}{2}\right) \sum_{k=1}^{n} C(k\varepsilon) &= \sum_{k=1}^{n} C\left((2k-1)\frac{\varepsilon}{2} + \frac{\varepsilon}{2}\right) S\left(\frac{\varepsilon}{2}\right) \\ &= \frac{1}{2} \sum_{k=1}^{n} \left[S\left((2k+1)\frac{\varepsilon}{2}\right) - S\left((2k-1)\frac{\varepsilon}{2}\right)\right] \\ &= \frac{1}{2} \left[S\left(x + \frac{\varepsilon}{2}\right) - S\left(\frac{\varepsilon}{2}\right)\right], \end{aligned}$$

where in the second equality we used the identity

$$S(a+b) - S(a-b) = 2\,C(a)\,S(b).$$

Now rewrite the sum in the form

$$\sum_{k=1}^{n} C(k\varepsilon)\,\varepsilon = \left[S\left(x + \frac{\varepsilon}{2}\right) - S\left(\frac{\varepsilon}{2}\right)\right] \frac{\varepsilon/2}{S(\varepsilon/2)}.$$

In the limit $n \to \infty$, $\varepsilon \to 0$, we have

$$\int_0^x C(u)\,du = S(x) \lim_{x \to 0} \frac{x}{S(x)}.$$

Since $C(0) = 1$, in a neighborhood $(0, x)$ of 0 with x sufficiently small, $C(u) > 0$ and $S(x) \neq 0$. The integral is then non-zero and hence the limit in the right-hand side exists and it is non-zero. In particular,

$$\lim_{x \to 0} \frac{S(x)}{x} = \lim_{x \to 0} \frac{S(x) - S(0)}{x - 0} = S'(0).$$

Therefore

$$\int_0^x C(u)du = \frac{S(x)}{S'(0)}.$$

The derivative of the integral in the left-hand side exists and therefore the derivative of the right-hand side exists, too, resulting in equation (7.19). □

Similarly, we can prove:

Theorem 7.8. *If $S(x)$ is continuous, then $C(x)$ is differentiable and*

$$C'(x) = -S'(0)\,S(x). \tag{7.20}$$

For $x = 0$, equations (7.19) and (7.20) give:

$$S'(0) = S'(0)\,C(0) = S'(0) \quad \text{and} \quad C'(0) = -S'(0)\,S(0) = 0.$$

That is, the derivative $S'(0)$ remains undefined and $C'(0) = 0$. Let's set $a = S'(0)$. Since $C(x)$ is strictly decreasing in $(0, \lambda)$, from $S(x)^2 + C(x)^2 = 1$, $S(x)$ must be strictly increasing near 0. Hence, $a > 0$.

We are now ready to identify the functions $S(x)$ and $C(x)$ with known functions. We can do it in at least two ways. The first is a little simpler and much cleaner but it assumes at least elementary knowledge of the theory of differential equations. The second is a bit longer and it contains implicit assumptions. However, it is very popular among scientists, and I thought it worthwhile to let the readers know about it. As I present it, the reader is asked to spot the secret assumptions and therefore identify the "weakness" of that approach.

<u>Method I</u>: The right-hand sides of equations (7.19) and (7.20) are differentiable. Therefore, $C(x)$ and $S(x)$ are twice-differentiable:

$$\begin{aligned} S''(x) &= a\,C'(x) = -a^2\,S(x), \\ C''(x) &= -a\,S'(x) = -a^2\,C(x). \end{aligned}$$

That is, the two functions $S(x)$, $C(x)$ satisfy the differential equation

$$y''(x) + a^2\,y(x) = 0,$$

which is solved by

$$C(x) = \cos(ax) \quad \text{and} \quad S(x) = \sin(ax).$$

<u>Method II</u>: The second way to identify the functions $S(x)$, $C(x)$ is through their Taylor series. Inductively, we can easily see that $C(x)$ and $S(x)$ have derivatives of any order. In particular

$$\begin{aligned} S^{(2n)}(x) &= (-1)^n\,a^{2n}\,S(x), \\ S^{(2n+1)}(x) &= (-1)^n\,a^{2n+1}\,C(x). \end{aligned}$$

Then

$$S^{(2n)}(0) = 0 \quad \text{for } S^{(2n+1)} = (-1)^n a^{2n+1},$$

and we can write the formal Taylor series for $S(x)$:

$$S(x) = \sum_{k=0}^{\infty} \frac{1}{k!}\,S^{(k)}(0)\,x^k = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!}\,(ax)^{2n+1}.$$

This series converges and it has the limit $\sin(ax)$, so $S(x) = \sin(ax)$. Similarly, we can show that

$$C(x) = \sum_{k=0}^{\infty} \frac{1}{k!}\,C^{(k)}(0)\,x^k = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}\,(ax)^{2n},$$

which also converges and has the limit $C(x) = \cos(ax)$.

Taking into account everything that has been discussed in this section, we can now state the following theorem:

Theorem 7.9. *The functional equation* (7.1) *with the conditions*

(a) $C(0) = 1$;

(b) $\lim_{x \to 0^+} C(x)$ *exists;*

(c) $\lim_{x \to 0^+} \frac{S(x)}{x} = 1$,

has a unique solution

$$S(x) = \sin x \quad \textit{and} \quad C(x) = \cos x.$$

Question. *Have you caught the problem with the second solution?*

Answer. The function

$$h(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

is smooth with all its derivatives at $x = 0$ being equal to 0. As a result, the functions $f(x)$ and $f(x) + h(x)$ have the same Taylor series at $x = 0$. Therefore, the claim of the second method that the solution to the original problem is $C(x) = \cos x$ and $S(x) = \sin x$ seems unfounded. Some additional work is necessary to fill the hole. More precisely: let the solution be $S(x) = \sin x + s(x)$, $C(x) = \cos x + c(x)$, where s, c are functions whose Taylor series vanish at $x = 0$. We must show that s, c vanish identically.

From equation (7.19) we find $s'(x) = c(x)$, while from equation (7.20) we find $c'(x) = -s(x)$. From these results we see that s, c satisfy the equation $y'' + y = 0$. Since they must be different from the sine and cosine functions, they vanish identically. Notice that, in trying to bridge the gap, we used the first method—that is, differential equations. One might thus claim that this renders the second method useless. Indeed, this might be the case in what I chose to present. I could have tried to establish functional equations for s and c and from them show that $c = s \equiv 0$. However, I mainly wanted to demonstrate that the second method can be made precise and, in some problems where no other method is available, it can be useful.

Returning to the analytic sine and cosine, we can now see that they result in a unique solution of (7.1): $S(x) = \sin x$, $C(x) = \cos x$. However, the conditions imposed are slightly stronger that the minimal conditions required.

7.2 D'Alembert–Poisson I Equation

Problem 7.10. *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ satisfying*

$$f(x+y) + f(x-y) = 2\,f(x)\,f(y) \quad \textit{for all } x, y \in \mathbb{R}. \tag{7.21}$$

Comment. Equation (7.21) is sometimes referred to as the **D'Alembert functional equation** [8], sometimes as the **Poisson functional equation** [34], and occasionally as the **cosine functional equation** [53]. I will call it the **D'Alembert–Poisson I functional equation**.

Solution. The equation (7.21) may be solved using the $\mathbb{NQR}$ method as described in Chapter 6. However, it admits trigonometric solutions such as $\cos(bx)$, and if we follow the $\mathbb{NQR}$ method verbatim we shall face difficulties that arise from the addition properties of $\cos(mx/n)$, for integer m, n. For this reason, Cauchy himself presented a modification of his $\mathbb{NQR}$ method that simplified the problem considerably. This is the method we follow here.

Setting $y = 0$ in (7.21), we find

$$2f(x) = 2f(x)f(0).$$

Therefore, either $f(x) \equiv 0$ or $f(0) = 1$. We shall investigate the second case. Since $f(x)$ is continuous and $f(0) = 1$, there exists $\varepsilon > 0$ such that $f(x) > 0$ for all $x \in [-\varepsilon, +\varepsilon]$.

Setting now $x = 0$ in (7.21), we find

$$f(y) + f(-y) = 2f(0)f(y) \;\Leftrightarrow\; f(-y) = f(y).$$

Therefore the function f is even and we only need to find its values for $x > 0$. Initially, we shall find the value of f for the points $\varepsilon, \varepsilon/2, \varepsilon/4, \ldots$ in the interval $[0, +\varepsilon]$. Toward this goal, let $y = x$ in (7.21):

$$f(x)^2 = \frac{1 + f(2x)}{2}. \tag{7.22}$$

At $x = \varepsilon$, we know that $f(\varepsilon) > 0$. There are two possibilities. Either $f(\varepsilon) \leq 1$ or $f(\varepsilon) > 1$. We shall examine the first case; the second is handled similarly. We define a number $\theta_0 \in [0, \pi/2]$ such that $f(\varepsilon) = \cos\theta_0$.

Setting $x = \varepsilon/2$ in (7.22), we find

$$f(\varepsilon/2)^2 = \frac{1 + f(\varepsilon)}{2} = \frac{1 + \cos\theta_0}{2} = \cos^2\frac{\theta_0}{2},$$

or, equivalently, $f(\varepsilon/2) = \cos(\theta_0/2)$, since both numbers are positive. Then, setting $x = \varepsilon/4$ in (7.22), we find

$$f(\varepsilon/4)^2 = \frac{1 + f(\varepsilon/2)}{2} = \frac{1 + \cos(\theta_0/2)}{2} = \cos^2\frac{\theta_0}{4},$$

or, equivalently, $f(\varepsilon/4) = \cos(\theta_0/4)$. Continuing inductively, we have

$$f\left(\frac{\varepsilon}{2^n}\right) = \cos\frac{\theta_0}{2^n} \quad \text{for all } n \in \mathbb{N}^*.$$

Now we shall evaluate the function f at the points $m\varepsilon/2^n$, $m, n \in \mathbb{N}^*$. This will allow us to have arguments for f out of the interval $[0, \varepsilon]$. To implement this, let's set $x = ny$ in (7.21) to find

$$f((n+1)y) = 2f(ny)f(y) - f((n-1)y). \tag{7.23}$$

In this equation we set $n = 2$, $y = \varepsilon/2^n$:

$$f\Big(\frac{3\varepsilon}{2^n}\Big) = 2f\Big(\frac{\varepsilon}{2^n}\Big)f\Big(\frac{\varepsilon}{2^{n-1}}\Big) - f\Big(\frac{\varepsilon}{2^n}\Big) = 2\cos\frac{\theta_0}{2^n}\cos\frac{\theta_0}{2^{n-1}} - \cos\frac{\theta_0}{2^n} = \cos\frac{3\theta_0}{2^n},$$

since

$$\begin{aligned}\cos(3x) = \cos(2x + x) &= \cos(2x)\cos x - \sin(2x)\sin x \\ &= \cos(2x)\cos x - 2\sin^2 x\cos x \\ &= \cos(2x)\cos x - 2\frac{1-\cos(2x)}{2}\cos x \\ &= 2\cos(2x)\cos x - \cos x.\end{aligned}$$

Inductively,

$$f\left(\frac{m\varepsilon}{2^n}\right) = \cos\frac{m\theta_0}{2^n}. \tag{7.24}$$

Let $x \in \mathbb{R}_+$. We take sequences $\{m_i\}$ and $\{n_i\}$ of natural numbers such that $\lim_{i\to\infty} m_i\varepsilon/2^{n_i} = x$; this is possible because the set of points of the form $m\varepsilon/2^n$ is dense in $\mathbb{R}$. Then $\lim_{i\to\infty} m_i/2^{n_i} = x/\varepsilon$, and since f is continuous,

$$f(x) = \cos\Big(\frac{\theta_0}{\varepsilon}x\Big) \quad \text{for } x \in \mathbb{R}_+.$$

We define $a = \theta_0/\varepsilon$ and, because f is even,

$$f(x) = \cos(ax) \quad \text{for } x \in \mathbb{R}.$$

The case $f(\varepsilon) > 1$ is studied similarly. We define a number ξ_0 such that $f(\varepsilon) = \cosh\xi_0$. Repeating the steps presented above, we obtain

$$f(x) = \cosh(ax) \quad \text{for } x \in \mathbb{R}. \qquad \square$$

7.3 D'Alembert–Poisson II Equation

Problem 7.11. *Find all continuous functions* $f : \mathbb{R} \to \mathbb{R}$ *which satisfy*

$$f(x+y)\,f(x-y) \;=\; f(x)^2 - f(y)^2 \quad \text{for all } x, y \in \mathbb{R}. \tag{7.25}$$

Comment. Occasionally this equation is called the **sine functional equation** [53]. To match my terminology of equation (7.21), I will refer to equation (7.25) as the **D'Alembert–Poisson II functional equation**.

Comment. Equation (7.25) may be written equivalently as

$$f(x)\,f(y) \;=\; f\left(\frac{x+y}{2}\right)^2 - f\left(\frac{x-y}{2}\right)^2 \quad \text{for all } x, y \in \mathbb{R}. \tag{7.26}$$

Solution. An obvious solution is $f(x) \equiv 0$. We shall search for solutions that do not vanish identically. Therefore, there exists $x_0 \in \mathbb{R}$ such that $f(x_0) \neq 0$. We can thus define

$$g(x) \;=\; \frac{f(x+x_0) - f(x-x_0)}{2f(x_0)}. \tag{7.27}$$

Defined like this, it seems that g depends on the point x_0 selected. However, this is not true. The function g is independent of x_0. To see this, let $\tilde{x}_0$ be another point for which $f(\tilde{x}_0) \neq 0$. We define

$$\tilde{g}(x) \;=\; \frac{f(x+\tilde{x}_0) - f(x-\tilde{x}_0)}{2f(\tilde{x}_0)}.$$

We will show that $g(x) \equiv \tilde{g}(x)$. Indeed, multiplying and dividing the right-hand side of the equation that defines $\tilde{g}(x)$ by $f(x_0)$ and then using equation (7.26) twice, we find

$$\begin{aligned}
\tilde{g}(x) \;&=\; \frac{f(x+\tilde{x}_0)f(x_0) - f(x-\tilde{x}_0)f(x_0)}{2f(\tilde{x}_0)f(x_0)} \\
&\overset{(7.26)}{=} \frac{f(\frac{x+\tilde{x}_0+x_0}{2})^2 - f(\frac{x+\tilde{x}_0-x_0}{2})^2 - f(\frac{x-\tilde{x}_0+x_0}{2})^2 + f(\frac{x-\tilde{x}_0-x_0}{2})^2}{2f(\tilde{x}_0)f(x_0)} \\
&=\; \frac{\left[f(\frac{x+x_0+\tilde{x}_0}{2})^2 - f(\frac{x+x_0-\tilde{x}_0}{2})^2\right] - \left[f(\frac{x-x_0+\tilde{x}_0}{2})^2 - f(\frac{x-x_0-\tilde{x}_0}{2})^2\right]}{2f(\tilde{x}_0)f(x_0)} \\
&\overset{(7.26)}{=} \frac{f(x+x_0)f(\tilde{x}_0) - f(x-x_0)f(\tilde{x}_0)}{2f(\tilde{x}_0)f(x_0)} \\
&=\; \frac{f(x+x_0) - f(x-x_0)}{2f(x_0)} \;=\; g(x).
\end{aligned}$$

The function $g(x)$ satisfies (7.21). To see this,

$$\begin{aligned} 2g(x)g(y) &= 2\left(\frac{f(x+x_0)-f(x-x_0)}{2f(x_0)}\right)\left(\frac{f(y+x_0)-f(y-x_0)}{2f(x_0)}\right) \\ &= \frac{1}{2f(x_0)^2}\,[f(x+x_0)f(y+x_0)-f(x+x_0)f(y-x_0) \\ &\qquad -f(x-x_0)f(y+x_0)+f(x-x_0)f(y-x_0)]\,. \end{aligned}$$

Let $u = \frac{1}{2}(x+y)$ and $v = \frac{1}{2}(x-y)$ or, equivalently, $x = u+v$ and $y = u-v$. Then

$$\begin{aligned} 2g(x)g(y) &= \frac{1}{2f(x_0)^2}\,[f(u+v+x_0)f(u-v+x_0)-f(u+v+x_0)f(u-v-x_0) \\ &\qquad -f(u+v-x_0)f(u-v+x_0)+f(u+v-x_0)f(u-v-x_0)] \\ &= \frac{1}{2f(x_0)^2}\,\big[f\big((u+x_0)+v\big)\,f\big((u+x_0)-v\big) \\ &\qquad -f\big(u+(v+x_0)\big)\,f\big(u-(v+x_0)\big) \\ &\qquad -f\big(u+(v-x_0)\big)\,f\big(u-(v-x_0)\big) \\ &\qquad +f\big((u-x_0)+v\big)\,f\big((u-x_0)-v\big)\big]\,. \end{aligned}$$

We can use (7.25) to rewrite the right-hand side of the last equation:

$$\begin{aligned} 2g(x)g(y) &= \frac{f(u+x_0)^2-f(v)^2-f(u)^2+f(v+x_0)^2}{2f(x_0)^2} \\ &\quad +\frac{-f(u)^2+f(v-x_0)^2+f(u-x_0)^2-f(v)^2}{2f(x_0)^2}. \end{aligned}$$

Now, we rearrange the terms on the right-hand side and use (7.25) once more:

$$\begin{aligned} &2g(x)g(y) \\ &=\frac{f(u+x_0)^2-f(u)^2+f(v+x_0)^2-f(v)^2-[f(v)^2-f(v-x_0)^2]-[f(u)^2-f(u-x_0)^2]}{2f(x_0)^2} \\ &=\frac{f(2u+x_0)\,f(x_0)+f(2v+x_0)\,f(x_0)-f(2v-x_0)\,f(x_0)-f(2u-x_0)\,f(x_0)}{2f(x_0)^2} \\ &=\frac{f(2u+x_0)+f(2v+x_0)-f(2v-x_0)-f(2u-x_0)}{2f(x_0)} \\ &=\frac{f(2u+x_0)-f(2u-x_0)}{2f(x_0)}+\frac{f(2v+x_0)-f(2v-x_0)}{2f(x_0)} = g(2u)+g(2v), \end{aligned}$$

or

$$2g(x)g(y) = g(x+y) + g(x-y).$$

Since g is continuous,

$$g(x) \equiv 0,\ g(x) = \cos(bx),\ g(x) = \cosh(bx).$$

Knowing $g(x)$, we can invert equation (7.27):

$$f(x+y) - f(x-y) = 2\, f(y)\, g(x)$$

for all $x \in \mathbb{R}$ and $y \in \mathbb{R}$ such that $f(y) \neq 0$.

For $g(x) \equiv 0$,

$$f(x+y) - f(x-y) = 0 \ \Rightarrow\ f(x+2y) = f(x) \ \Rightarrow\ f(2y) = f(0).$$

From (7.25) we can see that $f(0) = 0$ and therefore $f(x) \equiv 0$. This solution is not consistent with our assumptions.

For $g(x) = \cos(bx)$,

$$f(x+y) - f(x-y) = 2\, f(y)\, \cos(bx).$$

For $x = 0$, we find

$$f(y) - f(-y) = 2\, f(y) \ \Rightarrow\ f(-y) = -f(y).$$

Therefore,

$$f(y+x) + f(y-x) = 2\, f(y)\, \cos(bx).$$

This equation has been solved in Problem 4.6 and the solution is

$$f(x) = A\, \sin(bx) + B\, \cos(bx).$$

Since f must be odd,

$$f(x) = A\, \sin(bx)$$

is the final result.

For $g(x) = \cosh(bx)$, we find similarly

$$f(x) = A\, \sinh(bx). \qquad \square$$

7.4 Solved Problems

Problem 7.12. *Consider a function $\mathcal{C} : \mathbb{R} \to \mathbb{R}$ satisfying the conditions:*

(A) *It is continuous at all points $x \in \mathbb{R}$.*

(B) *It satisfies the functional equation*

$$\mathcal{C}(x+y) + \mathcal{C}(x-y) \;=\; 2\,\mathcal{C}(x)\,\mathcal{C}(y) \tag{7.28}$$

(C) *There is a λ that is the least nonnegative root of $\mathcal{C}(x) = 0$.*

(D) $\mathcal{C}(0) > 0$.

Prove that

(a) $\mathcal{C}(x) > 0$ *for all* $x \in (0, \lambda)$.

(b) $\mathcal{C}(0) = 1$.

(c) $\mathcal{C}(-x) = \mathcal{C}(x)$ *for all* $x \in \mathbb{R}$.

(d) $\mathcal{C}(x + 2\lambda) = -\mathcal{C}(x)$.

(e) $\mathcal{C}(2x) = 2\mathcal{C}(x)^2 - 1$ *for all* $x \in \mathbb{R}$.

(f) $\mathcal{C}(x/2) = \pm\sqrt{\frac{1+\mathcal{C}(x)}{2}}$ *for all* $x \in \mathbb{R}$.

(g) $|\mathcal{C}(x)| \leq 1$ *for all* $x \in \mathbb{R}$.

(h) *The function $\mathcal{C}(x)$ is periodic with least positive period 4λ.*

Solution. (a) Conditions (C) and (D) imply that $\lambda > 0$. Since λ is the least positive root of $\mathcal{C}(x)$, for any $x \in (0, \lambda)$, $\mathcal{C}(x) \neq 0$. The continuity of $\mathcal{C}(x)$ and the fact that $\mathcal{C}(0) > 0$ imply that $\mathcal{C}(x) > 0$ for any $x \in (0, \lambda)$.

(b) Setting $y = 0$ in the defining equation (7.28), we find $2\mathcal{C}(x) = 2\mathcal{C}(x)\,\mathcal{C}(0)$, and therefore

$$\mathcal{C}(x)\;(\mathcal{C}(0) - 1) \;=\; 0.$$

This equation is true for all x and, in particular, for $x \in (0, \lambda)$. In the latter case, $\mathcal{C}(x) \neq 0$ and thus $\mathcal{C}(0) = 1$.

(c) Setting $x = 0$ in equation (7.28), we find

$$\mathcal{C}(y) + \mathcal{C}(-y) \;=\; 2\mathcal{C}(0)\,\mathcal{C}(y).$$

Combining with the previous result, $\mathcal{C}(0) = 1$, it immediately follows that $\mathcal{C}(x)$ is even.

(d) In equation (7.28), we substitute $x + \lambda$ for x and λ for y:

$$\mathcal{C}(x + 2\lambda) + \mathcal{C}(x) \;=\; 2\mathcal{C}(x+\lambda)\,\mathcal{C}(\lambda) \;\Rightarrow\; \mathcal{C}(x + 2\lambda) + \mathcal{C}(x) \;=\; 0,$$

which is the result sought.

(e) For $y = x$, the defining equation (7.28) immediately gives $\mathcal{C}(2x) + \mathcal{C}(0) = 2\mathcal{C}(x)^2$ or $\mathcal{C}(2x) = 2\mathcal{C}(x)^2 - 1$.

(f) Solving for $C(x)$ the previous relation

$$\mathcal{C}(x) = \pm\sqrt{\frac{1+\mathcal{C}(2x)}{2}},$$

the relation sought is found upon replacing x with $x/2$.

(g) Let's assume that the statement $|\mathcal{C}(x)| \leq 1$, for all x, is not true. Then there is at least a number $a \in \mathbb{R}$ such that $|\mathcal{C}(a)| > 1$.

Let $x = \lambda + a$ and $y = \lambda - a$. Then equation (7.28) gives

$$\mathcal{C}(2\lambda) + \mathcal{C}(2a) = 2\mathcal{C}(\lambda + a)\,\mathcal{C}(\lambda - a). \tag{7.29}$$

The left-hand side can be rewritten as

$$[2\mathcal{C}(\lambda)^2 - 1] + [2\mathcal{C}(a)^2 - 1] = 2[\mathcal{C}(a)^2 - 1].$$

This is a positive number. The right-hand side can be rewritten as

$$2\mathcal{C}(\lambda + a)\mathcal{C}(2\lambda - (\lambda + a)) = -2\mathcal{C}(\lambda + a)\mathcal{C}(-(\lambda + a)) = -\mathcal{C}(\lambda + a)^2.$$

This is a negative number and therefore equation (7.29) is impossible. Therefore, we must have $|\mathcal{C}(x)| \leq 1$, for all x.

(h) That the function is periodic is easy to see by a double application of the result (d):

$$\mathcal{C}(4\lambda + x) = \mathcal{C}(2\lambda + (2\lambda + x)) = -\mathcal{C}(2\lambda + x) = \mathcal{C}(x).$$

It only remains to prove that 4λ is the least period. Let's assume that this is not true and therefore there is a period T such that $0 < T < 4\lambda$. Then

$$\mathcal{C}(T) = \mathcal{C}(0 + T) = \mathcal{C}(0) = 1.$$

We now apply equation (7.28) for $x = y = T/2$:

$$\mathcal{C}(T) + \mathcal{C}(0) = 2\,\mathcal{C}\left(\frac{T}{2}\right)^2 \Rightarrow \mathcal{C}\left(\frac{T}{2}\right) = \pm 1.$$

If $\mathcal{C}(T/2) = -1$, we set $x = y = T/4$ in equation (7.29) to find

$$\mathcal{C}\left(\frac{T}{2}\right) + \mathcal{C}(0) = 2\,\mathcal{C}\left(\frac{T}{4}\right)^2 \Rightarrow \mathcal{C}\left(\frac{T}{4}\right) = 0.$$

That is, $T/4$ is a positive root of $\mathcal{C}(x)$. Since $0 < T/4 < \lambda$, this is impossible. If $\mathcal{C}(T/2) = 1$, we set $x = \lambda + T/4$, $y = \lambda - T/4$ in equation (7.28) to find

$$\mathcal{C}(2\lambda)+\mathcal{C}\left(\frac{T}{2}\right) = 2\,\mathcal{C}\left(\lambda+\frac{T}{4}\right)\mathcal{C}\left(\lambda-\frac{T}{4}\right) \Rightarrow \mathcal{C}\left(\lambda+\frac{T}{4}\right)\mathcal{C}\left(\frac{\lambda-T}{4}\right) = 0, \tag{7.30}$$

since, from (a) and (b), $\mathcal{C}(2\lambda) = -1$. We also notice that

$$\mathcal{C}\left(\lambda+\frac{T}{4}\right) = \mathcal{C}\left(2\lambda-\left(\lambda-\frac{T}{4}\right)\right) = -\mathcal{C}\left(\lambda-\frac{T}{4}\right).$$

Equation (7.30) is thus equivalent to

$$\mathcal{C}\left(\lambda-\frac{T}{4}\right)^2 = 0.$$

However, since $0 < \lambda - T/4 < \lambda$, this is impossible, too. We thus conclude that 4λ is the least period. $\square$

Part III

GENERALIZATIONS

Chapter 8

The Pexider, Vincze and Wilson Equations

Cauchy's functional equations

$$\begin{aligned} f(x+y) &= f(x)+f(y), \\ f(x+y) &= f(x)\,f(y), \\ f(x\,y) &= f(x)+f(y), \\ f(x\,y) &= f(x)\,f(y), \end{aligned}$$

can be generalized as relations among three functions:

$$\begin{aligned} f(x+y) &= g(x)+h(y), \\ f(x+y) &= g(x)\,h(y), \\ f(x\,y) &= g(x)+h(y), \\ f(x\,y) &= g(x)\,h(y). \end{aligned}$$

Even more, one can generalize them to include an arbitrary number of functions:

$$\begin{aligned} f(x_1+x_2+\cdots+x_n) &= f_1(x_1)+f_2(x_2)+\cdots+f_n(x_n), \\ f(x_1+x_2+\cdots+x_n) &= f_1(x_1)\,f_2(x_2)\,\ldots\,f_n(x_n), \\ f(x_1\,x_2\,\ldots\,x_n) &= f_1(x_1)+f_2(x_2)+\cdots+f_n(x_n), \\ f(x_1\,x_2\,\ldots\,x_n) &= f_1(x_1)\,f_2(x_2)\,\ldots\,f_n(x_n). \end{aligned}$$

8.1 The First Pexider Equation

In order to understand fully Pexider's equations, let's first understand the original equation containing three functions. Then we will solve the most general version of it.

Problem 8.1. *Find all continuous functions $f, g, h : \mathbb{R} \to \mathbb{R}$ that satisfy the functional relation*

$$f(x+y) = g(x) + h(y) \quad \textit{for all } x, y. \tag{8.1}$$

Solution. Setting $x = y = 0$ in the defining equation, we see that

$$f(0) = g(0) + h(0). \tag{8.2}$$

Also, if $y = 0$ or $x = 0$,

$$f(x) = g(x) + h(0), \tag{8.3}$$
$$f(y) = g(0) + h(y). \tag{8.4}$$

Adding these two equations together and using the defining equation and (8.2), we find

$$f(x) + f(y) = f(x+y) + f(0),$$

which can be rewritten as

$$\big[f(x) - f(0)\big] + \big[f(y) - f(0)\big] = \big[f(x+y) - f(0)\big].$$

That is, the function $f(x) - f(0)$ satisfies the linear Cauchy equation (5.1) and therefore f is given by

$$f(x) = cx + a + b, \quad c \in \mathbb{R}.$$

In the above equation we set $g(0) = a$, $h(0) = b$ and thus $f(0) = a + b$. From (8.3) and (8.4), we find then

$$g(x) = cx + a, \quad h(x) = cx + b. \qquad \square$$

Having understood the simple case of three different functions we can now solve the problem in full generality.

Problem 8.2. *Find all continuous functions $f, f_i : \mathbb{R} \to \mathbb{R}$, $i = 1, 2, \ldots, n$, that satisfy the functional relation*

$$f(x_1 + x_2 + \cdots + x_n) = f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) \tag{8.5}$$

for all $x_1, x_2, \ldots, x_n$.

Comment. Equation (8.5) (and thus the simpler case (8.1) too) is known as the **first Pexider functional equation** or the **linear Pexider functional equation**.

Solution. Setting $x_1 = x_2 = \cdots = x_n = 0$ in the defining equation, we see that

$$f(0) = f_1(0) + f_2(0) + \cdots + f_n(0). \tag{8.6}$$

Also, we set successively $x_1 = x_2 = \cdots = x_{i-1} = x_{i+1} = \cdots = x_n = 0$, for $i = 1, 2, \ldots, n$:

$$\begin{aligned}
f(x_1) &= f_1(x_1) + f_2(0) + f_3(0) + \cdots + f_{n-1}(0) + f_n(0),\\
f(x_2) &= f_1(0) + f_2(x_2) + f_3(0) + \cdots + f_{n-1}(0) + f_n(0),\\
&\cdots\cdots\\
f(x_n) &= f_1(0) + f_2(0) + f_3(0) + \cdots + f_{n-1}(0) + f_n(x_n).
\end{aligned}$$

Adding these n equations and using the defining equation and (8.6), we find

$$f(x_1) + f(x_2) + \cdots + f(x_n) = f(x_1 + x_2 + \cdots + x_n) + (n-1)\, f(0),$$

or

$$\big[f(x_1) - f(0)\big] + \cdots + \big[f(x_n) - f(0)\big] = \big[f(x_1 + x_2 + \cdots + x_n) - f(0)\big].$$

Taking $x_3, \ldots, x_n = 0$ we obtain

$$\big[f(x_1) - f(0)\big] + \big[f(x_2) - f(0)\big] = \big[f(x_1 + x_2) - f(0)\big].$$

That is, the function $f(x) - f(0)$ satisfies the linear Cauchy equation (5.1). Therefore, $f(x) - f(0) = cx$. Setting $f_i(0) = a_i$, for all i, we then have

$$f(x) = cx + \sum_{i=1}^{n} a_i \quad \text{for } c \in \mathbb{R}.$$

Then

$$f_i(x) = cx + a_i \quad \text{for all } i = 1, 2, \ldots, n. \qquad \square$$

8.2 The Second Pexider Equation

Problem 8.3. *Find all continuous functions $f, f_i : \mathbb{R} \to \mathbb{R}$, $i = 1, 2, \ldots, n$, that satisfy the functional equation*

$$f(x_1 + x_2 + \cdots + x_n) = f_1(x_1)\, f_2(x_2) \ldots f_n(x_n) \tag{8.7}$$

for all $x_1, x_2, \ldots, x_n$, subject to the condition that none of the functions vanishes identically.

Comment. Equation (8.7) is known as the **second Pexider functional equation** or the **exponential Pexider functional equation**.

Solution. Setting $x_1 = x_2 = \cdots = x_n = 0$ in the defining equation, we see that

$$f(0) = f_1(0)\, f_2(0) \ldots f_n(0). \tag{8.8}$$

Also, we set successively $x_1 = x_2 = \cdots = x_{i-1} = x_{i+1} = \cdots = x_n = 0$, for $i = 1, 2, \ldots, n$:

$$\begin{aligned}
f(x_1) &= f_1(x_1)\, f_2(0)\, f_3(0) \ldots f_{n-1}(0)\, f_n(0),\\
f(x_2) &= f_1(0)\, f_2(x_2)\, f_3(0) \ldots f_{n-1}(0)\, f_n(0),\\
&\ldots\ldots\\
f(x_n) &= f_1(0)\, f_2(0)\, f_3(0) \ldots f_{n-1}(0)\, f_n(x_n).
\end{aligned}$$

Multiplying these n equations together and using the defining equation and (8.8), we find

$$f(x_1)\, f(x_2) \ldots f(x_n) = f(x_1 + x_2 + \cdots + x_n)\, f(0)^{n-1}.$$

If $f(0) = 0$, then $f_1(0)\, f_2(0) \ldots f_n(0) = 0$, so at least one of $f_1(0), f_2(0), \ldots,$ $f_n(0)$ will vanish, which in turn implies that $f(x) = 0$ for $x \in \mathbb{R}$. Since we are looking for solutions which include no identically vanishing functions, we have $f(0) \neq 0$ and

$$\frac{f(x_1)}{f(0)} \cdots \frac{f(x_n)}{f(0)} = \frac{f(x_1 + x_2 + \cdots + x_n)}{f(0)},$$

or, setting $x_3 = \cdots = x_n = 0$,

$$\frac{f(x_1)}{f(0)} \frac{f(x_2)}{f(0)} = \frac{f(x_1 + x_2)}{f(0)}.$$

That is, the function $f(x)/f(0)$ satisfies the exponential Cauchy equation (5.8), and therefore is of the form c^x for $c \in \mathbb{R}_+^*$. It follows that f is given by

$$f(x) = \left(\prod_{i=1}^{n} a_i\right) c^x \quad \text{for } c \in \mathbb{R}_+^*,$$

where we set $f_i(0) = a_i$ for all i. Then

$$f_i(x) = a_i\, c^x \quad \text{for } i = 1, 2, \ldots, n. \qquad \square$$

8.3 The Third Pexider Equation

Problem 8.4. *Find all continuous, nonconstant functions* $f, f_i : \mathbb{R}_+^* \to \mathbb{R}$, $i = 1, 2, \ldots, n$, *that satisfy the functional relation*

$$f(x_1 x_2 \ldots x_n) \;=\; f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n) \tag{8.9}$$

for all $x_1, x_2, \ldots, x_n$.

Comment. Equation (8.9) is known as the **third Pexider functional equation** or the **logarithmic Pexider functional equation**.

Solution. Setting $x_1 = x_2 = \cdots = x_n = 1$ in the defining equation, we see that

$$f(1) \;=\; f_1(1) + f_2(1) + \cdots + f_n(1). \tag{8.10}$$

Also, we set successively $x_1 = x_2 = \cdots = x_{i-1} = x_{i+1} = \cdots = x_n = 1$, for $i = 1, 2, \ldots, n$:

$$\begin{aligned}
f(x_1) &= f_1(x_1) + f_2(1) + f_3(1) + \cdots + f_{n-1}(1) + f_n(1),\\
f(x_2) &= f_1(1) + f_2(x_2) + f_3(1) + \cdots + f_{n-1}(1) + f_n(1),\\
&\;\;\cdots\cdots\\
f(x_n) &= f_1(1) + f_2(1) + f_3(1) + \cdots + f_{n-1}(1) + f_n(x_n).
\end{aligned}$$

Adding these n equations together and using the defining equation and (8.10), we find

$$f(x_1) + f(x_2) + \cdots + f(x_n) \;=\; f(x_1 x_2 \ldots x_n) + (n-1)\, f(1),$$

or

$$\big[f(x_1) - f(1)\big] + \cdots + \big[f(x_n) - f(1)\big] \;=\; \big[f(x_1 x_2 \ldots x_n) - f(1)\big],$$

or, setting $x_3 = \cdots = x_n = 1$,

$$\big[f(x_1) - f(1)\big] + \big[f(x_2) - f(1)\big] \;=\; \big[f(x_1 x_2) - f(1)\big].$$

That is, the function $f(x) - f(1)$ satisfies the logarithmic Cauchy equation (5.14), and therefore it equals either zero or $\log_\gamma x$, for some positive γ. Therefore, a nonconstant f is given by

$$f(x) \;=\; \log_\gamma x + \sum_{i=1}^{n} a_i,$$

where we set $f_i(1) = a_i$, for all i. Then

$$f_i(x) \;=\; \log_\gamma x + a_i \quad \text{for } i = 1, 2, \ldots, n. \qquad \square$$

8.4 The Fourth Pexider Equation

Problem 8.5. *Let* $f, f_i : \mathbb{R}_+^* \to \mathbb{R}$, $i = 1, 2, \ldots, n$, *be* $n+1$ *continuous functions that satisfy the functional relation*

$$f(x_1 x_2 \ldots x_n) = f_1(x_1)\, f_2(x_2) \ldots f_n(x_n), \tag{8.11}$$

for all $x_1, x_2, \ldots, x_n$. *Find all such nonconstant functions* f, f_i.

Comment. Equation (8.11) is known as the **fourth Pexider functional equation** or the **power Pexider functional equation**.

Solution. Setting $x_1 = x_2 = \cdots = x_n = 1$ in the defining equation, we see that

$$f(1) = f_1(1)\, f_2(1) \ldots f_n(1). \tag{8.12}$$

Also, we set successively $x_1 = x_2 = \cdots = x_{i-1} = x_{i+1} = \cdots = x_n = 1$, for $i = 1, 2, \ldots, n$:

$$\begin{aligned} f(x_1) &= f_1(x_1)\, f_2(1)\, f_3(1) \ldots f_{n-1}(1)\, f_n(1),\\ f(x_2) &= f_1(1)\, f_2(x_2)\, f_3(1) \ldots f_{n-1}(1)\, f_n(1),\\ &\cdots\cdots\\ f(x_n) &= f_1(1)\, f_2(1)\, f_3(1) \ldots f_{n-1}(1)\, f_n(x_n). \end{aligned}$$

Multiplying these n equations together and using the defining equation and (8.12), we find

$$f(x_1)\, f(x_2) \ldots f(x_n) = f(x_1 x_2 \ldots x_n)\, f(1)^{n-1}.$$

If $f(1) = 0$, then $f_1(1)\, f_2(1) \ldots f_n(1) = 0$. At least one of $f_1(1), f_2(1), \ldots, f_n(1)$ will vanish, which in turn implies that $f(x) = 0$, for all $x \in \mathbb{R}_+^*$. Since we are looking for solutions that include no identically vanishing functions, $f(1) \neq 0$ and

$$\frac{f(x_1)}{f(1)} \cdots \frac{f(x_n)}{f(1)} = \frac{f(x_1 x_2 \ldots x_n)}{f(1)},$$

or, setting $x_3 = \cdots = x_n = 1$,

$$\frac{f(x_1)}{f(1)}\, \frac{f(x_2)}{f(1)} = \frac{f(x_1 x_2)}{f(1)}.$$

That is, the function $f(x)/f(1)$ satisfies the power Cauchy equation, (5.20), and it is therefore either identically zero or of the form x^c, for a constant c. Therefore, any nonconstant f is given by

$$f(x) = \left(\prod_{i=1}^{n} a_i\right) x^c,$$

where we set $f_i(1) = a_i$ for all i. Then

$$f_i(x) = a_i\, x^c \quad \text{for } i = 1, 2, \ldots, n. \qquad \square$$

8.5 The Vincze Functional Equations

A further way to generalize the equations

$$f(x+y) = g(x) + h(y),$$
$$f(x+y) = g(x)\, h(y),$$

is the functional equation

$$f(x+y) \;=\; g_1(x)\, g_2(y) + h(y).$$

Similarly, we can generalize the equations

$$f(x\,y) = g(x) + h(y),$$
$$f(x\,y) = g(x)\, h(y)$$

to the functional equation

$$f(x\,y) \;=\; g_1(x)\, g_2(y) + h(y).$$

In this section we solve such equations.

Problem 8.6. *Find all continuous functions $f, g_1, g_2, h : \mathbb{R} \to \mathbb{R}$ that satisfy the functional relation*

$$f(x+y) \;=\; g_1(x)\, g_2(y) + h(y) \quad \textit{for all } x, y. \tag{8.13}$$

Comment. Equation (8.13) is known as the **first Vincze functional equation**, or Vincze I.

Solution. Setting $y = 0$ in the defining equation (8.13), we see that

$$f(x) \;=\; g_1(x)\, g_2(0) + h(0). \tag{8.14}$$

At this point we examine two separate cases: $g_2(0) = 0$ or $g_2(0) \neq 0$.

Case I: If $g_2(0) = 0$, then $f(x) = h(0)$, for all x. For $x = 0$, equation (8.13) then gives $h(y) = h(0) - g_1(0)g_2(y)$. Substituting this expression back into (8.13), we find $g_2(y)\,[g_1(x) - g_1(0)] = 0$. Therefore, either $g_2(x) = 0$ for all x or $g_1(x) = g_1(0)$ for all x. Therefore, the solutions of (8.13) in this case are:

$$\left\{\begin{array}{l} f(x) \;=\; a \\ g_1(x) \;=\; \text{arbitrary} \\ g_2(x) \;=\; 0 \\ h(x) = a \end{array}\right\} \quad \text{or} \quad \left\{\begin{array}{l} f(x) \;=\; a \\ g_1(x) \;=\; b \\ g_2(x) \;=\; \text{arbitrary}, \quad g_2(0) = 0 \\ h(x) = a - bg_2(x) \end{array}\right\},$$

where a and b are constants.

Case II: If $g_2(0) \neq 0$, then equation (8.14) can be solved for $g_1(x)$:

$$g_1(x) = \frac{f(x) - h(0)}{g_2(0)}. \tag{8.15}$$

Substituting this result in (8.13), we find

$$f(x+y) = f(x)\,\frac{g_2(y)}{g_2(0)} + h(y) - h(0)\,\frac{g_2(y)}{g_2(0)}.$$

We can rewrite this equation in a nicer form using the functions

$$\tilde{g}(x) = \frac{g_2(x)}{g_2(0)}, \quad H(y) = h(y) - h(0)\,\frac{g_2(y)}{g_2(0)}.$$

Then

$$f(x+y) = f(x)\,\tilde{g}(y) + H(y). \tag{8.16}$$

We have succeeded in reducing the initial equation (8.13) with four unknown functions to an equation (8.16) with three unknown functions. We can continue this reduction until we find an equation with only one unknown function. Setting $x = 0$ in (8.16), we find $H(y) = f(y) - f(0)\tilde{g}(y)$. When this is inserted back into (8.16), we get

$$f(x+y) = [f(x) - f(0)]\,\tilde{g}(y) + f(y);$$

that is, we discover an equation with two unknown functions. It is more convenient to work with the function

$$F(x) = f(x) - f(0).$$

Therefore

$$F(x+y) = F(x)\,\tilde{g}(y) + F(y). \tag{8.17}$$

We can now study two separate subcases: either $\tilde{g}(y) \equiv 1$ or $\tilde{g}(y) \not\equiv 1$.

Subcase IIa: $\tilde{g}(y) \equiv 1$ or, equivalently, $g_2(x) \equiv g_2(0)$. In this case, $F(x)$ satisfies (5.1):

$$F(x+y) = F(x) + F(y),$$

with solution $F(x) = cx$ or $f(x) = cx + f(0)$. Equation (8.15) then gives

$$g_1(x) = \frac{f(0) - h(0)}{g_2(0)} + \frac{c}{g_2(0)}\,x,$$

and (8.13) gives $h(x) = cx + h(0)$. Let's list the complete solution:

$$\left\{\begin{array}{l} f(x) = cx + a \\ g_1(x) = \frac{c}{b}\,x + \frac{a-d}{b} \\ g_2(x) = b \\ h(x) = cx + d \end{array}\right\},$$

where a, b, c, d are constants and $b \neq 0$.

Subcase IIb: $\tilde{g}(y) \not\equiv 1$. Then there exists $y_0 \in \mathbb{R}^*$ such that $\tilde{g}(y) \neq 1$. In equation (8.17), set $y = y_0$:

$$F(x + y_0) = F(x)\,\tilde{g}(y_0) + F(y_0).$$

In the same equation we now set $x = y_0$ and $y = x$:

$$F(y_0 + x) = F(y_0)\,\tilde{g}(x) + F(x).$$

Subtracting these two equations together, we find that

$$F(x) = c\,[\tilde{g}(x) - 1], \tag{8.18}$$

where $c = F(y_0)/[\tilde{g}(y_0) - 1]$. We now have two new subcases, $c = 0$ and $c \neq 0$.

Subsubcase IIb1: $c = 0$, so $f(x) = f(0)$ and $g_1(x) = [f(0) - h(0)]/g_2(0)$. The function g_2 is arbitrary, except for the facts that $g_2(0) \neq 0$ and that there is at least one point y_0 where g_2 has a value other than $g_2(0)$ and $h(x) = f(0) - g_2(x)\,[f(0) - h(0)]/g_2(0)$. The solution is thus

$$\left\{\begin{array}{l} f(x) = a \\ g_1(x) = b \\ g_2(x) = \text{arbitrary}, \quad g_2(0) \neq 0, \quad g_2(y_0) \neq g_2(0) \\ h(x) = a - b\,g_2(x) \end{array}\right\},$$

where a and b are constants.

Subsubcase IIb2: $c \neq 0$. Substituting (8.18) in (8.17), we find

$$\tilde{g}(x + y) = \tilde{g}(x)\,\tilde{g}(y);$$

i.e., the function $\tilde{g}(x)$ satisfies the exponential Cauchy functional equation (5.8). So either $\tilde{g}(x) \equiv 0$ (which is rejected since it does not satisfy the condition $\tilde{g}(0) = 1$) or $\tilde{g}(x) = a^x$ for some positive constant a. Therefore, $g_2(x) = g_2(0)\,a^x$ and $f(x) = c\,a^x - c + f(0)$. From (8.15) we have

$$g_1(x) = [c/g_2(0)]\,a^x + (f(0) - h(0) - c)/g_2(0).$$

Finally, $h(x) = (f(0) - c) - (f(0) - c - h(0))\,a^x$. Let's record the final result:

$$\left\{\begin{array}{l} f(x) = c\,a^x + k \\ g_1(x) = \frac{c}{d}\,a^x + \frac{b}{d} \\ g_2(x) = d\,a^x \\ h(x) = k - b\,a^x \end{array}\right\},$$

where a, b, c and d are constants satisfying $a > 0$, $c, d \neq 0$. $\square$

Problem 8.7. *Find all continuous functions* $f, g_1, g_2, h : \mathbb{R} \to \mathbb{R}$ *that satisfy the functional relation*

$$f(x\,y) \;=\; g_1(x)\,g_2(y) + h(y) \quad \textit{for all } x, y. \tag{8.19}$$

Comment. Equation (8.19) is known as the **second Vincze functional equation**, or Vincze II.

Solution. Setting $y = 1$ in the defining equation (8.19), we see that

$$f(x) \;=\; g_1(x)\,g_2(1) + h(1). \tag{8.20}$$

At this point we examine two separate cases: $g_2(1) = 0$ or $g_2(1) \neq 0$.

Case I: If $g_2(1) = 0$, then $f(x) = h(1)$, for all x. For $x = 1$, equation (8.19) then gives $h(y) = h(1) - g_1(1)g_2(y)$. Substituting this expression back in (8.19), we find $g_2(y)\,[g_1(x) - g_1(1)] = 0$. Therefore, either $g_2(x) = 0$ for all x or $g_1(x) = g_1(1)$ for all x. Therefore, the solutions of (8.19) in this case are:

$$\left\{\begin{array}{l} f(x) = a \\ g_1(x) \;=\; \text{arbitrary} \\ g_2(x) \;=\; 0 \\ h(x) = a \end{array}\right\} \quad \text{or} \quad \left\{\begin{array}{l} f(x) \;=\; a \\ g_1(x) \;=\; b \\ g_2(x) \;=\; \text{arbitrary}, \quad g_2(1) = 0 \\ h(x) = a - b g_2(x) \end{array}\right\},$$

where a and b are constants.

Case II: If $g_2(1) \neq 0$, then equation (8.20) can be solved for $g_1(x)$:

$$g_1(x) \;=\; \frac{f(x) - h(1)}{g_2(1)}. \tag{8.21}$$

Substituting this result in (8.19), we find

$$f(x\,y) \;=\; f(x)\,\frac{g_2(y)}{g_2(1)} + h(y) - h(1)\,\frac{g_2(y)}{g_2(1)}.$$

We can rewrite in a nicer form using the functions

$$\tilde{g}(x) \;=\; \frac{g_2(x)}{g_2(1)}, \quad H(x) \;=\; h(y) - h(1)\,\frac{g_2(y)}{g_2(1)}.$$

Then

$$f(x\,y) \;=\; f(x)\,\tilde{g}(y) + H(y). \tag{8.22}$$

We have succeeded in reducing the initial equation (8.19) with four unknown functions to an equation (8.22) with three unknown functions. We can continue this reduction until we find an equation with only one unknown function.

Setting $x = 1$ in (8.22) we find $H(y) = f(y) - f(1)\tilde{g}(y)$. When this is inserted back in (8.22),

$$f(x\,y) \;=\; [f(x) - f(1)]\,\tilde{g}(y) + f(y),$$

we discover an equation with two unknown functions. It is more convenient to work with the function

$$F(x) \;=\; f(x) - f(1).$$

Therefore,

$$F(x\,y) \;=\; F(x)\,\tilde{g}(y) + F(y). \tag{8.23}$$

We can now study two separate subcases: either $\tilde{g}(y) \equiv 1$ or $\tilde{g}(y) \not\equiv 1$.

Subcase IIa: $\tilde{g}(y) \equiv 1$ or, equivalently, $g_2(x) \equiv g_2(1)$. In this case, $F(x)$ satisfies (5.14):

$$F(x\,y) \;=\; F(x) \,+ F(y),$$

with solution $F(x) = \log_\gamma x$ or $f(x) = \log_\gamma x + f(1)$. Equation (8.21) then yields

$$g_1(x) = \frac{f(1) - h(1)}{g_2(1)} + \frac{1}{g_2(1)}\,\log_\gamma x,$$

and (8.19) gives that $h(x) = \log_\gamma x + h(1)$. Let's list the complete solution:

$$\left\{\begin{array}{l} f(x) \;=\; \log_\gamma x + a \\ g_1(x) \;=\; \frac{1}{b}\,\log_\gamma x + \frac{a-d}{b} \\ g_2(x) \;=\; b \\ h(x) = \log_\gamma x + d \end{array}\right\},$$

where a, b, d and γ are constants satisfying $b \neq 0$ and $\gamma > 0$.

Subcase IIb: $\tilde{g}(y) \not\equiv 1$. Then there exists $y_0 \in \mathbb{R}^*$ such that $\tilde{g}(y) \neq 1$. In equation (8.23), set $y = y_0$:

$$F(x\,y_0) \;=\; F(x)\,\tilde{g}(y_0) + F(y_0).$$

In the same equation we now set $x = y_0$ and $y = x$:

$$F(y_0\,x) \;=\; F(y_0)\,\tilde{g}(x) + F(x).$$

Subtracting these two equations together, we find that

$$F(x) \;=\; c\,[\tilde{g}(x) - 1], \tag{8.24}$$

where $c = F(y_0)/[\tilde{g}(y_0) - 1]$. We now have two new subcases, $c = 0$ and $c \neq 0$.

Subsubcase IIb1: $c = 0$, so $f(x) = f(1)$ and $g_1(x) = [f(1) - h(1)]/g_2(1)$. The function g_2 is arbitrary, except for the facts that $g_2(0) \neq 0$ and there is at least one point y_0 where g_2 takes a value other than $g_2(1)$ and $h(x) = f(1) - g_2(x)\,[f(1) - h(1)]/g_2(1)$. The solution is thus

$$\left\{\begin{array}{l} f(x) \,=\, a \\ g_1(x) \,=\, b \\ g_2(x) \,=\, \text{arbitrary}, \quad g_2(1) \neq 0, \quad g(y_0) \neq g_2(1) \\ h(x) = a - b\,g_2(x) \end{array}\right\},$$

where a and b are constants.

Subsubcase IIb2: $c \neq 0$. Substituting (8.24) in (8.23), we find

$$\tilde{g}(x\,y) \,=\, \tilde{g}(x)\,\tilde{g}(y);$$

i.e., the function $\tilde{g}(x)$ satisfies the power Cauchy functional equation (5.20). So either $\tilde{g}(x) \equiv 0$ (which is rejected since it does not satisfy the condition $\tilde{g}(1) = 1$) or $\tilde{g}(x) = x^a$ for some constant a. Therefore $g_2(x) = g_2(1)\,x^a$ and $f(x) = c\,x^a - c + f(1)$. From (8.21), $g_1(x) = [c/g_2(1)]\,a^x + (f(1) - h(1) - c)/g_2(1)$. Finally, $h(x) = (f(1) - c) - (f(1) - c - h(1))\,a^x$. Let's record the final result:

$$\left\{\begin{array}{l} f(x) \,=\, c\,x^a + k \\ g_1(x) \,=\, \frac{c}{d}\,x^a + \frac{b}{d} \\ g_2(x) \,=\, d\,x^a \\ h(x) = k - b\,x^a \end{array}\right\},$$

where a, b, c, d, k are constants satisfying $c, d \neq 0$. □

8.6 The Wilson Functional Equations

Wilson has generalized the (7.21) equation in a similar fashion to Pexider's generalization of the Cauchy equations.

Problem 8.8. *Find all continuous functions $f, g : \mathbb{R} \to \mathbb{R}$ that satisfy the functional equation*

$$f(x+y) + f(x-y) \,=\, 2\,f(x)\,g(y) \quad \textit{for all } x, y \in \mathbb{R}. \tag{8.25}$$

Comment. We will call equation (8.25) the **first Wilson functional equation**, or Wilson I. □

We will not present the solution, as it is relatively long. Instead we shall wait until Chapter 12, where stronger than continuity conditions may be imposed on the function, which simplifies the problem considerably. Here, we just present the answer.

Answer (Wilson). The most general continuous solutions of (8.25) are:

1. $f(x) \equiv 0,\ g(x) = \text{arbitrary};$
2. $f(x) = a\cos(\lambda x) + b\sin(\lambda x),\ g(x) = \cos(\lambda x);$
3. $f(x) = a\cosh(\lambda x) + b\sinh(\lambda x),\ g(x) = \cos(\lambda x);$
4. $f(x) = a + b\,x,\ g(x) = 1.$ □

Using this result, Wilson generalized the equation further:

Problem 8.9. *Find all continuous functions $e, f, g, h : \mathbb{R} \to \mathbb{R}$ that satisfy the functional equation*

$$e(x+y) + f(x-y) = 2\,h(x)\,g(y) \quad \textit{for all } x, y \in \mathbb{R}. \tag{8.26}$$

Comment. We will call equation (8.26) the **second Wilson functional equation**, or Wilson II. □

Again, we present only the solution to this equation.

Answer (Wilson). The most general continuous solutions of (8.26) are:

1. $e(x) = c,\ f(x) = -c,\ h(x) = 0,\ g(x) = \text{arbitrary};$
2. $e(x) = c,\ f(x) = -c,\ h(x) = \text{arbitrary},\ g(x) = 0;$
3. $e(x) = a\cos(\lambda x) + b\sin(\lambda x) + s,\ f(x) = c\cos(\lambda x) + d\sin(\lambda x) - s,$
   $g(x) = A\cos(\lambda x) + B\sin(\lambda x),\ h(x) = C\cos(\lambda x) + D\sin(\lambda x)$
   with $a + c = 2AC,\ c - a = 2BD,\ b + d = 2AD,\ b - d = 2BC;$
4. $e(x) = a\cosh(\lambda x) + b\sinh(\lambda x) + s,\ f(x) = c\cosh(\lambda x) + d\sinh(\lambda x) - s,$
   $g(x) = A\cosh(\lambda x) + B\sinh(\lambda x),\ h(x) = C\cosh(\lambda x) + D\sinh(\lambda x)$
   with $a + c = 2AC,\ a - c = 2BD,\ b + d = 2AD,\ b - d = 2BC;$
5. $e(x) = ax^2 + bx + c,\ f(x) = -ax^2 + b'x + c',\ g(x) = Ax + B,\ h(x) = Cx + D$
   with $a = AC/2,\ b + b' = 2BC,\ b - b' = 2AD,\ c + c' = 2BD.$ □

8.7 Solved Problems

As in the case of the Cauchy functional equations, many functional equations can be solved by reduction to one of the Pexider functional equations. In this section, we present several examples.

Problem 8.10. *Find the continuous solutions $f, g, h : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f\left(\frac{x+y}{2}\right) = \frac{g(x) + h(y)}{2} \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. We define the continuous function

$$F(x) = 2\,f\left(\frac{x}{2}\right).$$

Then the functional equation takes the form

$$F(x+y) = g(x) + h(y),$$

which has the solution

$$F(x) = cx + a + b, \quad g(x) = c\,x + a, \quad h(x) = cx + b,$$

where a, b, c are constants. Therefore

$$f(x) = cx + \frac{a+b}{2}, \quad g(x) = c\,x + a, \quad h(x) = cx + b. \qquad \square$$

Problem 8.11. *Find the continuous solutions $f, g, h : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) = \frac{g(x) + h(y)}{1 - g(x)\,h(y)/c^2} \quad \textit{for all } x, y \in \mathbb{R}, \tag{8.27}$$

where c is a constant.

Solution. We define new functions $F(x), G(x), H(x)$ by

$$f(x) = c\,\tan F(x), \quad g(x) = c\,\tan G(x), \quad h(x) = c\,\tan H(x),$$

in terms of which the given functional equation takes the form

$$\tan F(x+y) = \tan(G(x) + H(y)),$$

or

$$F(x+y) = G(x) + H(y) + m\,\pi \quad \text{for } m \in \mathbb{Z}.$$

Rearranging, we find

$$[F(x+y) - m\pi] = [G(x) - m\pi] + [H(y) - m\,\pi],$$

which is, of course, the linear Pexider equation, with solutions

$$F(x) = k\,x + a + b + m\pi, \quad G(x) = k\,x + a + m\pi, \quad H(x) = k\,x + b + m\pi.$$

Therefore,

$$f(x) = c\,\tan(k\,x + a + b), \quad g(x) = c\,\tan(k\,x + a), \quad h(x) = c\,\tan(k\,x + b),$$

where k, a, b, c are constants. $\square$

Comment. It should now be evident to the reader how to solve the functional equations

$$f(x+y) = \frac{f(x)+f(y)}{1+f(x)\,f(y)/c^2} \tag{8.28}$$

and

$$f(x+y) = \frac{f(x)\,f(y)-1}{f(x)\,f(y)},$$

(which admit the solutions $f(x) = c\tanh kx$ and $f(x) = \cot kx$, respectively) and their generalizations to three functions. □

Comment. Problem 8.11 with a single function, that is, the equation

$$f(x+y) = \frac{f(x)+f(y)}{1-f(x)\,f(y)/c^2} \quad \text{for all } x, y \in \mathbb{R}, \tag{8.29}$$

was proposed by Hungary for the IMO 1977 (in the particular case $c = 1$). However, the selection committee did not consider it as a viable candidate for the competition[1] — most probably because the problem is closely related to one widely known to physicists. The functional equation (8.28) is the rule for the relativistic addition of velocities:

$$v_{\text{rel}} = \frac{v_1+v_2}{1+v_1v_2/c^2},$$

where c is the speed of light. Physicists have for decades used the transformation

$$v = c\,\tanh\theta,$$

since the law becomes additive in the variable θ, which is known as the **rapidity**. Using the substitution $\theta \mapsto i\theta$, equation (8.28) is mapped to equation (8.29). Perhaps equations of the form (8.27) and (8.28) should be called the *Einstein–Lorentz equations*, to honor the founders of special relativity. □

Problem 8.12. *Find the continuous solutions $f, g, h : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(\sqrt[n]{x^n+y^n}) = g(x)\,h(y) \quad \textit{for all } x, y \in \mathbb{R},$$

where n is a constant.

Solution. For $y = 0$, the defining equation gives $f(x) = g(x)h(0)$. If $h(0) = 0$, then $f(x) \equiv 0$ and $g(x)$ is arbitrary.

For $x = 0$, the defining equation gives $f(y) = g(0)h(y)$. If $g(0) = 0$, then $f(x) \equiv 0$ and $h(x)$ is arbitrary.

[1] Both the longlist and the shortlist are given in [17].

We shall look for solutions in which none of the functions f, g, h vanishes identically. In this case, we define the new functions F, G, H by

$$F(u) = f(\sqrt[n]{u}), \quad G(u) = g(\sqrt[n]{u}), \quad H(u) = h(\sqrt[n]{u}).$$

Then the defining equation becomes

$$F(u+v) = G(u)\,H(v),$$

where $u = x^n$, $v = y^n$. The solution of this equation with all functions F, G, H non-vanishing is

$$F(u) = ab\,c^u, \quad G(u) = a\,c^u, \quad H(u) = b\,c^u,$$

with a, b any constants and c a positive constant. Then

$$f(x) = ab\,c^{x^n}, \quad g(x) = a\,c^{x^n}, \quad h(x) = b\,c^{x^n}. \qquad \square$$

Chapter 9

Vector and Matrix Variables

So far we have studied functional equations that include functions $f(x)$ of one independent variable x; that is, the domain of f is a subset of $\mathbb{R}$. However, one can study equations that include functions $f(x_1, x_2, \ldots, x_n)$ of many independent variables $x_1, x_2, \ldots, x_n$, where the domain of f is a subset of $\mathbb{R}^n$. One can view the collection of the independent variables as a vector $\vec{x} = (x_1, x_2, \ldots, x_n)$. Examples of equations of this category are

$$\begin{aligned} f(\vec{x}+\vec{y}) &= f(\vec{x}) + f(\vec{y}), \\ f(\vec{x}+\vec{y}) &= f(\vec{x})\, f(\vec{y}), \\ f(\vec{x},\vec{z}) &= f(\vec{x},\vec{y}) + f(\vec{y},\vec{z}). \end{aligned}$$

Even more, it is possible to study functions that take values in $\mathbb{R}^m$. Obviously, in these cases, the function itself is a vector. Examples of equations of this category are

$$\begin{aligned} \vec{f}(x\,y) &= \vec{f}(x) + \vec{f}(y), \\ f(x\,y) &= \vec{g}(x) \cdot \vec{g}(y), \\ \vec{f}(\vec{x}+\vec{y}) &= \vec{g}(\vec{x}) + \vec{h}(\vec{y}), \\ f(\vec{x}+\vec{y}) &= \vec{g}(\vec{x}) \cdot \vec{h}(\vec{y}), \\ \vec{f}(\vec{x},\vec{z}) &= \vec{f}(\vec{x},\vec{y}) + \vec{f}(\vec{y},\vec{z}). \end{aligned}$$

Often it is useful to consider functions from a subset of all matrices to another subset of all matrices. For example, let $\mathcal{M}(n)$ be the set of all nonsingular square matrices of order n with real elements and consider $f : \mathcal{M}(n) \to \mathbb{R}$ such that

$$f(\boldsymbol{X}\,\boldsymbol{Y}) = f(\boldsymbol{X})\, f(\boldsymbol{Y}) \quad \text{for all } \boldsymbol{X}, \boldsymbol{Y} \in \mathcal{M}(n).$$

Another example is the functional equation

$$F(\boldsymbol{X}\,\boldsymbol{Y}) = F(\boldsymbol{X})\, F(\boldsymbol{Y}),$$

for the function $F : \mathcal{M}(n) \to \mathcal{M}(m)$.

9.1 Equations of Cauchy and Pexider Type

Problem 9.1. *Find all continuous functions $f : \mathbb{R}^n \to \mathbb{R}$ that satisfy the functional relation*

$$f(x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) = f(x_1, x_2, \ldots, x_n) + f(y_1, y_2, \ldots, y_n).$$

Solution. In the defining equation we set $x_1 = \cdots = \hat{x}_k = \cdots = x_n = 0$ and $y_1 = \cdots = \hat{y}_k = \cdots = y_n = 0$ (where the hat above a variable means the variable is omitted). Then

$$f(0, \ldots, 0, x_k + y_k, 0 \ldots, 0) = f(0, \ldots, 0, x_k, 0 \ldots, 0) + f(0, \ldots, 0, y_k, 0 \ldots, 0).$$

Let $f_k(x) = f(0, \ldots, 0, x, 0 \ldots, 0)$. Then the previous equation reads

$$f_k(x + y) = f_k(x) + f_k(y),$$

with continuous solution $f_k(x) = c_k\, x$, and c_k a constant.

We can express the function $f(x_1, x_2, \ldots, x_n)$ in terms of $f_k(x)$ as follows:

$$\begin{aligned}
f(x_1, x_2, \ldots, x_n) &= f(x_1 + 0, 0 + x_2, \ldots, 0 + x_n) \\
&= f(x_1, 0, \ldots, 0) + f(0, x_2, \ldots, x_n) \\
&= f(x_1, 0, \ldots, 0) + f(0 + 0, x_2 + 0, 0 + x_3, \ldots, 0 + x_n) \\
&= f(x_1, 0, \ldots, 0) + f(0, x_2, 0, \ldots, 0) + f(0, 0, x_3, \ldots, x_n) \\
&= \ldots\ldots \\
&= f(x_1, 0, \ldots, 0) + f(0, x_2, 0, \ldots, 0) + \cdots + f(0, 0, \ldots, 0, x_n) \\
&= f_1(x_1) + f_2(x_2) + \cdots + f_n(x_n).
\end{aligned}$$

Therefore

$$f(x_1, x_2, \ldots, x_n) = c_1\, x_1 + c_2\, x_2 + \cdots + c_n\, x_n. \qquad \square$$

Comment. Above, we used "expanded" notation. In more compact notation, we just proved that the only continuous solution of the functional equation

$$f(\vec{x} + \vec{y}) = f(\vec{x}) + f(\vec{y})$$

is the linear function

$$f(\vec{x}) = \vec{c} \cdot \vec{x},$$

where $\vec{c}$ is a constant vector. $\square$

The next problem is a direct consequence of the previous one.

Problem 9.2. *Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuous symmetric function*[1] *that satisfies the functional equation*

$$f(x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) = f(x_1, x_2, \ldots, x_n) + f(y_1, y_2, \ldots, y_n),$$

and is such that

$$f(x, x, \ldots, x) = x.$$

Then f is the arithmetic mean of the variables

$$f(x_1, x_2, \ldots, x_n) = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Solution. Since $f(x_1, x_2, \ldots, x_n)$ is a symmetric function, all functions $f_k(x)$ defined previously are identical; let's use the common symbol $g(x)$ to indicate them. Then

$$f(x_1, x_2, \ldots, x_n) = g(x_1) + g(x_2) + \cdots + g(x_n),$$

and

$$x = f(x, x, \ldots, x) = n\, g(x).$$

Therefore $g(x) = x/n$, which immediately implies the result. □

Problem 9.3. *Find all continuous functions $\vec{f} : \mathbb{R}^n \to \mathbb{R}^m$ that satisfy the functional relation*

$$\vec{f}(\vec{x} + \vec{y}) = \vec{f}(\vec{x}) + \vec{f}(\vec{y}).$$

Solution. Let $f_\ell(\vec{x})$, $\ell = 1, 2, \ldots, m$ be the components of the function $\vec{f}$. The given vector equation results in m scalar equations:

$$f_\ell(\vec{x} + \vec{y}) = f_\ell(\vec{x}) + f_\ell(\vec{y}).$$

The continuous solutions of these equations were found in the previous problem:

$$f_\ell(\vec{x}) = \vec{c}_\ell \cdot \vec{x},$$

where $\vec{c}_\ell$ is a constant vector.

It might be more appealing to write the above equation in another form:

$$\begin{bmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix} = \begin{bmatrix} \vec{c}_1 \\ \vec{c}_2 \\ \vdots \\ \vec{c}_n \end{bmatrix} \cdot \vec{x}.$$

In other words, we have arranged the constant vectors in a matrix $\boldsymbol{C}$ such that $\vec{f} = \boldsymbol{C} \cdot \vec{x}$. □

[1] A symmetric function is a function that remains invariant under any permutation of its variables. (See page 204 for the corresponding definition of symmetric polynomials.)

Problem 9.4. *Find all continuous functions $\vec{f}, \vec{g}, \vec{h} : \mathbb{R}^n \to \mathbb{R}^m$ that satisfy the functional equation*

$$\vec{f}(\vec{x}+\vec{y}) = \vec{g}(\vec{x}) + \vec{h}(\vec{y}). \tag{9.1}$$

Solution. Setting $\vec{x} = \vec{0}$ and $\vec{g}(\vec{0}) = \vec{a}$, we get

$$\vec{f}(\vec{y}) = \vec{h}(\vec{y}) + \vec{a}.$$

Repeating the process for $\vec{y}$ and setting $\vec{h}(\vec{0}) = \vec{b}$ we get

$$\vec{f}(\vec{x}) = \vec{g}(\vec{x}) + \vec{b}.$$

Rearranging terms, we have

$$\begin{aligned} \vec{h}(\vec{y}) &= \vec{f}(\vec{y}) - \vec{a}, \\ \vec{g}(\vec{y}) &= \vec{f}(\vec{y}) - \vec{b}. \end{aligned}$$

Plugging this into the defining relation gives us

$$\vec{f}(\vec{x}+\vec{y}) = \vec{f}(\vec{x}) + \vec{f}(\vec{y}) - \vec{a} - \vec{b}.$$

Next we define $\vec{\jmath}(\vec{x}) = \vec{f}(\vec{x}) - \vec{a} - \vec{b}$, which gives us the relation

$$\vec{\jmath}(\vec{x}+\vec{y}) = \vec{\jmath}(\vec{x}) + \vec{\jmath}(\vec{y}),$$

the solution of which is

$$\vec{\jmath}(\vec{x}) = \boldsymbol{C} \cdot \vec{x},$$

where $\boldsymbol{C}$ is a constant matrix. Thus the general solution to (9.1) is

$$\begin{aligned} \vec{f}(\vec{x}) &= \boldsymbol{C} \cdot \vec{x} + \vec{a} + \vec{b}, \\ \vec{h}(\vec{x}) &= \boldsymbol{C} \cdot \vec{x} + \vec{b}, \\ \vec{g}(\vec{x}) &= \boldsymbol{C} \cdot \vec{x} + \vec{a}. \end{aligned}$$ □

9.2 Solved Problems

Problem 9.5 (Putnam 1959). *For each positive integer n, let f_n be a real-valued symmetric function of n variables. Suppose that for all n and for all real numbers $x_1, x_2, \ldots, x_{n+1}, y$ it is true that*

(a) $f_n(x_1+y, x_2+y, \ldots, x_n+y) = f_n(x_1, x_2, \ldots, x_n) + y$;

(b) $f_n(-x_1, -x_2, \ldots, -x_n) = -f_n(x_1, x_2, \ldots, x_n)$;

(c) $f_{n+1}\big(f_n(x_1, x_2, \ldots, x_n), \ldots, f_n(x_1, x_2, \ldots, x_n), x_{n+1}\big)$
$= f_{n+1}(x_1, x_2, \ldots, x_n, x_{n+1}).$

Find the functions f_n.

Solution. Let's first try to find expressions for $f_1(x)$ and $f_2(x, y)$ to help us make an educated guess about the general form of $f_n(x_1, x_2, \ldots, x_n)$.

In (a) we set $n = 1$, $x = 0$ to find $f_1(y) = f_1(0) + y$. From (b) we have $f_1(0) = 0$. Hence

$$f_1(x) = x.$$

In (a) we now set $n = 2$, $y = -x_1$, obtaining $f_2(0, x_2 - x_1) = f_2(x_1, x_2) - x_1$. By (b), this implies

$$-f_2(0, x_1 - x_2) = f_2(x_1, x_2) - x_1.$$

Since f_2 is a symmetric function,

$$-f_2(x_1 - x_2, 0) = f_2(x_1, x_2) - x_1.$$

Again in (a), we set $n = 2$, $y = -x_2$:

$$f_2(x_1 - x_2, 0) = f_2(x_1, x_2) - x_2.$$

Adding the last two equations, we find

$$f_2(x_1, x_2) = \frac{x_1 + x_2}{2}.$$

The previous results hint that

$$f_n(x_1, x_2, \ldots, x_n) = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

We shall prove this formula using induction.

For $n = 1, 2$ we have already shown it to be true. Assume it is true for $n = k$:

$$f_k(x_1, x_2, \ldots, x_k) = \frac{x_1 + x_2 + \cdots + x_k}{k}. \tag{9.2}$$

We must show that this implies that the formula is true for $n = k + 1$.

Let

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_{k+1}}{k + 1}$$

be the arithmetic mean of $x_1, x_2, \ldots, x_{k+1}$. Define the auxiliary variables $a_1, a_2, \ldots, a_{k+1}$ by $a_i = x_i - \bar{x}$. Obviously $\bar{a} = 0$. Equivalently, $a_1 + \cdots + a_k +$

$a_{k+1} = 0$ or $A = -a_{k+1}$, where we defined $A = a_1 + \cdots + a_k$. From condition (a) we have

$$\begin{aligned} f_{k+1}(x_1, x_2, \ldots, x_{k+1}) &= f_{k+1}(a_1 + \bar{x}, a_2 + \bar{x}, \ldots, a_{k+1} + \bar{x}) \\ &= f_{k+1}(a_1, a_2, \ldots, a_{k+1}) + \bar{x}. \end{aligned}$$

The proof will be complete if we show that

$$f_{k+1}(a_1, a_2, \ldots, a_{k+1}) = 0.$$

To this goal, notice that

$$\begin{aligned} f_{k+1}(a_1, a_2, \ldots, a_{k+1}) &= f_{k+1}(f_k(a_1, a_2, \ldots, a_k), \ldots, f_k(a_1, a_2, \ldots, a_k), a_{k+1}) \\ &= f_{k+1}\Big(-\frac{a_{k+1}}{k}, \ldots, -\frac{a_{k+1}}{k}, a_{k+1}\Big) \\ &\overset{(9.2)}{=} f_{k+1}(f_k(-a_{k+1}, 0, \ldots, 0), \ldots, f_k(-a_{k+1}, 0, \ldots, 0), a_{k+1}) \\ &\overset{(c)}{=} f_{k+1}(-a_{k+1}, 0, \ldots, 0, a_{k+1}). \end{aligned}$$

Since the function f_{k+1} is symmetric,

$$\begin{aligned} f_{k+1}(a_1, a_2, \ldots, a_{k+1}) &= f_{k+1}(a_{k+1}, 0, \ldots, 0, -a_{k+1}) \\ &\overset{(b)}{=} -f_{k+1}(-a_{k+1}, 0, \ldots, 0, a_{k+1}). \end{aligned}$$

In other words,

$$f_{k+1}(a_1, a_2, \ldots, a_{k+1}) = -f_{k+1}(a_1, a_2, \ldots, a_{k+1}),$$

which implies that $f_{k+1}(a_1, a_2, \ldots, a_{k+1}) = 0$, as sought. □

Comment. In the actual Putnam problem given to contestants, the expression of f_n was also given. That is, the contestants did not have to guess the functional form of f_n. □

Problem 9.6. *Find all continuous functions* $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C} : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}^{m\times m}$ *that satisfy the functional relation*

$$\boldsymbol{A}(\vec{x}, \vec{z}) = \boldsymbol{B}(\vec{x}, \vec{y}) \cdot \boldsymbol{C}(\vec{y}, \vec{z}), \tag{9.3}$$

where $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$ *are nonsingular square matrices.*

Solution. To begin, we set $\vec{y} = \vec{a}$ in (9.3), where $\vec{a}$ is some constant vector. So we have, after setting $\boldsymbol{B}(\vec{x}, \vec{a}) = \boldsymbol{M}(\vec{x})$ and $\boldsymbol{C}(\vec{a}, \vec{x}) = \boldsymbol{N}(\vec{x})$:

$$\begin{aligned} \boldsymbol{A}(\vec{x}, \vec{z}) &= \boldsymbol{M}(\vec{x}) \cdot \boldsymbol{N}(\vec{z}), \\ \boldsymbol{B}(\vec{x}, \vec{y}) \cdot \boldsymbol{C}(\vec{y}, \vec{z}) &= \boldsymbol{M}(\vec{x}) \cdot \boldsymbol{N}(\vec{z}). \end{aligned}$$

Now if we set $\vec{x} = \vec{b}$ and multiply the second equation on the left by $\boldsymbol{B}^{-1}(\vec{b}, \vec{y})$, we get

$$\boldsymbol{C}(\vec{y}, \vec{z}) = \boldsymbol{B}^{-1}(\vec{b}, \vec{y}) \cdot \boldsymbol{M}(\vec{b}) \cdot \boldsymbol{N}(\vec{z}) = \boldsymbol{L}(\vec{y}) \cdot \boldsymbol{N}(\vec{z}),$$

where $\boldsymbol{L}(\vec{y}) = \boldsymbol{B}^{-1}(\vec{b}, \vec{y}) \cdot \boldsymbol{M}(\vec{b})$. If we then plug this into the previous equation and multiply on the right by $\boldsymbol{N}^{-1}(\vec{z}) \cdot \boldsymbol{L}^{-1}(\vec{y})$, we get

$$\boldsymbol{B}(\vec{x}, \vec{y}) = \boldsymbol{M}(\vec{x}) \cdot \boldsymbol{L}^{-1}(\vec{y}).$$

Hence, for any nonsingular matrices $\boldsymbol{L}(\vec{y})$, $\boldsymbol{M}(\vec{x})$ and $\boldsymbol{N}(\vec{z})$, the solution is

$$\begin{aligned} \boldsymbol{A}(\vec{x}, \vec{z}) &= \boldsymbol{M}(\vec{x}) \cdot \boldsymbol{N}(\vec{z}), \\ \boldsymbol{B}(\vec{x}, \vec{y}) &= \boldsymbol{M}(\vec{x}) \cdot \boldsymbol{L}^{-1}(\vec{y}), \\ \boldsymbol{C}(\vec{y}, \vec{z}) &= \boldsymbol{L}(\vec{y}) \cdot \boldsymbol{N}(\vec{z}). \end{aligned}$$

□

Problem 9.7 (IMO 1981). *The function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{R}$ satisfies the following conditions:*

(a) $f(0, y) = y + 1$.

(b) $f(x + 1, 0) = f(x, 1)$.

(c) $f(x + 1, y + 1) = f(x, f(x + 1, y))$. *Find the value $f(4, 1981)$.*

Solution. We will compute successively $f(1, y)$, $f(2, y)$, $f(3, y)$, $f(4, y)$, for arbitrary y.

We notice that

$$\begin{aligned} f(1,0) &\overset{\text{(b)}}{=} f(0,1) \overset{\text{(a)}}{=} 2, \\ f(1,1) &\overset{\text{(c)}}{=} f(0, f(1,0)) = f(0,2) \overset{\text{(a)}}{=} 3, \\ f(1,2) &\overset{\text{(c)}}{=} f(0, f(1,1)) = f(0,3) \overset{\text{(a)}}{=} 4. \end{aligned}$$

We therefore suspect that

$$f(1, y) = y + 2. \tag{9.4}$$

We shall prove it using induction. Let it be true for $y = n$:

$$f(1, n) = n + 2.$$

Then, using conditions (a) and (c), it is true for $y = n + 1$:

$$f(1, n+1) \overset{\text{(c)}}{=} f(0, f(1, n)) = f(0, n+2) \overset{\text{(a)}}{=} n + 3 = (n+1) + 2.$$

Now we compute $f(2, y)$. We have:

$$f(2, y) \overset{\text{(c)}}{=} f(1, f(2, y-1)) \overset{(9.4)}{=} f(2, y-1) + 2.$$

The values $f(2, y)$ form an arithmetic progression with step 2. Hence

$$f(2, y) = 2y + f(2, 0).$$

Also

$$f(2, 0) \overset{\text{(b)}}{=} f(1, 1) \overset{(9.4)}{=} 3.$$

So finally

$$f(2, y) = 2y + 3. \tag{9.5}$$

Then we compute $f(3, y)$. We have:

$$f(3, y) \overset{\text{(c)}}{=} f(2, f(3, y-1)) \overset{(9.5)}{=} 2f(3, y-1) + 3.$$

The values $f(3, y)$ form a mixed progression with step 3 and ratio 2. Hence

$$f(3, y) = 2^y f(3, 0) + 3\,\frac{2^y - 1}{2 - 1}.$$

Also

$$f(3, 0) \overset{\text{(b)}}{=} f(2, 1) \overset{(9.5)}{=} 5,$$

and thus

$$f(3, y) = 2^{y+3} - 3.$$

Finally, we compute $f(4, y)$. We have

$$f(4, y) \overset{\text{(c)}}{=} f(3, f(4, y-1)) = 2^{f(4, y-1)+3} - 3.$$

This equation is exactly the recursion relation solved in Problem 3.32. Therefore

$$f(4, y) = 2^{2^{\cdot^{\cdot^{\cdot^{2^2}}}}} - 3,$$

where the number 2 appears $y + 3$ times. In particular,

$$f(4, 1981) = 2^{2^{\cdot^{\cdot^{\cdot^{2^2}}}}} - 3,$$

where the number 2 appears 1984 times. □

Chapter 10

Systems of Equations

Another generalization one might implement is to search for solutions in a system of functional equations. In this case, of course, the admissible solutions are those in the intersection of the individual sets of solutions. So, in principle, one might seek solutions for each equation in the system and then select the common ones.

Problem 10.1. *Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function that satisfies the Cauchy functional equations* (5.1) *and* (5.20). *Prove that $f(x) \equiv 0$ or $f(x) = x$.*

Solution. The set of solutions of equation (5.1) is $\{f(x) = a\,x,\ a \in \mathbb{R}\}$. The set of solutions of equation (5.20) is $\{f(x) \equiv 0,\ f(x) = x^b,\ b \in \mathbb{R}\}$. The intersection of the two sets is the set $\{f(x) \equiv 0,\ f(x) = x\}$. □

However, if one uses information from all of the equations in the system at once, the solution is often easier to reach, without solving explicitly each equation of the system. As an example, let's solve again the previous problem without relying on the solutions of (5.1) and (5.20).

Problem 10.2. *Let $f : \mathbb{R} \to \mathbb{R}$ be a function (not necessarily continuous) that satisfies the Cauchy functional equations* (5.1) *and* (5.20). *Prove that $f(x) \equiv 0$ or $f(x) = x$.*

Solution. As we have seen, (5.1) implies that

$$f(q) = f(1)\,q \quad \text{for all } q \in \mathbb{Q}.$$

At the same time, (5.20) implies that $f(x) \equiv 0$ or $f(x) > 0$ for all $x \in \mathbb{R}$. In the latter case, $f(1) = 1$. Therefore, either $f(x) \equiv 0$ or

$$f(q) = q \quad \text{for all } q \in \mathbb{Q}.$$

We show that f is strictly increasing. Indeed, for any two real numbers $x_1 > x_2$, (5.1) and the fact that f is strictly positive give

$$f(x_1) = f(x_2 + (x_1 - x_2)) = f(x_2) + f(x_1 - x_2) > f(x_2).$$

Now assume, for a contradiction, that $f(x) \not\equiv x$. Choose $x_0 \in \mathbb{R} \setminus \mathbb{Q}$ such that $f(x_0) \neq x_0$.

If $f(x_0) > x_0$, pick a rational number q_0 in the interval $(x_0, f(x_0))$. Then

$$x_0 < q_0 = f(q_0) < f(x_0).$$

But this contradicts the fact that f is strictly increasing.

If $f(x_0) < x_0$, pick a rational number q_0 in the interval $(f(x_0), x_0)$. Then

$$f(x_0) < f(q_0) = q_0 < x_0,$$

which also contradicts the fact that f is strictly increasing.

Therefore, $f(x_0) = x_0$; since x_0 was arbitrary, $f(x) \equiv x$. □

Comment. The result proved in the problem should not be underestimated. Once the monotonicity of f was established, the functional equations were bypassed in establishing the final result. More precisely: Once we knew that f is the identity function on $\mathbb{Q}$, there was only one monotonic extension of f to $\mathbb{R}$. □

Question. *Let $f : \mathbb{Q} \to \mathbb{R}$ be such that*

$$f(q) = \begin{cases} -1 & \text{if } q < -\sqrt{2}, \\ 0 & \text{if } -\sqrt{2} < q < +\sqrt{2}, \\ +1 & \text{if } q > +\sqrt{2}. \end{cases}$$

This function is nondecreasing, and thus monotonic. However, there are an infinite number of monotonic extensions of f to $\mathbb{R}$; one can choose any value of $f(\sqrt{2})$ in $(0,1)$. Why doesn't the previous comment apply here?

Answer. Although we allowed discontinuous solutions in the previous problem, the two equations combined enforced continuity. In Section 11.1, we show that various weaker conditions (one of them being monotonicity) for the linear Cauchy equation result in the same solutions—that is, they impose continuity on the solutions. And, if one tries to write down a formal proof for a unique extension, one discovers the following theorem: *Given two continuous functions f and g defined on $\mathbb{R}$ such that $f(q) = g(q)$ for all $q \in \mathbb{Q}$, $f(x) = g(x)$ for all $x \in \mathbb{R}$.* The theorem requires continuity, which has been implicit in the problem solved previously. However, the function f given above has no monotonic extension that is continuous at $f(\pm\sqrt{2})$.

Incidentally, we should point out that the theorem of the monotonic extension of a function can be extended beyond $\mathbb{Q}$ and $\mathbb{R}$: *Given two continuous functions f and g defined on the set A such that $f(q) = g(q)$ for all $q \in S$ where S is a dense subset of A, $f(x) = g(x)$ for all $x \in A$.* In other words, if a continuous function is known in a set S and all points of a larger set A are limits of sequences exclusively in S, then the function is uniquely extended in A.

In light of this, we can revisit the $\mathbb{NQR}$ method and the proof of the D'Alembert–Poisson I equation (7.21). The $\mathbb{NQR}$ method relies on the fact that $\mathbb{Q}$ is dense in $\mathbb{R}$. So, after establishing the function over $\mathbb{Q}$, we can then extend it to $\mathbb{R}$ easily and uniquely. Similarly, the set $D = \{m/2^n,\ m,n \in \mathbb{N}\}$, known as the set of **dyadic numbers**, is dense in $\mathbb{R}$. A function in D has a unique continuous extension in $\mathbb{R}$.

10.1 Solved Problems

Problem 10.3. *Find the continuous solutions $F, H, G, g, h, f : \mathbb{R} \to \mathbb{R}$ to the functional equation*

$$F(xy) = H(x)^{g(y)}\, G(y)^{h(x)},$$

given

$$f(xy) = h(x)\, g(y).$$

Solution. Using the second equation, we can rewrite the first in the form

$$F(xy)^{1/(g(y)h(x))} = F(xy)^{1/f(xy)} = H(x)^{1/h(x)}\, G(y)^{1/g(y)}.$$

This equation can be reduced to the power Pexider equation (8.11). If we set

$$f(x) = a\, b\, x^{\mu}, \quad g(x) = a\, x^{\mu}, \quad h(x) = b\, x^{\mu},$$

the original equation is equivalent to

$$A(x)\, B(y) = C(x\, y),$$

where

$$A(x) = H(x)^{1/h(x)}, \quad B(x) = G(x)^{1/g(x)}, \quad C(x) = F(x)^{1/f(x)}.$$

Therefore, $A(x), B(x), C(x)$ also satisfy (8.11) and they must be given by

$$C(x) = A\, B\, x^{\nu}, \quad A(x) = A\, x^{\nu}, \quad B(x) = B\, x^{\nu}.$$

From these expressions we can find $F(x), H(x), G(x)$. The system of equations is solved by the functions

$$\begin{aligned} f(x) &= a\, b\, x^{\mu}, & g(x) &= a\, x^{\mu}, & h(x) &= b\, x^{\mu}, \\ F(x) &= (A\, B\, x^{\nu})^{abx^{\mu}}, & G(x) &= (A\, x^{\nu})^{ax^{\mu}}, & H(x) &= (B\, x^{\nu})^{bx^{\mu}}. \end{aligned}$$

□

Problem 10.4. *Find the continuous solutions $F, H, G, g, h, f : \mathbb{R} \to \mathbb{R}$ to the functional equation*

$$F(x\,y) = g(y)\,H(x) + h(x)\,G(y)$$

given

$$f(xy) = h(x)\,g(y).$$

Solution. We can reduce to the preceding problem by exponentiation:

$$e^{F(x\,y)} = (e^{H(x)})^{g(y)}\,(e^{G(y)})^{h(x)}.$$

So the functional equation is solved by the functions

$$\begin{aligned} &f(x) = ab\,x^{\mu}, && g(x) = a\,x^{\mu}, && h(x) = b\,x^{\mu},\\ &F(x) = ab\,x^{\mu}\ln(A\,B\,x^{\nu}), && G(x) = ab\,x^{\mu}\ln(A\,x^{\nu}), && H(x) = ab\,x^{\mu}\ln(B\,x^{\nu}). \end{aligned}$$ □

Problem 10.5. *Find the continuous solutions $F, H, G, g, h, f : \mathbb{R} \to \mathbb{R}$ of the following systems of functional equations:*

(a) $\qquad F(x+y) = H(x)^{g(y)}\,G(y)^{h(x)}, \qquad f(x+y) = h(x)\,g(y).$

(b) $\qquad F(x+y) = g(y)\,H(x) + h(x)\,G(y), \quad f(x+y) = h(x)\,g(y).$

Solution. The solution proceeds as in the previous two problems. We obtain for (a) the solutions

$$\begin{aligned} &f(x) = a\,b\,c^{x}, && g(x) = a\,c^{x}, && h(x) = b\,c^{x},\\ &F(x) = (A\,B\,d^{x})^{abc^{x}}, && G(x) = (A\,d^{x})^{abc^{x}}, && H(x) = (B\,d^{x})^{abc^{x}}, \end{aligned}$$

and for (b) the solutions

$$\begin{aligned} &f(x) = ab\,c^{x}, && g(x) = a\,c^{x}, && h(x) = b\,c^{x},\\ &F(x) = ab\,c^{x}\ln(A\,B\,d^{x}), && G(x) = a\,c^{x}\ln(A\,d^{x}), && H(x) = b\,c^{x}\ln(B\,d^{x}). \end{aligned}$$ □

Problem 10.6. *Find all continuous solutions $f : \mathbb{R}^2 \to \mathbb{R}$ of the system*

$$\begin{aligned} f(x_1 + x_2, y) &= f(x_1, y) + f(x_2, y) \quad \textit{for all } x_1, x_2, y \in \mathbb{R},\\ f(x, y_1 + y_2) &= f(x, y_1) + f(x, y_2) \quad \textit{for all } x, y_1, y_2 \in \mathbb{R}. \end{aligned}$$

Solution. The first of the two equations is just the linear Cauchy equation in one variable — the first variable x of $f(x, y)$. All continuous solutions of this equation are of the form

$$f(x,y) = c(y)\,x,$$

where the constant here depends on y. Inserting this result in the second equation, we find

$$c(y_1 + y_2) = c(y_1) + c(y_2) \quad \text{for } y_1, y_2 \in \mathbb{R}.$$

This is again the linear Cauchy equation, with solution

$$c(y) = c\,y.$$

Therefore, the continuous solutions to the given system are the functions $f(x,y) = c\,xy$, with c a constant. □

Comment. Contrast this problem, which refers to a continuous function $f(x, y)$ satisfying *two* functional equations, with Problem 9.1, concerning a continuous function $F(x, y)$ satisfying a *single* functional equation

$$F(x_1 + x_2,\, y_1 + y_2) = f(x_1, y_1) + f(x_2, y_2) \quad \text{for all } x_1, x_2, y_1, y_2 \in \mathbb{R}.$$

The two results are quite different: $f(x,y) = c\,xy$ in the present case, versus $F(x,y) = c_1\,x + c_2\,y$ in the earlier problem. □

Part IV

CHANGING THE RULES

Chapter 11

Less Than Continuity

11.1 Imposing Weaker Conditions

Darboux was the first to show that the Cauchy functional equations can be solved by the same functions even if the condition of continuity was replaced by milder conditions. The four problems that follow discuss the linear Cauchy functional equation (5.1) with weaker conditions.

Problem 11.1. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

(a) $f(x) + f(y) = f(x+y)$ *for all* $x, y \in \mathbb{R}$, *and*

(b) f *is continuous at a point* $x_0 \in \mathbb{R}$.

Solution. We claim that condition (b) implies that f is continuous at every point $x \in \mathbb{R}$, i.e. $\lim_{y\to x} f(y) = f(x)$. To see this, let's set $h = x_0 + y - x$. As $y \to x$, we have $h \to x_0$ and

$$\begin{aligned} \lim_{y\to x} f(y) &= \lim_{y\to x} f\left(h + (x - x_0)\right) \\ &= \lim_{y\to x} \left(f(h) + f(x - x_0)\right) \\ &= \lim_{h\to x_0} f(h) + f(x - x_0) \\ &= f(x_0) + f(x - x_0) = f(x_0 + x - x_0) = f(x). \end{aligned}$$

The problem is thus equivalent to those of Section 5.1, with solution $f(x) = cx$, where c is an arbitrary constant. □

Problem 11.2. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

(a) $f(x) + f(y) = f(x+y)$ *for all* $x, y \in \mathbb{R}$, *and*

(b) *f is monotonically increasing.*

Solution. As we saw in Section 5.1, condition (a) implies that $f(q) = c\,q$ for all $q \in \mathbb{Q}$. Now for any $x \in \mathbb{R}$, we can always find an increasing sequence $\{a_n\}_{n\in\mathbb{N}}$ of rational numbers and a decreasing sequence $\{b_n\}_{n\in\mathbb{N}}$ of rational numbers, both converging to x. Then

$$a_n \le x \le b_n \quad \text{for all } n \in \mathbb{N}.$$

From condition (b), it follows that $f(a_n) \le f(x) \le f(b_n)$ and hence also $ca_n \le f(x) \le cb_n$, for all $n \in \mathbb{N}$. By taking the limit $n \to \infty$, we find $f(x) = c\,x$. □

Problem 11.3. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

(a) *$f(x) + f(y) = f(x + y)$ for all $x, y \in \mathbb{R}$, and*

(b) *there exists a positive number ε such that $f(x) \ge 0$ for all $x \in (0, \varepsilon)$.*

Solution. We will show that condition (b) implies that the function is monotonically increasing, reducing the problem to the previous one.

Let $x_1 > x_2$ such that $h = x_1 - x_2 \in (0, \varepsilon)$. Then $f(h) \ge 0$ and

$$f(x_1) = f(x_2 + h) = f(x_2) + f(h) \ge f(x_2) + 0 = f(x_2).$$

Using this relation in the intervals $[n\varepsilon/2, (n + 1)\varepsilon/2]$, for all $n \in \mathbb{Z}$, one can show easily that f is nondecreasing. □

Problem 11.4. *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that*

(a) *$f(x + y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$, and*

(b) *there exists a positive number ε such that f is bounded in $(0, \varepsilon)$.*

Solution. Condition (a) implies that

$$f(q\,x) = q\,x \quad \text{for all } q \in \mathbb{Q} \text{ and } x \in \mathbb{R}.$$

(See equation (5.7).)

Let $\{x_n\}$ be an arbitrary sequence of positive real numbers such that $\lim_{n\to\infty} x_n = 0$. Given $\{x_n\}$, we select a sequence of positive rational numbers $\{q_n\}$ as follows. For any n, the rational number q_n is chosen such that

$$\frac{1}{\sqrt{x_n}} < q_n < \frac{1}{\sqrt[3]{x_n}}.$$

Then $\lim_{n\to\infty} q_n = +\infty$. At the same time,

$$\sqrt{x_n} < q_n\, x_n < \sqrt[3]{x_n^2},$$

i.e., $\lim_{n\to+\infty}(q_n\, x_n) = 0$. Since the sequence $\{q_n x_n\}$ converges to zero, eventually all terms of the sequence will fall in the interval $(0, \varepsilon)$. That is, there exists $n_0 \in \mathbb{N}$ such that $q_n\, x_n \in (0, \varepsilon)$ for all $n > n_0$.

Condition (b) implies that there exists $M \in \mathbb{R}_+$ such that $|f(x)| \le M$ for all $x \in (0, \varepsilon)$. Therefore $|f(q_n\, x_n)| \le M$ for all $n > n_0$. Then

$$|f(x_n)| = \left| f\left(\frac{1}{q_n}\, q_n x_n\right)\right| = \frac{1}{q_n}\,|f(q_n\, x_n)| \le \frac{M}{q_n} \to 0 \quad \text{when} \quad n \to \infty.$$

Thus, for any sequence $\{x_n\}$ converging to zero, $\lim_{n\to+\infty} f(x_n) = 0 = f(0)$. Therefore, the function is continuous at $x = 0$. As we have seen, continuity at one point gives $f(x) = c\,x$ for all $x \in \mathbb{R}$. □

11.2 Discontinuous Solutions

Because of the results in the previous section, mathematicians initially thought that the solution to the linear Cauchy equation (5.1) always had to be $f(x) = c\,x$. However, Hamel proved in 1905 [50], using ideas from set theory, that this is not the case. His solution makes use of what is now called a *Hamel basis*. In this section we shall explain this result. The existence of a Hamel basis depends on the axiom of choice; there is no explicit construction that will lead to such a basis. Therefore, we will first use an artificial example to get some intuition for what follows.

An Analogy

Consider the set

$$A = \mathbb{Q} + \mathbb{Q}\sqrt{2} + \mathbb{Q}\sqrt{3} = \{r + s\sqrt{2} + t\sqrt{3} : r, s, t \in \mathbb{Q}\}.$$

Every number in A is uniquely expressed as a linear combination of elements in the set

$$H = \{1, \sqrt{2}, \sqrt{3}\};$$

we say that H is a *basis* for A, or that H *generates* A. A Hamel basis of $\mathbb{R}$ plays the same role as the set H plays for A.

Let's solve the linear Cauchy equation (5.1) for functions $f : A \to A$. We can repeat the method of Section 5.1. Omitting the details, we have

$$\begin{aligned}
&f\big((r_1+r_2+\cdots+r_n) + (s_1+s_2+\cdots+s_n)\sqrt{2} + (t_1+t_2+\cdots+t_n)\sqrt{3}\big)\\
&\ = f\big(r_1+r_2+\cdots+r_n\big) + f\big((s_1+s_2+\cdots+s_n)\sqrt{2}\big) + f\big((t_1+t_2+\cdots+t_n)\sqrt{3}\big),
\end{aligned}$$

from which we eventually arrive at

$$f(r + s\sqrt{2} + t\sqrt{3}) = r\,f(1) + s\,f(\sqrt{2}) + t\,f(\sqrt{3}).$$

Let's reexpress this result using the basis H. Let $\tilde{f} : H \to A$ be any mapping from H to A. Then the function f assigning to $x = r + s\sqrt{2} + t\sqrt{3}$ the value

$$f(x) = r\,\tilde{f}(1) + s\,\tilde{f}(\sqrt{2}) + t\,\tilde{f}(\sqrt{3})$$

is a solution of the linear Cauchy functional equation in A. Given all possible mappings $\tilde{f}$ of H, we get all possible solutions of the functional equation.

An example of $\tilde{f} : H \to A$ is given by

$$\tilde{f}(1) = q_0, \quad \tilde{f}(\sqrt{2}) = q_0\sqrt{2}, \quad \tilde{f}(\sqrt{3}) = q_0\sqrt{3},$$

for some fixed $q_0 \in \mathbb{Q}$. Then

$$f(x) = q_0\,x.$$

This is a linear function.

Another example of $\tilde{f} : H \to A$ is

$$\tilde{f}(1) = 1, \quad \tilde{f}(\sqrt{2}) = \tilde{f}(\sqrt{3}) = 0.$$

Then

$$f(r + s\sqrt{2} + t\sqrt{3}) = r.$$

This cannot be a linear function — for example, it vanishes at all points of the form $x = s\sqrt{2} + t\sqrt{3}$, for $s, t \in \mathbb{Q}$, yet it does not vanish everywhere.

Hamel's Solution

Definition 11.5. A (necessarily uncountable) subset H of $\mathbb{R}$ is a **Hamel basis** of $\mathbb{R}$ if for every real number x there exists a finite natural number n and elements $h_1, h_2, \ldots, h_n$ of H such that x is written *in a unique way* as a sum

$$x = \sum_{i=1}^{n} q_i\,h_i,$$

with rational coefficients q_i. We call this expression the **Hamel decomposition** of x in H.

The definition of a Hamel basis does not guarantee its existence. Hamel's major contribution to the problem was to prove, using the axiom of choice, the following theorem.

Theorem 11.6 (Hamel). *At least one Hamel basis exists.*

Since set theory is not the main focus of these introductory notes, we shall omit the proof. We are now in position to solve Cauchy's equations without any constraint on the functions f.

Theorem 11.7. *The most general solution to the linear Cauchy equation* (5.1) *is found by*

(i) *introducing a Hamel basis H in $\mathbb{R}$,*

(ii) *assigning arbitrary values for f at the points of the Hamel basis, and*

(iii) *setting $f(x) = \sum_i q_i f(h_i)$ if $x \in \mathbb{R}$ has a Hamel decomposition $x = \sum_i q_i h_i$.*

Proof. (Direct proposition) Given (5.1), one can show that

$$f\left(\sum_{i=1}^{n} y_i\right) = \sum_{i=1}^{n} f(y_i) \quad \text{for all } n \in \mathbb{N}^*, \text{ and } y_i \in \mathbb{R} \text{ for } i = 1, 2, \ldots, n,$$

and

$$f(q\,y) = q\,f(y) \quad \text{for all } y \in \mathbb{R} \text{ and } q \in \mathbb{Q}.$$

From these and the Hamel decomposition of any number $x = \sum_i q_i\, h_i$, we find

$$f(x) = f\left(\sum_i q_i\, h_i\right) = \sum_i f(q_i\, h_i) = \sum_i q_i\, f(h_i).$$

(Inverse proposition) Consider $x, x' \in \mathbb{R}$ with Hamel decompositions

$$x = \sum_{\{h_i\}} q_i h_i, \quad x' = \sum_{\{h'_j\}} q'_j h'_j,$$

where $\{h_i\}$, $\{h'_j\}$ stand for the set of Hamel elements in the decomposition of x, x' respectively. The two sets need not coincide or even have the same number of elements, but we can take the union $\{\tilde{h}_k\} = \{h_i\} \cup \{h'_j\}$ and write

$$x = \sum_{\{\tilde{h}_k\}} \tilde{q}_k \tilde{h}_k, \quad x' = \sum_{\{\tilde{h}_k\}} \tilde{q}'_k \tilde{h}_k,$$

by allowing terms with coefficient zero. Then

$$f(x) = \sum_{\{\tilde{h}_k\}} f(\tilde{h}_k)\, \tilde{q}_k, \quad f(x') = \sum_{\{\tilde{h}_k\}} f(\tilde{h}_k)\, \tilde{q}'_k,$$

and

$$f(x) + f(x') = \sum_{\{\tilde{h}_k\}} (\tilde{q}_k + \tilde{q}'_k)\, f(\tilde{h}_k).$$

On the other hand,

$$x + x' = \sum_{\{\tilde{h}_k\}} (\tilde{q}_k + \tilde{q}'_k)\, \tilde{h}_k \quad \text{and} \quad f(x + x') = \sum_{\{\tilde{h}_k\}} (\tilde{q}_k + \tilde{q}'_k)\, f(\tilde{h}_k),$$

so $f(x + x') = f(x) + f(x')$. □

Corollary. *The Hamel solution of* (5.1) *is continuous if and only if there exists a real number c such that $f(h) = c\,h$, for all elements h of the Hamel basis.*

Example 11.8. We construct one illustrative Hamel solution, assuming given a Hamel basis H. Let h_0 be any element of H. Define $f(h_0) = 1$ and $f(h) = 0$ for all $h \in H \setminus \{h_0\}$. Then the function

$$f(x) = \begin{cases} q_0 & \text{if } x = q_0 h_0 + \dots, \\ 0 & \text{otherwise,} \end{cases}$$

is a discontinuous solution of (5.1). □

With very little additional effort, we can find the most general solutions to the rest of the Cauchy equations.

Problem 11.9. *Give the most general solution to the exponential Cauchy functional equation* (5.8)*. Show how we can recover the solution $f(x) = a^x$ from the general one.*

Solution. As seen in Section 5.2, the function $g(x) = \ln f(x)$ satisfies the functional equation (5.1). Therefore, its most general solution is found by assigning arbitrary values $g(h)$ to the elements of the Hamel basis H, and for every $x \in \mathbb{R}$ with Hamel decomposition $x = \sum_i q_i h_i$, we write

$$g(x) = \sum_i q_i\, g(h_i).$$

Returning to function f, we have

$$\ln f(x) = \sum_i q_i \ln f(h_i) = \sum_i \ln f(h_i)^{q_i} = \ln \prod_i f(h_i)^{q_i},$$

and therefore

$$f(x) = \prod_i f(h_i)^{q_i},$$

where $f(h)$ is arbitrary for $h \in H$.

The function $g(x)$ will be continuous if there is a number c such that $g(h) = c\,h$ for all $h \in H$. This implies $\ln f(h) = c\,h$ or $f(h) = e^{ch} = a^h$ for all $h \in H$, and $a = e^c$. Therefore,

$$f(x) = \prod_i f(h_i)^{q_i} = \prod_i a^{h_i q_i} = a^{\sum_i h_i q_i} = a^x. \qquad \square$$

For the other two solutions, we will need a lemma.

Lemma 11.10. *There exists an uncountable set Δ of nonnegative elements $\delta \in \mathbb{R}_+^*$ such that any $x \in \mathbb{R}_+^*$ can be written as a finite product*

$$x = \prod_{i=1}^{n} \delta_i^{q_i},$$

for some $n \in \mathbb{N}^*$ and n rational numbers $q_1, q_2, \ldots, q_n$.

Proof. Let $x \in \mathbb{R}_+^*$. Then there exists a $u \in \mathbb{R}$ such that $x = e^u$. Using a Hamel basis, u can be expressed as a linear sum $u = \sum_{i=1}^n q_i h_i$, for some n and $q_i \in \mathbb{Q}$, $i = 1, 2, \ldots, n$. Then

$$x = e^u = e^{\sum_{i=1}^n q_i h_i} = \prod_{i=1}^{n} (e^{h_i})^{q_i}.$$

Therefore, the set $\Delta = \{\delta = e^h : h \in H\}$ has the required property:

$$x = \prod_{i=1}^{n} \delta_i^{q_i}. \qquad \square$$

Example 11.11. Although no explicit basis is known for $\mathbb{R}_+^*$, we can give a good example to convey the idea. If, instead of $\mathbb{R}_+^*$, we consider $\mathbb{Q}_+^*$, we can take for Δ the set of all prime numbers. Any $x \in \mathbb{Q}_+^*$ can be written as

$$x = \prod_{i=1}^{n} p_i^{k_i},$$

for some integer exponents k_i. $\square$

Problem 11.12. *Show that the most general solution to the logarithmic Cauchy equation* (5.14) *is found by*

(i) *introducing a basis Δ in $\mathbb{R}_+$,*

(ii) *assigning arbitrary values for f at the points of the basis, and*

(iii) *setting* $f(x) = \sum_i q_i f(\delta_i)$ *if* $x \in \mathbb{R}_+^*$ *has a decomposition* $x = \prod_i \delta_i^{q_i}$.

Solution. (Direct proposition) Given (5.14), one can show that

$$f\Big(\prod_{i=1}^{n} y_i\Big) = \sum_{i=1}^{n} f(y_i) \quad \text{for all } n \in \mathbb{N}^* \text{ and } y_i \in \mathbb{R} \text{ for } i = 1, 2, \ldots, n,$$

and

$$f(y^q) = q\, f(y_i) \quad \text{for all } y \in \mathbb{R} \text{ and } q \in \mathbb{Q}.$$

From these and the decomposition of any positive number $x = \prod_i \delta_i^{q_i}$, we find

$$f(x) = f\Big(\prod_i \delta_i^{q_i}\Big) = \sum_i f(\delta_i^{q_i}) = \sum_i q_i\, f(\delta_i).$$

(Inverse proposition) Consider $x, x' \in \mathbb{R}_+^*$ with decompositions

$$x = \prod_i \delta_i^{q_i}, \quad x' = \prod_j (\delta_j')^{q_j'},$$

where $\{\delta_i\}$, $\{\delta_j'\}$ stand for the sets of Hamel elements in the decomposition of x, x' respectively. As before, we take the union $\{\tilde{\delta}_k\} = \{\delta_i\} \cup \{\delta_j'\}$ and pad with zero coefficients to write

$$x = \prod_{\{\tilde{\delta}_k\}} \tilde{\delta}_k^{\tilde{q}_k}, \quad x' = \prod_{\{\tilde{\delta}_k\}} \tilde{\delta}_k^{\tilde{q}_k'}.$$

Then

$$f(x) = \sum_{\{\tilde{\delta}_k\}} f(\tilde{\delta}_k)\, \tilde{q}_k, \quad f(x') = \sum_{\{\tilde{\delta}_k\}} f(\tilde{\delta}_k)\, \tilde{q}_k',$$

and

$$f(x) + f(x') = \sum_{\{\tilde{\delta}_k\}} (\tilde{q}_k + \tilde{q}_k')\, f(\tilde{\delta}_k).$$

On the other hand,

$$x\, x' = \prod_{\{\tilde{\delta}_k\}} \tilde{\delta}_k^{\tilde{q}_k + \tilde{q}_k'}$$

and

$$f(x\, x') = \sum_{\{\tilde{h}_k\}} (\tilde{q}_k + \tilde{q}_k')\, f(\tilde{\delta}_k),$$

which shows that

$$f(x\, x') = f(x) + f(x').$$

The solution of (5.14) is continuous if and only if there exists a real number c such that $f(\delta) = c \ln \delta$ for all elements δ of Δ:

$$f(x) = \sum_i q_i f(\delta_i) = c \sum_i q_i \ln \delta_i = c \ln \prod_i \delta_i^{q_i} = c \ln x = \log_\gamma x,$$

where we have defined $\gamma = e^{1/c}$. $\square$

Problem 11.13. *Give the most general solutions to the power Cauchy functional equation* (5.20)*.*

Solution. As seen in Section 5.4, the function $g(x) = \ln f(x)$ satisfies the functional equation (5.14). Therefore, its most general solution is found by assigning arbitrary values $g(\delta)$ to the elements $\delta \in \Delta$, and by writing

$$g(x) = \sum_i q_i \, g(\delta_i)$$

for every $x \in \mathbb{R}$ with decomposition $x = \prod_i \delta_i^{q_i}$. Returning to the function f, we get

$$\ln f(x) = \sum_i q_i \ln f(\delta_i) = \sum_i \ln f(\delta_i)^{q_i} = \ln \prod_i f(\delta_i)^{q_i},$$

and therefore

$$f(x) = \prod_i f(\delta_i)^{q_i},$$

where the $f(\delta)$ are arbitrary for $\delta \in \Delta$.

The function $g(x)$ will be continuous if there is a number c such that $g(\delta) = c \ln \delta$ for all $\delta \in \Delta$. This implies $\ln f(\delta) = c \ln \delta$ or $f(\delta) = \delta^c$ for all $\delta \in \Delta$. Therefore,

$$f(x) = \prod_i f(\delta_i)^{q_i} = \prod_i (\delta_i^{q_i})^c = \left(\prod_i \delta_i^{q_i}\right)^c = x^c. \qquad \square$$

11.3 Solved Problems

Problem 11.14. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

(a) *$f(x+y) = f(\lambda x) + f(\lambda y)$ for all $x, y \in \mathbb{R}$, with λ some real number, and*

(b) *f is continuous at $x = 0$.*

Solution 1. Setting $x = y = 0$ in (a), we find that $f(0) = 0$. We shall examine the following cases:

(i) $\lambda = 1$: Then f satisfies (5.1) and is continuous at $x = 0$. Therefore $f(x) = cx$, where c is a real constant.

(ii) $\lambda = 0$: Then $f(x+y) = 2f(0) = 0$, i.e., $f(x) \equiv 0$.

(iii) $\lambda = -1$: Setting $y = 0$ in (a), we find $f(x) = f(-x)$. Also, setting $y = -x$, we find $f(-x) = -f(x)$. These two relations imply that $f(x) \equiv 0$.

(iv) $|\lambda| < 1$: Setting $y = 0$ in (a), we find $f(x) = f(\lambda x)$. Applying this equation n times yields

$$f(x) = f(\lambda^n x) \quad \text{for all } n \in \mathbb{N}^* \text{ and } x \in \mathbb{R}.$$

Since $|\lambda| < 1$, $\lim_{n\to\infty} \lambda^n = 0$. Then, by the continuity of f at $x = 0$,

$$0 = f(0) = f\Big(\lim_{n\to\infty} \lambda^n x\Big) = \lim_{n\to\infty} f(\lambda^n x) = \lim_{n\to\infty} f(x) = f(x).$$

(v) $|\lambda| > 1$: In this case $\mu = 1/\lambda$, with $|\mu| < 1$. Setting $y = 0$ and $x = \mu z$ in (a), we find $f(z) = f(\mu z)$. By working exactly as in case (iv), we arrive again at $f(x) \equiv 0$. □

Solution 2. Set x/λ in place of x and μx in place of y, where μ is a real number to be determined shortly:

$$f\left(\left(\frac{1}{\lambda} + \mu\right) x\right) = f(x) + f(\lambda\mu x).$$

Now select μ such that

$$\frac{1}{\lambda} + \mu = \lambda\mu, \quad \text{or, equivalently,} \quad \mu = \frac{1}{\lambda(\lambda - 1)}.$$

So when $\lambda \neq 0, 1$, the last equation implies $f(x) \equiv 0$.

For $\lambda = 0$, it is immediate that $f(x) \equiv 0$ and for $\lambda = 1$, we recover the continuous solutions of the linear Cauchy equation. □

Solution 3. If $\lambda = 0$, it is immediate that $f(x) \equiv 0$. If $\lambda \neq 0$, define a function g by $g(x) = f(\lambda x)$; in terms of g, the functional equation given in (a) becomes the linear Pexider equation:

$$f(x+y) = g(x) + g(y).$$

The functions f, g are continuous at $x = 0$. This can be used to show that they are continuous at all points, as we did with the linear Cauchy equation in Problem 11.1. Indeed, if we set $h = y - x$, then $\lim_{y\to x} h = 0$ and

$$\begin{aligned}\lim_{y\to x} f(y) &= \lim_{y\to x} f(h+x) = \lim_{y\to x} (g(h) + g(x)) \\ &= \lim_{h\to 0} g(h) + g(x) = g(0) + g(x) = f(x).\end{aligned}$$

We have established that the continuous solutions of the linear Pexider equation are $f(x) = cx + 2a$ and $g(x) = cx + a$, or

$$f(x) = \frac{c}{\lambda}x + a = cx + 2a \quad \text{for all } x \in \mathbb{R}.$$

Obviously, $a = 0$. If $\lambda = 1$, then $c =$ arbitrary; if $\lambda \neq 0, 1$, then $c = 0$. □

Problem 11.15 (Russia 1997). *For which value(s) of α does there exist a nonconstant function $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(\alpha(x + y)) = f(x) + f(y)?$$

Solution. Setting $x = y = 0$, we get $f(0) = 0$. Then if $\alpha = 0$, we set $y = 0$ to find $f(x) \equiv 0$. If $\alpha \neq 0$, we substitute x/α for x and y/α for y:

$$f(x + y) = f(\lambda x) + f(\lambda y),$$

where $\lambda = 1/\alpha$. From the previous problem we can see that a nonconstant solution exists only for the case $\alpha = 1$. (Since continuity or some other condition is not given, however, the solutions are not necessarily of the form $f(x) = cx$.) □

Problem 11.16. *Show that if the functions $f, g : \mathbb{R} \to \mathbb{R}$ satisfy the equation*

$$f(\lambda x, \lambda y) = g(\lambda)\, f(x, y) \quad \textit{for all } \lambda, x, y,$$

then g must be a solution of the power Cauchy equation.

Solution. Replacing λ by $\lambda\mu$ in the given functional equation, we find that

$$f(\lambda\mu x, \lambda\mu y) = g(\lambda\mu)\, f(x, y).$$

The left-hand side of this equation can be rewritten as follows:

$$f(\lambda\mu x, \lambda\mu y) = f(\lambda(\mu x), \lambda(\mu y)) = g(\lambda)\, f(\mu x, \mu y) = g(\lambda) g(\mu)\, f(x, y).$$

Combining the two equations, we conclude that

$$g(\lambda\mu) = g(\lambda)\, g(\mu).$$ □

Comment. If we require that g be continuous, then $g(x) = x^r$, and the given equation reduces to the Euler equation. (See Problem 2.20.) □

Problem 11.17 ([5], Problem 1690, February 2004). *Prove that there exist functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy*

$$f\,(x - f(y)) = f(x) + y \quad \text{for all } x, y \in \mathbb{R}, \tag{11.1}$$

and show how such functions can be constructed.

Solution. In the defining equation, we set $x = f(y)$:

$$f(0) = f^2(y) + y. \tag{11.2}$$

This implies that f is one-to-one, since

$$f(x) = f(y) \;\Rightarrow\; f^2(x) = f^2(y) \;\Rightarrow\; f(0) - x = f(0) - y \;\Rightarrow\; x = y.$$

Then we set $y = 0$ in the defining equation to find $f(x - f(0)) = f(x)$; therefore, $x - f(0) = x$ and finally,

$$f(0) = 0.$$

Returning to equation (11.2), we have thus found

$$f^2(y) = -y. \tag{11.3}$$

Finally, in the defining equation we replace y by $f(y)$, obtaining

$$f(x - f^2(y)) = f(x) + f(y),$$

which, with the help of (11.3), reduces to the linear Cauchy equation

$$f(x + y) = f(x) + f(y).$$

Inversely, if $f(x)$ satisfies the linear Cauchy equation and (11.3), then it satisfies (11.1). For $x = y = 0$, the linear Cauchy equation gives $f(0) = 0$. Then the same equation for $y = -x$ gives $f(-x) = -f(x)$. Finally, substituting $-f(y)$ for y in the linear Cauchy equation and using the last fact and (11.3), we find (11.1).

Therefore, the function f satisfies the original equation if and only if it satisfies the linear Cauchy equation and (11.3).

We can construct the solutions using Hamel's basis. Given any real number $x = \sum_i q_i h_i$, since f satisfies the linear Cauchy equation we have

$$f(x) = \sum_i q_i\, f(h_i),$$

and

$$f^2(x) = \sum_i q_i\, f^2(h_i).$$

To satisfy equation (11.3), we only need to consider the restriction of f to H. More precisely, we need to define values $f(h_i)$ such that $f^2(h_i) = -h_i$. To this end, we partition H into two disjoint sets H_+ and H_- that are in bijection, or one-to-one correspondence, with one another (again, this is possible by the axiom of choice). Thus, to each element $h_\alpha^+ \in H_+$, there corresponds $h_\alpha^- \in H_-$. Now define $f : H \to H$ so that

$$f(h_\alpha^+) = -h_\alpha^-, \quad f(h_\alpha^-) = +h_\alpha^+. \qquad \square$$

Problem 11.18 ([5], Problem 1626, June 2001). *Let $f, g, h : \mathbb{R} \to \mathbb{R}$ be functions such that*

$$f(g(0)) = g(f(0)) = h(f(0)) = 0, \tag{11.4}$$

and

$$f(x + g(y)) = g(h(f(x))) + y, \tag{11.5}$$

for all $x, y \in \mathbb{R}$. Prove that $h = f$ and that

$$g(x + y) = g(x) + g(y),$$

for all $x, y \in \mathbb{R}$.

Solution. Setting $x = y = 0$ in the defining equation (11.5), we find

$$f(g(0)) = g(h(f(0))) \Rightarrow g(0) = 0.$$

Then, setting only $x = 0$ in (11.5):

$$f(g(y)) = y. \tag{11.6}$$

In particular, for $y = 0$, this equation implies $f(0) = 0$. Similarly, setting $y = 0$, $x = g(y)$ in (11.5), we find:

$$f(g(y)) = g(h(f(g(y)))) \Rightarrow y = g(h(y)). \tag{11.7}$$

Using this result to rewrite the first term in the right-hand side of the original equation (11.5), we have

$$f(x + g(y)) = f(x) + y. \tag{11.8}$$

This equation produces both results sought. Upon substituting $y = h(z)$,

$$f(x + z) = f(x) + h(z).$$

For $x = 0$, the last equation implies that $f = h$ (and, therefore, f satisfies the linear Cauchy equation). Also, notice that equations (11.6) and (11.7) now imply that f and g are inverse to each other. If, instead, in equation (11.8), we set $x = g(z)$, we find that

$$f(g(z) + g(y)) = z + y \Rightarrow g(z) + g(y) = g(z + y). \qquad \square$$

The following problem given in the 1998 Romanian Olympiad is a duplication of Problem 11.2 but for the Vincze I functional equation (8.13). That is, instead of continuity, monotonicity is given for one of the unknown functions of (8.13).

Problem 11.19 (Romania 1998). *Find all functions $u : \mathbb{R} \to \mathbb{R}$ for which there exists a strictly monotonic function $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x+y) = f(x)u(y) + f(y) \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. For $y = 0$, $f(x)(1 - u(0)) = f(0)$. If $u(0) \neq 1$, then $f(x) = f(0)/(1 - u(0))$; that is, the function $f(x)$ will be constant and thus not strictly monotonic. Therefore, it must necessarily be $u(0) = 1$. This also implies $f(0) = 0$. If, in the defining equation, we set x in place of y and vice versa, we find

$$f(y+x) = f(y)u(x) + f(x).$$

Therefore,

$$f(x)u(y) + f(y) = f(y)u(x) + f(x).$$

Since, by the monotonicity of $f(x)$ and that $f(0) = 0$, $f(x) \neq 0$ for all $x \neq 0$, we can separate the terms depending on x from the terms depending on y (and thus the name **separation of variables**):

$$\frac{u(x) - 1}{f(x)} = \frac{u(y) - 1}{f(y)},$$

for all $x, y \neq 0$. Obviously, each term must be a constant, say a:

$$\frac{u(x) - 1}{f(x)} = a \Rightarrow u(x) = af(x) + 1.$$

If $a = 0$, $u(x) = 1$ and $f(x+y) = f(x) + f(y)$. The solution is given in Problem 11.2 on page 169: $f(x) = cx$, $x \in \mathbb{R}$. If $a \neq 0$, then

$$\begin{aligned} u(x+y) &= af(x+y) + 1 = a[f(x)u(y) + f(y)] + 1 \\ &= [af(x) + 1 - 1]u(y) + af(y) + 1 = [u(x) - 1]u(y) + u(y) \\ &= u(x)u(y). \end{aligned}$$

That is, the function $u(x)$ satisfies the exponential Cauchy equation, which is solved by $u(x) = b^x$ for $x \in \mathbb{Q}$. Since $u(x) = af(x) + 1$, if $f(x)$ is strictly monotonic, $u(x)$ is strictly monotonic, too. Using the same idea as in Problem 11.2, we conclude that $u(x) = b^x$ and $f(x) = (b^x - 1)/a$ for $x \in \mathbb{R}$. □

The reader of course noticed that the given monotonicity enforced continuity. This allowed us to extend the solutions from $\mathbb{Q}$ to $\mathbb{R}$ uniquely. It also points to an alternative approach which the reader might want to try: first prove that the functions f and u are continuous and then select from the continuous solutions of the Vincze I equation (8.13) those with the required properties.

The following problem, given by Korea in 1997, is really the same as the previous problem but for the Vincze II functional equation (8.19).

Problem 11.20 (Korea 1997). *Find all pairs of functions $f, u : \mathbb{R} \to \mathbb{R}$ such that*

(a) *if $x < y$, then $f(x) < f(y)$, and*

(b) *for all $x, y \in \mathbb{R}$, $f(xy) = f(x)u(y) + f(y)$.*

Solution. For $y = 0$, $f(x)u(0) = 0$. The function f cannot be constant, since it is strictly increasing; in particular, $f(x) \not\equiv 0$, so $u(0) = 0$.

If, in the defining equation, we set x in place of y and vice versa, we find

$$f(yx) = f(y)\,u(x) + f(x).$$

For $y = 0$, this gives $f(0)(1 - u(x)) = f(x)$. But $f(0)$ cannot vanish since this would imply that $f(x) \equiv 0$. Define $a = -1/f(0)$ and write

$$u(x) = af(x) + 1.$$

Then

$$\begin{aligned} u(xy) &= af(xy) + 1 = a[f(x)\,u(y) + f(y)] + 1 \\ &= [af(x) + 1 - 1]\,u(y) + af(y) + 1 = [u(x) - 1]\,u(y) + u(y) \\ &= u(x)\,u(y). \end{aligned}$$

That is, the function $u(x)$ satisfies the power Cauchy equation for all values of $x \in \mathbb{R}$. Using the $\mathbb{NQR}$ method, we can solve this equation for $x \in \mathbb{Q}_+^*$: $u(x) = x^c$. If f is strictly increasing, then $u(x) = af(x) + 1$ is strictly monotonic and thus $u(1) \neq 0$. At the same time, $u(1 \cdot 1) = u(1)u(1) \Rightarrow u(1) = 1$ and we conclude that $u(x)$ is strictly increasing (and thus $a > 0$). So finally $u(x) = x^c$ with $c > 0$ for $x \in \mathbb{Q}_+$. To find the values for negative rationals, we observe that $u(-x) < 0 < u(x)$, $x > 0$. Also $u(x^2) = u(-x)^2$ or $u(-x) = -\sqrt{u(x^2)} = -x^c$. Using the same idea as in Problem 11.2, we conclude that

$$u(x) = \begin{cases} x^c, & \text{if } x \geq 0, \\ -|x|^c, & \text{if } x < 0. \end{cases}$$

Also $f(x) = (u(x) - 1)/a$, with $a > 0$. $\square$

Chapter 12

More Than Continuity

So far, we have studied functional equations for functions that are continuous or satisfy weaker conditions. Often, however, the function may be required to satisfy stronger conditions. Probably the most common and most useful condition is differentiability. When a function is differentiable, it is possible to reduce the given functional equation to a differential equation, which can be solved more easily.

12.1 Differentiable Functions

Problem 12.1. *Find all differentiable solutions* $f : \mathbb{R} \to \mathbb{R}$ *of the linear Cauchy equation* (5.1).

Solution. Differentiating (5.1) with respect to x, we find

$$f'(x+y) = f'(x).$$

Now we set $x = 0$ and $f'(0) = c$. Then $f'(y) = c$, so

$$f(y) = cy + b.$$

From (5.1) we find that $f(0) = 0$ and therefore $b = 0$, giving the final result $f(x) = cx$. □

Comment. We notice that continuity or differentiability of $f(x)$ satisfying (5.1) leads to the same set of solutions for this equation. Therefore, one expects to be able to show — independently — that *any continuous solution of* (5.1) *must necessarily be differentiable.* Indeed, this can be done as shown below.

Since $f(x)$ is continuous in $\mathbb{R}$, it cannot approach $\pm\infty$ for any $x \in [0, 1]$ since, if it did, it would be discontinuous at that point. We thus assume that

$f(x)$ is integrable on the interval [0,1]. We then integrate equation (5.1) with respect to y from 0 to 1 to get

$$\int_0^1 f(x+y)\,dy = f(x) + \int_0^1 f(y)\,dy,$$

or

$$f(x) = \int_x^{1+x} f(u)\,du - \int_0^1 f(y)\,dy.$$

The second integral in the right-hand side is a constant, which we denote by $-C$:

$$f(x) = \int_x^{1+x} f(u)\,du + C.$$

We notice now that the right-hand side is a differentiable function; the left-hand side must therefore be differentiable, too. In particular,

$$f'(x) = f(x+1) - f(x) = f(1).$$

The observation that continuity implies differentiability for (at least some) functional equations is important since it allows us to convert functional equations to differential equations — a subject which allows systematic treatment of the problems posed. (For additional information, the reader should consult Section 12.3.) □

Problem 12.2. *Find all differentiable solutions $f : \mathbb{R} \to \mathbb{R}$ of the exponential Cauchy equation* (5.8)*.*

Solution. From (5.8) we find that $f(0) = 1$. Differentiating (5.8) with respect to x, we find that that

$$f'(x+y) = f'(x)\,f(y).$$

Now we set $x = 0$ and $f'(0) = c$. Then

$$f'(y) = c\,f(y) \;\Rightarrow\; \int_1^f \frac{df}{f} = c\int_0^y dy \;\Rightarrow\; f(y) = a^y,$$

where we set $a = e^c$. □

Problem 12.3. *Find all differentiable solutions $f : \mathbb{R} \to \mathbb{R}$ of the logarithmic Cauchy equation* (5.14)*.*

Solution. From (5.14), we find that $f(1) = 0$. Differentiating (5.14) with respect to x, we find

$$y\, f'(x\, y) = f'(x).$$

Now we set $x = 1$ and $f'(1) = c$. Then

$$y\, f'(y) = c \;\Rightarrow\; \int_0^f df = c \int_1^y \frac{dy}{y} \;\Rightarrow\; f(y) = c \ln y = \log_\gamma y,$$

where we set $c = 1/\ln\gamma$. □

Problem 12.4. *Find all differentiable solutions $f : \mathbb{R} \to \mathbb{R}$ of the power Cauchy equation* (5.20).

Solution. From (5.20), we find that $f(1) = 1$. Then differentiating (5.20) with respect to x, we find

$$y\, f'(x\, y) = f'(x)\, f(y).$$

Now we set $x = 1$ and $f'(1) = c$. Then

$$y\, f'(y) = c\, f(y) \;\Rightarrow\; \int_1^f \frac{df}{f} = c \int_1^y \frac{dy}{y} \;\Rightarrow\; \ln f(y) = c \ln y \;\Rightarrow\; f(y) = y^c.$$

□

Problem 12.5. *Find all differentiable solutions $f : \mathbb{R} \to \mathbb{R}$ of the Jensen functional equation* (5.27).

Solution. Differentiating (5.27) with respect to x, we find

$$\frac{1}{2} f'\left(\frac{x+y}{2}\right) = \frac{f'(x)}{2}.$$

Now we set $x = 0$ and $f'(0) = c$. Then $f'(y) = c$, and therefore

$$f(y) = c\, y + b.$$

□

Problem 12.6. *Find all differentiable solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) + f(x-y) = 2\, f(x). \tag{12.1}$$

Solution. Differentiating (12.1) with respect to x, we find

$$f'(x+y) + f'(x-y) = 2\,f'(x).$$

Differentiating (12.1) with respect to y, we find

$$f'(x+y) - f'(x-y) = 0.$$

Add the last two equations to get

$$f'(x+y) = f'(x).$$

Now if we set $x = 0$ and $f'(0) = c$ we finally find $f'(y) = c$, or

$$f(y) = c\,y + b. \qquad \square$$

Problem 12.7. *Find all differentiable solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) + f(x-y) = 2\,f(x)\,f(y). \tag{12.2}$$

Solution. Differentiating (12.2) with respect to x, we find

$$f'(x+y) + f'(x-y) = 2\,f'(x)\,f(y). \tag{12.3}$$

Note that the right-hand side is differentiable with respect to y. Then the left-hand side is, too. Therefore, f is twice differentiable.

Differentiating equation (12.3) with respect to x, we find

$$f''(x+y) + f''(x-y) = 2\,f''(x)\,f(y).$$

Differentiating (12.2) with respect to y twice, we find

$$f''(x+y) + f''(x-y) = 2\,f(x)\,f''(y).$$

The left-hand sides of the last two equations are equal. Then

$$f''(x)\,f(y) = f(x)\,f''(y).$$

Now we set $y = 0$. Then either $f(0) = 0$ or $f(0) \neq 0$. If $f(0) = 0$, we also have $f(x) \equiv 0$. If $f(0) \neq 0$, then set $f''(0)/f(0) = \pm\omega^2$. The function f is a solution of the differential equation

$$f''(x) \pm \omega^2\,f(x) = 0,$$

with solutions

$$\begin{aligned} f(x) &= A + B\,x, \\ f(x) &= A\,\cos\omega x + B\,\sin\omega x, \\ f(x) &= A\,\cosh\omega x + B\,\sinh\omega x. \end{aligned}$$

A and B can be determined easily using conditions found from the defining functional equation. Set $y = 0$ in (12.2) to find $f(x)(f(0) - 1) = 0$. If the function is not identically zero, it must satisfy $f(0) = 1$. Then, differentiating (12.2) with respect to y and setting $x = 0$, we find $f'(y) - f'(-y) = 2f'(y)$, which implies

$$f'(y) + f'(-y) = 0.$$

Then, setting $x = 0$ in (12.3), we find $0 = f(y)\,f'(0)$, i.e. $f'(0) = 0$. These conditions, $f(0) = 1$ and $f'(0) = 0$, require that $B = 0$ and $A = 1$. Therefore the solutions are $f(x) = 1$, $f(x) = \cos\omega x$, and $f(x) = \cosh\omega x$. $\square$

Problem 12.8 (Putnam 1963). *Find every twice differentiable real-valued function $f : \mathbb{R} \to \mathbb{R}$ satisfying the functional equation*

$$f(x+y)\,f(x-y) = f(x)^2 - f(y)^2 \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. The identically vanishing function $f(x) \equiv 0$ is an obvious solution. We shall search for solutions that are not identically vanishing. Therefore, there is an $x_0 \in \mathbb{R}$ such that $f(x_0) \neq 0$.

Differentiating the given equation with respect to x, we find

$$f'(x+y)\,f(x-y) + f(x+y)\,f'(x-y) = 2f(x)\,f'(x).$$

Then, differentiating with respect to y,

$$f''(x+y)\,f(x-y) - f(x+y)\,f''(x-y) = 0.$$

Now make the change of variables $u = x + y$ and $v = x - y$ to obtain

$$f''(u)\,f(v) = f(u)\,f''(v),$$

and set $v = x_0$. The quantity $f''(x_0)/f(x_0)$ may be positive, negative, or zero. We set it equal to $+b^2$, $-b^2$, or 0, as the case may be:

$$\begin{aligned} f''(u) - b^2 f(u) &= 0, \\ f''(u) + b^2 f(u) &= 0, \\ f''(u) &= 0. \end{aligned}$$

We can find respectively the solutions

$$\begin{aligned} f(u) &= A\,\sin(bu) + B\,\cos(bu), \\ f(u) &= A\,\sinh(bu) + B\,\cosh(bu), \\ f(u) &= A\,u + B. \end{aligned}$$

Setting $x = y = 0$ in the defining relation we find that $f(0) = 0$. This implies that $B = 0$. And the final solutions are

$$\begin{aligned} f(u) &= A\,\sin(bu), \\ f(u) &= A\,\sinh(bu), \\ f(u) &= A\,u. \end{aligned}$$

□

Problem 12.9. *Find twice differentiable real-valued functions $f, g : \mathbb{R} \to \mathbb{R}$ satisfying the functional equation*

$$f(x+y) + f(x-y) = 2\,f(x)\,g(y) \quad \textit{for all } x, y \in \mathbb{R}.$$

Solution. If $f(x) \equiv 0$, then $g(x)$ is arbitrary. If $g(x) \equiv 0$, then $f(x+y)+f(x-y) = 0$. Setting $y = 0$, we see that $f(x) \equiv 0$.

We shall search for solutions that do not vanish identically. Take $x_0 \in \mathbb{R}$ such that $f(x_0) \neq 0$ and $y_0 \in \mathbb{R}$ such that $g(y_0) \neq 0$.

Differentiating the given equation twice with respect to x and (separately) twice with respect to y, we find

$$\begin{aligned} f''(x+y) + f''(x-y) &= 2f''(x)\,g(y), \\ f''(x+y) + f''(x-y) &= 2f(x)\,g''(y). \end{aligned}$$

Comparing these two equations, we get

$$f''(x)\,g(y) = f(x)\,g''(y). \tag{12.4}$$

We now set $x = x_0$ and $y = y_0$:

$$\frac{f''(x_0)}{f(x_0)} = \frac{g''(y_0)}{g(y_0)}.$$

This ratio may be positive, negative, or zero. We set it to $+b^2$, $-b^2$, or 0, as the case may be.

Setting $x = x_0$ in (12.4), we find

$$\begin{aligned} g''(y) - b^2 g(y) &= 0, \\ g''(y) + b^2 g(y) &= 0, \\ g''(y) &= 0. \end{aligned}$$

We can find respectively the solutions

$$\begin{aligned} g(y) &= A_2 \sin(by) + B_2 \cos(by), \\ g(y) &= A_2 \sinh(by) + B_2 \cosh(by), \\ g(y) &= A_2\, y + B_2. \end{aligned}$$

Using $y = y_0$ in (12.4), we find similarly

$$\begin{aligned} f(x) &= A_1 \sin(bx) + B_1 \cos(bx), \\ f(x) &= A_1 \sinh(bx) + B_1 \cosh(bx), \\ f(x) &= A_1\, x + B_1. \end{aligned}$$

Inserting the results in the defining equation, we find the final solutions:

$$\begin{aligned} f(x) &= A_1 \sin(bx) + B_1 \cos(bx), & g(x) &= A_2 \sin(bx) + B_2 \cos(bx), \\ f(x) &= A_1 \sinh(bx) + B_1 \cosh(bx), & g(x) &= A_2 \sinh(bx) + B_2 \cosh(bx), \\ f(x) &= A_1\, x + B_1, & g(x) &= 1. \end{aligned}$$ □

Comment. The reader has recognized, of course, that the previously three functional equations we solved are the D'Alembert–Poisson I (7.21), D'Alembert–Poisson II (7.25) and the Wilson I equation (8.25). Here we have solved them by imposing stronger conditions than continuity on the solutions. Note, however, that the set of solutions is the same as those for continuous functions. We will return to this issue in Section 12.3. □

Problem 12.10. *Find all differentiable functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x + yf(x)) = f(x)\, f(y).$$

Solution. Setting $x = y = 0$, we find that either $f(0) = 0$ or $f(0) = 1$. If $f(0) = 0$, then setting $x = 0$ the defining equation implies that $f(x)$ vanishes identically. Therefore, we shall investigate the second case, $f(0) = 1$.

Differentiating the defining equation with respect to y, we find

$$f'(x + yf(x))\, f(x) = f(x)\, f'(y).$$

In this equation, let $x = x_0$, where x_0 is one of the points where f has a nonzero value. Then

$$f'(x_0 + yf(x_0)) = f'(y). \tag{12.5}$$

Differentiating the defining equation with respect to x, we find

$$f'(x + yf(x))\,(1 + yf'(x)) = f'(x)\, f(y).$$

In this equation, let $x = x_0$. Then

$$f'(x_0 + yf(x_0))\,(1 + yf'(x_0)) \;=\; f'(x_0)\,f(y).$$

Using (12.5), we find $f'(y)\,(1 + yf'(x_0)) \;=\; f'(x_0)\,f(y)$, or

$$f'(y) - \frac{a}{1+ay}\,f(y) \;=\; 0,$$

where we have set $a = f'(x_0)$. This equation may equivalently be written as

$$\frac{d}{dy}\left(\frac{f(y)}{1+ay}\right) \;=\; 0.$$

Upon integration we find

$$f(y) \;=\; 1 + a\,y. \qquad \square$$

12.2 Analytic Functions

Problem 12.11. *Find an analytic solution $f : [0,1) \to \mathbb{R}$ of the functional equation*

$$f(x) + f(x^2) \;=\; x \quad \textit{for all } x. \tag{12.6}$$

Solution. Since f is analytic, it has a convergent Taylor series for all $x \in [0,1)$. So let

$$f(x) \;=\; \sum_{n=0}^{\infty} a_n\,x^n.$$

Then

$$f(x^2) \;=\; \sum_{n=0}^{\infty} a_n\,x^{2n}.$$

Substituting in equation (12.6), we find

$$\begin{aligned} x &= \sum_{n=0}^{\infty} a_n\,x^n + \sum_{n=0}^{\infty} a_n\,x^{2n} \\ &= \sum_{k=0}^{\infty} a_{2k}\,x^{2k} + \sum_{k=0}^{\infty} a_{2k+1}\,x^{2k+1} + \sum_{n=0}^{\infty} a_n\,x^{2n}, \end{aligned}$$

or

$$\sum_{k=0}^{\infty}(a_{2k} + a_k)\,x^{2k} + (a_1 - 1)\,x + \sum_{k=1}^{\infty} a_{2k+1}\,x^{2k+1} \;=\; 0.$$

This equation will be true if

$$\begin{aligned} a_{2k} + a_k &= 0 \quad \text{for } k = 0, 1, 2, \dots, \\ a_{2k+1} &= 0 \quad \text{for } k = 1, 2, \dots, \\ a_1 &= 1. \end{aligned}$$

From the second equation we see that all coefficients with subscripts a_{2k+1} vanish. From the first equation we see that if k = odd, then $a_{2k} = a_k = 0$. If k = even = $2m$, then $a_{2^2 m} = a_{2k} = -a_k = -a_{2m} = a_m$. Now, a_m will vanish unless $m = 2m'$. Then $a_{2^3 m'} = a_{2m'} = -a_{m'}$. Inductively, $a_{2^\ell m''} = (-1)^\ell a_{m''}$. Therefore, we conclude that $a_{2^\ell} = (-1)^\ell$ for $\ell = 0, 1, \dots,$ and all other coefficients vanish; that is,

$$f(x) = \sum_{\ell=0}^{\infty} (-1)^\ell x^{2^\ell}. \qquad \square$$

This is the only analytic solution of (12.6). If we modify the domain of f slightly, the result can change drastically:

Problem 12.12. *Prove that there is no continuous function $f : [0,1] \to \mathbb{R}$ of the functional equation* (12.6).

Solution. Using the defining equation recursively, we find

$$\begin{aligned} f(x) &= x - f(x^2) \\ &= x - \big(x^2 - f(x^4)\big) \\ &= x - x^2 + \big(x^4 - f(x^8)\big) \\ &\dots \\ &= x - x^2 + x^4 - x^8 + \cdots + (-1)^n \big(x^{2^n} - f\big(x^{2^{n+1}}\big)\big). \end{aligned}$$

This result is true for all $x \in [0,1]$.

Setting $x = 0$ in equation (12.6), we find $f(0) = 0$. By the continuity of f, for any sequence $a_n \to 0$,

$$\lim_{n\to\infty} f(a_n) = 0.$$

For any $x \in [0, 1)$, we can take

$$a_n = x^{2^{n+1}}.$$

By taking the limit $n \to \infty$ in the equation of $f(x)$, we then conclude that

$$f(x) = x - x^2 + x^4 - x^8 + \cdots + (-1)^n x^{2^n} + \cdots,$$

for $x \in [0, 1)$ — the result of the last problem with a milder assumption.

It remains to examine the behavior of f at $x = 1$. Equation (12.6) can equivalently be written as

$$f(\sqrt{x}) + f(x) = \sqrt{x}.$$

Using this equation recursively, we find

$$\begin{aligned} f(x) &= \sqrt{x} - f(\sqrt{x}) \\ &= \sqrt{x} - \left(\sqrt[4]{x} - f(\sqrt[4]{x})\right) \\ &\cdots \\ &= \sqrt{x} - \sqrt[4]{x} + \sqrt[8]{x} + \cdots + (-1)^n\left(\sqrt[2^n]{x} - f\left(\sqrt[2^{n+1}]{x}\right)\right), \end{aligned}$$

or

$$f(x) + (-1)^n f\left(\sqrt[2^{n+1}]{x}\right) = \sqrt{x} - \sqrt[4]{x} + \sqrt[8]{x} + \cdots + (-1)^n \sqrt[2^n]{x}.$$

Setting $x = 1$ in (12.6), we find $f(1) = \frac{1}{2}$. By the continuity of f, for any sequence $b_n \to 1$,

$$\lim_{n\to\infty} f(b_n) = \tfrac{1}{2}.$$

For any $x \in (0, 1]$, we can take

$$b_n = \sqrt[2^{n+1}]{x}.$$

However, by attempting to take the limit, the term

$$(-1)^n f\left(\sqrt[2^{n+1}]{x}\right)$$

oscillates: the even subsequence b_{2n} gives $f(\frac{1}{2})$ and the odd subsequence b_{2n+1} gives $-f(\frac{1}{2})$. Hence we cannot construct a continuous function on all of $[0, 1]$. $\square$

Problem 12.13 (Putnam 1972). *Show that the power series representation for the series*

$$\sum_{k=0}^{\infty} \frac{x^k(x-1)^{2k}}{k!}$$

cannot have three consecutive zero coefficients.

Solution. We first observe that the given series represents an analytic function:

$$\sum_{k=0}^{\infty} \frac{(x(x-1)^2)^k}{k!} = e^{x(x-1)^2}.$$

We will set

$$f(x) = e^{g(x)}, \quad g(x) = x^3 - 2x^2 + x.$$

The power series of $f(x)$ sought is its Taylor series:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}\, x^k.$$

We must therefore show that, for any $k = 1, 2, \ldots$, the derivatives $f^{(k-1)}(0)$, $f^{(k)}(0)$, $f^{(k+1)}(0)$ are not simultaneously zero.

We have

$$\begin{aligned} f'(x) &= f(x)\, g'(x), \\ f^{(k)}(x) &= \frac{d^{k-1}}{dx^{k-1}} f'(x) = \frac{d^{k-1}}{dx^{k-1}} f(x)\, g'(x) = \sum_{\ell=0}^{k-1} \binom{k-1}{\ell} f^{(\ell)}(x)\, g^{(k-\ell)}(x). \end{aligned}$$

Since $g(x)$ is a polynomial of degree 3, its derivatives vanish from the fourth on. Thus

$$\begin{aligned} f^{(k)}(x) = {} & f^{(k-1)}(x)\, g'(x) \\ & + (k-1)\, f^{(k-2)}(x)\, g''(x) + \frac{(k-1)(k-2)}{2}\, f^{(k-3)}(x)\, g'''(x). \end{aligned}$$

So finally $f(0) = 1$, $f'(0) = 1$, $f''(0) = -3$ and

$$f^{(k)}(0) = f^{(k-1)}(0) - 4(k-1)\, f^{(k-2)}(0) + 3(k-1)(k-2)\, f^{(k-3)}(0)$$

for $k \geq 3$. We can now prove the statement as follows: If for any k all three derivatives $f^{(k-1)}(0)$, $f^{(k)}(0)$, $f^{(k+1)}(0)$ were zero, then, since the derivative $f^{(k+2)}(0)$ is a linear combination of them, we would have $f^{(k+2)}(0) = 0$, $f^{(k+3)}(x)(0) = 0$ and so on to infinity. That is, the Taylor series would be reduced to a finite sum (polynomial), which contradicts the initial expression containing powers x^k, with k arbitrarily large. We must therefore conclude that the three derivatives $f^{(k-1)}(0)$, $f^{(k)}(0)$, $f^{(k+1)}(0)$ cannot vanish simultaneously. $\square$

12.3 Stronger Conditions as a Tool

As mentioned in Section 12.1, even if we are searching for continuous solutions of a functional equation, it is useful to assume that the function is differentiable (one or more times) or even analytic. This allows us to use the vast number of tools from the theory of differential equations. In fact, it allows a uniform treatment of virtually all functional equations, a topic in which a universal approach is missing.

After differential (or analytic) solutions of a functional equation are found, they are examined for uniqueness. This idea is demonstrated in the next problem, where we solve again the functional equation (5.32). In other cases, one can show that the solutions must necessarily be differentiable (as many times as necessary). This idea has already been presented in Problem 12.1. It is reinforced with Problem 12.15 below.

Problem 12.14. *Find all continuous solutions $f : \mathbb{R} \to \mathbb{R}$ of the functional equation*

$$f(x+y) = f(x) + f(y) + a\,(1 - b^x)\,(1 - b^y), \tag{5.32}$$

where a, b are real constants and $b > 0$.

Solution. Assume that $f_p(x)$ is a doubly differentiable particular solution of (5.32). Differentiating (5.32) with respect to x and then with respect to y, we find

$$f_p''(x+y) = a\,(b^x \ln b)\,(b^y \ln b) = a\,b^{x+y}(\ln b)^2.$$

Setting $u = x + y$ and integrating twice, we find

$$f_p(x) = a\,b^x + c_1\,x + c_2,$$

with c_1, c_2 constants. For $x = y = 0$, equation (5.32) gives $f_p(0) = 0$ which, in turn, implies $c_2 = -a$. Therefore,

$$f_p(x) = a\,(b^x - 1) + c_1\,x.$$

Let $f(x)$ be any other continuous solution of the functional equation. Then the function $g(x) = f(x) - f_p(x)$ is a continuous function and satisfies

$$g(x+y) = g(x) + g(y).$$

We thus conclude that $g(x) = k\,x$ with k some real constant and

$$f(x) = g(x) + f_p(x) = a\,(b^x - 1) + c\,x,$$

with $c = k + c_1$. □

Comment. This method is more straightforward than the transformation (5.33) since it involves no guessing. In general, it is quite hard to guess the transformation that reduces a complicated equation to a simpler one. □

In Section 8.6, we wrote down without proof the continuous solutions for Wilson's first equation (8.25). In Section 12.1, we worked out the solution among twice differentiable functions. The results are identical. To use differentiation to prove the results of Section 8.6, one would have to establish that no continuous (but not twice differentiable) solution has been lost in the calculation.

Problem 12.15. *Prove that, for Wilson's first equation* (8.25), *any continuous solution is twice differentiable.*

Solution. Let

$$F(x) = \int_0^x f(u)\,du, \quad G(x) = \int_0^x g(u)\,du.$$

With this convention, $F(0) = G(0) = 0$. The functions $F(x)$ and $G(x)$ are differentiable[1] with $F'(x) = f(x)$ and $G'(x) = g(x)$.

We now integrate (8.25) with respect to y from 0 to y:

$$\int_0^y f(x+z)\,dz + \int_0^y f(x-z)\,dz = 2f(x) \int_0^y g(z)\,dz,$$

or

$$[F(x+y) - F(x)] + [F(x) - F(x-y)] = 2f(x)\,G(y).$$

This is of course

$$F(x+y) - F(x-y) = 2f(x)\,G(y).$$

Since the left-hand side of this equation is differentiable, the right-hand side is differentiable, too. This, in turn, implies that $f(x)$ is differentiable. We can thus differentiate the last equation with respect to x to find:

$$f(x+y) - f(x-y) = 2f'(x)\,G(y).$$

Again, since the left-hand side of this equation is differentiable with respect to x, the right-hand side is differentiable too, which implies that $f''(x)$ exists. Returning to the original equation, since $f(x)$ is twice differentiable, it is easily seen that $g(x)$ is also twice differentiable.

Thus the solution of (8.25) presented in Section 12.1 gives all continuous solutions. □

[1] See Section 19.1 for the fundamental theorem of integral calculus in use here.

Chapter 13

Functional Equations for Polynomials

13.1 Fundamentals

Polynomials hold a special place in mathematics. Functional equations satisfied by polynomials can be solved, of course, by all methods discussed so far and all those to be discussed in the following chapters. However, knowing that the solution to a functional equation is a polynomial can, as a rule, simplify the solution process since the requirement that the solution be a polynomial is a very restrictive one.

In particular, the following properties are quite useful:

Theorem 13.1. *If $P(x)$ is a polynomial of degree n and $Q(x)$ is a polynomial of degree m, the sum $P(x) + Q(x)$ is a polynomial of degree $max\{n, m\}$, the product $P(x)Q(x)$ is a polynomial of degree $n + m$, and the composition $P(Q(x))$ is a polynomial of degree nm.*

Proof. Straightforward. □

Theorem 13.2 (Binomial theorem). *For any x, y and $n \in \mathbb{N}^*$,*

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

Proof. Easy by induction. It can also be seen as a special case of equation (13.1) below. □

Theorem 13.3 (Fundamental theorem of algebra). *A polynomial $P(x)$ of degree $n \geq 1$ has at least one root in $\mathbb{C}$.*

The first proof of this theorem was given by D'Alembert. However, his demonstration used one assumption without proof: a continuous function defined on a closed and bounded set reaches a minimum somewhere in the set. It took another hundred years for this assumption to be resolved.

Let r be a root of $P(x)$, so $P(r) = 0$. If $P(x)$ is divided by $x - r$, which is of degree 1, the remainder R will be of degree 0:

$$P(x) = (x - r)\,Q(x) + R,$$

where the quotient $Q(x)$ is of degree $n - 1$. Setting $x = r$ in this equation, we see that $R = 0$ and

$$P(x) = (x - r)\,Q(x).$$

Therefore, we have shown the following:

Theorem 13.4. *A polynomial $P(x)$ is divisible by the binomial $x - r$ if and only if $P(r) = 0$.*

Theorem 13.5. *A polynomial $P(x)$ of degree $n \geq 1$ has exactly n roots in $\mathbb{C}$.*

Proof. As already seen, $P(x)$ has at least one root r_1, and it can be factorized in the form

$$P(x) = (x - r_1)\,P_{n-1}(x),$$

where $P_{n-1}(x)$ is of degree $n - 1$. If $n - 1 \geq 1$, then $P_{n-1}(x)$ has at least one root r_2, and it can be factorized in the form

$$P_{n-1}(x) = (x - r_2)\,P_{n-2}(x),$$

where $P_{n-2}(x)$ is of degree $n - 2$. Continuing this way, we finally reach a quotient polynomial $P_0(x)$ of degree 0, that is, one that is a constant:

$$P_1(x) = (x - r_n)\,P_0(x) = (x - r_n)\,A.$$

Combining all equations, we see that $P(x)$ factorizes in the form

$$P(x) = A\,(x - r_1)(x - r_2)\ldots(x - r_n).$$

Obviously, it has exactly n roots. □

Notice that the n roots may not be distinct. Instead, root r_1 might appear m_1 times, root r_2 might appear m_2 times, and so on. The numbers m_1, m_2, ... are known as **multiplicities** of the roots, and according to the previous theorem, $m_1 + m_2 + \cdots = n$. A root of multiplicity 1 is called **simple**.

Theorem 13.6. *A polynomial $P(x)$ of degree n is identically zero if it vanishes for (at least) $n + 1$ distinct values of x.*

Proof. Let $r_1, r_2, \ldots, r_n$ be n distinct values of x at which $P(x)$ vanishes. By Theorem 13.5,

$$P(x) = A(x - r_1)(x - r_2)\ldots(x - r_n).$$

Let r_{n+1} be the $(n+1)$-th value of x for which $P(x)$ vanishes. Then

$$P(r_{n+1}) = A(r_{n+1} - r_1)(r_{n+1} - r_2)\ldots(r_{n+1} - r_n).$$

Since r_{n+1} is distinct from $r_1, r_2, \ldots, r_n$, none of the numbers $r_{n+1} - r_i$ vanish, for $i = 1, 2, \ldots, r_n$. Therefore $A = 0$, which implies that $P(x) \equiv 0$. □

Theorem 13.7. *Two polynomials $P(x)$ and $Q(x)$ of degree n and $m \le n$ respectively are equal if they have equal values for (at least) $n+1$ distinct values of x.*

Proof. We set $R(x) = P(x) - Q(x)$. This polynomial is of degree n and it vanishes for (at least) $n+1$ distinct values of x. According to the previous theorem, it vanishes identically. Therefore $P(x) \equiv Q(x)$. □

Theorem 13.8. *The only periodic polynomial*

$$P(x + T) = P(x),$$

for some constant T, is the constant polynomial $P(x) = c$.

Proof. Let $P(0) = c$. By the periodicity of $P(x)$, it is true that

$$P(0) = P(T) = P(2T) = \cdots = c.$$

Therefore, the polynomial $P(x)$ and the constant polynomial $Q(x) = c$ take the same values at an infinite number of points. Therefore, they must be equal to each other. □

13.2 Symmetric Polynomials

For a quadratic polynomial

$$P(x) = a\,x^2 + b\,x + c,$$

the formulæ

$$r_1 + r_2 = -\frac{b}{a}, \quad r_1 r_2 = \frac{c}{a},$$

providing the sum and product of its two roots, are well known. They can be proved by writing

$$P(x) = a\,(x - r_1)\,(x - r_2),$$

expanding the product, and finally matching the coefficients with the coefficients of the original expression.

These formulæ can be generalized for any polynomial.

Theorem 13.9. *Given the roots $r_1, r_2, \ldots, r_n$ of a polynomial*

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

the following formulæ are true:

$$\sum_{i=1}^{n} r_i = -\frac{a_{n-1}}{a_n},$$

$$\sum_{\substack{i,j=1\\ i<j}}^{n} r_i r_j = +\frac{a_{n-2}}{a_n},$$

$$\sum_{\substack{i,j,k=1\\ i<j<k}}^{n} r_i r_j r_k = -\frac{a_{n-3}}{a_n},$$

$$\ldots$$

$$r_1 r_2 \ldots r_{n-1} r_n = (-1)^n \frac{a_0}{a_n}.$$

They are known[1] *as **Viète's formulæ**.*

Perhaps the proof is obvious to the reader. I give it in the following lines while I am introducing some additional material, too. I start by extending the binomial theorem to a product of different factors:

$$\begin{aligned} x+x_1 &= x+x_1, \\ (x+x_1)(x+x_2) &= x^2+(x_1+x_2)x+x_1x_2, \\ (x+x_1)(x+x_2)(x+x_3) &= x^3+(x_1+x_2+x_3)x^2+(x_1x_2+x_2x_3+x_3x_1)x+x_1x_2x_3, \\ &\ldots \end{aligned}$$

and, by induction,

$$\prod_{i=1}^{n}(x+x_i) = \sum_{i=0}^{n} e_i(x)\, x^{n-i}, \tag{13.1}$$

where

$$e_i(x_1, x_2, \ldots, x_n) = \sum_{1\le k_1<k_2<\cdots<k_i\le n} x_{k_1} x_{k_2} \ldots x_{k_i}$$

is known as the i-th **elementary symmetric polynomial** in n variables. (The term "symmetric" is defined formally below.) If $x_1 = x_2 = \cdots = x_n$, we recover the binomial theorem.

[1] After the French mathematician François Viète. Often he is mentioned by the Italian version of his name, Franciscus Vieta.

Viète's formulæ can be written as

$$e_i(r_1, r_2, \ldots, r_n) = (-1)^i \, \frac{a_{n-i}}{a_n},$$

and they are an immediate consequence of equation (13.1) and the factorization of the polynomial $P(x)$:

$$P(x) = a_n \, (x - r_1)(x - r_2) \ldots (x - r_n).$$

Example 13.10. It is well known that the quadratic polynomial $P(x) = ax^2 + bx + c$ does not have real roots if $b^2 < 4ac$. Imagine that there was no explicit formula for the roots of the quadratic polynomial. Could we have derived the same conclusion in such a case? The answer is yes. From the formulæ

$$r_1 + r_2 = -\frac{b}{a}, \quad r_1 r_2 = \frac{c}{a},$$

we see that

$$(r_1 + r_2)^2 - 2r_1 r_2 = \frac{b^2 - 2ac}{a^2},$$

or

$$r_1^2 + r_2^2 = \frac{b^2 - 2ac}{a^2}.$$

If the roots are real, the left-hand side is nonnegative. Therefore, if $b^2 < 2ac$, there cannot be real roots. However, this inequality is not the most strict. We can compute[2]

$$(r_1 - r_2)^2 = \frac{b^2 - 4ac}{a^2}.$$

From this we conclude that there cannot be real roots if $b^2 < 4ac$; this improves the previous result.

You can use this idea in higher-order polynomials. Just to get experience, let's do it for the cubic polynomial:

$$P(x) = ax^3 + bx^2 + cx + d.$$

Then

$$r_1 + r_2 + r_3 = -\frac{b}{a} \quad r_1 r_2 + r_2 r_3 + r_3 r_1 = \frac{c}{a},$$

and

$$r_1^2 + r_2^2 + r_3^2 = \frac{b^2 - 2ac}{a},$$

$$(r_1 - r_2)^2 + (r_2 - r_3)^2 + (r_3 - r_1)^2 = \frac{2b^2 - 6ac}{a^2}.$$

If $b^2 < 3ac$, a cubic polynomial cannot have three real roots. □

[2] We see that, for a quadratic polynomial, its discriminant is given in terms of the difference of its roots: $D = a^2(r_1 - r_2)^2$. In general, for a polynomial of degree n, we define its discriminant by $D = a_n^{2(n-1)} \prod_{i=1}^{n} \prod_{j=i+1}^{n} (r_i - r_j)^2$.

Using the ideas above, the solution to the following problem should now be obvious to you.

Problem 13.11 (USA 1983). *Prove that the roots of*

$$x^5 + bx^4 + cx^3 + dx^2 + ex + f = 0$$

cannot all be real if $2b^2 < 5c$.

In the previous example we used the expression $P(x, y, z) = x^2 + y^2 + z^2$, which is a polynomial itself with the additional property that it remains unchanged under exchanges of its variables. Such a polynomial is called *symmetric*:

Definition 13.12. A **symmetric polynomial** $S(x_1, x_2, \ldots, x_n)$ in n variables is a polynomial of the n variables that satisfies

$$S(x_1, x_2, \ldots, x_n) = S(x_{\sigma_1}, x_{\sigma_2}, \ldots, x_{\sigma_n}),$$

for any permutation $(\sigma_1, \sigma_2, \ldots, \sigma_n)$ of $(1, 2, \ldots, n)$.

Theorem 13.13. *Any symmetric polynomial can be expressed as a polynomial in the elementary symmetric polynomials.*

From what has been discussed above, it is immediate that

Corollary. *Any symmetric polynomial* $S(r_1, r_2, \ldots, r_n)$ *of the roots of a polynomial* $P(x)$ *can be expressed in terms of the coefficients of the polynomial* $P(x)$.

The previous corollary explains why, in Example 13.10, I used symmetric polynomials

$$P(x, y, z) = x^2 + y^2 + z^2,$$
$$P(x, y, z) = (x - y)^2 + (y - z)^2 + (z - x)^2,$$

to draw conclusions about the roots and not polynomials of the form

$$P(x, y, z) = (x - y)^2 + y^2 + z^2,$$

for example. The former are written easily in terms of the coefficients of the polynomial through Viète's formulæ.

13.3 More on the Roots of Polynomials

Theorem 13.14. (a) *If r is a simple root of the polynomial $P(x)$, then it is not a root of its derivative $P'(x)$.*

(b) *If r is a root of multiplicity m, where $m \geq 2$, then it is a root of $P'(x)$ with multiplicity $m-1$.*

Proof. Let

$$P(x) = (x-r)^m \, Q(x),$$

with $m \geq 1$. Since m gives the maximal exponent for the factor $(x-r)^m$ that divides $P(x)$, $x-r$ does not divide $Q(x)$ and, therefore, $Q(r) \neq 0$.

If we differentiate the preceding equation, we find:

$$P'(x) = (x-r)^{m-1} \left(m \, Q(x) + (x-r) \, Q'(x) \right).$$

Obviously, r is a root of $P'(x)$ with multiplicity at least $m-1$. Now, $x-r$ does not divide the quotient $q(x) = m \, Q(x) + (x-r) \, Q'(x)$ since $q(r) = m \, Q(r) \neq 0$. Therefore, r is a root of $P(x)$ with multiplicity $m-1$. In particular, if $m=1$, then it has multiplicity 0 for $P'(x)$—that is, it is not a root. □

Rolle's theorem provides a useful tool that can be used to draw conclusions about the existence of roots for a polynomial.

Theorem 13.15. *If a and b are two distinct roots of the polynomial $P(x)$, then $P'(x)$ has a simple root in the interval (a,b).*

Proof. Since $P(x)$ is continuous and differentiable in all $\mathbb{R}$, it satisfies the conditions of Rolle's theorem in the interval $[a,b]$. Therefore, there exists an ξ such that $P'(\xi) = 0$, that is a root of $P'(x)$.

To prove that such a root is simple, we will do simple counting. Imagine that $P(x)$ has n real roots $r_1, \ldots, r_k$ with multiplicities $m_1, \ldots m_k$ (the multiplicities may be 1). Then $m_1 + \cdots + m_k = n$. Its derivative $P'(x)$ is a polynomial of degree $n-1$ and has, at least, the following roots: $r_1, \ldots, r_k$ with multiplicities $m_1 - 1, \ldots m_k - 1$ (some multiplicities may be 0) and $\xi_1, \xi_2, \ldots, \xi_{k-1}$ in the intervals (r_1, r_2), (r_2, r_3), $\ldots$, (r_{k-1}, r_k). Therefore, $P'(x)$ has at least $(m_1 - 1) + \cdots + (m_k - 1) + k - 1 = n - k + k - 1 = n - 1$ roots. But this is exactly the numbers of roots $P'(x)$ must have. We thus conclude that all ξ_i, $i = 1, 2, \ldots, k-1$ are simple roots. □

Problem 13.16 (BWMC 1991). *Let a, b, c, d, e be distinct real numbers. Prove that the equation*

$$\begin{aligned}&(x-a)(x-b)(x-c)(x-d)\\+&(x-a)(x-b)(x-c)(x-e)\\+&(x-a)(x-b)(x-d)(x-e)\\+&(x-a)(x-c)(x-d)(x-e)\\+&(x-b)(x-c)(x-d)(x-e)=0\end{aligned}$$

has four distinct real solutions.

Solution. The left-hand side is the derivative $P'(x)$ of the polynomial

$$P(x) = (x-a)(x-b)(x-c)(x-d)(x-e),$$

which has five distinct real roots a, b, c, d, e. Without loss of generality, we will assume that $a < b < c < d < e$. As proved above, $P'(x)$ has a simple root in each of the intervals (a, b), (b, c), (c, d), (d, e) — that is, $P'(x)$ four distinct roots. □

Let's return to the USA 1983 Problem 13.11 to demonstrate how the previous idea can be used in "reverse".

Solution. We set $P(x) = x^5 + bx^4 + cx^3 + dx^2 + ex + f$. Now let's assume that $P(x)$ has five real roots. According to the argument of the previous problem, the polynomial

$$P_1(x) = P'(x) = 5x^4 + 4bx^3 + 3cx^2 + 2dx + e$$

will have four real roots.

However, we can repeat the argument for the polynomial $P_1(x)$. Since it has four real roots, the polynomial

$$P_2(x) = P_1'(x) = 20x^3 + 12bx^2 + 6cx + 2d$$

will have three real roots.

And once more: Since $P_2(x)$ has three real roots, the polynomial

$$P_3(x) = P_2'(x) = 60x^2 + 24bx + 6c = 6\,(10x^2 + 4bx + c)$$

will have two real roots. This is a quadratic polynomial and we can decide if it has real roots by computing its discriminant. For the polynomial inside the parenthesis,

$$D = 16b^2 - 40c = 8(2b^2 - 5c).$$

Therefore, if $2b^2 < 5c$, $D < 0$ and the polynomial does not have real roots. Therefore, the initial assumption that $P(x)$ has five real roots is not correct. □

We can easily generalize the previous problem:

Problem 13.17. *The roots of the polynomial*

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

cannot all be real if

$$(n-1)\, a_{n-1}^2 < 2n\, a_n\, a_{n-2}.$$

Solution. If $P(x)$ had n real roots then

$$\begin{aligned}\frac{d^{n-2}}{dx^{n-2}}P(x) &= \frac{n!\, a_n}{2}\, x^2 + (n-1)!\, a_{n-1}\, x + a_{n-2}\, (n-2)! \\ &= (n-2)!\left(\frac{(n-1)n\, a_n}{2}\, x^2 + (n-1)\, a_{n-1}\, x + a_{n-2}\right)\end{aligned}$$

would have two real roots. The discriminant of the quadratic polynomial is

$$D = (n-1)\left((n-1)a_{n-1}^2 - 2na_na_{n-2}\right).$$

If $D < 0$ or, equivalently, $(n-1)a_{n-1}^2 < 2na_na_{n-2}$, then the quadratic polynomial cannot have real roots and therefore $P(x)$ cannot have all real roots. □

The following two lemmas will be useful in what follows.

Lemma 13.18. *A polynomial $P(x)$ has the same number of positive roots or one more than its derivative $P'(x)$.*

Proof. Let $p_1, p_2, \ldots p_\ell$ be distinct positive roots of $P(x)$, with multiplicities $m_1, m_2, \ldots, m_\ell$. Then $p_1, p_2, \ldots p_\ell$ are positive roots of $P'(x)$, with multiplicities $m_1 - 1, m_2 - 1, \ldots, m_\ell - 1$, and in each of the intervals (p_1, p_2), (p_2, p_3), $\ldots (p_{\ell-1}, p_\ell)$, by Rolle's theorem, there is a root of $P'(x)$. This gives

$$(m_1 - 1) + (m_2 - 1) + \cdots + (m_\ell - 1) + \ell - 1 = m_1 + m_2 + \cdots + m_\ell - 1$$

positive roots for $P'(x)$.

There are three possibilities:

(a) If $P(x)$ has no other roots, then $P'(x)$ has no additional positive roots.

(b) If $x = 0$ is a root of $P(x)$, then there is one more root of $P'(x)$ in $(0, p_1)$.

(c) If $P(x)$ has a largest negative root r, then $P'(x)$ will have a root in (r, p_1), which may or may not be positive.

In any case, the conclusion of the theorem is satisfied. □

Lemma 13.19. *If a_k is the first nonvanishing coefficient of polynomial of degree n with ℓ positive roots, then*

$$\operatorname{sgn} a_k = (-1)^{\ell} \operatorname{sgn} a_n.$$

Proof. For the polynomial

$$P(x) = a_n x^n + \cdots + a_{k+1} x^{k+1} + a_k x^k = x^k \,(a_n x^{n-k} + \cdots + a_{k+1} x + a_k),$$

let $p_1, \ldots p_\ell$ be its positive roots and $r_{\ell+1}, \ldots, r_{n-k}$ the negative ones. Then

$$P(x) = a_n \, x^k \,(x - p_1) \cdots (x - p_\ell)(x - r_{\ell+1}) \cdots (x - r_{n-k}).$$

From this equation we find

$$a_k = a_n (-1)^{\ell} p_1 \cdots p_\ell \,(-r_{\ell+1}) \cdots (-r_{n-k}),$$

from which the advertised relation follows. □

Theorem 13.20 (Descartes's law of signs). *If all the coefficients and roots of the polynomial $P(x)$ are real, then the number of positive roots (multiplicities taken into account) is equal to the number of sign changes in the sequence of coefficients of $P(x)$.*

Proof. Without loss of generality, we assume that the coefficient a_n in

$$P(x) = a_n x^n + \cdots + a_0$$

is positive. Then we notice that if all coefficients up to and including a_{k-1} vanish,

$$P(x) = a_n x^n + \cdots + a_k x^k,$$

we can write the polynomial in the form

$$P(x) = x^k \,(a_n x^{n-k} + \cdots + a_{k+1} x + a_k).$$

The factor x^k contributes neither positive nor negative roots. Therefore $P(x)$ has the same number of positive roots as

$$Q(x) = a_n x^{n-k} + \cdots + a_{k+1} x + a_k$$

and the same sign changes in the sequence of coefficients. However, $Q(x)$ has a nonvanishing constant term. Therefore, without loss of generality, we can assume a nonvanishing constant term $a_0 \neq 0$ for the polynomial $P(x)$.

Given $a_0 \neq 0$ and $a_n > 0$, we will prove the theorem by induction.

Base of induction. For a polynomial of degree 1, $P(x) = a_1x + a_0$, $a_0 \neq 0$, $a_1 > 0$, there is a real root $-a_0/a_1$, which is positive if a_0 is negative. Therefore, there is one sign change in the coefficients of $P(x)$ and the theorem is true.

Induction step. Assume that the assertion is true for any polynomial of degree $n-1$ with a nonvanishing constant term and a positive coefficient for the $(n-1)$-th term. We must show it is true for a polynomial $P(x)$ of degree n with a nonvanishing constant term and a positive coefficient for the n-th term. Consider the derivative of $P(x)$:

$$P'(x) = na_nx^{n-1} + (n-1)a_{n-1}x^{n-2} + \cdots + a_1.$$

Again, without loss of generality, we can assume that $a_1 \neq 0$. $P'(x)$ is of degree $n-1$ and the theorem applies. That is, the number of sign changes in its coefficients equals the number ℓ of its positive roots.

We have seen that $P(x)$ has either the same number of positive roots or one more.

Case 1: P and P' have the same number of positive roots. In this case, the first nonvanishing coefficient of each polynomial has sign $(-1)^\ell$, where ℓ is the number of positive roots. For $P(x)$ this coefficient is a_0 and for $P'(x)$ it is a_1. Looking at the sequences of coefficients for $P(x)$ and $P'(x)$,

$$\begin{array}{cccccc} a_n, & a_{n-1}, & \ldots, & a_2, & a_1, & a_0, \\ na_n, & (n-1)a_{n-1}, & \ldots, & 2a_2, & a_1, & \end{array}$$

we see that the number of sign changes in the coefficients of $P(x)$ is the same with the number of sign changes in the coefficients of $P'(x)$. Therefore, the theorem is valid for a polynomial of degree n in this case.

Case 2: P' has one less positive root than P. In this case, the first nonvanishing coefficient of $P'(x)$, a_1, has sign $(-1)^\ell$, and a_0 has sign $(-1)^{\ell+1}$. Looking at the sequences of coefficients for $P(x)$ and $P'(x)$,

$$\begin{array}{cccccc} a_n, & a_{n-1}, & \ldots, & a_2, & a_1, & a_0, \\ na_n, & (n-1)a_{n-1}, & \ldots, & 2a_2, & a_1, & \end{array}$$

we see that in the first sequence there is an additional change of sign between a_1 and a_0. Therefore, $P(x)$ has $\ell+1$ positive roots and the same number of sign changes in the sequence of its coefficients. This concludes the proof. □

Comment. If $P(x)$ also has complex roots, since they will appear in conjugate pairs, the number of positive roots will be equal to the number of sign changes minus an even number. However, this statement is not automatic. A proof is required, which you can provide easily. Finally, notice that if we do not know anything about the roots of the polynomial, then Descartes's law of signs should be stated as follows: *If the coefficients of a polynomial $P(x)$ are real, the number of its positive roots is equal to the number of sign changes in the sequence of coefficients of $P(x)$ or less than this number by an even integer.* □

Problem 13.21 ([62]). *Assuming that the continued radical*

$$\sqrt{1+\sqrt{7+\sqrt{1+\sqrt{7+\cdots}}}}$$

converges, find its value.

Solution. Let

$$\ell = \sqrt{1+\sqrt{7+\sqrt{1+\sqrt{7+\cdots}}}}.$$

Then we notice that

$$\ell = \sqrt{1+\sqrt{7+\ell}},$$

or

$$\ell^4 - 2\ell^2 - \ell - 6 = 0.$$

The polynomial $P(x) = x^4 - 2x^2 - x - 6$ has real coefficients and the number of sign changes in the sequence of its coefficients is 1. Therefore, $P(x)$ has only one positive root. It is easily seen that $P(2) = 0$ and thus $\ell = 2$. □

Theorem 13.22 (Budan). *If all the coefficients and roots of the polynomial $P(x)$ are real, then the number of positive roots with multiplicity in the interval (a, b) is equal to the number of sign changes in the sequence of coefficients of $P(x + a)$ minus the number of sign changes in the sequence of coefficients of $P(x + b)$.*

Proof. For the polynomial $Q(x) = P(x + a)$, the number of positive roots N_a will be given by the number of sign changes in the sequence of coefficients of $Q(x)$. But these are exactly the roots of $P(x)$ which are greater than a. Similarly, the number of roots N_b of $P(x)$ which are greater than b will be given by the number of sign changes in the sequence of coefficients of $P(x+b)$. The difference $N_a - N_b$ is exactly the number of positive roots of $P(x)$ in the interval (a, b). □

Comment. If $P(x)$ also has complex roots, the number of positive roots in the interval (a, b) will be equal to the above difference minus an even number. □

So far, despite all the stated theorems, we have not presented any that answers the questions:

1. Given a polynomial $P(x)$, does it have a real root?
2. How many real roots does it have?
3. How many real roots in the interval (a, b)?

These questions are answered by our last theorem, proved by Charles Sturm. Toward that goal, we must introduce[3] some simple notation and state some auxiliary results.

First, we deal with the possibility of multiple roots. We will show that this case can be reduced to the case of simple roots. Let

$$P(x) = (x - s_1)(x - s_2) \cdots (x - s_\ell)(x - r_1)^{m_1} (x - r_2)^{m_2} \cdots (x - r_k)^{m_k},$$

where $s_1, s_2, \ldots, s_\ell$ are simple roots of $P(x)$ and $r_1, r_2, \ldots, r_k$ roots of multiplicities $m_1, m_2, \ldots, m_k$ respectively greater than 1. As we have seen, the derivative of $P'(x)$ will be of the form

$$P'(x) = (x-a_1)(x-a_2) \cdots (x-a_{\ell-1})(x-r_1)^{m_1-1} (x-r_2)^{m_2-1} \cdots (x-r_k)^{m_k-1},$$

with $a_1, \ldots, a_{k+\ell}$ simple roots different from $s_1, \ldots, s_\ell$. Therefore,

$$G(x) = (x - r_1)^{m_1-1} (x - r_2)^{m_2-1} \cdots (x - r_k)^{m_k-1}$$

is the greatest common divisor of $P(x)$ and $P'(x)$. If we factor $P(x)$ in the form

$$P(x) = G(x)\, p(x),$$

the equation $P(x) = 0$ is equivalent to

$$G(x) = 0 \quad \text{or} \quad p(x) = 0.$$

The polynomial $p(x) = 0$ only has simple roots. $G(x)$ may have multiple roots. By repeating the procedure as many times as necessary, we can reduce the initial equation to a sequence of equations involving only simple roots.

For Sturm's theorem, we thus assume that $P(x)$ only has simple roots. Then the greatest common divisor of $P(x)$ and $P'(x)$, which we will indicate by $P_1(x)$, is a constant number $c \neq 0$. Using Euclid's algorithm for the

[3]The material on Sturm's theorem is based on Dörrie's book [18], which lists this theorem among the top 100 mathematical gems that can be understood with elementary mathematics. The fundamental theorem of algebra and Abel's result on the impossibility of solving algebraically fifth-degree equations and higher also make the list. Although I do not fully agree with all items in the list, these three results are certainly rightfully there and [18] is a great book to own and read.

polynomials $P(x)$ and $P_1(x)$, we will denote the quotients in the successive divisions by $q_i(x)$ and the corresponding remainders by $-P_{i+1}(x)$:

$$\begin{aligned}
P(x) &= P_1(x)\, q_1(x) - P_2(x), && \deg P_2 < \deg P_1 = n-1,\\
P_1(x) &= P_2(x)\, q_2(x) - P_3(x), && \deg P_3 < \deg P_2,\\
P_2(x) &= P_3(x)\, q_3(x) - P_4(x), && \deg P_4 < \deg P_3,\\
&\dots\\
P_{s-1}(x) &= P_s(x)\, q_s(x), && \deg P_s = 0.
\end{aligned}$$

The algorithm terminates at some step, say the s-th step, and $P_s(x) = c$ is the greatest common divisor of $P(x)$ and $P'(x)$. In this way we construct a sequence of polynomials with decreasing degrees,

$$P(x), P_1(x), P_2(x), P_3(x), \dots, P_s(x).$$

This sequence is called the **Sturm chain** for the polynomial $P(x)$. The following three lemmas discuss the properties of the Sturm chain.

Lemma 13.23. *Two successive polynomials in the Sturm chain do not vanish simultaneously at any point.*

Proof. If, for some x, $P_i(x) = P_{i+1}(x) = 0$, then from the i-th equation of Euclid's algorithm we see that $P_{i+2}(x) = 0$, which then implies $P_{i+3}(x) = 0$ and so on until the last equation, $P_s(x) = c = 0$. But this is impossible. □

Lemma 13.24. *If r is a root of $P_i(x)$, then $P_{i-1}(r)$ and $P_{i+1}(r)$ have opposite signs.*

Proof. Using the $(i-1)$-th equation, $P_{i-1}(x) = P_i(x)\, q_i(x) - P_{i+1}(x)$, we see immediately that, at $x = r$, $P_{i-1}(r) = -P_{i+1}(r)$. □

Lemma 13.25. *If r is a root of $P(x)$, then there is a neighborhood of r such that $P_1(x)$ maintains the same sign for all points in this neighborhood.*

Proof. If r is a root of $P(x)$, then $P_1(r) \neq 0$. Since $P_1(x)$ is continuous, there is a neighborhood in which it is positive or negative. □

Now, we select a point x such that it is not a root for any of the polynomials in the Sturm chain. Then we create the sequence of signs

$$\operatorname{sgn} P(x),\ \operatorname{sgn} P_1(x),\ \operatorname{sgn} P_2(x),\ \operatorname{sgn} P_3(x),\ \dots,\ \operatorname{sgn} P_s(x),$$

which is known as a **Sturm sign chain**, and let $\sigma(x)$ stand for the number of sign changes[4] in a Sturm sign chain.

[4]Notice that sign changes are the same in the original Sturm chain or in a chain that is formed by the original chain but with the polynomials multiplied by arbitrary positive numbers. As a result, we can always avoid fractional coefficients in the calculations.

Theorem 13.26 (Sturm). *The number of real roots of $P(x)$ in the interval (a, b) is equal to the difference $\sigma(a) - \sigma(b)$.*

Proof. Changes in the value of the function $\sigma(x)$ can happen when one of the polynomials of the chain changes sign. But this means that x passes through a zero of one of the polynomials. Say that x_0 is a zero of $P_i(x)$. Then there is a neighborhood such that

1. $P_i(x)$ does not vanish except at the point x_0 and
2. the functions $P_{i-1}(x)$ and $P_{i+1}(x)$ maintain their sign if $i \neq 0$, or just the function $P_1(x)$ maintains its sign if $i = 0$.

Let's consider the two points $a = x_0 - \epsilon$ and $b = x_0 + \epsilon$, with $\epsilon > 0$, belonging to this neighborhood and look at the signs of the polynomials under discussion.

Case $(i \neq 0)$. P_i changes sign as it goes through the point x_0. Therefore, $P_i(a)$ and $P_i(b)$ have opposite signs. At the same time, $P_{i-1}(a)$ and $P_{i+1}(b)$ have the same sign but exactly opposite of the sign of $P_{i+1}(a)$ and $P_{i-1}(b)$. Consequently, each of the sign chains

$$\begin{gathered}\operatorname{sgn} P_{i-1}(a), \ \operatorname{sgn} P_i(a), \ \operatorname{sgn} P_{i+1}(a), \\ \operatorname{sgn} P_{i-1}(b), \ \operatorname{sgn} P_i(b), \ \operatorname{sgn} P_{i+1}(b)\end{gathered}$$

has only one sign change and the net contribution to $\sigma(a) - \sigma(b)$ is zero.

Case $(i = 0)$. Again, P_0 changes sign as it goes through the point x_0. Therefore $P_0(a)$ and $P_0(b)$ have opposite signs. At the same time, $P_1(a)$ and $P_1(b)$ have the same sign. Consequently, among the two pairs

$$\operatorname{sgn} P_0(a), \ \operatorname{sgn} P_1(a), \quad \operatorname{sgn} P_0(b), \ \operatorname{sgn} P_1(b),$$

one has identical signs and one has opposite signs. In fact, it is always the first pair that has the opposite signs: when $P_1 > 0$, P must be strictly increasing and thus $P(a) < 0$; when $P_1 > 0$, P must be strictly decreasing and thus $P(a) > 0$. The net contribution to $\sigma(a) - \sigma(b)$ is one.

Therefore, $\sigma(x)$ changes values only when x goes through zeroes of $P(x)$. In particular, it changes by one every time it passes through one such zero. From this, the statement of the theorem follows. □

Example 13.27. For the polynomial $P(x) = x^5 - 3x - 1$, the corresponding Sturm chain is

$$P(x) = x^5 - 3x - 1, \quad P_1(x) = 5x^4 - 3, \quad P_2(x) = 12x + 5, \quad P_3(x) = 1.$$

We will consider the following Sturm sign chains.

1. At $x = -M$, $M \to +\infty$: $\qquad -1, \ +1, \ -1, \ +1.$

2. At $x = -2$: $\quad -1, \; +1, \; -1, \; +1.$
3. At $x = -1$: $\quad +1, \; +1, \; -1, \; +1.$
4. At $x = 0$: $\quad -1, \; -1, \; +1, \; +1.$
5. At $x = 1$: $\quad -1, \; +1, \; +1, \; +1.$
6. At $x = 2$: $\quad +1, \; +1, \; +1, \; +1.$
7. At $x = M$, $M \to +\infty$: $\quad +1, \; +1, \; +1, \; +1.$

The polynomial $P(x)$ has only three real roots since $\sigma(-\infty) - \sigma(+\infty) = 3$. In fact, there is one root in each of the intervals $(-2,-1)$, $(-1,0)$, $(1,2)$ since $\sigma(a) - \sigma(b) = 1$ for any of them. $\square$

13.4 Solved Problems

Polynomials can be useful in finding the values of finite sums and products, either by evaluation or by the use of Viète's formulæ. This idea is demonstrated in the next problem.

Problem 13.28. *Prove that*

$$\prod_{k=1}^{n-1} \sin\frac{k\pi}{n} = \frac{n}{2^{n-1}} \quad \textit{for any } n \geq 2.$$

Solution. We start with the polynomial

$$P(x) = x^n - 1,$$

having roots

$$\omega_k = e^{2k\pi i/n} \quad \text{for } k = 0, 1, \ldots, n-1.$$

Since

$$P(x) = (x-1)(x^{n-1} + x^{n-2} + \cdots + x + 1),$$

the polynomial

$$Q(x) = x^{n-1} + x^{n-2} + \cdots + x + 1$$

has roots ω_k, $k = 1, \ldots, n-1$. That is,

$$Q(x) = (x-\omega_1)(x-\omega_2)\ldots(x-\omega_{n-1}).$$

Then we notice that

$$\begin{aligned} 1 - \omega_k &= 1 - \cos\frac{2\pi k}{n} - i\sin\frac{2\pi k}{n} \\ &= 2\sin^2\frac{\pi k}{n} - i2\sin\frac{\pi k}{n}\cos\frac{\pi k}{n} = -2ie^{k\pi i/n}\sin\frac{\pi k}{n}. \end{aligned}$$

Setting $x = 1$ in $Q(x)$, we find

$$\prod_{k=1}^{n-1}(1-\omega_k) = n,$$

which is exactly the identity sought. (Since the roots appear in conjugate pairs, only the 2's and the sine parts of the roots survive. Alternatively, take the magnitude of the two sides.) $\square$

Incidentally, from the previous problem, since $P(\omega_k) = 0$, we also see that, if ω is any of the n-th roots of 1 different from $\omega_0 = 1$,

$$1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0,$$

another well-known identity.

Problem 13.29. *Find all polynomials $P(x)$ such that*

$$x\,P(x-1) = (x-3)\,P(x) \quad \textit{for all } x.$$

Solution. Substituting $x = 0, 1, 2$ successively in the given equation, we find that these values are roots of $P(x)$. For example, $0\,P(-1) = -3\,P(0)$ implies $P(0) = 0$, after which $1\,P(0) = -2\,P(1)$ implies $P(1) = 0$, and $2\,P(1) = -1\,P(2)$ implies $P(2) = 0$. Let's set

$$P(x) = x(x-1)(x-2)\,Q(x),$$

where $Q(x)$ is some polynomial. Substituting in the given equation we get

$$x(x-1)(x-2)(x-3)Q(x-1) = x(x-1)(x-2)(x-3)Q(x),$$

or

$$Q(x-1) = Q(x) \quad \text{for all } x.$$

This equation implies that $Q(x)$ is a constant, say c; thus

$$P(x) = c\,x(x-1)(x-2). \qquad \square$$

Problem 13.30 (USA 1975). *If $P(x)$ denotes a polynomial of degree n such that*

$$P(k) = \frac{k}{k+1} \quad \textit{for } k = 0, 1, \ldots, n, \tag{13.2}$$

determine $P(n+1)$.

Solution. The given conditions may be written as

$$(k+1)\,P(k) - k = 0 \quad \text{for } k = 0, 1, \ldots, n.$$

This implies that the polynomial

$$Q(x) = (x+1)\,P(x) - x$$

of degree $n+1$ has $0, 1, \ldots, n$ as its $n+1$ roots. Therefore:

$$Q(x) = a\,x(x-1)(x-2)\ldots(x-n).$$

To find a, notice that the definition of $Q(x)$ in terms of $P(x)$ implies that $Q(-1) = 1$, while the previous equation gives

$$Q(-1) = a\,(-1)^{n+1}(n+1)!\,.$$

We thus conclude that

$$a = \frac{(-1)^{n+1}}{(n+1)!}$$

and

$$P(x) = \left(\frac{(-1)^{n+1}}{(n+1)!}\,(x-1)(x-2)\ldots(x-n) + 1\right)\frac{x}{x+1}.$$

Then

$$P(n+1) = \left(\frac{(-1)^{n+1}}{n+1} + 1\right)\frac{n+1}{n+2},$$

or

$$P(n+1) = \begin{cases} \dfrac{n}{n+2} & \text{if } n \text{ is even,} \\ 1 & \text{if } n \text{ is odd.} \end{cases}$$

□

Problem 13.31. *Find all polynomials $P(x)$ such that*

$$P(F(x)) = F(P(x)) \quad \textit{and} \quad P(0) = 0,$$

where F is a function defined on $\mathbb{R}$ that satisfies

$$F(x) > x \quad \textit{for all } x \geq 0.$$

Solution. Define $a_0 = F(0)$. Setting $x = 0$ in the functional equation gives

$$P(a_0) = a_0.$$

Starting with a_0, we now define the sequence

$$a_{n+1} = F(a_n),$$

which is strictly increasing, since $F(a_n) > a_n$. In the defining functional equation, we set $x = a_0$. Then we get $P(F(a_0)) = F(P(a_0))$, which implies $P(a_1) = F(a_0)$, that is,

$$P(a_1) = a_1.$$

We thus suspect that

$$P(a_n) = a_n \quad \text{for all } n. \tag{13.3}$$

Suppose this is true for $n = k$: $P(a_k) = a_k$. Then we set $x = a_k$ in the defining functional equation to find $P(F(a_k)) = F(P(a_k))$, hence $P(a_{k+1}) = F(a_k)$, hence $P(a_{k+1}) = a_{k+1}$. That is, it is also true for $k+1$. By induction, (13.3) is true.

We have thus shown that the polynomial $Q(x) = P(x) - x$ has an infinite number of distinct roots: a_n, with $n = 0, 1, 2, \ldots$. Therefore, it must be the zero polynomial. This implies that

$$P(x) = x. \qquad \square$$

Problem 13.32. *Find all polynomials $P(x)$ for which there is a polynomial in two variables $Q(u, v)$ such that*

$$P(xy) = Q(x, P(y)).$$

Solution 1. Any polynomial of degree zero $P(x) = a_0$ satisfies the conditions of the problem. We will thus seek polynomials of degree $n \geq 1$.

Let $Q(u, v)$ be a polynomial of degree n_1 with respect to u and degree n_2 with respect to v. Then $Q(x, P(y))$ is of degree n_1 with respect to x and of degree nn_2 with respect to y. Therefore, the given equation requires that $n = n_1$ and $n = nn_2$ or $n_1 = n$ and $n_2 = 1$. We can write the polynomial $Q(u, v)$ in the form

$$Q(u, v) = q_1(u) + q_2(u)\, v,$$

where the polynomials $q_1(u), q_2(u)$ are of degree n. The given functional equation then takes the form of a Vincze II functional equation (8.19):

$$P(xy) = q_1(x) + q_2(x)\, P(y),$$

with solution

$$P(x) = ax^n + b, \quad Q(u, v) = b\,(1 - u^n) + u^n\, v. \qquad \square$$

If we do not want to use the result of the Vincze II equation, we can proceed as follows:

Solution 2. Let

$$P(x) = \sum_{k=0}^{n} a_k x^k,$$

$$q_1(u) = \sum_{k=0}^{n} b_k u^k,$$

$$q_2(u) = \sum_{k=0}^{n} c_k u^k.$$

Then the given equation translates to

$$\sum_{k=0}^{n} a_k x^k y^k = \sum_{k=0}^{n} b_k x^k + \sum_{k=0}^{n} \sum_{\ell=0}^{n} c_k a_\ell x^k y^\ell,$$

or

$$a_k y^k = b_k + c_k \sum_{\ell=0}^{n} a_\ell y^\ell \quad \text{for } k = 0, 1, \ldots, n. \tag{13.4}$$

The last equation, for $k = n$, gives

$$a_n y^n = b_n + c_n \sum_{\ell=0}^{n} a_\ell y^\ell,$$

and therefore

$$\begin{aligned} a_n &= c_n\, a_n, \\ 0 &= c_n\, a_{n-1} = c_n\, a_{n-2} = \cdots = c_n\, a_1, \\ 0 &= b_n + c_n\, a_0. \end{aligned}$$

The equation $a_n(c_n - 1) = 0$ requires that either $a_n = 0$ or $c_n = 1$. If $a_n = 0$, then $P(x)$ is of degree $n - 1$. Repeating the same argument, we conclude that either $a_{n-1} = 0$ or $c_{n-1} = 1$. And this continues until we reach a constant polynomial. We thus conclude that, if we wish to find nonconstant polynomials, we must have $c_n = 1$ and $a_n \neq 0$. Then

$$0 = a_{n-1} = a_{n-2} = \cdots = a_1,$$

and

$$b_n = -a_0.$$

Returning to equation (13.4),

$$a_k y^k = b_k + c_k \left(a_0 + a_n\, y^n\right) \quad \text{for } k = 0, 1, \ldots, n, \tag{13.5}$$

we see that

$$0 = c_{n-1} = c_{n-2} = \cdots = c_1 = c_0,$$

and

$$b_0 = a_0, \quad b_1 = b_2 = \cdots = b_{n-1} = 0.$$

All this gives, again, the result that $P(x) = a_n x^n + a_0$ and $q_1(u) = a_0(1 - u^n)$ and $q_2(u) = u^n$. □

Problem 13.33. *Find all polynomials $P(x)$ for which there exists a polynomial of two variables $Q(u, v)$ such that*

$$P(x + y) = Q(x, P(y)).$$

Solution 1. Any polynomial of degree zero $P(x) = a_0$ satisfies the conditions of the problem. We will thus seek polynomials of degree $n \geq 1$.

Let $Q(u, v)$ be a polynomial of degree n_1 with respect to u and degree n_2 with respect to v. Then $Q(x, P(y))$ is of degree n_1 with respect to x and of degree nn_2 with respect to y. Therefore, the given equation requires that $n = n_1$ and $n = nn_2$ or $n_1 = n$ and $n_2 = 1$. We can write the polynomial $Q(u, v)$ in the form

$$Q(u, v) = q_1(u) + q_2(u)\, v,$$

where the polynomials $q_1(u), q_2(u)$ are of degree n. The given functional equation then takes the form of a Vincze I functional equation (8.13):

$$P(x + y) = q_1(x) + q_2(x)\, P(y),$$

with solution

$$P(x) = ax + b, \quad Q(u, v) = a\, u + v. \qquad □$$

As with the previous problem, we could finish the solution of this problem without any reference to the Vincze II equation:

Solution 2. Let

$$P(x) = \sum_{k=0}^{n} a_k x^k,$$

$$q_1(u) = \sum_{k=0}^{n} b_k u^k,$$

$$q_2(u) = \sum_{k=0}^{n} c_k u^k.$$

Then the given equation translates to

$$\sum_{k=0}^{n} a_k\, (x + y)^k = \sum_{k=0}^{n} b_k\, x^k + \sum_{k=0}^{n} \sum_{\ell=0}^{n} c_k a_\ell\, x^k y^\ell. \tag{13.6}$$

The right-hand side has all terms $x^k y^\ell$ but the left-hand side has only terms $x^k y^\ell$ with $k+\ell \le n$. Therefore, the coefficients of the terms $x^k y^\ell$ with $k+\ell > n$ on the right-hand side must vanish:

$$\begin{aligned} c_n\, a_\ell &= 0 \quad \text{for } \ell = 1, 2, \ldots, n, \\ c_{n-1}\, a_\ell &= 0 \quad \text{for } \ell = 2, 3, \ldots, n, \\ &\ldots \\ c_1\, a_\ell &= 0 \quad \text{for } \ell = n. \end{aligned}$$

The last equation, $c_1 a_n = 0$, requires either $c_1 = 0$ or $a_n = 0$. If $a_n = 0$, then $P(x)$ is of degree $n-1$. Restarting for the new polynomial, we would find $c_1 a_{n-1} = 0$ and either $a_{n-1} = 0$ or $c_1 = 0$. Assuming that $a_{n-1} = 0$, $P(x)$ reduces to a polynomial of degree $n-2$. So, in order not to reduce $P(x)$ to a constant polynomial, we must assume $c_1 = 0$ at some stage. So, without loss of generality, we assume that that's the case in the equations written above and formally treat $a_n \neq 0$ in these equations. Therefore,

$$c_1 = c_2 = \cdots = c_n = 0.$$

Equation (13.6) now becomes

$$a_0 + a_1(x+y) + \sum_{k=2}^{n} a_k\,(x+y)^k = \sum_{k=0}^{n} b_k\, x^k + c_0 \sum_{\ell=0}^{n} a_\ell\, y^\ell. \tag{13.7}$$

Only the left-hand side has mixed terms $x^k y^\ell$. Therefore, it must be

$$a_2 = a_3 = \cdots = a_n = 0.$$

This leaves all terms x^k with $k \ge 2$ unbalanced. Consequently,

$$b_2 = b_3 = \cdots = b_n = 0.$$

Finally, equating the coefficients of the remaining terms in (13.7),

$$\begin{aligned} a_0 &= b_0 + c_0\, a_0, \\ a_1 &= b_1, \\ a_1 &= c_0 a_1. \end{aligned}$$

From the third equation, $a_1(1 - c_0) = 0$. To have a nonconstant polynomial, a_1 must be nonzero. Then $c_0 = 1$. The first equation then requires $b_0 = 0$. Therefore we have arrived at the result

$$P(x) = a_1\, x + a_0, \quad q_1(u) = a_1\, x, \quad q_2(u) = 1,$$

as in the previous solution. □

Chapter 14

Conditional Functional Equations

14.1 The Notion of Conditional Equations

Consider this modified problem involving Jensen's functional equation, where the domain is the interval $[0, 1]$ instead of $\mathbb{R}$:

Problem 14.1. *Find the continuous solutions $f : [0, 1] \to \mathbb{R}$ of the functional equation*

$$f\left(\frac{x+y}{2}\right) = \frac{f(x)+f(y)}{2} \quad \textit{for all } x, y \in [0, 1].$$

As we know, a different domain might change the set of solutions. However, one might still want to proceed in the same way we solved the original problem (see page 100). For $y = 0$, the equation gives

$$f\left(\frac{x}{2}\right) = \frac{f(x)+f(0)}{2} = \frac{f(x)+b}{2},$$

where $b = f(0)$. In this relation we substitute $x = y + z$:

$$f\left(\frac{y+z}{2}\right) = \frac{f(y+z)+b}{2}.$$

However, the left-hand side can be rewritten from the defining relation and thus

$$f(y+z) = f(y) + f(z) - b,$$

which can be rewritten in the form

$$g(x+y) = g(x) + g(y)$$

if we define $g(x) = f(x) - b$. The last equation looks deceptively similar to the linear Cauchy equation (5.1) with $x, y \in [0,1]$. However, it is not exactly the same. The equation is not valid for all $x, y \in [0,1]$; it is valid for all $x, y \in [0,1]$ whose sum $x + y$ also belongs in [0,1]. We cannot apply the equation, for example, to $x = y = 1$. Such a functional equation is called a **conditional** (or **restricted**) functional equation.

Quite interestingly, conditional functional equations are not encountered frequently in mathematical competitions (for a recent problem given in IMO 2008, see Problem 20.152). This does not take into account the fact that certain conditions given for functions may be interpreted as conditional functional equations. For example, the condition (13.2) on the polynomial $P(x)$,

$$P(x) = \frac{x}{x+1} \quad \text{for all } x \in \{0, 1, \ldots, n\},$$

may be thought of as a conditional functional equation since it is valid only for a subset of the domain. This point of view makes the differentiation of equations into regular and conditional a little vague. I personally prefer to avoid using this terminology but the reader must be aware of it since mathematicians have studied many equations that have been labeled as conditional.

14.2 An Example

In this section, we present just a single example of a conditional equation. We will assume that the interested reader will search the literature for additional information and problems.

Problem 14.2 ([7], Problem 81-1, January 1981). *Let $f : \mathbb{R}^k \to \mathbb{R}$ be a continuous function satisfying the relation*[1]

$$f(\vec{u} + \vec{v}) = f(\vec{u}) + f(\vec{v}) \quad \textit{whenever } \vec{u} \perp \vec{v}.$$

Show that there exist a constant c and a vector $\vec{a} \in \mathbb{R}^k$ such that

$$f(\vec{u}) = c\,\|\vec{u}\|^2 + \vec{a} \cdot \vec{u},$$

where $\|\vec{u}\|$ is the magnitude of $\vec{u}$.

Solution. The solution given below follows M. St. Vincent [66].

Let

$$f(\vec{u}) = \frac{f(\vec{u}) + f(-\vec{u})}{2} + \frac{f(\vec{u}) - f(-\vec{u})}{2} \equiv f_{\text{even}}(\vec{u}) + f_{\text{odd}}(\vec{u}).$$

Obviously, the even and odd parts must satisfy the same equation. So we shall study them independently.

[1] The condition $\vec{u} \perp \vec{v}$ is read "$\vec{u}$ is perpendicular to $\vec{v}$" and it is equivalent to $\vec{u} \cdot \vec{v} = 0$.

Case 1: The even part. Given any $\vec{u}, \vec{v}$ with equal magnitudes $\|\vec{u}\| = \|\vec{v}\|$, we can define

$$\vec{x} = \frac{\vec{u}+\vec{v}}{2}, \quad \vec{y} = \frac{\vec{u}-\vec{v}}{2}.$$

These vectors are perpendicular, i.e. $\vec{x} \cdot \vec{y} = 0$. Also

$$f_{\text{even}}(\vec{u}) = f_{\text{even}}(\vec{x}+\vec{y}) = f_{\text{even}}(\vec{x}) + f_{\text{even}}(\vec{y}),$$
$$f_{\text{even}}(\vec{v}) = f_{\text{even}}(\vec{x}+(-\vec{y})) = f_{\text{even}}(\vec{x}) + f_{\text{even}}(\vec{y}),$$

or

$$f_{\text{even}}(\vec{u}) = f_{\text{even}}(\vec{v}).$$

The function f_{even} has equal values for vectors of equal lengths. In other words, it cannot depend on the direction of its argument:

$$f_{\text{even}}(\vec{u}) = f_{\text{even}}(\|\vec{u}\|).$$

For simplicity we shall write $\|\vec{u}\| = u$. In the given functional equation, since $\vec{u} \perp \vec{v}$, $\|\vec{u}+\vec{v}\|^2 = \|\vec{u}\|^2 + \|\vec{v}\|^2 = u^2 + v^2$. Therefore,

$$f_{\text{even}}\big(\sqrt{u^2+v^2}\,\big) = f_{\text{even}}(u) + f_{\text{even}}(v) \quad \text{for all } u, v \in \mathbb{R}_+^*.$$

We have thus found an unrestricted equation for the length of the vectors, which has been solved in Problem 5.14:

$$f_{\text{even}}(u) = c\,u^2,$$

where c is a constant.

Case 2: The odd part. First we will establish that

$$f(\lambda\vec{u}) = \lambda\, f(\vec{u}) \quad \text{for all } \lambda \in \mathbb{R}. \tag{14.1}$$

We do so via the $\mathbb{NQR}$ method. We first show that

$$f(n\vec{u}) = n\, f(\vec{u}) \quad \text{for all } n \in \mathbb{N}. \tag{14.2}$$

Since f_{odd} is an odd continuous function, $f_{\text{odd}}(\vec{0}) = 0$. For $n = 1$, the equality is trivially true. Suppose it is true for some positive integer k. Given any $\vec{u} \neq \vec{0}$, consider the vector $\vec{v}$ such that $\vec{u} \cdot \vec{v} = 0$ and having length $\|\vec{v}\| = \sqrt{k}\,\|\vec{u}\|$. This vector has the property that

$$(k\vec{u} - \vec{v}) \perp (\vec{u} + \vec{v}).$$

Therefore, from the defining equation,

$$f_{\text{odd}}(k\vec{u} - \vec{v}) + f_{\text{odd}}(\vec{u}+\vec{v}) = f_{\text{odd}}((k+1)\vec{u}).$$

However, each term on the left-hand side can be rewritten using that equation too, since the arguments are sums of perpendicular vectors. Finally,

$$(k+1)\, f_{\text{odd}}(\vec{u}) \;=\; f_{\text{odd}}((k+1)\vec{u}).$$

That is, equation (14.2) is true for $k+1$ and thus, by induction, it is valid for all positive integers. In this equation, we now replace $\vec{u}$ by $(m/n)\vec{u}$, where $m, n \in \mathbb{N}$, to find

$$f_{\text{odd}}\left(\frac{m}{n}\vec{u}\right) \;=\; \frac{m}{n}\, f_{\text{odd}}(\vec{u}).$$

Therefore, we have shown that

$$f_{\text{odd}}(q\vec{u}) \;=\; q\, f_{\text{odd}}(\vec{u}) \quad \text{for all } q \in \mathbb{Q}.$$

Using the continuity of f_{odd} and a sequence of rational numbers $\{q_n\}$ that converges to a real number λ, we finally arrive at the advertised result.

Now we shall show that

$$f_{\text{odd}}(\vec{u}+\vec{v}) \;=\; f_{\text{odd}}(\vec{u}) + f_{\text{odd}}(\vec{v}) \quad \text{for } \vec{u}, \vec{v}. \tag{14.3}$$

If $\vec{u} = \vec{v} = \vec{0}$, this equation is trivially true. So, without loss of generality, given the vectors $\vec{u}, \vec{v}$, we will assume that $\vec{u} \neq \vec{0}$ and define $\mu = \vec{u} \cdot \vec{v}/\|\vec{u}\|^2$. Then

$$\vec{u} \perp (\vec{v} - \mu\vec{u}).$$

Therefore, from the defining equation,

$$\begin{aligned}
f_{\text{odd}}(\vec{u}+\vec{v}) &= f_{\text{odd}}((1+\mu)\vec{u}) + f_{\text{odd}}(\vec{v}-\mu\vec{u}) \\
&= (1+\mu)\, f_{\text{odd}}(\vec{u}) + f_{\text{odd}}(\vec{v}-\mu\vec{u}) \\
&= f_{\text{odd}}(\vec{u}) + \mu\, f_{\text{odd}}(\vec{u}) + f_{\text{odd}}(\vec{v}-\mu\vec{u}) \\
&= f_{\text{odd}}(\vec{u}) + f_{\text{odd}}(\mu\vec{u} + \vec{v} - \mu\vec{u}) \\
&= f_{\text{odd}}(\vec{u}) + f_{\text{odd}}(\vec{v}),
\end{aligned}$$

as desired. Equation (14.3) was solved in Problem 9.1, the solution being

$$f_{\text{odd}}(\vec{u}) \;=\; \vec{a} \cdot \vec{u},$$

where $\vec{a}$ is a constant vector. □

Chapter 15

Functional Inequalities

15.1 Useful Concepts and Facts

So far, we have studied functional relations that contained only equalities between the various terms. Some inequalities did appear but they were not the central piece in any problem except Problem 2.9 on page 47. In this chapter, we turn to some theory and problems involving functions satisfying inequalities.

Definition 15.1. A function $f : \mathbb{R} \to \mathbb{R}$ or $f : I \to \mathbb{R}$, where I is an interval in $\mathbb{R}$, is called

- **superadditive** if $f(x+y) \geq f(x) + f(y)$ for all x, y;
- **subadditive** if $f(x+y) \leq f(x) + f(y)$ for all x, y;
- **weakly convex** if $f\left(\dfrac{x+y}{2}\right) \leq \dfrac{f(x)+f(y)}{2}$ for all x, y;
- **strongly convex**, or just **convex**, if

  $$f(tx + (1-t)y) \leq t\,f(x) + (1-t)\,f(y) \quad \text{for all } x, y \text{ and all } 0 \leq t \leq 1;$$

- (weakly/strongly) **concave** if $-f$ is (weakly/strongly) convex.

Example 15.2. The function $f(x) = |x|$ with $x \in \mathbb{R}$ is subadditive since $|x+y| \leq |x| + |y|$. The function $f(x) = x^2$ with $x \in \mathbb{R}_+^*$ is superadditive since $(x+y)^2 = x^2 + y^2 + 2xy \geq x^2 + y^2$. Finally, the functions $f(x) = -\ln x$, $f(x) = 1/x$ defined in $\mathbb{R}_+^*$ are convex and, therefore, $f(x) = \ln x$, $f(x) = -1/x$ are concave. $\square$

Comment. Convex and concave functions are important means in proving old and new inequalities. However, this seems to me a topic more appropriate for a book in inequalities; see [24], for example. Here I will present only a quick example.

If f is convex then, from the defining equation, one can show that

$$\frac{1}{n}\sum_{k=1}^{n} f(x_i) \geq f\left(\frac{1}{n}\sum_{k=1}^{n} x_i\right)$$

for any n and real numbers $x_1, x_2, \ldots, x_n$. This inequality (as its special case for $n = 2$) is known as **Jensen's inequality**.

For the convex function $f(x) = -\ln x$ and n positive numbers $x_1, x_2, \ldots, x_n$, Jensen's inequality gives

$$\frac{1}{n}\sum_{k=1}^{n} x_i \geq \left(\prod_{k=1}^{n} x_i\right)^{1/n},$$

which is known as **Cauchy's inequality**. □

The following statements are true:

- A convex function is weakly convex (obvious), but the converse need not be true.
- A weakly convex function that has a point of discontinity is discontinuous everywhere. Equivalently, a weakly convex function that is continuous at one point is continuous everywhere.
- Any continuous, weakly convex function is convex.
- Any convex function is continuous.

Theorem 15.3. *If $f : (0, +\infty) \to \mathbb{R}$ is differentiable, $f(x)/x$ is decreasing if and only if $f'(x) \leq f(x)/x$ on $(0, +\infty)$, and $f(x)/x$ is increasing if and only if $f'(x) \geq f(x)/x$ on $(0, +\infty)$.*

Proof. Since $x > 0$, we have the chain of equivalences

$$f'(x) \leq \frac{f(x)}{x} \iff \frac{xf'(x) - f(x)}{x^2} \leq 0 \iff \frac{d}{dx}\left(\frac{f(x)}{x}\right) \leq 0;$$

but the last inequality means that $f(x)/x$ is decreasing. The opposite case is similar. □

Theorem 15.4. *Let $f : (0, +\infty) \to \mathbb{R}$ be a function. If $f(x)/x$ is decreasing, then f is subadditive. If $f(x)/x$ is increasing, then f is superadditive.*

Proof. If $f(x)/x$ is decreasing, then

$$f(x+y) = x\,\frac{f(x+y)}{x+y} + y\,\frac{f(x+y)}{x+y} \leq x\,\frac{f(x)}{x} + y\,\frac{f(y)}{y} = f(x) + f(y);$$

that is, $f(x)$ is subadditive. The opposite case is similar. □

Combining the previous two theorems gives a sufficient condition for subadditivity (superadditivity):

Theorem 15.5. *Let $f : (0, +\infty) \to \mathbb{R}$ be differentiable. If $f'(x) \leq f(x)/x$ in $(0, +\infty)$, then f is subadditive. If $f'(x) \geq f(x)/x$ in $(0, +\infty)$, then f is superadditive.*

We say that f is a **Lipschitz function at** x_0, or that it satisfies a **Lipschitz condition of order** α **at** x_0, if there exist a constant $M > 0$ (that may depend on x_0) and an interval $(x_0 - s, x_0 + s)$ such that

$$|f(x) - f(x_0)| \leq M\,|x - x_0|^\alpha \quad \text{for all } x \in (x_0 - s, x_0 + s).$$

Theorem 15.6. *Let $\alpha > 0$. If f satisfies the Lipschitz condition of order α at x_0, then f is continuous at x_0 (but not necessarily differentiable).*

Proof. Let $\varepsilon > 0$ be arbitrary, and choose $\delta(\varepsilon) = (\varepsilon/M)^{1/\alpha} > 0$. Without loss of generality, we shall take $\varepsilon < s^\alpha M$, where s is the radius of the neighborhood of the point satisfying the Lipschitz condition. This implies that $\delta(\varepsilon) < s$ and all points x satisfying $|x - x_0| < \delta$ are points in $(x_0 - s, x_0 + s)$. For these points, the Lipschitz condition gives

$$|f(x) - f(x_0)| \leq M\,|x - x_0|^\alpha \leq M\,\delta^\alpha = \varepsilon,$$

that is, $\lim_{x\to x_0} f(x) = f(x_0)$. □

Theorem 15.7. *Let $\alpha > 1$. If f satisfies the Lipschitz condition of order α at x_0, then f is differentiable at x_0, and $f'(x_0) = 0$.*

Proof. We know that f is continuous at x_0 by the previous theorem. We must show that the limit

$$\lim_{x\to x_0} g(x) = \frac{f(x) - f(x_0)}{x - x_0}$$

exists and equals 0. But

$$|g(x)| = \frac{|f(x) - f(x_0)|}{|x - x_0|} \leq M\,|x - x_0|^{\alpha - 1}$$

by the Lipschitz condition. But $\alpha - 1 > 0$; therefore, $|g(x)|$ does tend to 0 as x approaches x_0, and so must $g(x)$. □

15.2 Solved Problems

Problem 15.8. *We say that f satisfies the* ***uniform Lipschitz condition of order*** *α on the interval I if there exists a constant $M > 0$ such that*

$$|f(x) - f(y)| \leq M\,|x - y|^{\alpha}$$

for all $x, y \in I$. Show that, if $\alpha > 1$, then f is constant on I.

Solution. Using Theorem 15.7, we conclude that f is differentiable on I and $f'(y) = 0$ for all $y \in I$; that is, f is constant. □

Problem 15.9 (China 1983; Hungary 1987). *A function f defined on the interval $[0, 1]$ satisfies $f(0) = f(1)$ and*

$$|f(x_2) - f(x_1)| < |x_2 - x_1| \quad \textit{for } x_1 \neq x_2.$$

Prove that

$$|f(x_2) - f(x_1)| < \tfrac{1}{2} \quad \textit{for all } x_1, x_2.$$

Solution. There are two possibilities: either $|x_2 - x_1| \leq \frac{1}{2}$ or $|x_2 - x_1| > \frac{1}{2}$. If $|x_2 - x_1| \leq \frac{1}{2}$, then, from the given inequality,

$$|f(x_2) - f(x_1)| < |x_2 - x_1| \leq \tfrac{1}{2}.$$

If $|x_2 - x_1| > \frac{1}{2}$, with $x_2 > x_1$ (say), then $x_2 - x_1 > \frac{1}{2}$ and

$$\begin{aligned} |f(x_2) - f(x_1)| &= |f(x_2) - f(1) + f(0) - f(x_1)| \\ &\leq |f(x_2) - f(1)| + |f(0) - f(x_1)| \\ &< |x_2 - 1| + |0 - x_1| \\ &= 1 - x_2 + x_1 \ = \ 1 - (x_2 - x_1) \\ &< 1 - \tfrac{1}{2} = \tfrac{1}{2}. \end{aligned}$$

□

Problem 15.10 (IMO 1972). *Let f and g be real-valued functions defined for all real values of x and y, and satisfying the equation*

$$f(x + y) + f(x - y) \ = \ 2f(x)g(y)$$

for all x, y. Prove that if $f(x)$ is not identically zero, and if $|f(x)| \leq 1$ for all x, then $|g(y)| \leq 1$ for all y.

Solution. Assume, as a contradiction, that $|g(y_0)| = \lambda > 1$ for some y_0. Since $f(x)$ is not identically zero, there is also an x_0 such that $f(x_0) \neq 0$. Now consider x_1 to be that point of the set $\{x_0 + y_0, x_0 - y_0\}$ that gives the greater absolute value for $f(x)$. From the defining equation and the properties of the absolute value,

$$\begin{aligned}2|f(x_1)| &\geq |f(x_0+y_0)| + |f(x_0-y_0)| \geq |f(x_0+y_0) + f(x_0-y_0)| \\ &= 2\,|f(x_0)|\,|g(y_0)|\end{aligned}$$

or

$$|f(x_1)| \geq \lambda\,\mu$$

where we set $|f(x_0)| = \mu < 1$.

Now let x_2 be that point in the set $\{x_1 + y_0, x_1 - y_0\}$ that gives the greater absolute value for $f(x)$. As before,

$$\begin{aligned}2|f(x_2)| &\geq |f(x_1+y_0)| + |f(x_1-y_0)| \geq |f(x_1+y_0) + f(x_1-y_0)| \\ &= 2\,|f(x_1)|\,|g(y_0)|\end{aligned}$$

or

$$|f(x_2)| \geq \lambda^2\,\mu.$$

Continuing in the same fashion, one can define a sequence of points x_n such that

$$|f(x_n)| \geq \lambda^n\,\mu.$$

Since $\lambda > 1$, there exists n_0 such that $\lambda^{n_0}\,\mu > 1$, and we have the desired contradiction. □

Problem 15.11 (USA 2000). *Call a real-valued function f* ***very convex*** *if*

$$\frac{f(x)+f(y)}{2} \geq f\left(\frac{x+y}{2}\right) + |x-y|$$

for all real numbers x and y. Prove that no very convex function exists.

Solution. Assume, as a contradiction, that such an f exists, and take two numbers $y \geq x$. Divide the interval (x, y) into three equal pieces of size $\varepsilon = (y-x)/3$, by setting

$$x_0 = x, \quad x_1 = x + \varepsilon, \quad x_2 = x + 2\varepsilon, \quad x_3 = x + 3\varepsilon = y.$$

Apply the given functional inequality to the pairs of points (x_0, x_2) and (x_1, x_3) to obtain

$$\frac{f(x_0)+f(x_2)}{2} \geq f(x_1) + 2\varepsilon \quad \text{and} \quad \frac{f(x_1)+f(x_3)}{2} \geq f(x_2) + 2\varepsilon.$$

Adding together these inequalities, we find that

$$\frac{f(x_0)+f(x_3)}{2} \geq \frac{f(x_1)+f(x_2)}{2} + 4\varepsilon.$$

It then follows that

$$\frac{f(x_0)+f(x_3)}{2} \geq f\left(\frac{x_1+x_2}{2}\right) + 5\varepsilon$$

or

$$\frac{f(x)+f(y)}{2} \geq f\left(\frac{x+y}{2}\right) + \frac{5}{3}(y-x).$$

Therefore, if the function f satisfies the inequality given with $|x-y|$ on the right-hand side, it will also satisfy it with $\frac{5}{3}|x-y|$. By iteration, the function satisfies the inequality

$$\frac{f(x)+f(y)}{2} \geq f\left(\frac{x+y}{2}\right) + \left(\frac{5}{3}\right)^n |x-y|$$

for any n. However the right-hand side increases without limit as n increases and, therefore, the inequality cannot be satisfied. □

Part V

EQUATIONS WITH NO PARAMETERS

Chapter 16

Iterations

Functional equations for functions of a single variable and no parameters[1] have appeared in this book in several places. In particular, most of the functional equations in Chapter 4 were of this type. In general, given a function f, a functional equation satisfied by f is much harder to solve when no parameters enter into the functional equation. The related equations that we have studied so far were simple enough and we did not have to deal with many of the rich and interesting features encountered in the theory of functional equations with no parameters. In this and the following chapter, we will highlight some of these features. However, an exhaustive presentation of the topic is beyond the scope of this book. The interested reader should consult [40] or, at a higher level, Kuczma's book [28] for a thorough and deeper coverage of the topic by a highly regarded expert.

16.1 The Need for New Methods

Some important functional equations in this category are:

$$f^2(x) = 0, \tag{16.1}$$

$$f^2(x) = x, \tag{16.2}$$

$$f(g(x)) = f(x) + a \quad \text{for } a \neq 0, \tag{16.3}$$

$$f(g(x)) = \lambda\, f(x) \quad \text{for } \lambda \neq 0, \tag{16.4}$$

$$f(g(x)) = (f(x))^p \quad \text{for } p \neq 1. \tag{16.5}$$

Equation (16.3) is known as the **Abel equation**; equation (16.4) is known as the **Schröder equation**, and equation (16.5) is known as the **Böttcher equation**.

[1] See page 42 for this terminology.

If we set $x = y$ in the four Cauchy functional equations studied in Chapter 5, they become

$$f(2x) = 2f(x), \quad f(2x) = f(x)^2, \quad f(x^2) = 2f(x), \quad f(x^2) = f(x)^2,$$

which are special cases of the Schröder (first and third) and Böttcher (second and fourth) equations, with the obvious solutions

$$f(x) = a\,x, \quad f(x) = a^x, \quad f(x) = \log_a x, \quad f(x) = x^a.$$

One would like to know under what conditions the Cauchy equations above *uniquely* characterize the linear, the exponential, the logarithmic and the power functions, respectively.

Equations (16.1) and (16.2) are special cases of the equation

$$f^n(x) = h(x),$$

where $h(x)$ is a known function and $n > 1$ is a given integer. The solutions $f(x)$ are called the **iterative n-th roots** of h. In particular, the solutions of (16.2) (the iterative square roots of the identity function) are called **involutions**.

Unfortunately, the methods we have presented so far are not adequate to solve functional equations with no parameters. To demonstrate this claim, let's work out the solution of equation (16.1), which is the simplest possible example.

Problem 16.1. *Let $I = [a, b]$, with $a \leq 0 \leq b$. Find all functions $f : I \to I$ such that $f^2(x) = 0$ and $f \not\equiv 0$.*

Solution. Define

$$K = \{x \in I : f(x) = 0\},$$
$$N = \{x \in I : f(x) \neq 0\}.$$

We shall show that f satisfies the condition of the problem if and only if $K \neq \varnothing$, $f(N) \subset K$, and $0 \in K$.

Necessity: Since $f \not\equiv 0$, there exists an $x_0 \in I$ such that $f(x_0) \neq 0$. Therefore, the set N is nonempty.

We set $y_0 = f(x_0)$. Then $f(y_0) = 0$ since $f(f(x_0)) = 0$ by definition. The point y_0 belongs to K, which is thus nonempty, too.

Similarly, for any $x' \in N$ we can set $y' = f(x')$ and by the defining equation $f(y') = 0$, that is, $y' \in K$. Therefore $f(N) \subset K$.

The point 0 must be in K, that is, $f(0) = 0$. If this is not true, then for any point $k \in K$, $f(f(k)) = f(0) \neq 0$, which contradicts the defining equation.

Sufficiency: Let K and N be two nonempty sets such that $0 \in K$, $K \cap N = \varnothing$ and $K \cup N = I$. Let $\tilde{f}$ be any function $\tilde{f} : N \to K$. The extension of $\tilde{f}$ to a function $f : I \to I$ by

$$f(x) = \begin{cases} 0 & \text{if } x \in K, \\ \tilde{f}(x) & \text{if } x \in N \end{cases}$$

is easily verified to satisfy $f^2(x) = 0$ for all $x \in I$. □

Question. *What are the continuous solutions?*

Answer. By continuity, the image $I' = f(I)$ is an interval $I' = [a', b']$ with $a \le a' < 0 < b' \le b$. And since $f^2(x) = 0$ for all x, $f(I') = \{0\}$. In particular, $f(a') = f(b') = 0$. Therefore, $f(x) = 0$ for all $x \in [a', b']$, and in the intervals $[a, a')$ and $(b', b]$, the function can take any value (consistent with continuity) in the interval $[a', b']$. (See Figure 16.1.)

Comment. Notice that $[a', b']$ is not necessarily equal to the set K, which may include additional points.

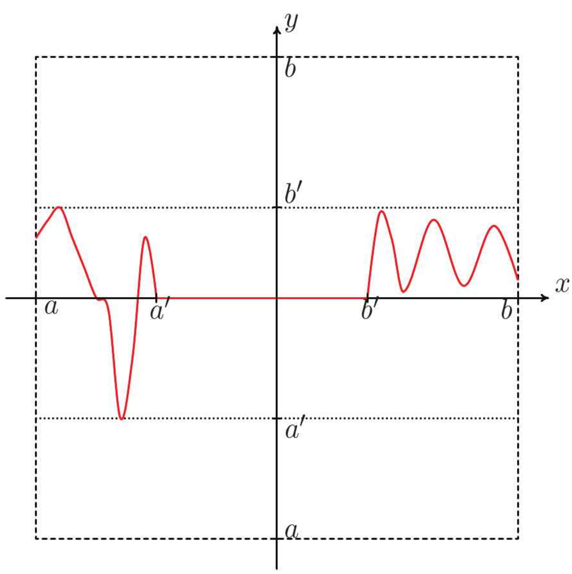


Figure 16.1: The graph of a continuous function that satisfies $f^2(x) = 0$.

In the previous example, we found all solutions of the functional equation. Often, we are interested only in identifying a particular subset of solutions, not necessarily all solutions. This simplifies the task considerably. For example, we can find solutions of the equation (16.2) relatively easily. It is immediate that reflections and inversions are solutions:

$$f_1(x) = -x, \quad f_2(x) = \frac{1}{x}.$$

However, these are not the only solutions. With a little more effort, one can discover that, given an invertible function g,

$$f(x) = g^{-1}(-g(x))$$

is an involution. We have thus discovered a family of solutions parametrized by the invertible function g but, of course, we have not shown that these are the only solutions. (For a complete solution see Problem 16.15.)

16.2 Iterates, Orbits, Fixed Points, and Cycles

The iterates of a function f with domain X have already been defined in Section 1.3:

$$f^2 = f \circ f, \qquad f^n = f \circ f^{n-1} \quad \text{for } n > 2.$$

For convenience we may often define $f^0 = \mathrm{id}_X$. It is noteworthy that any two iterates of f commute:

$$f^n \circ f^m = f^m \circ f^n = f^{n+m} \quad \text{for all } n, m \in \mathbb{N}.$$

The sequence of points

$$\{x_n = f^n(x)\}_{n \in \mathbb{N}}$$

is called the **splinter**[2] of x.

Consider two points $x, y \in X$. We call them **equivalent points** under iteration of f and write $x \sim y$ (more exactly $x \sim_f y$ but we will omit the subscript f when there is no ambiguity) if there exist $n, m \in \mathbb{N}$ such that $f^n(x) = f^m(y)$. This relation between points satisfies the three properties of an equivalence relation:

1. *Reflexivity*: for all $x \in X$, $x \sim x$.
2. *Symmetry*: for all $x, y \in X$, if $x \sim y$, then $y \sim x$ too.
3. *Transitivity*: for all $x, y, z \in X$, if $x \sim y$ and $y \sim z$, then $x \sim z$, too.

Therefore, $\sim$ is an equivalence relation. We call the corresponding equivalence classes **orbits**. That is, the orbit O_x of a point x is

$$O_x = \{y \in X : x \sim y\}.$$

[2] The term was introduced by J. S. Ullian and Gy. Targoński and followed by Kuczma et al. [28].

The orbits partition X into a union of pairwise disjoint sets.

If $x_0 \in X$ is such that

$$f^k(x_0) = x_0 \quad \text{and} \quad f^\ell(x_0) \neq x_0 \quad \text{for } \ell = 1, \ldots, k-1,$$

then x_0 is called a **fixed point of order** k of f. Fixed points x_* of order 1 are characterized by

$$f(x_*) = x_*$$

and they are simply called **fixed points**. Clearly, a fixed point of order k is a fixed point of the function $g = f^k$.

If $x_0 \in X$ is a fixed point of order k, then all of the points in the sequence defined by

$$x_n = f^n(x_0) \quad \text{for } n \in \mathbb{N}$$

are also fixed points of order k, since $(f^n)^k = (f^k)^n$. There are exactly k distinct points in this sequence, and they form a set called a **cycle of order** k or, simply, a k**-cycle**. For this reason, x_0 is also called a **periodic point** of period k.

Example 16.2. Consider the function $f(x) = x^2 - 2$ with domain $[-2, 2]$. How many fixed points of order n (if any) does it have?

It is actually easy to see that there will be at least one for each n. Looking back at Problem 1.55, we proved that the equation $f^n(x) = x$ has exactly 2^n real roots. Since a fixed point of order n should not be a fixed point of any lower order, and there are

$$2^0 + 2^1 + 2^2 + \cdots + 2^{n-1} = 2^n - 1$$

total roots for $f^k(x) = x$, $k = 1, 2, \ldots, n-1$, the equation $f^n(x) = x$ has at least one root different from all roots of the previous equations. To exactly count the fixed points of order n is a simple exercise in number theory but, unfortunately, requires knowledge of at least one important piece of number theory. For the reader's curiosity and satisfaction, I present the counting at the end of the example. Here, to uncover the pattern, for illustrative purposes, I shall work by example. (Any calculations are automatic from the results of Problem 1.55.)

- The fixed points (of order 1) are

$$x_* = 2\cos 0, \quad x_{**} = 2\cos\frac{2\pi}{3}.$$

These two points, satisfying $f(x) = x$, will also satisfy $f^n(x) = x$ for any $n > 1$. Therefore, they must be removed from the solutions of $f^n(x) = x$, which are the candidates for fixed points of order n.

• The roots of $f^2(x) = x$ and thus the candidates for fixed points of order 2 are

$$x_* = 2\cos 0, \quad x_{**} = 2\cos\frac{2\pi}{3}, \quad a_2 = 2\cos\frac{2\pi}{5}, \quad b_2 = 2\cos\frac{4\pi}{5}.$$

The first two solutions are those to be excluded. The remaining two solutions, however, are genuine fixed points of second order. These points, satisfying $f^2(x) = x$, will also satisfy $x = f^{2k}(x)$, for any $k > 0$. We also notice that they are members of a 2-cycle:

$$f(a_2) = 2\cos\frac{4\pi}{5} = b_2, \quad f(b_2) = 2\cos\frac{8\pi}{5} = 2\cos\left(2\pi - \frac{2\pi}{5}\right) = a_2.$$

In other words, the splinter of any of the two roots, say

$$a_2,\ f(a_2),\ f^2(a_2),\ f^3(a_2),\ f^4(a_2),\ f^5(a_2),\ f^6(a_2),\ f^7(a_2),\ f^8(a_2),\ \ldots,$$

has only two distinct points:

$$a_2,\ b_2,\ a_2,\ b_2, \ldots.$$

• The roots of $f^3(x) = x$ and thus the candidates for fixed points of order 3 are

$$x_* = 2\cos 0, \quad a_3 = 2\cos\frac{2\pi}{7}, \quad b_3 = 2\cos\frac{4\pi}{7}, \quad c_3 = 2\cos\frac{6\pi}{7},$$
$$\alpha_3 = 2\cos\frac{2\pi}{9}, \quad \beta_3 = 2\cos\frac{4\pi}{9}, \quad x_{**} = 2\cos\frac{2\pi}{3}, \quad \gamma_3 = 2\cos\frac{8\pi}{9}.$$

Excluding x_* and x_{**}, the remaining six solutions are genuine fixed points of third order. These points, satisfying $f^3(x) = x$, will also satisfy $x = f^{3k}(x)$ for any $k > 0$.

We also notice that a_3, b_3, c_3 and $\alpha_3, \beta_3, \gamma_3$ form two 3-cycles respectively:

$$f(a_3) = 2\ \cos\frac{4\pi}{7} = b_3, \quad f(b_3) = 2\ \cos\frac{8\pi}{7} = c_3, \quad f(c_3) = 2\ \cos\frac{16\pi}{7} = a_3,$$

and

$$f(\alpha_3) = 2\ \cos\frac{4\pi}{9} = \beta_3, \quad f(\beta_3) = 2\ \cos\frac{8\pi}{9} = \gamma_3, \quad f(\gamma_3) = 2\ \cos\frac{16\pi}{9} = \alpha_3.$$

• There are 16 candidates for fixed points of order 4. From them we must remove the fixed points of order 1 (two points) and order 2 (another two points). We are left with 12 fixed points of order 4:

$$a_4 = 2\ \cos\frac{2\pi}{15}, \quad b_4 = 2\ \cos\frac{4\pi}{15}, \quad c_4 = 2\ \cos\frac{8\pi}{15}, \quad d_4 = 2\ \cos\frac{14\pi}{15},$$
$$\alpha_4 = 2\ \cos\frac{2\pi}{17}, \quad \beta_4 = 2\ \cos\frac{4\pi}{17}, \quad \gamma_4 = 2\ \cos\frac{8\pi}{17}, \quad \delta_4 = 2\ \cos\frac{16\pi}{17},$$
$$\aleph_4 = 2\ \cos\frac{6\pi}{17}, \quad \beth_4 = 2\ \cos\frac{12\pi}{17}, \quad \gimel_4 = 2\ \cos\frac{10\pi}{17}, \quad \daleth_4 = 2\ \cos\frac{14\pi}{17},$$

which form three 4-cycles. We have labeled the points of these cycles by Latin, Greek, and Hebrew letters respectively.

The general pattern should be obvious now. For exact counting of the fixed points of order n, see the following paragraph that uses a result from number theory.

Counting the n-periodic points: Consider the arithmetic functions f and g defined on $\mathbb{N}$ and assume that they satisfy the relation

$$g(n) = \sum_{d|n} f(d)$$

for $n \geq 1$. The notation $d|n$ stands for "d divides n"; thus the sum is over all divisors d of n. This relation can be solved for $f(n)$. The result is

$$f(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) g(d),$$

where $\mu(n)$ is the Möbius function defined in Example 3.6. This formula is known as the **Möbius inversion formula**. You will find the proof in any text of number theory, such as [13]. Also, inverting the latter formula gives the former.

Now consider any function $f(x)$ for which the equation $f^n(x) = x$ has 2^n solutions. A fixed point of order d satisfying $x = f^d(x)$ will also be a solution of $x = f^{qd}(x)$ for any integer q. From this we see that, if $d|n$, the solutions of $x = f^d(x)$ are also solutions of $x = f^n(x)$. Let $N(d)$ be the number of fixed points of order d. Our discussion implies that

$$\sum_{d|n} N(d) = 2^n,$$

which we can invert to find

$$N(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) 2^d.$$

For example, for $n = 4$ we have $d = 1, 2, 4$ and thus

$$N(4) = \mu(4)\, 2 + \mu(2)\, 2^2 + \mu(1)\, 2^4 = 0 \cdot 2 - 1 \cdot 4 + 1 \cdot 16 = 12,$$

as found previously. □

16.3 Fixed Points: Discussion

Consider a function f and a point x_0 in its domain. If x_0 is a fixed point, then its splinter is a constant sequence $x_0, x_0, x_0, \ldots$. But what happens if x_0 is

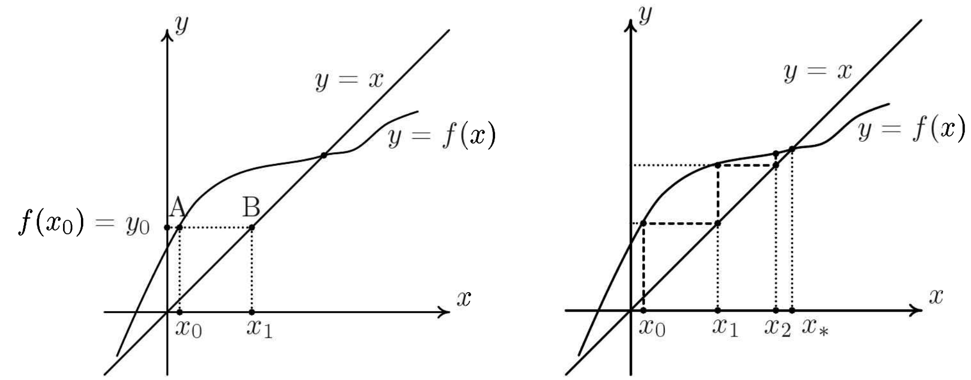


Figure 16.2: Left: Graphical representation of the map $x_0 \mapsto x_1 = f(x_0)$. First draw the graph Γ of the function $y = f(x)$ and the diagonal $y = x$. Given the point x_0, we raise a vertical line to find the intersection point A with the graph Γ. Point A has coordinates $(x_0, f(x_0))$. We then draw a horizontal line through A that intersects the diagonal at point B with coordinates $(f(x_0), f(x_0))$. Projecting this point on the x-axis determines the point x_1 with value $f(x_0)$. Right: Graphical construction of the splinter of x_0. Using the procedure described in the left picture, we can construct the successive images $x_0 \mapsto x_1 \mapsto x_2 \mapsto \cdots$. In the particular case shown in this picture, the splinter of x_0 converges to the fixed point x_*.

not a fixed point? To answer this question, we will first demonstrate how to construct the splinter graphically. This is best done in Figure 16.2, which the reader should consult to understand the construction.

For the graph Γ given in Figure 16.2, we see that the splinter of any point x_0 (apart from the fixed point x_*) creates an increasing sequence of points converging to x_*. In fact, we can easily prove the following theorem:

Theorem 16.3. *If f is continuous and the splinter converges to a limit ℓ, then ℓ must be a fixed point.*

Proof. Since any subsequence of a converging sequence converges to the same limit, we have:

$$\begin{aligned}\ell &= \lim_{n\to\infty} f^n(x_0) = \lim_{n\to\infty} f^{n+1}(x_0) \\ &= \lim_{n\to\infty} f(f^n(x_0)) = f\big(\lim_{n\to\infty} f^n(x_0)\big) = f(\ell). \qquad \square\end{aligned}$$

Figure 16.3 illustrates two very special but important cases of fixed points. In the case on the left, the splinter of any point converges to x_*. In the case on the right, the splinter of any point moves further away from x_*. This motivates a definition:

Definition 16.4. A fixed point x_* of a function f is called **attractive** or **stable** if there is a neighborhood $N(x_*)$ of x_* such that for any $x \in N(x_*)$ the splinter of x converges to x_*.

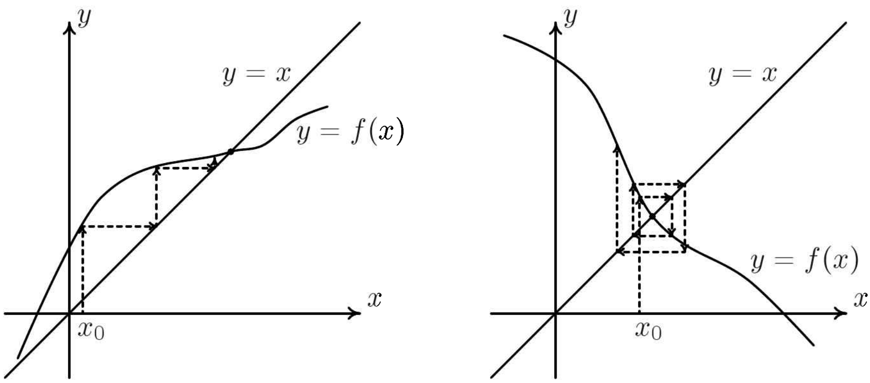


Figure 16.3: Left: An attractive fixed point. Right: A repulsive fixed point.

A fixed point x_* of f is called **repulsive** or **unstable** if there is a neighborhood $N(x_*)$ of x_* such that for any $x \in N(x_*) \setminus \{x_*\}$ the splinter of x contains a point not in $N(x_*)$.

Consider the set F of all attractive and repulsive fixed points of a function f. We show that each such point is isolated in F; that is, for each $x_* \in F$, there is a neighborhood $N(x_*)$ such that $x'_* \notin N(x_*)$ for all $x'_* \in F \setminus \{x_*\}$. If there were at least one $x'_* \in N(x_*)$, its splinter would be the constant sequence $x'_*, x'_*, x'_*, \ldots$ which neither converges to x_* nor has a point outside $N(x_*)$.

We can now establish simple criteria that allow us to check for fixed points of a function f and their nature.

Theorem 16.5. *Let f be a continuous function such that $f(a) > a$ and $f(b) < b$ at some points a and b. Then f has a fixed point x_* between a and b.*

Proof. We assume $a < b$ (the case $b < a$ is studied analogously). Then for the continuous function $g(x) = f(x) - x$, we have $g(a) > 0$ and $g(b) < 0$. Therefore, by Bolzano's theorem, there must be a point $x_* \in (a, b)$ such that $g(x_*) = 0$ or $f(x_*) = x_*$. □

Corollary (Brouwer's fixed point theorem). *Any continuous map $f : [a, b] \to [a, b]$ has a fixed point $x_* \in [a, b]$.*

Proof. If $f(a) = a$ or $f(b) = b$, we are done. If that's not the case, then it must be the case that $f(a) > a$ and $f(b) < b$. By the previous theorem, there is an $x_* \in (a, b)$ such that $f(x_*) = x_*$. □

Comment. Following a popular trend, I have called the last result *Brouwer's fixed point theorem.* However, Brouwer proved the corresponding theorem in the (much harder) case of dimension two. Obviously, in one dimension it is just a restatement of Bolzano's theorem.

Question. *The function* $f : [1, +\infty) \to [1, +\infty)$ *with*

$$x \mapsto f(x) = x + \frac{1}{x}$$

has no fixed point $x_* \in (1, +\infty)$. *How is this possible? Has any implicit assumption been introduced in the previous proof?*

Answer. Theorem 16.3 applies only when the domain is a closed, bounded interval. If any of these properties is removed, the statement may not be valid.

Theorem 16.6. *Let* f *be a continuous function and let* x_* *be a fixed point of* f.

(a) *A sufficient condition for* x_* *to be an attractive fixed point is that*

$$\left|\frac{f(x) - f(x_*)}{x - x_*}\right| < 1 \quad \textit{for all points } x \textit{ in some neighborhood of } x_*.$$

(b) *A sufficient condition for* x_* *to be a repulsive fixed point is that*

$$\left|\frac{f(x) - f(x_*)}{x - x_*}\right| > 1 \quad \textit{for all points } x \textit{ in some neighborhood of } x_*.$$

Proof. We will prove the first statement; the second statement is proved similarly with the appropriate changes.

In the relation $|f(x) - f(x_*)| < |x - x_*|$, we replace x by the points $x_n = f^n(x)$ in the splinter of x. We obtain

$$|x_{n+1} - x_*| = |f(x_n) - f(x_*)| < |x_n - x_*|$$

for any $n > 1$. Therefore the sequence

$$|x_1 - x_*|, \ |x_2 - x_*|, \ \ldots, \ |x_n - x_*|, \ \ldots$$

is bounded and decreasing and, hence, converges. Let $\lim_{n\to\infty} |x_n - x_*| = \ell$. Obviously, $\ell \geq 0$. We will show that, in fact, $\ell = 0$. Toward this goal, we will assume that $\ell > 0$ and show that it leads to a contradiction.

In the given condition

$$|f(x) - x_*| < |x - x_*| \ ,$$

we set $x = x_* - \ell$ to find

$$|f(x_* - \ell) - x_*| < \ell \ \Rightarrow \ x_* - \ell < f(x_* - \ell) < \ell + x_*.$$

For simplicity we set $J = (x_* - \ell, x_* + \ell)$. Since f is continuous, according to the definition of continuity on page 17, there exists a neighborhood $I_- = (x_* - \ell - \varepsilon_-, x_* - \ell + \varepsilon_-)$ of $x_* - \ell$ such that $f(I_-) \subset J$.

Similarly, if we set $x = x_* + \ell$ in the given condition, we conclude that there exists a neighborhood $I_+ = (x_* + \ell - \varepsilon_+, x_* + \ell + \varepsilon_+)$ of $x_* + \ell$ such that $f(I_+) \subset J$.

As $n \to \infty$, there will be some n such that $x_n \in I_-$ or $x_n \in I_+$. Then $x_{n+1} = f(x_n) \in J$ and thus $|x_{n+1} - x_*| < \ell$. But it is impossible. □

Corollary. *Let f be a continuous function and let x_* be a fixed point of f at which f is differentiable.*

(a) *A sufficient condition for x_* to be an attractive fixed point is that*

$$\left|f'(x_*)\right| < 1.$$

(b) *A sufficient condition for x_* to be a repulsive fixed point is that*

$$\left|f'(x_*)\right| > 1.$$

Proof. Since

$$f'(x_*) = \lim_{x \to x_*} \frac{f(x) - f(x_*)}{x - x_*}$$

for a sufficiently small neighborhood of x_*, the conditions in (a) and (b) imply the corresponding ones in Theorem 16.6. □

Obviously, if $|f'(x_*)| = 1$ — we call x_* a **marginal point** — we need to look at higher-order derivatives to decide about its behavior. Also, if $f'(x_*) = 0$, we call x_* **superstable**.

Attractive and repulsive fixed points are very special. Look, for example, at the two functions

$$f(x) = x, \quad f(x) = -x.$$

For the former, all points are fixed points, while for the latter, $x_* = 0$ is the unique fixed point. However, none of these fixed points is attractive or repulsive. The splinter of any point in the neighborhood of a fixed point neither converges nor moves away from the fixed point.

16.4 Cycles: Discussion

Consider again the function $f(x) = -x$, with which we closed the previous section. The splinter of any point $x_0 \neq 0$,

$$x_0, \ -x_0, \ x_0, \ -x_0, \ x_0, \ -x_0, \ \ldots,$$

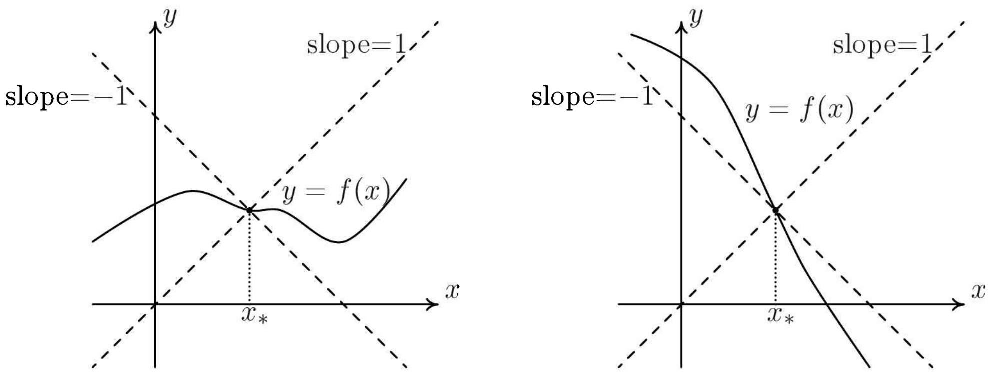


Figure 16.4: Geometric interpretation of the criteria that characterize attractiveness and repulsiveness of a fixed point. The quotient $\big(f(x)-f(x_*)\big)/(x-x_*)$ is the slope of the line that joins the fixed point (x_*, x_*) with the nearby point $(x, f(x))$. Therefore, for an attractive fixed point, all such secants of the graph of $f(x)$ have slopes in the interval $(-1, 1)$ (left picture) while, for a repulsive fixed point, all such secants have slopes in $(-\infty, -1) \cup (1, +\infty)$ (right picture). A similar statement is true for the slope of the tangent at the fixed point (if the derivative at x_* exists).

is a periodic sequence with period 2. The points x_0 and $-x_0$ form a 2-cycle.

Do N-cycles exist for arbitrary N? Sure; the function f given by

$$f(x) = \begin{cases} i-1 & \text{if } x \in [i-\frac{1}{2}, i+\frac{1}{2}), \text{ with } i = 1, 2, \ldots, N-1, \\ N-1 & \text{if } x \in [-\frac{1}{2}, \frac{1}{2}), \\ 0 & \text{otherwise}, \end{cases} \tag{16.6}$$

has an N-cycle

$$0,\ N-1,\ N-2,\ \ldots,\ 2,\ 1.$$

This function is discontinuous, but can be made continuous while preserving this N-cycle. (It is enough to leave f undisturbed at the points $0, 1, \ldots, N-1$.)

However, unlike the discontinuous example in (16.6), which has no cycles of other orders (apart from fixed points), any continuous example will have cycles of other orders as well, if $N > 2$:

Theorem 16.7 (Sharkovskii). *Let $f : I \to I$ be a continuous function defined on a closed interval I. If f has an m-cycle, then it has an n-cycle for any n such that $m \prec n$, where $\prec$ is the total order relation in $\mathbb{N}^*$ given by*

$$3 \prec 5 \prec 7 \prec \cdots \prec 2\cdot 3 \prec 2\cdot 5 \prec 2\cdot 7 \prec \cdots \prec 2^2\cdot 3 \prec 2^2\cdot 5 \prec 2^2\cdot 7 \prec \cdots$$
$$\cdots \prec 2^n\cdot 3 \prec 2^n\cdot 5 \prec 2^n\cdot 7 \prec \cdots \prec 2^m \prec 2^{m-1} \prec 2^n \cdots \prec 8 \prec 4 \prec 2 \prec 1;$$

that is, first come all odd numbers (in the usual order), then come their multiples by 2, then come their multiples by $4 = 2^2$, then come their multiples by $8 = 2^3$, and so on, and finally come the powers of two in decreasing order.

Thus, if f has a 3-cycle, it must have a cycle of every order! On the other hand, the presence of a 2-cycle doesn't imply the presence of cycles of other orders, except fixed points (consider the example, already seen, of the function $f(x) = -x$ on $[-1, 1]$).

We will make no attempt to prove this deep theorem, which is considered by many as one of the most important theorems in modern mathematics.[3] Suffice it to say that the proof uses results from a vast swath of mathematics.

Exercise 16.8. *Let's return to Example 16.2. There we showed that the function on $[-2, 2]$ defined by $f(x) = x^2 - 2$ has 3-cycles. Therefore, according to Sharkovskii's theorem, it must have a cycle of every order. Using the results we have found for this function, can you demonstrate the existence of n-cycles for some values of n other than 3? Notice that I have already written down the 4-cycles.* □

In the previous section, we showed that if the splinter of a continuous function converges, then it must converge to a fixed point. This established only a necessary condition, not a sufficient one. The following theorem improves on this situation. For the rest of this chapter, let I be a closed interval.

Theorem 16.9. *Let $f : I \to I$ be a continuous function. The splinter of any $x \in I$ converges to some fixed point if and only if f has only fixed points and no other cycles.*

The next theorem extends the result to a function with specific cycles.

Theorem 16.10. *Let $f : I \to I$ be a continuous function with cycles of order $1, 2, 2^2, \ldots, 2^n$ only. The splinter of any point $x \in I$ converges to some fixed point or cycle.*

The last theorem suggests that, as in the case of fixed points, there are special cycles that might attract or repel splinters. So we introduce the following definition.

Definition 16.11. Let $f : I \to I$ be a differentiable function.[4] A k-cycle of f is called

(a) **attractive** or **stable** if at least one point of the cycle is an attractive fixed point of f^k, and

[3] Yet, Sharkovskii's results [59, 60] went unnoticed for many years. The fact that his original arguments are quite formidable and that he published his work in Russian did not help. A simplified proof was found by Štefan in 1977 and was published in English [64]. After that, several other proofs were found. For the most recent proof, as well as a review of the history of the theorem, the reader should look at [47].

[4] I adopt differentiability to simplify somewhat my discussion and be able to present simple proofs for the theorems that follow.

(b) **repulsive** or **unstable** if at least one point of the cycle is a repulsive fixed point of f^k.

Lemma 16.12. *Fix $x_0 \in I$, and let*[5]

$$D = \frac{d}{dx} f^n(x)\Big|_{x=x_0}.$$

If $x_1 = f(x_0)$, $x_2 = f(x_1)$, $\dots$, $x_{n-1} = f(x_{n-2})$, then

$$D = f'(x_0) f'(x_1) \dots f'(x_{n-1}).$$

Proof. For $n = 2$, the statement is true by the well-known theorem on the differentiation of a composite function,

$$\frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x), \tag{16.7}$$

applied to $g = f$, $x = x_0$. Now, let it be true for $n = k$:

$$\frac{d}{dx} f^k(x)\Big|_{x=x_0} = f'(x_0)\, f'(x_1) \dots f'(x_{k-1}).$$

We will show that it is also true for $n = k + 1$. In equation (16.7), we set $g = f^k$:

$$\frac{d}{dx} f^{k+1}(x) = f'(f^k(x))\, \frac{d}{dx} f^k(x).$$

For $x = x_0$ the last equation gives

$$\frac{d}{dx} f^{k+1}(x)\Big|_{x=x_0} = f'(x_0)\, f'(x_1) \dots f'(x_{k-1}) f'(x_k). \qquad \square$$

The next two theorems are useful criteria to identify attractive and repulsive cycles.

Theorem 16.13. *Let f be a differentiable function. Every point of an attractive k-cycle of f is an attractive fixed point of f^k. Every point of a repulsive k-cycle of f is a repulsive fixed point of f^k.*

Proof. Let $g = f^k$, and let the points in the k-cycle be $a_1, a_2, \dots, a_k$. The previous lemma gives

$$g'(a_1) = f'(a_1) f'(a_2) \dots f'(a_k).$$

[5] I have used the symbol "D" as a shorthand for *discriminant.* A discriminant, as that of a quadratic polynomial, is an instrument that allows us to differentiate between two or more choices.

But if we start with a_2 instead of a_1, we get

$$g'(a_2) = f'(a_2)\dots f'(a_k)f'(a_1),$$

because $a_1 = f(a_k)$; therefore, $g'(a_2) = g'(a_1)$. Similarly, $g'(a_{i+1}) = g'(a_i)$ for each i. Thus all the points of an attractive or repulsive cycle behave the same. □

As an immediate corollary of this proof we have:

Theorem 16.14. *Let $f : I \to I$ be a differentiable function with $a_1, a_2, \dots, a_k$ a k-cycle of f. Set $D = f'(a_1)f'(a_2)\dots f'(a_k)$. If $|D| < 1$, the cycle is attractive. If $|D| > 1$, the cycle is repulsive.*

16.5 From Iterations to Difference Equations

Functional equations that involve a function $f : \mathbb{R} \to \mathbb{R}$ and some of its iterations are intimately related to difference equations. This is useful when the corresponding difference equation can be solved more easily than the original functional equation.

To illustrate the idea, consider the functional equation

$$f^3(x) - f^2(x) - 8f(x) = -12x.$$

The point x can be arbitrary, so we can take it to be $x = f^n(y)$:

$$f^{n+3}(y) - f^{n+2}(y) - 8f^{n+1}(y) = -12f^n(y).$$

Now consider the splinter of y,

$$y, \quad f(y), \quad f^2(y), \quad \dots, f^n(y), \quad f^{n+1}(y), \quad f^{n+2}(y), \dots,$$

which we can identify as a sequence

$$a_0, \quad a_1, \quad a_2, \quad \dots, a_n, \quad a_{n+1}, \quad a_{n+2}, \dots.$$

Then, the given functional equation becomes a difference equation (one for every point y)

$$a_{n+3} - a_{n+2} - 8a_{n+1} = -12a_n,$$

with the fundamental set of solutions 2^n, $n2^n$, $(-3)^n$. Therefore, the sequences[6]

$$a_n = A\,2^n,$$
$$a_n = C\,(-3)^n,$$

[6]Notice that I neglect the sequence $a_n = Bn2^n$. Can you explain why?

solve the difference equation, from which we recover the two solutions

$$y = a_0 = A, \quad a_1 = f(y) = A\,2,$$
$$y = a_0 = C, \quad a_1 = f(y) = C\,(-3);$$

that is

$$f(y) = 2y \quad \text{or} \quad f(y) = -3y$$

for the original functional equation. The issue of uniqueness remains open unless additional conditions are specified. (See the solved Problem 16.19 and the unsolved Problem 20.135.)

The reverse method is also useful: starting from a difference equation, one might consider iterates of a function to draw conclusions. See, for instance, the solved Problem 16.18.

16.6 Solved Problems

Problem 16.15. *Find all functions $f : I \to I$ with $I = (a, b)$, $a \le 0 \le b$ that satisfy equation* (16.2).

Solution. The function f is injective. Indeed, if two images are equal, say $y_1 = y_2$, then $f(x_1) = f(x_2) \Rightarrow f^2(x_1) = f^2(x_2) \Rightarrow x_1 = x_2$.

The defining functional equation implies that all points of I are fixed points of f^2; that is, they are fixed points of order 1 or 2. Therefore, either $f(x) = x$ or $f(x) \neq x$ and $f^2(x) = x$. Define the sets

$$F = \{x \in I : f(x) = x\},$$
$$L = \{x \in I : f(x) < x\},$$
$$G = \{x \in I : f(x) > x\}.$$

These are obviously pairwise disjoint and their union equals I. All fixed points belong to F. All remaining points form 2-cycles $(x, f(x))$ and they belong to $L \cup G$. The two points of a cycle (ℓ, ℓ') belong to different sets since, if they were in the same set (say L), there would be a contradiction ($\ell' = f(\ell) < \ell$ and $\ell = f(\ell') < \ell'$.) Therefore, the 2-cycles define a bijective map between L and G. If f_L is the restriction of f to L, $f_L : L \to G$, and f_G is the restriction of f to G, $f_G : G \to L$, then $f_L = f_G^{-1}$.

Inversely, given a partition of I to three sets F, L, G and $g : L \to G$ a bijective function, the function

$$f(x) = \begin{cases} g(x), & \text{if } x \in L, \\ x, & \text{if } x \in F, \\ g^{-1}(x), & \text{if } x \in G, \end{cases}$$

satisfies equation (16.2). □

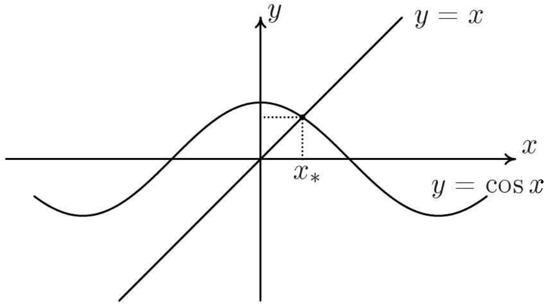


Figure 16.5: The cosine function has a unique fixed point x_*, whose approximate value is 0.739.

Exercise 16.16. *What are the continuous solutions of* (16.2)*?* □

Problem 16.17 (Putnam 1952)**.** *Given any real number N_0, if $N_{j+1} = \cos N_j$, prove that $\lim_{j\to\infty} N_j$ exists and is independent of N_0.*

Solution. (See Figure 16.5.) The sequence of points $\{N_j\}$ is the splinter of the cosine function $f(x) = \cos x$ generated by the point N_0. The cosine function has a fixed point x_* defined by $\cos x_* = x_*$. At the fixed point, $f'(x_*) = -\sin x_*$ and thus $|f'(x_*)| < 1$, that is, x_* is an attractive fixed point, which implies that $\lim_{j\to\infty} N_j$ exists and is independent of N_0. □

Problem 16.18 (Putnam 1947)**.** *If $\{a_n\}$ is a sequence of numbers such that for $n \geq 1$,*

$$(2 - a_n)\, a_{n+1} = 1,$$

prove that $\lim a_n$ as $n \to \infty$ exists and is equal to one.

Solution. The sequence defined by the given recursion relation

$$a_{n+1} = \frac{1}{2 - a_n}$$

can be written as the splinter[7] of the function

$$f(x) = \frac{1}{2 - x}$$

[7]This problem may be solved more easily by simpler techniques. However, the use of iterations helps us to demonstrate two ideas: how we can employ a function when a sequence is given, and how to modify the techniques presented in theory to verify the existence of an attractive fixed point when the theorems we have proved fail.

generated by the point $x = a_1$. As in the previous problem, we expect that $f(x)$ has an attractive fixed point, and indeed this is so:

$$\frac{1}{2-x_*} = x_* \quad \Longrightarrow \quad (x_* - 1)^2 = 0 \quad \Longrightarrow \quad x_* = 1.$$

However, the criteria to confirm that it is an attractive point fail. At $x_* = 1$, $f'(x_*) = 1$ and the ratio $|f(x)-1|/|x-1|$ is less than 1 only on one side of x_*. So, to establish that the splinter converges to 1, additional work is required.

If $x = 1$, then the splinter is the constant sequence 1, 1, 1, ..., that is obviously convergent to 1. If $a_1 \neq 1$ we must consider three cases.

Case I: $x < 1$. When $x < 1$,

$$\frac{|f(x)-1|}{|x-1|} = \frac{1}{|2-x|} < 1,$$

and, therefore, $f^n(x) \to 1$ as $x \to 1^-$.

Case II: $x > 2$. In this case $f(x) < 0$ and the first point in the splinter of x goes back to the previous case.

Case III: $2 > x > 1$. In this case we have $(x-1)^2 > 0$, so $1 > x(2-x)$, so

$$\frac{1}{2-x} > x;$$

that is, the splinter of x is an increasing sequence. As such, at some value of n, the iteration will produce a point $a_n > 2$, which takes us back to case II. (One can check that $f(x) > 2$ when $x > \frac{3}{2}$.) □

Problem 16.19 (Putnam 1988). *Prove that there exists a unique function f from the set $\mathbb{R}_+^*$ of positive real numbers to $\mathbb{R}_+^*$ such that*

$$f(f(x)) = 6x - f(x)$$

and $f(x) > 0$ for all $x > 0$.

Solution. Setting $f^{n-1}(x)$ in place of x and using the notation $a_n = f^n(x)$, $n = 0, 1, 2, \ldots$, we find the recursion relation

$$a_{n+1} + a_n - 6a_{n-1} = 0.$$

We will seek solutions $a_n = \lambda^n$, which leads to the characteristic equation

$$\lambda^2 + \lambda - 6 = 0,$$

with solutions $\lambda_\pm = 2, -3$. Then, 2^n and $(-3)^n$ give a fundamental set of solutions. Since the function sought must be positive, only the first solution is acceptable, and thus $f(x) = 2x$. □

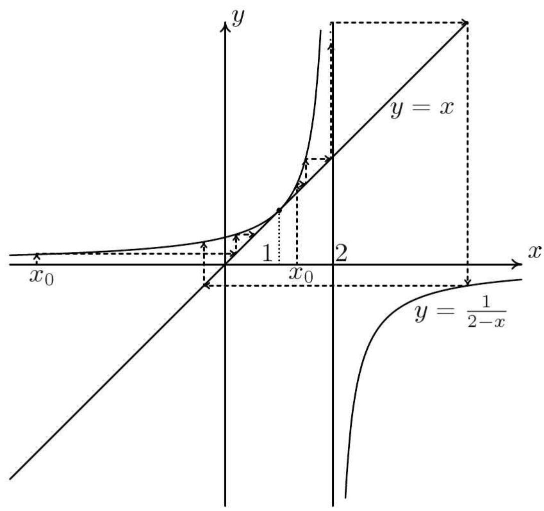


Figure 16.6: The graph of $f(x) = 1/(2-x)$. Also shown is part of the splinter of a point x_0 in the two cases $x_0 < 1$ and $1 < x_0 < 2$.

Problem 16.20 (Putnam 1979). *Establish necessary and sufficient conditions on the constant k for the existence of a continuous real-valued function $f(x)$ satisfying*

$$f(f(x)) = kx^9$$

for all real x.

Comment. Such an f is an iterative square root of the function $x \mapsto kx^9$.

Solution. If $k \geq 0$, we can see immediately that $f(x) = k^{1/4}x^3$ is a solution. Therefore, $k \geq 0$ is a sufficient condition to produce a solution.

We show that this condition is also necessary. Assume $k < 0$. It is a simple general fact that, if $f \circ f$ is injective, so is f, because

$$f(x) = f(y) \Rightarrow f(f(x)) = f(f(y)) \Rightarrow kx^9 = ky^9.$$

Since $x \mapsto kx^9$ is injective, so is f. Since f is also continuous it must be strictly increasing or strictly decreasing; in either case, the second iterate f^2 must be strictly increasing. But this is impossible if $k < 0$. □

Problem 16.21 (IMO 1976). *Let $f(x) = x^2 - 2$ and $f^j(x) = f(f^{j-1}(x))$ for $j = 2, 3, \ldots$. Show that for any positive integer n, the roots of the equation $f^n(x) = x$ are real and distinct.*

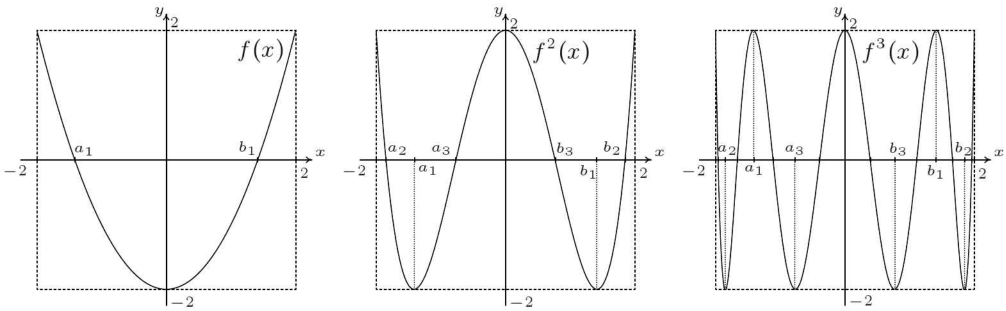


Figure 16.7: The graphs of $f(x) = x^2 - 2$, $x \in [-2, 2]$, and its first two iterations $f^2(x)$ and $f^3(x)$.

Solution. If $f(x)$ is a polynomial of degree 2, then $f^n(x)$ is a polynomial of degree 2^n. Therefore, we must prove that $f^n(x)$ has 2^n real and distinct fixed points (not necessarily of order n). Actually, since f is even, f^n is easily seen to be an even polynomial and we can restrict ourselves to the interval $(-\infty, 0]$ since a similar reasoning applies for the interval $[0, +\infty)$. We thus need to prove that f^n has 2^{n-1} negative fixed points[8].

Now we show that, if $x > 2$, then $f^n(x) > x$. For $n = 1$, if $x > 2$, it is easy to see that $f(x) = x^2 - 2 > x^2 - x = x(x-1) > x$. Let it be true for $n = k$. That is, let $f^k(x) > x$ for $x > 2$. Then set $y = f(x)$, with $x > 2$. Obviously $y = f(x) > 2$ and therefore $f^k(y) > y$. But the last inequality is equivalent to $f^{k+1}(x) > f(x)$, and since $f(x) > x$, also $f^{k+1}(x) > x$. This concludes the proof. By symmetry, $f^n(x) > x$ for $x < -2$.

Now, let's show that f has two fixed points. Of course, by solving the quadratic equation $f(x) = x$, we can immediately find $x = 2$ or $x = -1$, that is, one positive and one negative fixed point. However, I am going to use an alternative procedure as an example of how to prove the similar statement for the iterations of f. Since $f(a) > a$ for any $a < -2$ and $f(0) = -2 < 0$, by Theorem 16.5, f has a fixed point in $(a, 0)$. Therefore, f has $1 = 2^0$ negative fixed points.

Since $f(-2)f(0) < 0$, by Bolzano's theorem, there is a point a_1 in $(-2, 0)$ such that $f(a_1) = 0$. In the next iteration, $f^2(a_1) = f(0) = -2$. Also, $f^2(0) = f(-2) = 2$. Therefore, the graph of $f^2(x)$ looks as shown in Figure 16.7 (middle plot). In each of the two intervals (a, a_1) and $(a_1, 0)$, f^2 has a fixed point by the use of Theorem 16.5. Therefore, it has 2^1 fixed points in $(-\infty, 0)$.

By Bolzano's theorem, f^2 also has one root in each of the two intervals (a, a_1) and $(a_1, 0)$. Let them be a_2 and a_3. In the next iteration $f^3(a_2) =$

[8] Have I implied that if x_* is a fixed point, then $-x_*$ is also a fixed point? Answer yes or no and provide an explanation. As a specific example, look at the point $x = 2$.

$f(0) = -2$ and similarly for a_3. Also, $f^3(0) = f(2) = 2$. Therefore, the graph of $f^3(x)$ looks as shown in Figure 16.7 (right plot). In each of the intervals (a, a_2), (a_2, a_1), (a_1, a_3) and $(a_3, 0)$, f^3 has a fixed point by the use of Theorem 16.5. Therefore, it has 2^2 negative fixed points.

We can continue inductively to find that $f^n(x)$ achieves the value -2 at 2^{n-2} points and the value $+2$ at $2^{n-2}+1$ points, which splits the domain $[+2, 0]$ into 2^{n-1} subintervals in each of which the function f^n has a fixed point. (Here we used the fact that the splinters of ± 2 are $2 \mapsto f(2) = 2 \mapsto 2 \mapsto 2 \mapsto \ldots$ and $-2 \mapsto f(-2) = 2 \mapsto 2 \mapsto 2 \mapsto \ldots$. Also, if a is a root of f^k, its splinter — starting at the k-th iteration — will be $a \mapsto 0 \mapsto -2 \mapsto 2 \mapsto 2 \mapsto \ldots$.) $\square$

Now that you have understood the previous solution, I propose for you the following variation of Problem 1.59, from the 1998 Turkey Olympiad:

Problem 16.22. *Consider the function $f : [0, 1] \to [0, 1]$ defined by*

$$f(x) = 4x(1 - x).$$

How many distinct roots does the equation $f^{1992}(x) = x$ have?

Figure 16.8 shows the first iterations of $f(x)$. After trying the problem (and hopefully solving it), read Section 16.7.

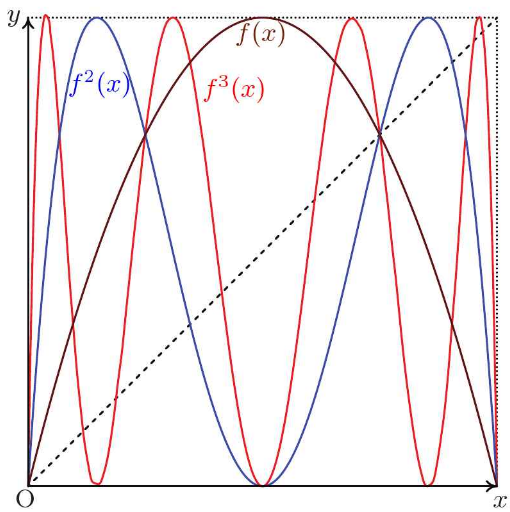


Figure 16.8: The function $f(x) = 4x(1 - x)$, $x \in [0, 1]$, and its first two iterations f^2 and f^3.

Problem 16.23 (Putnam 1971). *Determine all polynomials $P(x)$ such that*

$$P(x^2 + 1) = P(x)^2 + 1,$$

and $P(0) = 0$.

Solution. The point $x_0 = 0$ is a fixed point of $P(x) = 0$. If x_k is a fixed point, then the defining equation requires that

$$P(x_k^2 + 1) = x_k^2 + 1;$$

that is, $x_{k+1} = x_k^2 + 1$ is a fixed point, too. Therefore, the infinite sequence

$$0,\ 1,\ 2,\ 5,\ 26,\ \ldots$$

is a sequence of fixed points for any polynomial sought. In other words, the polynomial $Q(x) = P(x) - x$ vanishes for an infinite number of values and, hence, must vanish identically. Therefore, the only polynomial that satisfies the given conditions is

$$P(x) = x.$$

□

16.7 A Taste of Chaos

I tried to avoid writing this section as it is slightly more distant from the topic of mathematical competitions and the goals of this book. However, it is closely related to the topic of iterations, and eventually I felt like adding it. But I stay within what has been discussed in the previous sections and I only tease you. Hopefully, this brief encounter with chaos will whet your appetite and make you seek further information elsewhere[9].

The logistic function

$$f_\lambda(x) = \lambda\, x\,(1 - x) \quad \text{for } x \in [0, 1]$$

is the standard example[10]. The properties of this function depend on the value of the constant λ; so, we have used an additional index λ to differentiate functions with different values of the constant. If $\lambda \in [0, 1]$, then f maps $[0, 1]$ into $[0, 1]$. We will assume that this is the case.

[9] There are many great books written at various levels. For beginners I recommend [41] for a nonquantitative introduction, [36] for a more quantitative, entertaining and illuminating introduction of a broad coverage of related topics, and [42] for a pedagogical course-like presentation of the subject.

[10] The first pedagogical presentation of the logistic function (and other related functions) was given in an influential article by Robert May [55]. May's presentation uses biology's population growth perspective and summarizes all results without proofs. It is a concise article that the reader should find approachable and useful. In his article, at the end of his opening section, May writes: "The review ends with an evangelical plea for the introduction of these difference equations into elementary mathematics courses, so that students' intuition may be enriched by seeing the wild things that simple nonlinear equations can do." His plea has been successful: the logistic function can be found even in first-semester calculus courses and, of course, in any book more or less related to the topic.

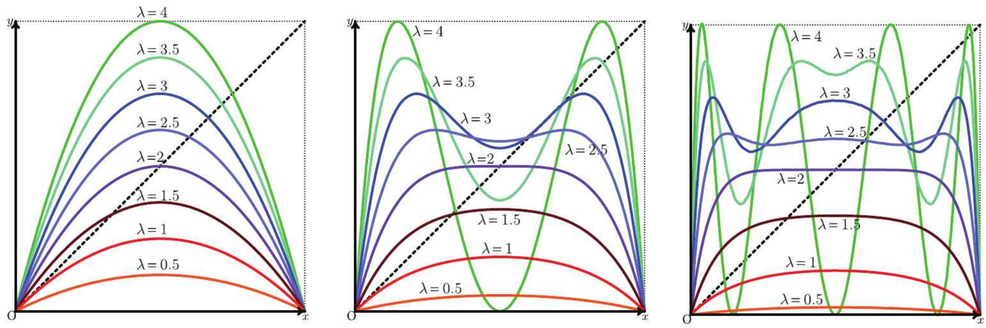


Figure 16.9: The graph of the logistic function f_λ and its first two iterations, f_λ^2 and f_λ^3, for various values of the constant λ. The diagonal line is the graph of the function $y = x$.

In Figure 16.9, I have plotted the logistic function for the values $\lambda = 0.5$, 1, 1.5, 2, 2.5, 3, 3.5, 4 of the constant (left picture), its first iterate $f_\lambda^2(x)$ for the same values of the constant (middle picture) and its second iterate $f_\lambda^3(x)$ for the same values of the constant (right picture). From the plots we notice the following feature: the iterates develop multiple "humps" that are less or more pronounced depending on the value of the constant λ. As a result, both the number of fixed points at each iteration and the slope of the tangent at the fixed points (which characterizes attractiveness and repulsivness) are strongly dependent on λ. In the remaining section, I will attempt to explain the precise dependence of the fixed points on the constant λ. To proceed, the reader should have, at least, fully understood Example 16.2.

Fixed Points of Order 1

We start with the fixed points of $f_\lambda(x)$. These can be found easily:

$$f_\lambda(x_*) = x_* \implies x_* \left(\lambda - 1 - \lambda x_*\right) = 0 \implies x_*^{(1)} = 0 \ \text{ or } \ x_*^{(2)} = 1 - \frac{1}{\lambda}.$$

For $\lambda < 1$, the second solution lies outside the domain, but for $\lambda > 1$ it is an actual fixed point. For $\lambda_c^{(0)} = 1$, the two fixed points coincide. We thus consider this value of λ as a **critical value** that separates two different behaviors. We say that, at $\lambda_c^{(0)} = 1$, the fixed point $x_* = 0$ **bifurcates** (splits) into two fixed points $x_*^{(1)} = 0$ and $x_*^{(2)} = 1 - \frac{1}{\lambda}$.

The derivative of $f_\lambda(x)$ is

$$f_\lambda'(x) = \lambda(1 - 2x)$$

and

$$f_\lambda'(0) = \lambda, \quad f_\lambda'\left(1 - \frac{1}{\lambda}\right) = -\lambda + 2.$$

By Corollary 16.3, the point $x_*^{(1)}$ is attractive for $\lambda < 1$ and repulsive for $\lambda > 1$. The behavior for the critical value $\lambda_c^{(0)} = 1$ must be inferred by other means. Since the splinter of any point converges to $x_*^{(1)} = 0$, it is still an attractive point. The point $x_*^{(2)}$ is attractive for $\lambda < 3$ and repulsive for $\lambda > 3$. The value $\lambda_c^{(1)} = 3$ is thus another critical value. The behavior of $x_*^{(2)}$ for $\lambda_c^{(1)}$ must be inferred by other means.

2-Cycles

Consider the points a and b such that

$$f(a) = b, \quad f(b) = a.$$

Therefore, they must be solutions of the equation $f^2(x) = x$,

$$\lambda^2 x(1-x)\,[1 - \lambda x(1-x)] = x,$$

and not be fixed points of f. For $\lambda \leq 1$, we cannot have such cycles due to Theorem 16.9. Therefore, we must have $\lambda > 1$. Since the two fixed points of order 1 must necessarily be solutions to the previous equation, it can be written in the form

$$x\left[x - \left(1 - \frac{1}{\lambda}\right)\right]\,(Ax^2 + Bx + C) = 0,$$

with the quadratic factor the one determining possible new solutions. Comparing the polynomials term by term, we conclude that the equation that determines the possible 2-cycles is

$$\lambda^2 x^2 - \lambda(\lambda+1)x + (\lambda+1) = 0,$$

with discriminant

$$D = \lambda^2(\lambda+1)(\lambda-3).$$

This discriminant is positive for $\lambda > 3$ (2-cycles exist) and negative for $\lambda < 3$ (2-cycles do not exist). When the 2-cycles exist, their points are

$$a, b = \frac{\lambda + 1 \pm \sqrt{\lambda^2 - 2\lambda - 3}}{2\lambda}.$$

The value $\lambda_c^{(1)} = 3$ is the critical value encountered previously. For it, $D = 0$ and we have a double root — a single fixed point, not a 2-cycle:

$$x_* = \frac{2}{3} = 1 - \frac{1}{\lambda_c^{(1)}},$$

that is, the point $x_*^{(2)}$. For $\lambda > 3$, the fixed point $x_*^{(2)}$ bifurcates into two points that are not fixed points of f but they constitute a 2-cycle. The discriminant for the 2-cycle is

$$D_{2-cycle} \;=\; f'_\lambda(a)\, f'_\lambda(b) \;=\; \lambda^2\,(1-2a)\,(1-2b) \;=\; -\lambda^2 + 2\,\lambda + 4,$$

and it takes the value $D_{2-cycle} = -1$ for $\lambda_c^{(2)} = 1+\sqrt{6}$. If $\lambda < 1+\sqrt{6}$, the 2-cycle is attractive; if $\lambda > 1+\sqrt{6}$, the 2-cycle is repulsive. Therefore, we discovered a new critical point $\lambda_c^{(2)} = 1+\sqrt{6} \simeq 3.4495$.

2^n-Cycles

Continuing in the same fashion as above, we can search for 4-cycles, 8-cycles, and so on. Since the calculations become exceedingly harder, however, analytical results are harder to obtain, so one relies on the help of computers. In this way, we can discover that for $\lambda > 1+\sqrt{6}$, a 4-cycle emerges. This 4-cycle is attractive for $\lambda < \lambda_c^{(3)} \simeq 3.5441$ and is repulsive for $\lambda > \lambda_c^{(3)}$.

For $\lambda > \lambda_c^{(3)}$, an 8-cycle emerges. This 8-cycle is attractive for $\lambda < \lambda_c^{(4)} \simeq 3.5644$ and is repulsive for $\lambda > \lambda_c^{(4)}$.

For $\lambda > \lambda_c^{(4)}$, a 16-cycle emerges. This 16-cycle is attractive for $\lambda < \lambda_c^{(5)} \simeq 3.5688$ and is repulsive for $\lambda > \lambda_c^{(5)}$.

This procedure of **period doubling** continues indefinitely and creates the increasing sequence

$$\lambda_c^{(0)} < \lambda_c^{(1)} < \lambda_c^{(2)} < \lambda_c^{(3)} < \cdots \leq 4.$$

Since the sequence is bounded, it converges to a limit

$$\lim_{n\to\infty} \lambda_c^{(n)} \;=\; \lambda_c \simeq 3.5699.$$

Bifurcation Diagram

To prepare the reader for a more complicated plot, we have plotted the bifurcations of the logistic function f_λ up to 4-cycles in Figure 16.10.

The vertical axis in this figure gives the value of the fixed point (of any order) versus the value of the constant λ (horizontal axis). A change in color indicates a bifurcation, that is, the appearance of a new cycle with double period. The dashed lines indicate repulsive points and cycles.

The bifurcations for the cycles come faster and faster; that is, the **bifurcation distances** $\Delta\lambda_n = \lambda_c^{(n+1)} - \lambda_c^{(n)}$ decrease:

$$\Delta\lambda_1 > \Delta\lambda_2 > \Delta\lambda_3 > \ldots .$$

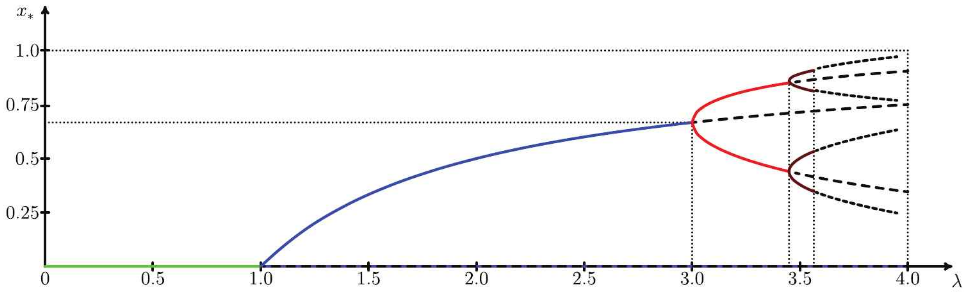


Figure 16.10: Partial bifurcation diagram for logistic function f_λ. When the dashed lines are omitted, this is called an **orbit diagram**.

Furthermore, Feigenbaum observed, the decrease is almost similar to that of a geometric progression: $\Delta\lambda_n \simeq \delta\,\Delta\lambda_{n+1}$. This approximate equation can become exact,

$$\lim_{n\to\infty} \frac{\Delta\lambda_n}{\Delta\lambda_{n+1}} = \delta,$$

where $\delta \simeq 4.669$. The number δ is thus known as **Feigenbaum's constant.**

Similarly, we can define a decreasing sequence of **bifurcation widths** Δw_n,

$$\Delta w_1 > \Delta w_2 > \Delta w_3 > \dots,$$

that measures the widths of the lower forks at the bifurcation points. Then it can be proved that this sequence is also similar to that of a geometric progression:

$$\lim_{n\to\infty} \frac{\Delta w_n}{\Delta w_{n+1}} = \alpha,$$

where $\alpha \simeq 2.5029$.

3-Cycles

Now, let's return to the 3-cycles. Consider the points a, b and c such that

$$f(a) = b, \quad f(b) = c, \quad f(c) = a.$$

Therefore, they must be solutions of the equation $f^3(x) = x$,

$$\lambda^3 x(1-x)\,[1-\lambda x(1-x)]\left\{1-\lambda^2 x(1-x)\,[1-\lambda x(1-x)]\right\} = x.$$

Since the fixed points of f will necessarily be roots of this equation, it should factorize in the form

$$x\left(x-1+\frac{1}{\lambda}\right) P(x) = 0,$$

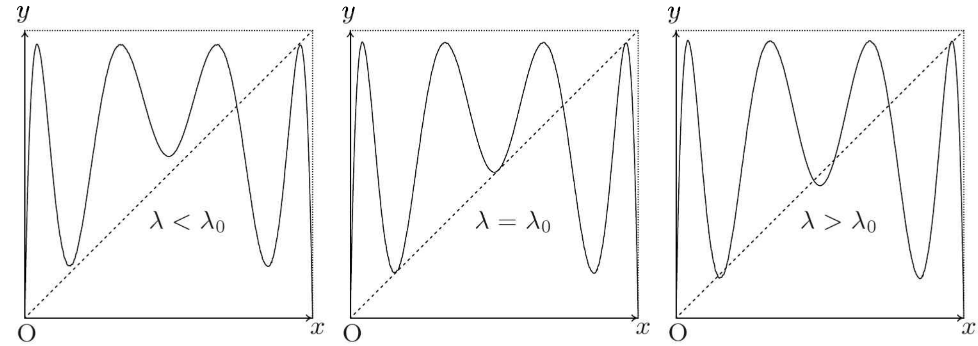


Figure 16.11: Varying the parameter λ to find the critical value λ_0 at which 3-cycles start to appear. At the critical value, f_λ^3 is tangent to the diagonal at three points. Below the critical value the curve pulls away from these points, while above the critical value each contact point bifurcates to create two new points.

where $P(x)$ is a polynomial of degree 6.

For some values λ, the polynomial $P(x)$ has no real roots; for some other values, it has six real roots. In the critical case, $\lambda = \lambda_0$ that separates these two domains, the polynomial should have three real double roots. (See Figure 16.11.) Therefore we write

$$P(x) = [(x-\alpha)(x-\beta)(x-\gamma)]^2 = (x^3 - Ax^2 + Bx - C)^2,$$

where

$$A = \alpha + \beta + \gamma, \quad B = \alpha\beta + \beta\gamma + \gamma\alpha, \quad C = \alpha\beta\gamma.$$

Comparing the two forms of the polynomial, we can find A, B, C as functions of λ:

$$\begin{aligned} A &= \frac{3\lambda+1}{2\lambda}, \\ B &= \frac{(3\lambda+1)(\lambda+3)}{8\lambda^2}, \\ C &= \frac{-\lambda^3 + 7\lambda^2 + 5\lambda + 5}{16\lambda^3}. \end{aligned}$$

In addition, we find the three conditions

$$g_1(\lambda)(\lambda^2 - 2\lambda - 7) = g_2(\lambda)(\lambda^2 - 2\lambda - 7) = g_3(\lambda)(\lambda^2 - 2\lambda - 7) = 0,$$

where g_1, g_2, g_3 are some functions for which the exact form is not important. Obviously, the last conditions admit the common solution $\lambda^2 - 2\lambda - 7 = 0$, or $\lambda = 1 \pm \sqrt{8}$. From the two values of λ thus obtained, only the value $\lambda_0 = 1 + \sqrt{8} \simeq 3.8284$ is acceptable. Given this value of λ, one can then find easily A, B, C and, from them after solving a cubic equation, the roots α, β, γ.

From Sharkovskii's theorem we know that, if there are 3-cycles, there will be cycles of any period. One can proceed to find them by the same techniques used for the 2^n-cycles.

Chaos

The word *chaos* is derived from the Greek word $\chi\acute{\alpha}o\varsigma$, meaning *absolute disorder.* It was introduced in the influential paper of Li and Yorke [54] entitled *Period Three Implies Chaos.* It really means the existence of nonperiodic orbits, for which Li and Yorke established a criterion. The term became very popular and today, *chaos theory*[11] is a term used even by laymen.

I will avoid precise definitions in this subsection. Instead I will use a loose, heuristic discussion. So, given the function $f : I \to I$, consider an uncountable set $S \subset I$ of points of f which satisfies the following two conditions:

- The splinter of a point belonging to S does not approach asymptotically any periodic orbit.
- The splinters of two points belonging to S do not approach each other asymptotically but they can come arbitrarily close.

Such a set S is called a **scrambled set**.

If a function f has such a **scrambled set**, it is called **Li–Yorke chaotic**. Then, Li and Yorke showed that the condition "f has a 3-cycle" is a sufficient one for f to be Li–Yorke chaotic. After Li and Yorke's work, their result was extended and generalized. In particular, one can actually show that it suffices for f to have a $(2m+1)2^n$-cycle for some m, n, $m \geq 1$ to be Li–Yorke chaotic. Furthermore, a function with 2^n-cycles of any order but no $(2m+1)2^n$-cycles with $m \neq 0$ can also be Li–Yorke chaotic.

It turns out that the Li–Yorke chaos provides only a limited aspect of the study of chaos in a system. The reason is that, if all scrambled sets are of measure zero, their probability of choosing an initial point in these sets is zero. Therefore, although the Li–Yorke chaos may be present, *it is unobservable* in such a scenario. Therefore, other criteria and definitions of chaos — in particular, criteria for *observable* chaos — which are not necessarily equivalent to Li–Yorke chaos have been proposed. Such a criterion is, for example, the *Lyapunov exponent* that measures sensitivity on the initial conditions. Given two nearby points x and y separated by distance d_0, under n iterations they will find themselves separated by distance d_n. If we write

$$d_n = d_0 \, e^{nL_n},$$

the number L_n is the **Lyapunov exponent** for the n-th iterate. A positive L implies that the two points will separate exponentially fast and it is considered a footprint of chaos.

[11] I have always found the term *chaos* very misleading and inappropriate since it implies lack of control and knowledge. Yet, the subject is about deterministic equations.

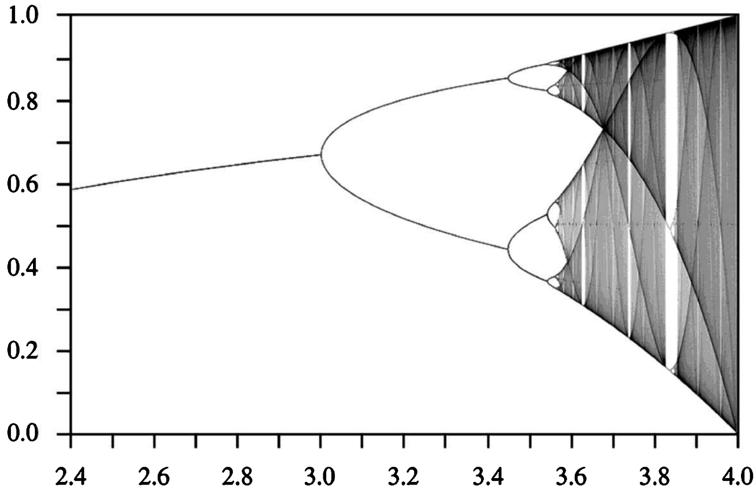


Figure 16.12: The orbit diagram of the logistic function $f_\lambda(x)$.

Exercise 16.24. *Assuming that the limit* $\lim_{n\to\infty} L_n$ *exists and it is* L, *show that*

$$L = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} \ln |f'(x_k)|,$$

and that for attractive cycles $L < 0$. □

Orbit Diagram, Self-Similarity, and Universality

Figure 16.12 shows the orbit diagram (that is, only the attracting points) of the logistic function. I will make some comments to reveal the mixture of order and chaos present in this diagram.

The most evident feature is the "unshaded" regions known as the **periodic windows**. There is a countable but infinite number of such windows and they have a finite width. From numerical analysis, it is known that the sum of the widths of all windows is about 10% of $4 - \lambda_c$.

It appears as if, inside the periodic windows, the splinter of any point x approaches a stable cycle (that is, we have "order"), although outside these windows the splinter behaves chaotically, taking any possible value in $[0, 1]$. However, the situation is more complicated (and exciting). Imagine that we magnify a part of the orbit diagram, say the upper branch from $\lambda = 3.2$ to $\lambda = 3.65$. Then the emerging picture would be indistinguishable from the original orbit diagram of Figure 16.12. As such, we can magnify again to find another identical part, and so on ad infinitum. We have uncovered an amazing self-similar pattern.

To be more concrete, let's, for example, look at the window that starts at $\lambda_0 \simeq 3.8284$. This is the *period-three window*, which is unique with this period and has the largest width among windows. Given a λ in the window and *almost* any point x, the splinter of x will approach asymptotically the 3-cycle. The word "almost" is used here since there are points x for which their splinters do not approach asymptotically the 3-cycle. This can be understood as follows. Inside the period-three window, there is a copy of the original orbit diagram. In the same way that period doubling led to the stable periodic orbits of period 2^n in the original orbit diagram, period doubling inside the period-three window leads to stable periodic orbits of periods $3 \cdot 2^n$. Then, inside the period-three window there is a period-nine window, which in the magnification will appear as a period-three window. And so on. The nonperiodic orbits that are found inside the period-three window provide a "small" Li–Yorke chaos in this window but they have measure zero and thus they cannot be observed.

Feigenbaum discovered that the qualitative features of the function (that is everything we have concluded except the numerical values) are *identical* for all functions (not only the logistic function) that are concave with a single maximum (known as **unimodal** functions). Examples of such functions are

$$\begin{aligned} f_a(x) &= a \sin(\pi x), \\ f_b(x) &= b - x^2. \end{aligned}$$

Although numerical values (of the bifurcation points, for instance) vary from function to function, the constants α and δ are the same for all such functions. This means that Feigenbaum's constant plays the same role in the theory of unimodal functions as Archimedes' constant does in the theory of the circle.

Based on the previous observations and a mathematical framework borrowed from physics[12], Feigenbaum built a beautiful mathematical formalism to explain the appearance of the constants δ and α. Unfortunately, any attempt to explain more on the topic would take us very deeply into mathematics and physics. Hopefully, the reader has appreciated the rich behavior of even simple functions and the deep connections among various areas of modern mathematics and physics.

Problem 16.25. *The functional equation*

$$\alpha f^2(x) = f(\alpha x), \quad \alpha \neq 0$$

is called the ***Feigenbaum equation****. It is encountered in Feigenbaum's theory for the explanation of the universal properties of unimodal functions. Find as*

[12]This framework is known as **renormalization** and it plays a fundamental role in the description and explanation of critical phenomena in condensed matter physics and the unification of forces in high energy physics.

many continuous solutions $f : \mathbb{R} \to \mathbb{R}$ as you can. (*For such solutions, see* [56].)

A Quiz

First attempt to solve the following problem:

Problem 16.26 (USA 1974)**.** *Let a, b, c denote three distinct integers and let $P(x)$ denote a polynomial with integer coefficients. Show that it is impossible that $P(a) = b$, $P(b) = c$, and $P(c) = a$.*

That is, $P(x)$ does not have a 3-cycle. In fact, you can try to explore the validity of the statement for any k-cycle.

Then, notice that the function

$$f(x) = 4x(1-x) = -4x^2 + 4x + 0$$

is polynomial with integer coefficients. Moreover, we have shown that it has cycles of any order. Was the problem incorrectly given before the publication of Li and Yorke's paper in 1975 and knowledge of Sharkovskii's theorem? Explain.

Chapter 17

Solving by Invariants and Linearization

In this chapter we consider functional equations of the form

$$f(h(x, f(x))) = H(x, f(x)), \tag{17.1}$$

where $f(x)$ is the unknown function and $h(u, v), H(u, v)$ are given functions of two variables. This equation includes as special cases all equations we have discussed in the last chapter. In particular, the Abel (16.3), Schröder (16.4), and Böttcher (16.5) equations are of the form

$$f(h(x)) = H(f(x)), \tag{17.2}$$

which is known as the **conjugacy equation**. When there is an injective function $f(x)$ such that the above equation is true, we call the functions h, H **conjugate**.

17.1 Constructing Solutions

Characteristic Function

Consider the graph $\Gamma_f \subset \mathbb{R}^2$ of the function $f : A \to A$ that solves equation (17.1):

$$\Gamma_f = \{(x, f(x)),\ x \in A\}.$$

If $x \in A$, then $h(x, f(x)) \in A$ too. So if the point $(x, f(x))$ belongs to the graph, so does the point $(h(x, f(x)), f(h(x, f(x))))$. Using (17.1), we can write the latter point as $(h(x, f(x)), H(x, f(x)))$.

With this motivation, we define a function $\chi : \mathbb{R}^2 \to \mathbb{R}^2$ such that $(x, y) \mapsto (x', y')$, where

$$x' = h(x, y), \quad y' = H(x, y).$$

Thus defined, this function maps Γ_f to itself and it is known as the **characteristic function** of the equation (17.1). For the conjugacy equation (17.2), the characteristic function is

$$(x, y) \to (h(x), H(y)).$$

Invariants

Definition 17.1. Given the function $\chi : \mathbb{R}^2 \to \mathbb{R}^2$ and a function $\Phi(u, v)$ of two variables, the function Φ is called an **invariant** of χ if $\Phi(u, v) = \Phi(\chi(u, v))$ for all $(u, v) \in \mathbb{R}^2$.

Definition 17.2. Given the function $\chi : \mathbb{R}^2 \to \mathbb{R}^2$ and a subset A of the plane, A is called an **invariant set** with respect to χ if $\chi(A) \subset A$.

Some immediate comments are:

1. If Φ is an invariant and c is a number, then the locus

$$A_c = \{(u, v) \in \mathbb{R}^2 : \Phi(u, v) = c\}$$

is an invariant set.

2. If $\Phi_1(u, v)$ and $\Phi_2(u, v)$ are two invariants of the same function χ, then $F(\Phi_1(u, v), \Phi_2(u, v))$ is also an invariant of χ for any function $F(u, v)$.
3. If A_1 and A_2 are two invariant sets of the same function χ, then their intersection $A_1 \cap A_2$ is an invariant of χ too.

There exists a theory for the construction of invariants, but it is beyond the goal of this book. We will rely on simple constructions only. For example, if the function χ is **cyclic**, that is, if $\chi^k = \mathrm{id}$ for some k, we can construct invariants as follows. Select any function $\phi(u, v)$ and define

$$\phi_i(u, v) = \phi(\chi^i(u, v)) \quad \text{for } i = 1, 2, \ldots, k-1.$$

Then

$$\Phi(u, v) = \phi(u, v)\phi_1(u, v)\phi_2(u, v) \ldots \phi_{k-1}(u, v)$$

is an invariant of χ. In some other cases, we can construct invariants by simple inspection.

Finding Solutions

Given a functional equation (17.1), we may write down its characteristic function and construct an invariant $\Phi(x, y)$. Then the set of points $\Phi(x, y) = c$ may give a solution to the functional equation that can be found by solving for $y = f(x)$. We say "may" since an equation may have an invariant but may not have a solution.

The procedure of using invariants to find solutions can be understood better with an example:

Example 17.3. Let's construct at least one solution of the equation

$$f(f(x) - 2x) = 2f(x) - 3x.$$

The corresponding characteristic function χ is given by

$$\chi(x, y) = (y - 2x,\, 2y - 3x).$$

It's easy to check that this function is cyclic: its (composition) square is the identity. So, let's pick the function $\phi(x, y) = x$, from which we can construct the invariant

$$\Phi(x, y) = x(y - 2x).$$

Equating $\Phi(x, y)$ to a number c results in $x(y - 2x) = c$, or

$$y = \frac{c}{x} + 2x.$$

By substitution into the original equation, we can verify that the function

$$f(x) = \frac{c}{x} + 2x$$

is indeed a solution. □

17.2 Linear Equations

A **linear functional equation** is one of the form

$$f(g(x)) = h(x)\, f(x) + F(x), \tag{17.3}$$

where $f : I \to I$ is the unknown function and $g, h, F : I \to I$ are known functions with $g(I) \subset I$. When $F(x) \not\equiv 0$, the equation is called **nonhomogeneous**; when $F(x) \equiv 0$,

$$f(g(x)) = h(x)\, f(x), \tag{17.4}$$

and it is called **homogeneous**.

The Schröder equation (16.4) is a homogeneous linear equation with $h(x)$ the constant function equal to λ. The Abel equation (16.3) is a nonhomogeneous linear equation with $h(x) = 1$ and $F(x) = a$.

A linear homogeneous equation has the important property that, given any two solutions, their sum is also a solution. The following theorem provides an alternative way to construct additional solutions to (17.4), given that one solution is already known.

Theorem 17.4. *Let $f_0(x)$ be a solution of* (17.4). *If $\Phi(x)$ is an invariant of its characteristic function that depends only on the first variable, then the product*

$$f(x) = \Omega(\Phi(x))\, f_0(x),$$

where Ω is an arbitrary function, is also a solution of (17.4).

Proof. Since $\Phi(x)$ is an invariant, we have $\Phi(g(x)) = \Phi(x)$ and therefore $\Omega(\Phi(g(x))) = \Omega(\Phi(x))$. So

$$\begin{aligned} f(g(x)) &= \Omega(\Phi(g(x)))\, f_0(g(x)) && \text{(definition of } f) \\ &= \Omega(\Phi(x))\, f_0(g(x)) && \text{(invariance of } \Phi) \\ &= \Omega(\Phi(x))\, h(x)\, f_0(x) && (f_0 \text{ solves (17.4))} \\ &= h(x)\, f(x), && \text{(definition of } f) \end{aligned}$$

as required. □

As with linear difference equations, the solutions of the nonhomogeneous equation (17.3) are obtained from those of the homogeneous one by the addition of a particular solution:

Theorem 17.5. *Let $f_p(x)$ be a particular solution of the nonhomogeneous equation* (17.3).

(a) *If $f_0(x)$ is a solution of the homogeneous equation* (17.4), *the function $f(x) = f_p(x) + f_0(x)$ is a solution of the nonhomogeneous equation* (17.3).

(b) *Every solution of the nonhomogeneous equation* (17.3) *can by obtained from $f_0(x)$ in the way stated in part* (a).

Proof. (a) It is easy to verify this statement by direct substitution on the right-hand side of (17.3):

$$\begin{aligned} h(x)\,(f_p(x) + f_0(x)) + F(x) &= [h(x)\, f_p(x) + F(x)] + [h(x)\, f_0(x)] \\ &= f_p(g(x)) + f_0(g(x)). \end{aligned}$$

(b) Let $f_p(x)$ be some solution of the nonhomogeneous equation (17.3) and let $f(x)$ be any other solution. Then

$$f(g(x)) = h(x)\, f(x) + F(x), \quad f_p(g(x)) = h(x)\, f_p(x) + F(x).$$

By subtracting the two equations we find

$$(f - f_p)(g(x)) = h(x)\,(f - f_p)(x);$$

that is, the difference $f - f_p$ must be a solution to the homogeneous equation (17.4). □

17.3 The Abel and Schröder Equations

The Abel and Schröder equations play central roles in the theory of functional equations without parameters. They are the simplest linear equations, and their solutions are used in the construction of solutions of more complicated equations. This will become obvious in the remainder of the chapter.

Usually, we choose $a = 1$ in Abel's equation:

$$f(g(x)) = f(x) + 1. \tag{17.5}$$

Assume that f, g are defined on some subset A of $\mathbb{R}$.

Theorem 17.6. *Abel's equation has a solution if and only if the function $g(x)$ has neither fixed points nor cycles. In that case, it has infinitely many solutions.*

Proof. Setting $g(x)$ in place of x, we find that

$$f(g^2(x)) = f(g(x)) + 1 = f(x) + 2.$$

Repeating this procedure, we get

$$f(g^n(x)) = f(x) + n \quad \text{for } n \in \mathbb{N}^*. \tag{17.6}$$

If g has a cycle of order n ($n = 1$ being the case of a fixed point) for the point x_0, then $f(g^n(x_0)) = f(x_0)$ and we thus arrive at the contradiction $0 = n$.

Now consider the orbits generated by the iteration of g. (See Section 16.2.) For each orbit O_a we select a representative $a \in O_a$. Also, for simplicity, let $b = f(a)$. For any point $a' \in O_a$, there exist $n, n' \in \mathbb{N}^*$ such that $g^n(a) = g^{n'}(a')$. Then, using equation (17.6), we get

$$f(g^{n'}(a')) = f(g^n(a)),$$

which implies $f(a') = b + n - n'$. That is, the function f is known on all points of the orbit if it is known at one point. From this we conclude that the function f is fully defined when one value b is given for one point a of each of the orbits generated by g. Since there are an infinite number of ways to assign $a \mapsto b$, the theorem is proved. $\square$

Now let's turn our attention to the Schröder equation (16.4). We see immediately that for $\lambda > 0$, $\lambda \neq 1$, if we exponentiate Abel's equation (17.5)

$$\lambda^{f(g(x))} = \lambda\, \lambda^{f(x)},$$

any solution $f(x)$ of Abel's equation gives a solution $\lambda^{f(x)}$ for the Schröder equation too. Of course, there may be additional solutions not obtained in this way and, obviously, this construction does not provide any solution for $\lambda < 0$ (if there are such solutions).

17.4 Linearization

Imagine that a (nonlinear) function $g(x)$ is given. Consider also that we can find a continuous monotonic solution $\sigma(x)$ of the Schröder equation

$$\sigma(g(x)) = \lambda\,\sigma(x) \quad \text{for } \lambda \neq -1, 0, 1.$$

We can use the above data to simplify equation (17.3). We introduce the change of variables

$$y = \sigma(x), \quad \tilde{f}(y) = f(x).$$

Since σ is monotonic, it has an inverse σ^{-1} and $x = \sigma^{-1}(y)$. Also, if we replace x by $g(x)$ in the equation $\tilde{f}(\sigma(x)) = f(x)$, we get $\tilde{f}(\sigma(g(x))) = f(g(x))$, hence $\tilde{f}(\lambda\sigma(x)) = f(g(x))$, hence $\tilde{f}(\lambda y) = f(g(x))$. Substituting these results into (17.3), we find

$$\tilde{f}(\lambda y) = h(\sigma^{-1}(y))\,\tilde{f}(y) + F(\sigma^{-1}(y)).$$

We can make this equation more appealing by defining new functions

$$\tilde{h}(y) = h(\sigma^{-1}(y)), \quad \tilde{F}(y) = F(\sigma^{-1}(y)).$$

Then,

$$\tilde{f}(\lambda y) = \tilde{h}(y)\,\tilde{f}(y) + \tilde{F}(y). \tag{17.7}$$

This has the same form as (17.3), but a simpler function $\tilde{g}(y) = \lambda y$ appears on the left-hand side. Thus it may be simpler to solve than the original equation.

Similarly, if a monotonic solution $\sigma(x)$ of Abel's equation is used, we can make a similar change of variables in (17.3) to arrive at a simpler equation:

$$\tilde{f}(y+1) = \tilde{h}(y)\,\tilde{f}(y) + \tilde{F}(y), \tag{17.8}$$

with $\tilde{g}(y) = y + 1$.

Example 17.7. Consider the equation

$$f((x-a)^2 + a) = b\,f(x),$$

with a, b constant and $b > 0$.

Instead of x, it is more convenient to use $X = x - a$:

$$f(X^2 + a) = b\,f(X + a).$$

The function $g(X) = X^2 + a$ is quadratic. So we consider a monotonic solution of the Schröder equation

$$\sigma(X^2) = 2\sigma(X),$$

with solution $\sigma(X) = \log X$. We introduce the change of variables

$$y = \ln X, \quad \tilde{f}(y) = f(X + a).$$

Since $X = e^y$, we see that $\tilde{f}(y) = f(e^y + a)$ and the original equation transforms to

$$\tilde{f}(2y) = b\,\tilde{f}(y).$$

This equation has an obvious solution:

$$\tilde{f}(y) = y^{\log_2 b}.$$

Returning to the original notation, we obtain

$$f(x) = (\ln(x - a))^{\log_2 b}. \qquad \square$$

It is tempting to state that this method of reduction to a simpler equation, known as *linearization*, can be applied to the more general functional equation

$$F(x, f(x), f(g(x))) = 0,$$

where F, g are known functions and f is the unknown one. Assume that one can find bijective functions σ and τ such that

$$\sigma(g(x)) = \tilde{g}(\sigma(x)),$$
$$\tau(F(x, y, z)) = \tilde{F}(x, \tau(y), \tau(z)).$$

The first equation is the conjugacy equation (17.2) and it replaces the function g by its conjugate $\tilde{g}$ which is, hopefully, a simpler function. The second equation is a generalization of the conjugacy equation to a function of more variables — notice the way the arguments are treated based on their interpretation. Upon the change of variables

$$x' = \sigma(x), \quad \tilde{f} = \tau \circ f \circ \sigma^{-1},$$

the original functional equation transforms to

$$\tilde{F}(\sigma^{-1}(x'), \tilde{f}(x'), \tilde{f}(\tilde{g}(x'))) = 0,$$

which might be easier to solve. This equation is a **linearization** of the original equation if $\tilde{F}, \tilde{g}$ are linear functions.

17.5 Solved Problems

The two problems that follow are taken from [40] and they complete, more or less, the necessary ideas the reader should know about the topic at this introductory level.

Problem 17.8. *Find at least one solution of the equation*

$$f(x^2 - 2f(x)) = f(x)^2.$$

Solution. The characteristic function χ of the equation is given by

$$\chi(x, y) = (x^2 - 2y,\, y^2).$$

We write down the function

$$\Phi(x, y) = \frac{x^2}{y}.$$

This function is not invariant under χ, but transforms as

$$\Phi(\chi(x, y)) = \frac{(x^2 - 2y)^2}{y^2} = (\Phi(x, y) - 2)^2.$$

This leads us to look for the fixed points of the function

$$g(x) = (x - 2)^2$$

which are

$$x^*_\pm = 1, 4.$$

This allows us to define two invariant sets:

$$A_+ = \{(x, y) : \Phi(x, y) = 4\} = \{(x, y) : x^2/y = 4\},$$
$$A_- = \{(x, y) : \Phi(x, y) = 1\} = \{(x, y) : x^2/y = 1\}.$$

The set A_+ is the graph of the function

$$y = \frac{x^2}{4},$$

while the set A_- is the graph of the function

$$y = x^2.$$

It can be verified by direct substitution that both functions are solutions of the given functional equation.

The previous construction brings up an interesting question: if a function $\Phi(x, y)$ is not invariant but has cycles, could we use them to find solutions? In particular, $g(x) = (x-2)^2$ has a 2-cycle. Can it lead to a solution of the given equation? As we demonstrate below, the answer is affirmative.

To find the 2-cycle, we search for fixed points of $g^2(x) = [(x-2)^2 - 2]^2$:

$$g^2(x) = x.$$

Let a_* be a solution of this equation. (Show that there is such a point.) Then we define $b_* = g(a_*) = (a_* - 2)^2$. The two numbers a_* and b_* are the distinct points of a 2-cycle. Now consider the set

$$A = \{(x, y) : (\Phi(x, y) - a_*)(\Phi(x, y) - b_*) = 0\}.$$

This defines an invariant set. To see this, if a point belongs to A, then $\Phi(x, y)$ takes one of the two values of the 2-cycle, say a_*. So $\Phi(x, y) = a_*$ and

$$\Phi(\chi(x, y)) - b_* = (\Phi(x, y) - 2)^2 - b_* = (a_* - 2)^2 - b_* = 0;$$

that is, $\Phi(\chi(x, y))$ also belongs to A. We thus find the solution

$$f(x) = \begin{cases} x^2/a_* & \text{if } x \geq 0, \\ x^2/b_* & \text{if } x < 0, \end{cases}$$

or the solution

$$f(x) = \begin{cases} x^2/b_* & \text{if } x \geq 0, \\ x^2/a_* & \text{if } x < 0. \end{cases} \qquad \square$$

Problem 17.9. *Find a solution $f : (1, +\infty) \to \mathbb{R}_+^*$ of the functional equation*

$$f(x^a) = \lambda\, x^b\, f(x),$$

where $a, \lambda \in \mathbb{R}_+^*$ and $b \in \mathbb{R}$.

Solution. Linearization: Consider the Schröder equation

$$\sigma(x^a) = a\, \sigma(x),$$

which admits the solution $\sigma(x) = \ln x$. Upon the change of variables

$$x' = \ln x, \quad \tilde{f}(x') = f(x),$$

the original functional equation transforms to

$$\tilde{f}(ax') = \lambda\, e^{bx'}\, \tilde{f}(x').$$

To solve this new equation, we try the method of invariants in a modified version that can often be useful.

Solution by invariants: We write the function $\tilde{f}(x')$ as the ratio of two other functions:

$$\tilde{f}(x') = \frac{A(x')}{B(x')}.$$

Then

$$\frac{A(ax')}{B(ax')} = \frac{\lambda}{e^{-bx'}}\frac{A(x')}{B(x')}.$$

Finding a solution $\tilde{f}(x')$ can now be reduced to finding solutions to two simpler equations:

$$A(ax') = \lambda\, A(x'), \tag{17.9}$$

$$B(ax') = e^{-bx'}\, B(x'). \tag{17.10}$$

We solve each equation using the method of invariants.

Equation for A: The characteristic function of equation (17.9) is

$$\chi(x', y) = (ax', \lambda\, y) =: (x'', y').$$

Instead of using χ directly, we modify it by applying a suitable function. So, we apply the logarithmic function to find

$$\begin{aligned} \ln x'' &= \ln a + \ln x', \\ \ln y' &= \ln \lambda + \ln y. \end{aligned}$$

We now see that the functions

$$\begin{aligned} \Phi_1(x', y) &= \frac{\ln x'}{\ln a} - \frac{\ln y}{\ln \lambda}, \\ \Phi_2(x', y) &= \Omega\left(\frac{\ln x'}{\ln a}\right), \end{aligned}$$

where Ω is a periodic function with period 1, are invariants. Setting $\Phi_1(x, y) = 0$, we find the solution

$$A_0(x') = (x')^{\log_a \lambda}.$$

We can combine this solution with the second invariant to construct other solutions of the equation according to Theorem 17.4:

$$A(x') = (x')^{\log_a \lambda}\,\Omega\left(\frac{\ln x'}{\ln a}\right).$$

Equation for B: We will solve equation (17.10) along similar lines. Its characteristic function is

$$\chi(x', y) = (a\, x', e^{-bx'}\, y) =: (x'', y').$$

This we modify as follows:

$$\begin{aligned} x'' &= a\,x', \\ y' &= -b\,x' + \ln y. \end{aligned}$$

From this, we can see that

$$\Phi(x', y) \;=\; \ln y + \frac{b}{a-1}\,x'$$

is an invariant. Setting $\Phi(x', y) = 0$ gives the solution

$$B(x') \;=\; e^{bx'/(1-a)}.$$

A solution to the original equation: Using the functions A and B, we now have a solution of the linearized equation:

$$\tilde{f}(x') \;=\; (x')^{\log_a \lambda}\, e^{bx'/(1-a)}\, \Omega(\log_a x').$$

Returning to the original notation and the initial problem, the function

$$f(x) \;=\; (\ln x)^{\log_a \lambda}\, x^{b/(a-1)}\, \Omega(\log_a \ln x),$$

where Ω, a periodic function with period 1, is a solution of the given functional equation. □

Chapter 18

More on Fixed Points

Although we defined the concept of a fixed point for a function in one variable, the concept is much deeper and permeates all topics in mathematics. The concept really belongs to the area of topology[1] — a branch of mathematics that originated with the work of Poincaré dealing with and characterizing the geometric properties of any set.

Fixed point theorems are used as a tool to prove existence theorems in differential, integral, and other equations. They are used in dynamical systems and their applications to game theory, economics, biology, etc.

The reader may look to Shashkin's book [37] for a nice and short review of the use of fixed points. In this chapter, we only demonstrate how fixed points can be used to solve functional equations that contain parameters.

18.1 Solved Problems

Problem 18.1 (IMO 1983). *Find all functions f defined on the set of positive real numbers which take positive real values and satisfy the conditions*

(a) $f(xf(y)) = y\,f(x)$, *for all positive* x, y;

(b) $f(x) \to 0$ *as* $x \to +\infty$.

Solution. The function $f(x) \equiv 0$ is an obvious solution. To search for other solutions, assume the existence of $x_0 \in \mathbb{R}_+^*$ for which $f(x_0) \neq 0$.

Setting $y = x$ in condition (a) gives $f(xf(x)) = xf(x)$, so $xf(x)$ is a fixed point of f for all $x \in \mathbb{R}_+^*$. To finish the problem, we must therefore determine the set F of fixed points of f.

We observe that the set F has the following properties:

---

[1]A beautiful elementary introduction to topology may be found in [20]; Chapter 10 contains a discussion of fixed point theorems in topology.

- If $a, b \in F$, then $ab \in F$. To see this, we set $x = a$ and $y = b$ in condition (a); then $f(af(b)) = bf(a)$, hence $f(ab) = ab$, since $f(a) = a$ and $f(b) = b$.

- If $a \in F$, then $a^{-1} \in F$. To see this, we set $x = a^{-1}$ and $y = a$ in condition (a); then $f(a^{-1}f(a)) = af(a^{-1})$, hence $f(1) = af(a^{-1})$, hence $f(a^{-1}) = a^{-1}$.

That is, F is a multiplicative subgroup of $\mathbb{R}_+^*$. As such, it contains 1.

We shall show that 1 is the only one. Let $a \in F$ such that $a \neq 1$. Thus all powers a^n of a are also fixed points:

$$f(a^n) = a^n.$$

Without loss of generality, we can assume $a > 1$ (otherwise we replace it with a^{-1}). In the limit $n \to \infty$, the last equation gives

$$\lim_{x \to +\infty} f(x) = +\infty,$$

which contradicts condition (b). Therefore, F cannot have any fixed point distinct from 1.

We conclude that $xf(x) = 1$, so

$$f(x) = \frac{1}{x}. \qquad \square$$

Problem 18.2 (IMO 1994). *Let S be the set of real numbers strictly greater than -1. Find all functions $f : S \to S$ satisfying the conditions*

(a) *$f(x + f(y) + xf(y)) = y + f(x) + yf(x)$ for all $x, y \in S$;*

(b) *$f(x)/x$ is strictly increasing on each of the intervals $-1 < x < 0$ and $0 < x$.*

Solution. Setting $y = x$ in condition (a) gives

$$f(x + f(x) + xf(x)) = x + f(x) + xf(x), \tag{18.1}$$

so $x + f(x) + xf(x)$ is a fixed point for all $x \in S$. To finish the problem, we must therefore determine the set F of fixed points of f.

For any x, we can choose the value of y so that the right-hand side of the defining equation, $y(1 + f(x)) + f(x)$, vanishes. So, let x_0 such that $f(x_0) = 0$. Then, in the defining equation, we set $x = y = x_0$. This gives $x_0 = 0$, showing that 0 is a fixed point of f. We show that it is the only one.

Consider an arbitrary fixed point $a \neq 0$. According to equation (18.1), the points

$$\begin{aligned} a_1 &= a + f(a) + a\,f(a) \;=\; (a+1)^2 - 1, \\ a_2 &= a_1 + f(a_1) + a_1\,f(a_1) \;=\; (a_1+1)^2 - 1, \\ &\dots \end{aligned}$$

are also fixed points and, in turn, the sequence

$$\frac{f(a)}{a}, \quad \frac{f(a_1)}{a_1}, \quad \frac{f(a_2)}{a_2}, \quad \dots,$$

is the constant sequence $1,\ 1,\ 1, \dots$. However, we notice that if $a \in (-1, 0)$, then $a_1 \in (-1, 0)$ and so on; if $a \in (0, +\infty)$ then $a_1 \in (0, +\infty)$ and so on. In other words, the sequence $\{f(a_n)/a_n\}$ contradicts condition (b) and therefore we cannot have a fixed point in $(-1, 0) \cup (0, +\infty)$.

Hence $x + f(x) + xf(x) = 0$, which implies

$$f(x) = -\frac{x}{x+1}. \qquad \square$$

Problem 18.3 (IMO 1996). *Find all functions f defined on $\mathbb{N}$ and taking their values in $\mathbb{N}$ such that*

$$f(m + f(n)) = f(f(m)) + f(n) \quad \textit{for all } m, n \textit{ in } \mathbb{N}.$$

Solution. The function $f(x) \equiv 0$ is an obvious solution. To search for other solutions, assume the existence of $n_0 \in \mathbb{N}$ for which $f(n_0) \neq 0$.

Setting $m = n = 0$ in the defining equation, we find $f(f(0)) = f(f(0)) + f(0)$, so $f(0) = 0$. Then, setting $m = 0$ in the defining equation, we find

$$f(f(n)) = f(n). \tag{18.2}$$

Using this result, we may write the original functional equation as

$$f(m + f(n)) = f(m) + f(n). \tag{18.3}$$

Equation (18.2) says that $N = f(n)$ is a fixed point of f for all $n \in \mathbb{N}$. The set F of fixed points of F contains at least 0 and $f(n_0)$. Let's denote by N_0 the smallest nonzero integer in F. Now, let a, b be two fixed points $f(a) = a$, $f(b) = b$. Setting $m = a, n = b$ in (18.3), we find $f(a + b) = a + b$; the sum of any fixed points is also a fixed point. Therefore $k\,N_0$, for any $k \in \mathbb{N}$, is a fixed point.

If N is any fixed point, we use Euclid's algorithm to write $N = q\,N_0 + r$, with $0 \le r < N_0$. Then

$$\begin{aligned} q\,N_0 + r \;=\; N \;=\; f(N) &= f(r + qN_0) \\ &= f(r + f(qN_0)) \quad (qN_0 \text{ is a fixed point}) \\ &= f(r) + f(qN_0) \quad (\text{using equation (18.3)}) \\ &= f(r) + q\,N_0 \qquad (qN_0 \text{ is a fixed point}), \end{aligned}$$

or $f(r) = r$. The residue r is a fixed point smaller than N_0. Therefore, it can only be $r = 0$; that is, $N = q\,N_0$. The fixed points are precisely the multiples of N_0:

$$F = \{\; k\,N_0,\;\; k \in \mathbb{N} \;\}.$$

Since $f(n)$ is a fixed point for any n, we conclude that $f(n)$ is a multiple of N_0 for any n. To define which multiple, we use again Euclid's algorithm to write $n = q\,N_0 + r$, $0 \le r < N_0$. Then

$$f(n) \;=\; f(q\,N_0 + r) \;=\; f(q\,N_0) + f(r) \;=\; q\,N_0 + f(r).$$

$f(r)$ must be a multiple of N_0:

$$f(0) = 0, \qquad f(r) = n_r\,N_0 \quad \text{for } r = 1, \ldots, N_0 - 1,$$

where the n_r's are $N_0 - 1$ undetermined numbers.

Therefore, there is an infinite number of nontrivial solutions to the given functional equation. Each solution is characterized by a nonzero natural number N_0 and a function $\tilde{f} : \{0, 1, 2, \ldots, N_0 - 1\} \to N_0\mathbb{N}$ such that $\tilde{f}(0) = 0$. $\square$

Part VI

GETTING ADDITIONAL EXPERIENCE

Chapter 19

Miscellaneous Problems

In this chapter, we study miscellaneous problems that belong in the following categories:

- Functional equations involving integrals (integral functional equations).
- Problems that are not intrinsically about functional relations, but can be solved by reduction to a functional relation.
- An assortment of problems that do not fit well with any of the ideas previously discussed.

19.1 Integral Functional Equations

For problems that include integral functional equations, the (first) fundamental theorem of integral calculus [12] often proves useful.

Theorem 19.1. *For an integrable function f on $[a, b]$, define*

$$F(x) = \int_a^x f(t)dt,$$

if $x \in [a, b]$. Then $F(x)$ is continuous on $[a, b]$. Moreover, if f is continuous at x, the derivative $F'(x)$ exists. In particular,

$$\frac{d}{dx}\int_a^x f(t)\,dt = f(x).$$

In the solution of the following problems we use this result without explicit reference, assuming that the function f has the required properties. Sometimes this is easily seen by the way the functional equation is written (as in the case of the next problem) but sometimes it must added in the assumptions of the problem (as in the case of Problem 19.3).

Problem 19.2. *Find the function* $f : \mathbb{R} \to \mathbb{R}$ *such that*

$$f(x) = \frac{1}{2\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} f(t)\,dt \quad \textit{for all } \varepsilon > 0.$$

Solution. First we write

$$f(x) = \frac{1}{2\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} f(t)\,dt = \frac{1}{2\varepsilon} \left(\int_0^{x+\varepsilon} f(t)\,dt - \int_0^{x-\varepsilon} f(t)\,dt \right).$$

Differentiating with respect to x, we find that

$$f'(x) = \frac{1}{2\varepsilon} \left(f(x+\varepsilon) - f(x-\varepsilon) \right). \tag{19.1}$$

Since the first derivative exists, we see that the second derivative also exists:

$$f''(x) = \frac{1}{2\varepsilon} \left(f'(x+\varepsilon) - f'(x-\varepsilon) \right).$$

Differentiating (19.1) with respect to ε yields

$$0 = \frac{\left(f'(x+\varepsilon) + f'(x-\varepsilon)\right)\varepsilon - \left(f(x+\varepsilon) - f(x-\varepsilon)\right)}{2\varepsilon^2}.$$

Using the previous equation and (19.1), we get $f'(x+\varepsilon) + f'(x-\varepsilon) = 2\,f'(x)$. Differentiating with respect to ε once more, we get

$$f''(x+\varepsilon) = f''(x-\varepsilon).$$

Setting $x + \varepsilon$ for x and then $x = 0$, we obtain the answer:

$$f''(2\varepsilon) = f''(0) \equiv a, \quad \text{hence} \quad f(y) = \frac{a}{2}\,y^2 + b\,y + c,$$

where $y = 2\varepsilon > 0$. Substituting this result in the defining equation, we find that $a = 0$. One can also show that the solution is valid when $y \leq 0$. □

Problem 19.3 (Putnam 1940). *Find* $f(x)$ *such that*

$$\int f(x)^n\,dx = \left(\int f(x)\,dx \right)^n. \tag{19.2}$$

Comment. The statement of this problem is incomplete. Besides the conditions of the function f (for which we assume that $f : \mathbb{R} \to \mathbb{R}$ is continuous and integrable, and similarly for its n-th power), no information is given about the constant n. We shall thus assume that n is an integer different from 1.

Solution. We set

$$F(x) = \int^x f(t)\,dt, \qquad \text{so} \quad F'(x) = f(x), \tag{19.3}$$

$$G(x) = \int^x f(t)^n\,dt, \qquad \text{so} \quad G'(x) = f(x)^n = F'(x)^n. \tag{19.4}$$

The given equation (19.2) thus becomes $G(x) = F(x)^n$. Upon differentiation, $G'(x) = n\,F(x)^{n-1}\,F'(x)$. Using (19.4), we then get

$$F'(x)^n = n\,F(x)^{n-1}\,F'(x);$$

that is,

$$F'(x)\,\left(F'(x)^{n-1} - n\,F(x)^{n-1}\right) = 0.$$

This last equation implies that, for any x,

$$F'(x) = f(x) = 0 \quad \text{or} \quad F'(x)^{n-1} - n\,F(x)^{n-1} = 0.$$

The function $F'(x) = f(x) \equiv 0$ is obviously a solution. To find others, assume there is x_0 such that $F'(x_0) = f(x_0) \neq 0$. Since $f(x)$ is continuous, there exists a neighborhood $(x_0-\delta,\, x_0+\delta)$ of x_0 (with $\delta > 0$ to be determined) such that $F'(x) = f(x) \neq 0$ for all points x in the neighborhood. Therefore, for all points in the neighborhood,

$$F'(x)^{n-1} = n\,F(x)^{n-1}.$$

If $n = 2k$, then

$$F'(x) = \sqrt[2k-1]{2k}\,F(x),$$

and if $n = 2k+1$, then

$$F'(x) = \pm\sqrt[2k]{2k+1}\,F(x).$$

From the last two equations we can determine the function F (and thus f) easily:

$$F(x) = c\,e^{\sqrt[2k-1]{2k}\,x} \;\Rightarrow\; f(x) = c\,\sqrt[2k-1]{2k}\,e^{\sqrt[2k-1]{2k}\,x}$$

in the even case and

$$F(x) = c\,e^{\pm\sqrt[2k]{2k+1}\,x} \;\Rightarrow\; f(x) = \pm c\,\sqrt[2k]{2k+1}\,e^{\pm\sqrt[2k]{2k+1}\,x}$$

in the odd case. We now see that the radius of the neighborhood can be made arbitrarily large: the functions are defined in all of $\mathbb{R}$. □

Problem 19.4 (Putnam 1963). *Find every real-valued function satisfying the condition*

$$\int_0^x f(t)\,dt = x\sqrt{f(0)f(x)} \quad \textit{for all } x \geq 0. \tag{19.5}$$

Solution. Again we make the assumption that f is continuous. If $f(0) = 0$ we have

$$\int_0^x f(t)dt = 0 \quad \text{for all } x \geq 0,$$

so f vanishes identically. If $f(0) < 0$, we have $f(x) < 0$ for $x \in [0, \varepsilon]$ by continuity for some $\varepsilon > 0$; then (19.5) gives a contradiction for $x = \varepsilon$ (left-hand side negative, right-hand side positive). Therefore, $f(0) > 0$.

Differentiating equation (19.5), we get

$$\frac{d}{dx}\int_0^x f(t)dt = \frac{d}{dx}\left(x\sqrt{f(0)f(x)}\right),$$

and the chain rule and the fundamental theorem of calculus then give

$$f(x) = \sqrt{f(0)f(x)} + x\,\frac{d}{dx}\sqrt{f(0)f(x)};$$

hence

$$\frac{f(x)}{f(0)} = \sqrt{\frac{f(x)}{f(0)}} + x\,\frac{d}{dx}\sqrt{\frac{f(x)}{f(0)}}.$$

In terms of the function

$$g(x) = \sqrt{\frac{f(x)}{f(0)}},$$

this differential equation becomes

$$g(x)^2 = g(x) + x\,\frac{dg(x)}{dx},$$

which can be solved by separation of variables:

$$\int \frac{dx}{x} = \int \frac{dg}{g(g-1)} = \int dg\left(\frac{1}{g-1} - \frac{1}{g}\right),$$

or

$$\ln x = \ln|g-1| - \ln|g| - \ln c,$$

where c is a positive constant. The last equation is equivalent to

$$c\,x = \left|1 - \frac{1}{g}\right|.$$

When the expression inside the absolute value is positive, we get

$$g(x) = \frac{1}{1-cx}.$$

In this case, the function is defined for $x \in [0, 1/c)$. When the expression inside the absolute value is negative, we get

$$g(x) = \frac{1}{1+cx}.$$

In this case the function is defined for $x \in [0, +\infty)$.

Therefore, the solutions of the differential equation are

$$\begin{aligned} f(x) &= \frac{a}{(1-cx)^2} \quad \text{for } x \in [0, 1/c), \quad \text{with } a > 0,\ c > 0, \\ f(x) &= \frac{a}{(1+cx)^2} \quad \text{for } x \in [0, +\infty), \quad \text{with } a \geq 0,\ c \geq 0. \end{aligned}$$ □

Problem 19.5. *Find all integrable functions on* $(0, a)$ *satisfying*

$$0 \leq f(x) \leq K \int_0^x f(t)\, dt$$

for all $0 \leq x \leq a$ *with* $K > 0$.

Solution. We set

$$F(x) = \int_0^x f(t)\, dt \geq 0,$$

and thus $F'(x) = f(x)$. Also $F(0) = 0$. From the given inequality we have

$$F'(x) - K\, F(x) \leq 0 \;\Rightarrow\; \frac{d}{dx}(F(x)e^{-Kx}) \leq 0.$$

Integrating this inequality over the interval $(0, x)$ yields

$$F(x)e^{-Kx} \leq 0.$$

Therefore $F(x) = 0$, from which follows that $f(x) = 0$. □

19.2 Problems Solved by Functional Relations

Perhaps the reader has met continued radicals and continued fractions. We saw one in Problem 13.21. Two additional examples are:

$$\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\dots}}}}}}$$

and

$$\cfrac{1}{2+\cfrac{1}{2+\cfrac{1}{2+\cfrac{1}{2+\cfrac{1}{2+\cfrac{1}{2+\cdots}}}}}}$$

Such expressions, for which the terms have a periodic pattern, are easy to handle. If we call them A and B respectively, we notice that they require

$$A = \sqrt{2+A},$$

and

$$B = \frac{1}{2+B},$$

which are easy to solve to find

$$A = 2 \quad \text{for } B = \sqrt{2}-1.$$

Continued radicals and continued fractions that are not periodic are harder to handle. Also, sometimes, even if an expression is periodic, the algebraic equation to solve might not be very "friendly', so we may wish to apply different methods (for such a case, see Problem 19.6).

Problem 19.6 (IMO 1969 Longlist). *Prove that for $a > b^2$,*

$$\sqrt{a-b\sqrt{a+b\sqrt{a-b\sqrt{a+b\sqrt{a-\dots}}}}} = \sqrt{a-\frac{3b^2}{4}}-\frac{b}{2}.$$

Comment. The continued radical of the problem is obviously periodic. If we denote its value by C, then

$$C = \sqrt{a-b\sqrt{a+bC}}.$$

On the right-hand side, C appears under two nested radicals. If we remove them, we will end up with a quartic algebraic equation. With little effort (and the known procedure for quartic equations) it can be solved, but I did not try to do so. The solution presented below uses a different approach.

Solution. We define

$$f(x) \equiv \sqrt{a - x\sqrt{a + x\sqrt{a - x\sqrt{a + x\sqrt{a - \ldots}}}}},$$

where the domain will be set below. Obviously $f(x) > 0$ with $f(0) = \sqrt{a}$. From the definition of $f(x)$,

$$f(-x) = \sqrt{a + x\sqrt{a - x\sqrt{a + x\sqrt{a - x\sqrt{a + \ldots}}}}},$$

and thus

$$f(-x) = \sqrt{a + x\,f(x)},$$

or

$$f(-x)^2 - x\,f(x) = a.$$

Substituting $-x$ for x, we also have

$$f(x)^2 + x\,f(-x) = a.$$

Adding and subtracting the last two equations gives

$$f(-x)^2 + f(x)^2 + x\,[f(-x) - f(x)] = 2a, \tag{19.6}$$
$$f(x)^2 - f(-x)^2 + x\,[f(-x) + f(x)] = 0. \tag{19.7}$$

Equation (19.7) can be simplified as follows. First we write it in the form

$$f(x)^2 - f(-x)^2 + x\,[f(-x) + f(x)] = 0;$$

factoring the left-hand side and grouping gives

$$[f(x) - f(-x) + x]\,[f(x) + f(-x)] = 0.$$

Since the function f is positive for all values in the domain of its definition, $f(x) + f(-x) > 0$ for all x. Therefore, we must necessarily have

$$f(x) - f(-x) + x = 0.$$

Squaring this equation and subtracting from (19.6) also gives

$$-f(-x)\,f(x) \;=\; -a + x^2.$$

Since the left-hand side is negative, we must restrict the domain of f to those values of x which satisfy $x^2 < a$. Looking at the last two equations, we interpret them as Viète's formulæ for the roots $f(x)$ and $-f(-x)$ of the quadratic polynomial

$$y^2 + x\,y + (x^2 - a) \;=\; 0,$$

which, of course, we can solve easily to find

$$f(x) \;=\; -\frac{x}{2} + \sqrt{a - \frac{3x^2}{4}}. \qquad \square$$

Problem 19.7 (Putnam 1966). *Prove*

$$\sqrt{1 + 2\sqrt{1 + 3\sqrt{1 + 4\sqrt{1 + \ldots}}}} \;=\; 3.$$

Comment. This problem was first posed by Ramanujan in 1911 ([4], Question No. 289, April 1911). Problems for the Putman mathematical competition are not always original. Often they are known interesting ideas encountered in various areas of mathematics. To learn how Putnam problems are written, read the article [57] by a mathematician who served on the Putnam Problems Subcommittee.

Solution. We define

$$f(x) \;\equiv\; \sqrt{1 + x\sqrt{1 + (x+1)\sqrt{1 + (x+2)\sqrt{1 + \ldots}}}}.$$

In this definition, a priori it seems that x may be any real number, but this is not the case. We leave it as an exercise to our reader to discuss what happens for all values of x; we shall focus on $x \geq 1$.

From the definition,

$$f(x+1) \;\equiv\; \sqrt{1 + (x+1)\sqrt{1 + (x+2)\sqrt{1 + (x+3)\sqrt{1 + \ldots}}}},$$

and therefore the function $f(x)$ satisfies the functional relation

$$f(x) \;=\; \sqrt{1 + x\,f(x+1)}.$$

This may be written in the form

$$f(x+1) = \frac{f(x)^2 - 1}{x}. \tag{19.8}$$

We shall use this relation shortly.

First, we find a lower bound for $f(x)$:

$$\begin{aligned}\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{1+\ldots}}}} &\geq \sqrt{x\sqrt{x\sqrt{x\sqrt{\ldots}}}}\\ &= x^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots}\\ &= x \;\geq\; \frac{x+1}{2}.\end{aligned}$$

Then we find an upper bound for $f(x)$:

$$\begin{aligned}&\sqrt{1+x\sqrt{1+(x+1)\sqrt{1+(x+2)\sqrt{1+\ldots}}}}\\ &\qquad \leq \sqrt{(1+x)\sqrt{2(x+1)\sqrt{3(x+1)\sqrt{\ldots}}}}\\ &\qquad = \sqrt{1\sqrt{2\sqrt{3\sqrt{4\ldots}}}}\,(x+1)^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots}\\ &\qquad < \sqrt{1\sqrt{2\sqrt{4\sqrt{8\ldots}}}}\,(x+1)^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots}\\ &\qquad = \sqrt{2^{\frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\cdots}}\,(x+1)^{\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots} = 2(x+1).\end{aligned}$$

Therefore

$$\frac{x+1}{2} \leq f(x) \leq 2(x+1).$$

Substituting $x+1$ for x in the above relation gives

$$\frac{x+2}{2} \leq f(x+1) \leq 2(x+2).$$

Using equation (19.8) now, we can write it in the form

$$\frac{x(x+2)}{2} + 1 \leq f(x)^2 \leq 2x(x+2) + 1,$$

or in the stronger form

$$\frac{x(x+2)}{2} + \frac{1}{2} < f(x)^2 < 2x(x+2) + 2.$$

Notice that the lower bound equals $(x+1)^2/2$ and the upper bound equals $2(x+1)^2$. The last inequality implies for $f(x)$:

$$\frac{(x+1)}{\sqrt{2}} < f(x) < \sqrt{2}(x+1).$$

We can repeat the above procedure of substituting $x+1$ for x a second time to find

$$\frac{(x+1)}{\sqrt[2^2]{2}} < f(x) < \sqrt[2^2]{2}(x+1),$$

a third time to find

$$\frac{(x+1)}{\sqrt[2^3]{2}} < f(x) < \sqrt[2^3]{2}(x+1),$$

etc., up to n times:

$$\frac{(x+1)}{\sqrt[2^n]{2}} < f(x) < \sqrt[2^n]{2}(x+1).$$

In the limit $n \to \infty$, we find that

$$(x+1) \le f(x) \le (x+1) \;\; \Rightarrow \;\; f(x) = x+1,$$

since $\sqrt[2^n]{2} \to 1$.

The number given in the problem equals $f(2) = 3$. $\square$

Ramanujan discovered the above result while he was a student. His method then was a repeated application of the identity

$$n+2 = \sqrt{1+(n+1)(n+3)}.$$

That is,

$$n(n+2) = n\sqrt{1+(n+1)(n+3)} = n\sqrt{1+(n+1)\sqrt{1+(n+2)(n+4)}} = \cdots.$$

Isn't this easier than the method presented above? Of course, it is! But it ignores verifying that the passage from the finite sum to the infinite one is allowed. This is a non-trivial issue. For example, here is an alternative solution of Ramanujan's problem (taken from [51]) and which gives a different (incorrect) answer:

$$4 = \sqrt{1+2\,\frac{15}{2}} = \sqrt{1+2\sqrt{1+3\,\frac{221}{12}}} = \cdots = \sqrt{1+2\sqrt{1+3\sqrt{1+\cdots}}}.$$

More generally, Ramanujan discovered [33] that if

$$f(x) \equiv \sqrt{ax+(n+a)^2+x\sqrt{a(x+n)+(n+a)^2+(x+n)\sqrt{a(x+2n)+(n+a)^2+(x+2n)\sqrt{\dots}}}},$$

then

$$f(x) = x + n + a.$$

Now try the following problem, proposed by Ramanujan at the same time as the previous one:

Problem 19.8. *Evaluate the continued radical*

$$S = \sqrt{6+2\sqrt{7+3\sqrt{8+4\sqrt{9+\cdots}}}}.$$

Answer. $S = 4$.

Let's return once more to the IMO 1976 problem and see one outstanding connection with continued radicals. Toward this goal, we present a beautiful formula ([38], Problem 195) that generalizes the well-known results

$$\frac{\sqrt{2}}{2} = \sin\frac{\pi}{4}, \quad \frac{\sqrt{2-\sqrt{2}}}{2} = \sin\frac{\pi}{8}, \quad \frac{\sqrt{2+\sqrt{2}}}{2} = \cos\frac{\pi}{8} = \sin\frac{3\pi}{8}, \quad \dots .$$

Lemma 19.9. *For $\alpha_i \in \{-1, 1\}$, $i = 0, 1, \dots, n$,*

$$2\sin\left[\left(\alpha_0 + \frac{\alpha_0\alpha_1}{2} + \cdots + \frac{\alpha_0\alpha_1\dots\alpha_n}{2^n}\right)\frac{\pi}{4}\right] = \alpha_0\sqrt{2+\alpha_1\sqrt{2+\alpha_2\sqrt{\cdots+\alpha_n\sqrt{2}}}}.$$

The proof is easy by induction. Given the continued radical

$$\alpha_0\sqrt{2+\alpha_1\sqrt{2+\alpha_2\sqrt{2+\dots}}},$$

according to this lemma, its partial sums are given by

$$x_n = 2\sin\left[\left(\alpha_0 + \frac{\alpha_0\alpha_1}{2} + \cdots + \frac{\alpha_0\alpha_1\dots\alpha_{n-1}}{2^{n-1}}\right)\frac{\pi}{4}\right].$$

The series

$$\alpha_0 + \frac{\alpha_0\alpha_1}{2} + \cdots + \frac{\alpha_0\alpha_1\dots\alpha_{n-1}}{2^{n-1}} + \cdots$$

is absolutely convergent (since the absolute value of the summands are the terms of a geometric progression with ratio $\frac{1}{2}$) and thus it converges to some number α. Therefore, the original continued radical converges to the number

$$2\sin\frac{\alpha\pi}{4}.$$

Problem 19.10 (IMO 1976). *Let $f(x) = x^2 - 2$ and $f^j(x) = f(f^{j-1}(x))$ for $j = 2, 3, \ldots$. Show that for any positive integer n, the roots of the equation $f^n(x) = x$ are real and distinct.*

Solution. The equation $f^j(x) = x$ can be written $f(f^{j-1}(x)) = x$ or $f^{j-1}(x)^2 - 2 = x$. From this,

$$f^{j-1}(x) = \pm\sqrt{2+x}.$$

By repeating this process,

$$f^{j-2}(x) = \pm\sqrt{2 \pm \sqrt{2+x}},$$

and inductively after n steps, starting with $j = n$,

$$x = \pm\sqrt{2 \pm \sqrt{2 \pm \sqrt{2 \pm \ldots \sqrt{2+x}}}},$$

where we have n nested radicals. There are 2^n choices in the signs. Given a combination of signs $\alpha_0, \alpha_1, \ldots, \alpha_{n-1}$,

$$x = \alpha_0\sqrt{2 + \alpha_1\sqrt{2 + \alpha_2\sqrt{2 + \cdots + \alpha_{n-1}\sqrt{2+x}}}},$$

by repetitive substitution of the x in the right-hand side, x is given by a periodic continued radical

$$x = \alpha_0\sqrt{2 + \alpha_1\sqrt{2 + \alpha_2\sqrt{2 + \ldots}}} \quad \text{for } \alpha_{n+k} = \alpha_k, \quad k = 0, 1, \ldots, n-1.$$

This corresponds to a real number $2\sin(\alpha\pi/4)$. Therefore, each choice in the combination of signs corresponds to one root; there are 2^n real solutions. $\square$

To help you understand how this works and compare this result to the result of Problem 1.55, I have listed the roots of $f^3(x) = x$ and the corresponding choices of signs in Table 19.1. In doing so, I have omitted the calculations. You may try to carry out the summation of the series to prove the result of Problem 1.55. If you fail, look at [48].

The following problem I wrote is in direct analogy with Ramanujan's continued radical.

α_0	α_1	α_2	$\sin\frac{\alpha\pi}{4}$	$\cos\left(\frac{\pi}{2}-\frac{\alpha\pi}{2}\right)$
-1	-1	-1	$\sin\left(-\frac{\pi}{6}\right)$	$\cos\frac{2\pi}{3}$
-1	-1	$+1$	$\sin\left(-\frac{\pi}{14}\right)$	$\cos\frac{4\pi}{7}$
-1	$+1$	-1	$\sin\left(-\frac{5\pi}{14}\right)$	$\cos\frac{6\pi}{7}$
-1	$+1$	$+1$	$\sin\left(-\frac{7\pi}{18}\right)$	$\cos\frac{8\pi}{9}$
$+1$	-1	-1	$\sin\left(\frac{3\pi}{14}\right)$	$\cos\frac{2\pi}{7}$
$+1$	-1	$+1$	$\sin\left(\frac{\pi}{18}\right)$	$\cos\frac{4\pi}{9}$
$+1$	$+1$	-1	$\sin\left(\frac{5\pi}{18}\right)$	$\cos\frac{2\pi}{9}$
$+1$	$+1$	$+1$	$\sin\left(\frac{\pi}{2}\right)$	$\cos 0$

Table 19.1: Roots of $f^3(x) = x$ using the method of continued radicals. The last column gives the roots as the cosine of an angle. These answers are exactly those of Problem 1.55. Also, we see that when the signs are chosen such that $\alpha_0\alpha_1\alpha_2 = 1$, the continued radical provides the roots corresponding to the angles $\theta_k = \frac{2\pi k}{2^3-1}$, and when the signs are chosen such that $\alpha_0\alpha_1\alpha_2 = -1$, the continued radical provides the roots corresponding to the angles $\theta_k = \frac{2\pi k}{2^3+1}$.

Problem 19.11. *Evaluate the continued fraction*

$$\cfrac{1}{2+\cfrac{1}{3+\cfrac{1}{4+\cfrac{1}{5+\cfrac{1}{6+\cfrac{1}{7+\cdots}}}}}}.$$

Solution. In order to better understand the pattern, we will define the function

$$f(x) = \cfrac{1}{x+\cfrac{1}{(x+1)+\cfrac{1}{(x+2)+\cfrac{1}{(x+3)+\dots}}}} = \cfrac{1}{1+\cfrac{\frac{1}{x(x+1)}}{1+\cfrac{\frac{1}{(x+1)(x+2)}}{1+\cfrac{\frac{1}{(x+2)(x+3)}}{1+\dots}}}}.$$

Imagine that we have found functions $f_n(x)$, $n = 0, 1, 2, \dots$ and $k_n(x)$, $n = 1, 2, \dots$ such that

$$f_{n-1} - f_n = k_n f_{n+1} \quad \text{for } n = 1, 2, \dots.$$

Then

$$\frac{f_{n-1}}{f_n} - 1 = k_n \frac{f_{n+1}}{f_n} \Rightarrow \frac{f_{n-1}}{f_n} = 1 + k_n \frac{f_{n+1}}{f_n} \Rightarrow \frac{f_n}{f_{n-1}} = \frac{1}{1 + k_n \frac{f_{n+1}}{f_n}}.$$

Applying this equation repeatedly yields

$$\frac{f_1}{f_0} = \frac{1}{1+k_1\frac{f_2}{f_1}} = \cfrac{1}{1+k_1\cfrac{1}{1+k_2\frac{f_3}{f_2}}} = \cfrac{1}{1+k_1\cfrac{1}{1+k_2\cfrac{1}{1+k_3\frac{f_4}{f_3}}}} = \cfrac{1}{1+\cfrac{k_1}{1+\cfrac{k_2}{1+\cfrac{k_3}{1+\cdots}}}}.$$

Therefore it remains to find the functions f_n such that

$$f_{n-1} - f_n = \frac{1}{(x+n-1)(x+n)} f_{n+1}.$$

With a little effort (trial and error), one can discover that

$$f_0(x) = 1 + \frac{1}{x}\frac{1}{1!} + \frac{1}{x(x+1)}\frac{1}{2!} + \frac{1}{x(x+1)(x+3)}\frac{1}{3!} + \cdots,$$

and $f_n(x) = f_{n-1}(x+1)$, $n = 1, 2, 3, \ldots$ Then

$$f(x) = \frac{f_0(x+1)}{f_0(x)}.$$

In particular, the continued fraction of the problem has the value

$$f(2) = \frac{f_0(3)}{f_0(2)},$$

with

$$f_0(2) = \sum_{k=0}^{\infty} \frac{1}{k!}\frac{1}{(k+1)!} \quad \text{and} \quad f_0(3) = 2\sum_{k=0}^{\infty} \frac{1}{k!}\frac{1}{(k+2)!}. \qquad \square$$

You might be bothered by the fact that the final result is still in terms of infinite series. Well, you are not alone in this. However, you may use this problem—besides as an example on how to use functions in deriving results for continued fractions—as a reminder that often in mathematics, results in closed form may not be possible. Instead, we present them in terms of well known special functions. Readers with experience in this topic might have recognized the special function

$${}_0F_1(b, z) = 1 + \frac{1}{b}\frac{z}{1!} + \frac{1}{b(b+1)}\frac{z^2}{2!} + \frac{1}{b(b+1)(b+2)}\frac{z^3}{3!} + \cdots,$$

which is known as the (0,1) **hypergeometric function**. The indices 0 and 1 count the number of parameters in the numerator and denominator of each term (no parameter and one parameter respectively for ${}_0F_1$). In general, one can define the ${}_nF_m(a_1, \ldots, a_n, b_1, \ldots, b_m, z)$ with n parameters in the numerator and m parameters in the denominator of each term. For example, the ${}_1F_1$ is:

$$ {}_1F_1(a,b,z) = 1 + \frac{a}{b}\frac{z}{1!} + \frac{a(a+1)}{b(b+1)}\frac{z^2}{2!} + \frac{a(a+1)(a+2)}{b(b+1)(b+2)}\frac{z^3}{3!} + \cdots . $$

There is an extensive theory for hypergeometric functions. In particular, the method I used in the solution of the problem is due to Gauss, who studied the hypergeometric functions systematically. If you wish to gain additional experience, try the following problem.

Problem 19.12. *Prove that*

$$ \tanh z = \cfrac{z}{1 + \cfrac{z^2}{3 + \cfrac{z^2}{5 + \cfrac{z^2}{7 + \cdots}}}}. $$

Hint. It is true that

$$ \cosh z = {}_0F_1\left(\frac{1}{2}, \frac{z^2}{4}\right) \quad \text{and} \quad \frac{\sinh z}{z} = {}_0F_1\left(\frac{3}{2}, \frac{z^2}{4}\right). $$ □

In Chapter 15 we mentioned that convex and concave functions are useful for proving inequalities although we did not elaborate extensively. Functions and appropriate functional relations (beyond the Jensen inequality) can be very helpful in proving inequalities. Such is the following example.

Problem 19.13 (BWMC 1992). *Let* $a = \sqrt[1992]{1992}$. *Which of the following two numbers is greater?*

$$ 1992 \quad vs. \quad a^{a^{a^{\cdot^{\cdot^{\cdot^{a}}}}}}, $$

where the a appears 1992 times in the second expression.

Solution. We will consider the function $f(x) = a^x$ since

$$ f^n(x) = a^{a^{a^{\cdot^{\cdot^{\cdot^{a^x}}}}}}, $$

where the a appears n times. Notice that $x = 1992$ is a fixed point of $f(x)$,

$$ f(1992) = 1992, $$

and therefore $f^n(1992) = 1992$ for all n as we can easily verify.

Then we notice that

$$a = \sqrt[1992]{1992} = e^{\frac{\ln 1992}{1992}} > 1 + \frac{\ln 1992}{1992} > 1,$$

where we used the exponential inequality $e^x > 1 + x$ for $x \neq 0$. And since $a > 1$, the function $f(x) = a^x$ is strictly increasing. For $x > y$ it is then $f(x) > f(y)$. By repeated application of this relation,

$$x > y \;\Rightarrow\; f^n(x) > f^n(y).$$

We now apply the last functional relation for $x = 1992$, $y = a$, and $n = 1991$:

$$f^{1991}(1992) > f^{1991}(a) \;\Rightarrow\; 1992 > a^{a^{a^{\cdot^{\cdot^{\cdot^{a}}}}}}. \qquad \square$$

19.3 Assortment of Problems

Problem 19.14 (IMO 1990). *Construct a function $f : \mathbb{Q}_+^* \to \mathbb{Q}_+^*$ such that*

$$f(xf(y)) = \frac{f(x)}{y} \quad \textit{for all } x, y \in \mathbb{Q}_+^*. \tag{19.9}$$

Solution. Let $x = 1/f(y)$. Then

$$f\left(\frac{1}{f(y)}\right) = y\,f(1). \tag{19.10}$$

From this equation we see that f is injective:

$$f(a) = f(b) \;\Rightarrow\; f\left(\frac{1}{f(a)}\right) = f\left(\frac{1}{f(b)}\right) \;\Rightarrow\; a\,f(1) = b\,f(1) \;\Rightarrow\; a = b.$$

Next, let $y = 1$ in (19.9):

$$f(xf(1)) = f(x).$$

Since f is injective, we conclude that $xf(1) = x$, or[1]

$$f(1) = 1.$$

Setting $x = 1$ now in the defining equation (19.9), we find

$$f(f(y)) = \frac{1}{y}. \tag{19.11}$$

[1]Clearly, the point $x = 1$ is a fixed point but this interpretation is not important in this problem.

Finally, we can prove that f is multiplicative. Replacing y by $1/f(y)$ in (19.9) and making use of (19.10), we find that f satisfies the power Cauchy equation:

$$f(x\,y) = f(x)\,f(y).$$

Any $x \in \mathbb{Q}_+^*$ can be uniquely expanded in a product of primes

$$x = \prod_i p_i^{n_i},$$

where $n_i \in \mathbb{Z}^*$. From the power Cauchy equation,

$$f(x) = f\left(\prod_i p_i^{n_i}\right) = \prod_i (f(p_i))^{n_i}$$

and

$$f(f(x)) = f\left(\prod_i (f(p_i))^{n_i}\right) \overset{(19.11)}{\Rightarrow} \frac{1}{x} = \prod_i (f(f(p_i)))^{n_i}$$

or

$$\prod_i (f(f(p_i)))^{n_i} = \prod_i p_i^{-n_i}.$$

The two sides will agree for any function f whose values at the prime numbers are chosen arbitrarily, but otherwise to satisfy

$$f(f(p_i)) = \frac{1}{p_i} \quad \text{for all } p_i.$$

For example, we can partition the set of primes into two sets Q and Q' and order their elements: $Q = \{q_1, q_2, q_3, \ldots, \ldots\}$ and $Q' = \{q_1', q_2', q_3', \ldots, \ldots\}$. Then define f such that $f(q_i) = q_i'$ and $f(q_i') = 1/q_i$ for all i. □

Problem 19.15 (IMO 1986). *Find all functions $f : \mathbb{R}_+ \to \mathbb{R}_+$ such that*

(a) $f(2) = 0$,

(b) $f(x) \neq 0$ *for* $0 \leq x < 2$,

(c) $f(xf(y))\,f(y) = f(x+y)$ *for all* $x, y \in \mathbb{R}_+$.

Solution. Setting $y = 2$ in (c) and using (a), we find that

$$f(x+2) = f(xf(2))\,f(2) = 0.$$

Since $x \geq 0$ in (c), this result implies that

$$f(x) = 0 \quad \text{for all } x \geq 2.$$

Setting $x = y = 0$ in (c), we find that $f(0) = 0, 1$. But $f(0) \neq 0$ from (b) and hence $f(0) = 1$.

Now, set $x = 2 - y$, $0 < y < 2$ in (c):

$$f((2-y)f(y))\, f(y) \;=\; f(2) \;=\; 0.$$

Since the condition (b) is given, $f((2-y)f(y)) = 0$. This necessarily implies that $(2-y)\, f(y) \geq 2$ or

$$f(y) \geq \frac{2}{2-y} \quad \text{for } 0 < y < 2. \tag{19.12}$$

In fact, we will now show that

$$f(x) \;=\; \frac{2}{2-x} \quad \text{for } 0 < x < 2.$$

Assume that the above is not the case. Then there are points in $(0, 2)$ that satisfy the strict inequality of (19.12). Let x_0 be the *first* point in $(0, 2)$ such that $f(x_0) > 2/(2 - x_0)$. Then define $x_0' \neq x_0$ such that

$$f(x_0') \;=\; \frac{2}{2-x_0}.$$

If $x_0' < x_0$, we must have

$$f(x_0') \;=\; \frac{2}{2-x_0'}.$$

This, in conjunction with the definition of x_0', implies that $x_0 = x_0'$, which is not possible. Therefore, we must assume that $x_0' > x_0$. Now let $x_0'' = 2 - x_0'$. The condition $x_0' > x_0$ gives $2 - x_0' < 2 - x_0 \Rightarrow x_0'' < 2 - x_0$ or

$$x_0'' + x_0 \;<\; 2.$$

In (c), set $x = x_0''$ and $y = x_0$:

$$f(x_0''\, f(x_0')) \;=\; f(x_0'' + x_0') \;=\; f(2) \;=\; 0 \;\Rightarrow\; x_0''\, f(x_0') \geq 2 \;\Rightarrow\; x_0''\, \frac{2}{2-x_0} \geq 2,$$

or

$$x_0'' + x_0 \geq 2.$$

However, this inequality contradicts the previous one. So we must have

$$f(x) \;=\; \begin{cases} \dfrac{2}{2-x} & \text{if } x \in [0, 2), \\ 0 & \text{if } x \in [2, +\infty). \end{cases} \qquad \square$$

Problem 19.16 (Greece 1996). *Let $f : (0, +\infty) \to \mathbb{R}$ be a function such that*

(a) *f is strictly increasing;*

(b) *$f(x) > -\frac{1}{x}$ for all $x > 0$;*

(c) *$f(x)\, f(f(x) + \frac{1}{x}) = 1$ for all $x > 0$.*

Find $f(1)$.

Solution. From (c), we see immediately that $f(x) \neq 0$ for all $x > 0$. Now let $y = f(x) + \frac{1}{x} > 0$ for $x > 0$. In (b), we substitute x with y:

$$f(y) > -\frac{1}{y}.$$

At the same time, (c) gives

$$f(x) = \frac{1}{f(y)}. \tag{19.13}$$

Also, in (c), we replace x by y, obtaining

$$f(y)\, f\left(f(y) + \frac{1}{y}\right) = 1 \Rightarrow \frac{1}{f(y)} = f\left(f(y) + \frac{1}{y}\right).$$

By (19.13), this gives

$$f(x) = f\left(\frac{1}{f(x)} + \frac{1}{y}\right).$$

Since f is strictly increasing, it is injective. From the equation we found above we thus conclude that

$$x = \frac{1}{f(x)} + \frac{1}{y}$$

or

$$x = \frac{1}{f(x)} + \frac{1}{f(x) + \dfrac{1}{x}}.$$

We can easily rewrite this equation as a quadratic one in $f(x)$:

$$x^2 f(x)^2 - x f(x) - 1 = 0,$$

with solutions

$$f(x) = \frac{\phi}{x} \quad \text{for } f(x) = -\frac{1}{\phi\, x},$$

where

$$\phi = \frac{1 + \sqrt{5}}{2}$$

is the golden ratio. Only one solution is strictly increasing:

$$f(x) = -\frac{1}{\phi\, x}.$$

So, finally, $f(1) = -1/\phi$. □

Chapter 20

Additional Problems

A note of warning to the reader: the separation of the problems to categories in this chapter is artificial. I only do it in case one wishes to find and work on a group of problems containing (but not focusing exclusively) on a particular idea.

20.1 Functions

Problem 20.1. *Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by*

$$f(x) = \lim_{n\to\infty} \lim_{m\to\infty} \cos(n!\pi x)^{2m}, \qquad n, m \in \mathbb{N}^*,$$

is the Dirichlet function.

Problem 20.2 (Thomæ's ruler function). *If $x \in \mathbb{Q}$, let $x = m/n$, where $n > 0$ and m, n have no common divisor greater than 1. Then we define*

$$f(x) = \begin{cases} \frac{1}{n} & \text{if } x \in \mathbb{Q}, \\ 0 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Show that f is discontinuous at every rational number in [0,1] and continuous at every irrational number in [0,1].

Problem 20.3 ([61]). *Let $g(x)$ be a function with domain [0,2] defined as follows:*

$$g(x) = \begin{cases} 0 & \text{if } x \in [0, \frac{1}{3}] \cup [\frac{5}{3}, 2], \\ 3x - 1 & \text{if } x \in [\frac{1}{3}, \frac{2}{3}], \\ 1 & \text{if } x \in [\frac{2}{3}, \frac{4}{3}], \\ -3x + 5 & \text{if } x \in [\frac{4}{3}, \frac{5}{3}]. \end{cases}$$

We then define the extension of g to all of $\mathbb{R}$ by

$$g(x+2) = g(x).$$

Finally, we define the function $f : [0,1] \to \mathbb{R}^2$ by $t \mapsto (x,y)$ such that

$$x(t) = \sum_{n=1}^{\infty} \frac{1}{2^n}\, g\big(3^{2n-2}t\big),$$

$$y(t) = \sum_{n=1}^{\infty} \frac{1}{2^n}\, g\big(3^{2n-1}t\big).$$

Prove that the $f([0,1]) = [0,1] \times [0,1]$.

Comment. A curve such as the one given in the above problem is called a **space filling curve**. Peano was the first to consider such curves.

Problem 20.4. *Prove that there is no continuous injective function $f : [0,1] \to [0,1] \times [0,1]$ such that $f([0,1]) = [0,1] \times [0,1]$.*

Problem 20.5 ([12], Problem 4.17). *The function $f\big|[0,1]$ is defined as follows:*

$$f(x) = \begin{cases} x & \text{if } x \text{ is rational,} \\ 1-x & \text{if } x \text{ is irrational.} \end{cases}$$

Prove that:

(a) *$f^2(x) = x$, for all $x \in [0,1]$.*
(b) *$f(x) + f(1-x) = 1$, for all $x \in [0,1]$.*
(c) *f is continuous only at $x = 1/2$.*
(d) *f assumes every value between 0 and 1.*
(e) *$f(x+y) - f(x) - f(y)$ is rational for all $x, y \in [0,1]$.*

Problem 20.6. *Let $f : \mathbb{R} \to \mathbb{R}$ be a function that satisfies the exponential Cauchy equation*

$$f(x+y) = f(x)\,f(y) \quad \text{for all } x, y \in \mathbb{R},$$

and for which there is a function $g : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) = 1 + x\,g(x) \quad \text{for all } x, y \in \mathbb{R}.$$

If

$$\lim_{x\to 0} g(x) = 1,$$

show that the derivative $f'(x)$ exists for all $x \in \mathbb{R}$, and that

$$f'(x) = f(x).$$

Problem 20.7 ([5], Problem 1795, April 2008). *Find a function $f : [0,1] \to [0,1]$ such that for each non-trivial interval $I \subset [0,1]$, we have $f(I) = [0,1]$.*

Problem 20.8 ([2], Problem 886, November 2008). *Call a function f **good** if $f^{2008}(x) = -x$ for all $x \in \mathbb{R}$, where f^{2008} denotes the 2008-th iterate of f. Prove the following:*

(a) *Every good function is bijective and odd and cannot be monotonic.*

(b) *If f is good and $x_0 \neq 0$, there exist infinitely many 5-tuples*

$$(p_1, p_2, p_3, p_4, p_5)$$

of distinct positive integers whose sum is a multiple of 5 and for which, with $q_k = f^{p_k}(x_0)$, $q_1 \neq q_i$ for $i = 2,3,4,5$ and $q_i \neq q_{i+1}$ for $i = 2,3,4$.

Problem 20.9 (Greece 1983). *If the function $f : [0, +\infty) \to \mathbb{R}$ satisfies the relation*

$$f(x)\, e^{f(x)} = x,$$

for all x in its domain, prove that:

(a) *f is monotonic over its entire domain;*

(b) $\lim\limits_{x\to+\infty} f(x) = +\infty$*;*

(c) $\lim\limits_{x\to+\infty} \frac{f(x)}{\ln x} = 1$.

Problem 20.10 (Greece 1983). *The function $f : \mathbb{R} \to \mathbb{R}$ is defined by $f(x) = x^5 + x - 1$.*

(a) *Prove that f is bijective.*

(b) *Show that $f(1001^{999}) < f(1001^{1000})$.*

(c) *Determine the roots of the equation $f(x) = f^{-1}(x)$.*

Problem 20.11 (LMO 1989). *How many real solutions does the following equation have?*

$$\sin(\sin(\sin(\sin(\sin(x))))) = \frac{x}{3}.$$

Problem 20.12 (Russia 1998). *Let $f(x) = x^2 + ax + b\cos x$. Find all values of a, b for which the equations $f(x) = 0$ and $f(f(x)) = 0$ have the same (non-empty) set of real roots.*

Problem 20.13 (Mongolia 2000). *A function $f : \mathbb{R} \to \mathbb{R}$ satisfies the following conditions:*
(a) $|f(a) - f(b)| \le |a - b|$ *for any real numbers $a, b \in \mathbb{R}$.*
(b) $f^3(0) = 0$.
Prove that $f(0) = 0$.

Problem 20.14 (IMO 1970 Longlist). *Suppose that f is a real function defined for $0 \le x \le 1$ having the first derivative f' for $0 \le x \le 1$ and the second derivative f'' for $0 < x < 1$. Prove that if*

$$f(0) = f'(0) = f'(1) = f(1) - 1 = 0,$$

there exists a number $0 < y < 1$ such that $|f''(y)| \ge 4$.

Problem 20.15 (Iran 1997). *Suppose that $f : \mathbb{R} \to \mathbb{R}$ has the following properties:*
(a) $f(x) \le 1$ *for all $x \in \mathbb{R}$,*
(b) $f\left(x + \frac{13}{42}\right) + f(x) = f\left(x + \frac{1}{6}\right) + f\left(x + \frac{1}{7}\right)$ *for all $x \in \mathbb{R}$.*
Prove that f is periodic.

Problem 20.16 (IMO 1978 Longlist). *Assume the functions $f_1, f_2, \ldots, f_n : I \to \mathbb{R}_+$ are concave. Prove that their geometric mean*

$$f = \sqrt[n]{f_1 f_2 \cdots f_n}$$

is also a concave function.

Problem 20.17 (Turkey 1998). *Let $A = \{1, 2, 3, 4, 5\}$. Find the number of functions from the set of non-empty subsets of A to A for which $f(B) \in B$ for every $B \subset A$ and $f(B \cup C) \in \{f(B), f(C)\}$ for every $B, C \subset A$.*

Problem 20.18 ([1], Problem 11472, December 2009). *Let n be a non-negative integer, and let f be a $(4n+3)$-times continuously differentiable function on $\mathbb{R}$. Show that there is a number ξ such that, at $x = \xi$,*

$$\prod_{k=0}^{4n+3} \frac{d^k f(x)}{dx^k} \geq 0.$$

Problem 20.19 ([1], Problem 11215, April 2006). *A car moves along the real line from $x = 0$ at $t = 0$ to $x = 1$ at $t = 1$, with differentiable position function $x(t)$ and differentiable velocity function $v(t) = x'(t)$. The car begins and ends the trip at a standstill; that is, $v = 0$ at both the beginning and the end of the trip. Let V be the maximum velocity attained during the trip. Prove that at some time between the beginning and end of the trip, $|v'| > V^2/(V-1)$.*

Problem 20.20 ([1], Problem 11221, April 2006). *Give an example of a function g from $\mathbb{R}$ into $\mathbb{R}$ such that g is differentiable everywhere, g' is differentiable on one dense subset of $\mathbb{R}$, and g' is discontinuous on another dense subset of $\mathbb{R}$.*

Problem 20.21 ([1], Problem 11232, June/July 2006). *Let f be a continuous function from $\mathbb{R}$ into $\mathbb{R}$ that is lower bounded. Show that there exists a real number x_0 such that $f(x_0) - f(x) < |x - x_0|$ holds for all x other than x_0.*

Problem 20.22 ([5], Problem 1732, December 2005). *For each positive integer n, define the function $f_n : [0,1]^n \to [0,1]^2$ by*

$$f(x_1, x_2, \ldots, x_n) = \left(\frac{x_1 + x_2 + \cdots + x_n}{n}, \sqrt[n]{x_1 x_2 \ldots x_n} \right),$$

and let $I(n)$ be the image of f_n. Determine

$$\bigcup_{n=1}^{\infty} I(n).$$

Problem 20.23 (Putnam 1974). *In the standard definition, a real-valued function of two real variables $g : \mathbb{R}^2 \to \mathbb{R}$ is continuous if, for every point $(x_0, y_0) \in \mathbb{R}^2$ and every $\varepsilon > 0$, there is a $\delta > 0$ such that $\sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$ implies $|g(x,y) - g(x_0,y_0)| < \varepsilon$.*

By contrast, $f : \mathbb{R}^2 \to \mathbb{R}$ is said to be continuous in each variable separately if, for each fixed value y_0 of y, the function $f(x, y_0)$ is continuous in the usual *sense as a function of* x*, and, similarly,* $f(x_0, y)$ *is continuous as a function of* y *for each fixed* x_0*.*

Let $f : \mathbb{R}^2 \to \mathbb{R}$ *be continuous in each variable separately. Show that there exists a sequence of continuous functions* $g_n : \mathbb{R}^2 \to \mathbb{R}$ *such that*

$$f(x, y) = \lim_{n\to\infty} g_n(x, y) \quad \text{for all } (x, y) \in \mathbb{R}^2.$$

Problem 20.24 ([1], Problem 11009, April 2003)**.** *Consider the function* f *defined by*

$$f(x) = \frac{a^x - b^x}{c^x - d^x},$$

where $a > b \geq c > d > 0$*, with the removable singularity at zero filled in. Prove that* f *is convex on* $\mathbb{R}$*, and that* $\log f$ *is either convex or concave on* $\mathbb{R}$*. Determine the values of* a, b, c, d *for which* $\log f$ *is convex on* $\mathbb{R}$*.*

Problem 20.25 ([3], Problem 2700A, December 2001)**.** *Show that the function* e^{-xn^2} *can be written in the following form:*

$$e^{-xn^2} = \sum_{k=0}^{n-1} (-1)^k \frac{n^{2k} x^k}{k!} + (-1)^n \frac{n^{2n} x^n}{n!}\, \phi_x(n),$$

where

$$\phi_x(n) = 1 - \int_0^{xn^2} e^{-t} \left(1 - \frac{t}{xn^2}\right)^n dt.$$

Determine the leading large n *behavior of* $\phi_x(n)$*, and show that*

$$\lim_{n\to\infty} n\phi_x(n) = \frac{1}{x}.$$

Problem 20.26 (Ukraine 1998)**.** *Let a function* $f : [0, 1] \to [0, 1]$ *such that*

$$f\big(f(x) + y\big) = f(x) + f(y),$$

for all $x, y \in [0, 1]$*. It is also known that there is a* $\lambda \in (0, 1)$ *such that* $f(\lambda) \neq 0, \lambda$*.*

(a) *Give an example of such a function.*

(b) *Prove that for any* $x \in [0, 1]$*,*

$$f^{19}(x) = f^{98}(x).$$

Problem 20.27 (HIMC 1999). *The function*

$$f(x,y,z) = \frac{x^2+y^2+z^2}{x+y+z}$$

is defined for every x, y, z such that $x+y+z \neq 0$. Find a point (x_0, y_0, z_0) such that

$$0 < x_0^2 + y_0^2 + z_0^2 < \frac{1}{1999}$$

and

$$1.999 < f(x_0, y_0, z_0) < 2.$$

20.2 Problems That Can Be Solved Using Functions

Problem 20.28 (Bulgaria 2000). *Prove that for any two real numbers a and b there exists a real number $c \in (0,1)$ such that*

$$\left| ac + b + \frac{1}{c+1} \right| > \frac{1}{24}.$$

Problem 20.29 (ROM 1996). *Prove the inequality*

$$2^{a_1} + 2^{a_2} + \cdots + 2^{a_{1996}} \leq 1995 + 2^{a_1+a_2+\cdots+a_{1996}},$$

for any real non-positive numbers a_1, a_2, $\ldots$, a_{1996}.

20.3 Arithmetic Functions

Problem 20.30. *Let a sequence of real numbers a_n satisfying the condition*

$$a_n = \lambda\, a_{n-1} + \omega\, n\mu^n, \quad n = 1, 2, \ldots,$$

with $\lambda \neq 0$. Find the expression of a_n in terms of a_0.

Problem 20.31 (Putnam 1963). *Let f be a function with the following properties:*

(a) *$f(n)$ is defined for every positive integer n;*
(b) *$f(n)$ is a positive integer;*
(c) *$f(2) = 2$;*
(d) *$f(mn) = f(m)f(n)$, for all relatively prime integers m and n;*
(e) *f is strictly increasing.*

Prove that $f(n) = n$ for $n = 1, 2, 3 \ldots$.

Problem 20.32 (Australia 1984). *A non-negative integer $f(n)$ is assigned to each positive integer n in such a way that the following conditions are satisfied:*
(a) *$f(mn) = f(m) + f(n)$, for all positive integers m, n;*
(b) $f(n) = 0$, whenever the final (right-hand) decimal digit of n is 3;
(c) $f(10) = 0$.
Calculate $f(1984)$ and $f(1985)$.

Problem 20.33 (Austria 1986). *For n a given positive integer, determine all functions $F : \mathbb{N} \to \mathbb{R}$ such that*

$$F(x + y) = F(xy - n) \quad \text{for all } x, y \in \mathbb{N} \text{ with } xy > n.$$

Problem 20.34 (Estonia 2000; Netherlands TST 2008). *Find all functions $f : \mathbb{N}^* \to \mathbb{N}^*$ such that*

$$f^3(n) + f^2(n) + f(n) = 3n,$$

for all $n \in \mathbb{N}^*$.

Problem 20.35 (BMO 2009). *Denote by $\mathbb{N}^*$ the set of all positive integers. Find all functions $f : \mathbb{N}^* \to \mathbb{N}^*$ such that*

$$f\left(f^2(m) + 2f^2(n)\right) = m^2 + 2n^2,$$

for all $m, n \in \mathbb{N}^*$.

Problem 20.36 (IMO 1988). *A function f is defined on the positive integers by*

$$\begin{aligned} f(1) = 1, \quad f(3) &= 3, \\ f(2n) &= f(n), \\ f(4n + 1) &= 2f(2n + 1) - f(n), \\ f(4n + 3) &= 3f(2n + 1) - 2f(n), \end{aligned}$$

for all positive integers n. Determine the number of positive integers n, less than or equal to 1988, for which $f(n) = n$.

Problem 20.37 (IMO 1988 Shortlist). *Let $f(n)$ be a function defined on the set of all positive integers and having its values in the same set. Suppose that*

$$f(f(n) + f(m)) = m + n,$$

for all positive integers n, m. Find the possible value for $f(1988)$.

Problem 20.38 (IMO 1998). *Consider all functions f from the set $\mathbb{N}^*$ of all positive integers into itself satisfy the equation*

$$f\left(t^2 f(s)\right) = f(t)^2 s,$$

for all s and t in $\mathbb{N}^*$. Determine the least possible value for $f(1998)$.

Problem 20.39 (Putnam 1957). *If facilities for division are not available, it is sometimes convenient in determining the decimal expansion of $1/A$, $A > 0$ to use the iteration*

$$X_{k+1} = X_k (2 - AX_k), \quad k = 0, 1, 2, \ldots,$$

where X_0 is a selected "starting" value. Find the limitations, if any, on the starting value X_0 in order that the above iteration converges to the desired value $1/A$.

Problem 20.40 (Vietnam 1998). *Let $a \geq 1$ be a real number, and define the sequence $x_1, x_2, \ldots$ by $x_1 = a$ and*

$$x_{n+1} = 1 + \ln\left(\frac{x_n(x_n^2 + 3)}{3x_n^2 + 1}\right).$$

Prove that this sequence has a finite limit, and determine it.

Problem 20.41 (CSM 1998). *Find all functions $f : \mathbb{N}^* \to \mathbb{N}^* \smallsetminus \{1\}$ such that, for all $n \in \mathbb{N}^*$,*

$$f(n) + f(n+1) = f(n+2)\, f(n+3) - 168.$$

Problem 20.42 (SPCMO 2001). *Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ such that*

$$f(x + y + f(y)) = f(x) + 2y$$

for all integers x, y.

Problem 20.43 (RMM 2008). *Prove that every bijective function $f : \mathbb{Z} \to \mathbb{Z}$ can be written as $f = u + v$, where $u, v : \mathbb{Z} \to \mathbb{Z}$ are bijective functions.*

Problem 20.44 (Iran 1997). *Find all functions $f : \mathbb{N}^* \to \mathbb{N} \setminus \{1\}$ such that*

$$f(n+1) + f(n+3) = f(n+5)\, f(n+7) - 1375,$$

for all $n \geq 0$.

Problem 20.45 (IMO 1989 Longlist). *For $n \in \mathbb{N}^*$, let $X = \{1, 2, \ldots, n\}$ and let k be an integer such that $n/2 \leq k \leq n$. Determine, with proof, the number of all functions $f : X \to X$ that satisfy the following conditions:*
(a) $f^2 = f$;
(b) *the number of elements in the image of f is k;*
(c) *for each y in the image of f, the number of all points in X such that $f(x) = y$ is at most 2.*

Problem 20.46 (IMO 1978). *Let $\{f(n)\}$ be a strictly increasing sequence of positive integers:*

$$0 < f(1) < f(2) < f(3) < \cdots .$$

Of the positive integers not belonging to the sequence, the n-th in order is of magnitude $f^2(n) + 1$. Determine $f(240)$.

Problem 20.47 (IMO 1971 Shortlist). *Let $T_k = k - 1$ for $k = 1, 2, 3, 4$ and*

$$T_{2k-1} = T_{2k-2} + 2^{k-2}, \quad T_{2k} = T_{2k-5} + 2^k,$$

for $k \geq 3$. Show that

$$\begin{aligned} 1 + T_{2k-1} &= \left\lfloor \frac{12}{7}\, 2^{n-1} \right\rfloor, \\ 1 + T_{2k} &= \left\lfloor \frac{17}{7}\, 2^{n-1} \right\rfloor, \end{aligned}$$

for all k.

Problem 20.48 (IMO 2010). *Determine all functions $g : \mathbb{N}^* \to \mathbb{N}^*$ such that*

$$(g(m) + n)\,(g(n) + m)$$

is a perfect square for all $m, n \in \mathbb{N}^*$.

Problem 20.49 (Bulgaria 2001). *A sequence $\{a_n\}$ is defined by*

$$a_{n+2} = 3a_{n+1} - 2a_n,$$

for $n \geq 1$ and $a_1 = \lambda$, $a_2 = 5\lambda - 2$, where λ is a real number.

(a) *Find all values of λ such that the sequence $\{a_n\}$ is convergent.*

(b) *Prove that if $\lambda = 1$, then*

$$a_{n+2} = \left\lfloor \frac{7a_{n+1}^2 - 8a_n a_{n+1}}{1 + a_n + a_{n+1}} \right\rfloor.$$

Problem 20.50 ([2], Problem 890, November 2008). *Let a be any positive integer, and let H_n be the sequence recursively defined by $H_0 = 0$, $H_1 = 1$, and*

$$H_{n+2} = a\,H_{n+1} + H_n \quad \text{for all } n \geq 0.$$

(For example, for $a = 1$ these are the Fibonacci numbers and for $a = 2$ the Pell numbers.) Let k be a non-negative integer and let G_n be the sequence defined by

$$G_n = H_n\,H_{n+k}.$$

Find a positive integer m and constants $c_1, c_2, \ldots, c_m$ (depending only on a and k) such that

$$G_n = \sum_{i=1}^{m} c_i\,G_{n-i},$$

for all $n \geq m$ or prove that, for all m, such constants cannot exist.

Problem 20.51 (Netherlands 2009). *Find all functions $f : \mathbb{N}^* \to \mathbb{N}^*$ that satisfy the following two conditions:*

(a) *$f(n)$ is a perfect square for all $n \in \mathbb{N}^*$;*

(b) *$f(m + n) = f(m) + f(n) + 2mn$, for all $m, n \in \mathbb{N}^*$.*

Problem 20.52 (Netherlands TST 2009). *Find all functions $f : \mathbb{Z} \to \mathbb{Z}$ that satisfy the condition*

$$f(m + n) + f(mn - 1) = f(m)\,f(n) + 2$$

for all $m, n \in \mathbb{Z}$.

Problem 20.53 (Putnam 1966). *Let $0 < x_1 < 1$ and $x_{n+1} = x_n(1 - x_n)$, $n = 1, 2, \ldots$. Show that*

$$\lim_{n\to\infty} n x_n = 1.$$

Problem 20.54 (USA TST 2004). *Define the function $f : \mathbb{N} \to \mathbb{Q}$ by $f(0)$ and*

$$f(3n+k) = -\frac{3}{2} f(n) + k, \quad k = 0, 1, 2, \quad n \in \mathbb{N}.$$

Prove that f is injective and find $f(\mathbb{N})$.

20.4 Functional Equations With Parameters

Problem 20.55 ([14], Problem 477). *Show that there is only one value of the constant b for which there exists a real function defined for all real numbers with the property that, for all real x and y,*

$$f(x-y) = f(x) - f(y) + bxy.$$

Problem 20.56. *Show that there are two continuous functions $f : \mathbb{R} \to \mathbb{R}$ that solve the functional relation*

$$f(x) + f(y) = f(x)\, f(y) - xy - 1 \quad \text{for all } x, y \in \mathbb{R}.$$

Problem 20.57. *Find all functions $f : \mathbb{R}_+^* \to \mathbb{R}_+^*$ such that, for all $x, y \in \mathbb{R}_+^*$,*

$$f(x+y) + f(x)\, f(y) = f(x) + f(y) + f(xy).$$

Problem 20.58. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that, for all $x, y \in \mathbb{R}$,*

$$f(x+y) + f(x)\, f(y) = f(xy) + 1.$$

Problem 20.59 (IMO 2005 Shortlist). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that, for all $x, y \in \mathbb{R}$,*

$$f(x+y) + f(x)\, f(y) = f(xy) + 2xy + 1.$$

Problem 20.60. *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that, for all $x, y \in \mathbb{R}$,*

$$f(x+y) + f(x)\,f(y) = f(xy) + g(x,y),$$

where g is a given function symmetric in x, y and $g(0,0) = 1$.

Problem 20.61. *Let $f, g_1, g_2, h_1, h_2 : \mathbb{R} \to \mathbb{R}$ be continuous functions that satisfy the functional relation*

$$f(x+y) = g_1(x)\,g_2(y) + h_1(x)\,h_2(y),$$

for all x, y. Find all such functions.

Problem 20.62 (CSM 1999). *Find all functions $f : (0, +\infty) \to \mathbb{R}$ such that*

$$f(x) - f(y) = (y - x)\,f(xy),$$

for all $x, y > 1$.

Problem 20.63. *Let S^1 be the 1-dimensional circle, that is, the unit interval with end points identified. Find all continuous functions $\Phi : S^1 \to S^1$ such that*

$$\Phi(\theta_1 + \theta_2) = \Phi(\theta_1) + \Phi(\theta_2),$$

for all $\theta_1, \theta_2 \in S^1$.

Problem 20.64 (NMC 1998). *Find all functions from the rational numbers to the rational numbers satisfying*

$$f(x+y) + f(x-y) = 2f(x) + 2f(y),$$

for all rational x and y.

Problem 20.65 (Iran 1997). *Suppose $f : \mathbb{R}_+^* \to \mathbb{R}_+^*$ is a decreasing function such that, for all $x, y \in \mathbb{R}_+^*$,*

$$f(x+y) + f\big(f(x) + f(y)\big) = f\Bigg(f\big(x + f(y)\big) + f\big(y + f(x)\big) \Bigg).$$

Prove that

$$f(x) = f^{-1}(x).$$

Problem 20.66 (BMO 2007). *Find all real functions defined on* $\mathbb{R}$ *such that*

$$f(f(x)+y) = f(f(x)-y)+4f(x)\,y,$$

for all real numbers x, y.

Problem 20.67 (IMO 2004 Shortlist). *Find all functions* $f:\mathbb{R}\to\mathbb{R}$ *satisfying the equation*

$$f\left(x^2+y^2+2f(xy)\right) = f(x+y)^2$$

for all $x, y \in \mathbb{R}$.

Problem 20.68 (IMO 1992). *Find all functions* $f:\mathbb{R}\to\mathbb{R}$ *such that*

$$f\left(x^2+f(y)\right) = y+f(x)^2$$

for all $x, y \in \mathbb{R}$.

Problem 20.69 (IMO 1999). *Determine all functions* $f:\mathbb{R}\to\mathbb{R}$ *such that*

$$f(x-f(y)) = f(f(y))+xf(y)+f(x)-1$$

for all real numbers x, y.

Problem 20.70 (Iran 1999; BH TST 2008). *Find all functions* $f:\mathbb{R}\to\mathbb{R}$ *satisfying*

$$f(f(x)+y) = f(x^2-y)+4f(x)y,$$

for all $x, y \in \mathbb{R}$.

Problem 20.71 (Italy 1999). (a) *Find all the strictly monotonic functions* $f:\mathbb{R}\to\mathbb{R}$ *such that*

$$f(x+f(y)) = f(x)+y,$$

for all $x, y \in \mathbb{R}$.

(b) *Prove that for every integer* $n > 1$ *there do not exist strictly monotonic functions* $f:\mathbb{R}\to\mathbb{R}$ *such that*

$$f(x+f(y)) = f(x)+y^n,$$

for all $x, y \in \mathbb{R}$.

Problem 20.72 (Vietnam 1999). *Let $f(x)$ be a continuous function defined on $[0,1]$ such that $f(0) = f(1) = 0$ and*

$$2f(x) + f(y) = 3f\left(\frac{2x+y}{3}\right),$$

for all $x, y \in [0,1]$. Prove that $f(x) = 0$ for all $x, y \in [0,1]$.

Problem 20.73 (India 2000). *Suppose $f : \mathbb{Q} \to \{0,1\}$ is a function with the property that for $x, y \in \mathbb{Q}$,*

$$\text{if } f(x) = f(y) \text{ then } f(x) = f\left(\frac{x+y}{2}\right) = f(y).$$

If $f(0) = 0$ and $f(1) = 1$, show that $f(q) = 1$ for all rational numbers q greater than or equal to 1.

Problem 20.74 (Korea 2000). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying*

$$f(x^2 - y^2) = (x - y)\,(f(x) + f(y))$$

for all $x, y \in \mathbb{R}$.

Problem 20.75 (APMO 2002). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that there are only finitely many $s \in \mathbb{R}$ for which $f(s) = 0$ and*

$$f(x^4 + y) = x^3 f(x) + f^2(y)$$

for all $x, y \in \mathbb{R}$.

Problem 20.76 (India 2005). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x^2 + y\,f(z)) = x\,f(x) + z\,f(y),$$

for all $x, y, z \in \mathbb{R}$.

Problem 20.77 (Poland 1990). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$(x - y)\,f(x + y) - (x + y)\,f(x - y) = 4xy\,(x^2 - y^2),$$

for all $x, y \in \mathbb{R}$.

Problem 20.78 (Ukraine TST 2007). *Find all functions $f : \mathbb{Q} \to \mathbb{Q}$ such that*

$$f(x^2 + y + f(xy)) = 3 + (x + f(y) - 2)\, f(x)$$

for all $x, y \in \mathbb{Q}$.

Problem 20.79 (Ukraine TST 2008). *Find all functions $f : \mathbb{R}_+^* \to \mathbb{R}_+^*$ such that*

$$f(x + f(y)) = f(x + y) + f(y),$$

for all pairs of positive reals x and y.

Problem 20.80 (Japan 2004). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(xf(x) + f(y)) = f(x)^2 + y$$

for all $x, y \in \mathbb{R}$.

Problem 20.81 (Japan 2006). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x)^2 + 2yf(x) + f(y) = f(f(x) + y)$$

for all $x, y \in \mathbb{R}$.

Problem 20.82 (Japan 2008). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x + y)\, f(f(x) - y) = x\, f(x) - y\, f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 20.83 (Japan 2009). *Find all functions $f : \mathbb{R}_+ \to \mathbb{R}_+$ such that*

$$f(x^2) + f(y) = f(x^2 + y + xf(4y))$$

for all $x, y \in \mathbb{R}_+$.

Problem 20.84 (IMO 1969 Longlist). *Find all functions f defined for all x that satisfy the condition*

$$xf(y) + yf(x) = (x+y)f(x)f(y)$$

for all x and y. Prove that exactly two of them are continuous.

Problem 20.85 (IMO 1976 Longlist). *Find the function $f(x)$ defined for all real values of x that satisfies the conditions*
(a) $f(x+2) - f(x) = x^2 + 2x + 4$, *for all x,*
(b) $f(x) = x^2$, *if $x \in [0,2)$.*

Problem 20.86 (IMO 1979 Shortlist). *Prove that the functional equations*

$$f(x+y) = f(x) + f(y)$$

and

$$f(x+y+xy) = f(x) + f(y) + f(xy)$$

are equivalent.

Problem 20.87 (Austria 2001). *Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that, for all real numbers x and y, the functional equation*

$$f(f(x)^2 + f(y)) = x\,f(x) + y$$

is satisfied.

Problem 20.88 (IMO 1978 Longlist). *(Version 1) Let $c, s : \mathbb{R}^* \to \mathbb{R}$ be non-constant in any interval and satisfy*

$$c\left(\frac{x}{y}\right) = c(x)\,c(y) - s(x)\,s(y) \quad \text{for all } x, y \in \mathbb{R}^*.$$

Prove that:
(a) $c(1/x) = c(x)$, $s(1/x) = -s(x)$, *for any $x \neq 0$;*
(b) $c(1) = 1$, $s(1) = s(-1) = 0$;
(c) *c, s are either both even or both odd functions;*
(d) *Find functions c and s that also satisfy $c(x) + s(x) = x^n$ for all $x \neq 0$, where n is a given positive integer.*

(Version 2) Given a non-constant function $f : \mathbb{R}_+^* \to \mathbb{R}$ such that $f(xy) = f(x)f(y)$ for any $x, y \neq 0$, find functions $c, s : \mathbb{R}_+^* \to \mathbb{R}$ that satisfy

$$c\left(\frac{x}{y}\right) = c(x)\,c(y) - s(x)\,s(y) \quad \textit{for all } x, y \in \mathbb{R}^*$$

and

$$c(x) + s(x) = f(x) \quad \textit{for all } x \in \mathbb{R}^*.$$

Problem 20.89 (IMO 1989 Longlist). *Let $f : \mathbb{Q} \to \mathbb{R}_+$ satisfy the following conditions:*
(a) $f(0) = 0$;
(b) $f(ab) = f(a)\,f(b)$;
(c) $f(a+b) \leq f(a) + f(b)$;
(d) $f(m) \leq 1989$ *for all* $m \in \mathbb{Z}$.
Prove that

$$f(a+b) = \max\{f(a),\ f(b)\}, \quad \textit{if } f(a) \neq f(b).$$

Problem 20.90 (IMO 2002 Shortlist). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(f(x) + y) = 2x + f(f(y) - x) \quad \textit{for all } x, y \in \mathbb{R}.$$

Problem 20.91 (IMO 2002). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$[f(x) + f(z)]\,[f(y) + f(t)] = f(xy - zt) + f(xt + yz) \quad \textit{for all } x, y, z, t \in \mathbb{R}.$$

Problem 20.92 (IMO 2010). *Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that the equality*

$$f(\lfloor x \rfloor y) = f(x)\,\lfloor f(y) \rfloor$$

holds for all $x, y \in \mathbb{R}$.

Problem 20.93 ([1], Problem 11345, February 2008). *Find all increasing functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x + f(y)) = f^2(x) + f(y),$$

for all real x and y.

Problem 20.94 (Ukraine 2001). *Does there exist a function $f : \mathbb{R} \to \mathbb{R}$ such that, for all $x, y \in \mathbb{R}$, the following equality holds?*

$$f(xy) = \max\{f(x), y\} + \min\{f(y), x\}.$$

Problem 20.95 (BH 1997). *Let $f : A \to \mathbb{R}$, $A \subset \mathbb{R}$, be a function with the following characteristic:*

$$f(x + y) = f(x)\,f(y) - f(xy) + 1,$$

for all $x, y \in A$.

(a) *If $f : A \to \mathbb{R}$, $\mathbb{N} \subset A \subset \mathbb{R}$, is such a function, prove that the following is true:*

$$f(n) = \begin{cases} \frac{c^{n+1}-1}{c-1} & \text{if } n \in \mathbb{N}, \quad c \neq 1, \\ n + 1 & \text{if } n \in \mathbb{N}, \quad c = 1, \end{cases}$$

with $c = f(1) - 1$.

(b) *Solve the given functional equation for $A = \mathbb{N}$.*

(c) *If $A = \mathbb{Q}$, find all the functions f that satisfy the given equation and the condition $f(1997) \neq f(1998)$.*

20.5 Functional Equations with No Parameters

Problem 20.96 ([3], Problem 2828, April 2003). *Find all functions $f : \mathbb{R} \setminus \{1\} \to \mathbb{R}$ satisfying the functional equation*

$$f(x) + a\,f\left(\frac{x + b}{x - 1}\right) = c - x,$$

where $a \neq \pm 1$, $b \neq -1$ are constants.

Problem 20.97 (Turkey TST 1999). *Determine all functions $f : \mathbb{R} \to \mathbb{R}$ such that the set*

$$\left\{\frac{f(x)}{x} \;:\; x \neq 0 \quad \text{and} \quad x \in \mathbb{R}\right\}$$

is finite, and for all $x \in \mathbb{R}$,

$$f(x - 1 - f(x)) = f(x) - x - 1.$$

Problem 20.98 (Switzerland 1999). *Determine all functions $f : \mathbb{R}^* \to \mathbb{R}$ satisfying*

$$\frac{1}{x} f(-x) + f\left(\frac{1}{x}\right) = x,$$

for all $x \in \mathbb{R}^*$.

Problem 20.99 (Putnam 1971). *Find all functions $f : \mathbb{R} \setminus \{0, 1\} \to \mathbb{R}$ satisfying the functional equation*

$$f(x) + f\left(\frac{x-1}{x}\right) = 1 + x.$$

Problem 20.100 (Austria 1985). *Determine all functions $f : \mathbb{R} \to \mathbb{R}$ satisfying the functional equation*

$$x^2 f(x) + f(1-x) = 2x - x^4 \quad \text{for all } x \in \mathbb{R}.$$

Problem 20.101 (APMO 1989). *Determine all nondecreasing functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x) + g(x) = 2x \quad \text{for all } x,$$

and where g is defined by

$$f(g(x)) = g(f(x)) = x.$$

Problem 20.102. *Find all the solutions $f : I \to I$ of the functional equation*

$$f^3(x) = 0.$$

Problem 20.103. *Find all the solutions of the functional equation*

$$f^3(x) = x.$$

Problem 20.104. *Is there a continuous solution of the functional equation*

$$f^4(x) = x$$

that is not a solution of $f^2(x) = x$? If no, why? If yes, give an example.

Problem 20.105 (Putnam 1996). *Let $c \geq 0$ be a constant. Give a complete description, with proof, of the set of all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$f(x) = f\left(x^2 + c\right),$$

for all $x \in \mathbb{R}$.

Problem 20.106 (Putnam 2000). *Let $f(x)$ be a continuous function such that*

$$f\left(2x^2 - 1\right) = 2x\, f(x)$$

for all x. Show that $f(x) = 0$ for $-1 \leq x \leq 1$.

Problem 20.107 (Korea 1999). *Suppose that, for any real x with $|x| \neq 1$, a function $f(x)$ satisfies*

$$f\left(\frac{x-3}{x+1}\right) + f\left(\frac{3+x}{1-x}\right) = x.$$

Find all possible $f(x)$.

Problem 20.108 (India 1992). *Determine all functions $f : \mathbb{R} \setminus [0,1] \to \mathbb{R}$ such that*

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)}.$$

Problem 20.109 (Israel 1995). *For a given real number α, find all functions $f : (0, +\infty) \to (0, +\infty)$ satisfying*

$$\alpha\, x^2 f\left(\frac{1}{x}\right) + f(x) = \frac{x}{x+1}.$$

Problem 20.110 (Poland 1993; India 1994). *Find all real-valued functions f on the reals such that*
(a) $f(-x) = -f(x)$, *for all x;*
(b) $f(x+1) = f(x) + 1$, *for all x;*
(c) $f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2}$, *for $x \neq 0$.*

Problem 20.111 (Poland 1992). *Find all functions $f : \mathbb{Q}_+^* \to \mathbb{Q}_+^*$ such that*

$$f(x+1) = f(x) + 1$$

and

$$f(x^3) = f(x)^3,$$

for all x.

Problem 20.112 (IMO 1979 Shortlist). *For all rational x satisfying $0 \le x \le 1$, f is defined by*

$$f(x) = \begin{cases} \frac{f(2x)}{4} & \text{if } 0 \le x \le \frac{1}{2}, \\ \frac{3+f(2x-1)}{4} & \text{if } \frac{1}{2} \le x \le 1. \end{cases}$$

Given that $x = 0.b_1b_2b_3\ldots$ is the binary representation of x, find $f(x)$.

Problem 20.113 (Korea 1993). *Let n be a given natural number. Find all continuous functions $f(x)$ satisfying*

$$\sum_{k=0}^{n} \binom{n}{k} f\left(x^{2^k}\right) = 0.$$

Problem 20.114 (Croatia 1995). *Let $t \in (0,1)$ be a given number. Find all functions $f : \mathbb{R} \to \mathbb{R}$ continuous at $x = 0$ satisfying*

$$f(x) - 2f(tx) + f(t^2x) = x^2 \quad \text{for all } x \in \mathbb{R}.$$

Problem 20.115 (IMO 1979 Longlist). *Let E be the set of all bijective functions from $\mathbb{R}$ to $\mathbb{R}$ satisfying*

$$f(t) + f^{-1}(t) = 2t \quad \text{for all } t \in \mathbb{R}.$$

Find all elements of E that are monotonic functions.

20.6 Fixed Points and Cycles

Problem 20.116. *Let $f : [a,b] \to [a,b]$ be a monotonic function, not necessarily continuous. Prove that f has a fixed point.*

Problem 20.117. *Let $f : [a, b] \to [a, b]$ be a continuous surjective map. Prove that the function $f^2(x)$ has at least two fixed points.*

Problem 20.118 ([2], Problem 841, January 2007)**.** *Assume that the quadratic polynomial $f(x) = a\,x^2 + b\,x + c$, $a \neq 0$, has two distinct fixed points x_1 and x_2. If 1 and -1 are two fixed points of the function $f^2(x)$, but not of $f(x)$, find the exact values of x_1 and x_2.*

Problem 20.119 ([5], Problem Q982, June 2008)**.** *It is well known that if $f : [0, 1] \to [0, 1]$ is a continuous function, then f must have at least one fixed point. However, the example $f(x) = x^2$, which only has 0 and 1 as fixed points, shows that the set of fixed points does not need to be an interval.*

Let $f : [0, 1] \to [0, 1]$ be a function with $|f(x) - f(y)| \leq |x - y|$ for all $x, y \in [0, 1]$. Prove that the set of all fixed points of f is either a single point or an interval.

Problem 20.120 (IMSUC 2007)**.** *Let*

$$T = \{(tq, 1 - t) \in \mathbb{R}^2 \mid t \in [0, 1],\ q \in \mathbb{Q}\}.$$

Prove that each continuous function $f : T \to T$ has a fixed point.

Problem 20.121 (IMO 1976 Shortlist)**.** *Let $I = (0, 1]$. For a given number $a \in (0, 1]$, we define the map $T : I \to I$ as follows:*

$$T(x) = \begin{cases} x + (1 - a) & \text{if } 0 < x \leq a, \\ x - a & \text{if } a < x \leq 1. \end{cases}$$

Show that, for every interval $J \subset I$, there exists an integer $n > 0$ such that $T^n(J) \cap J \neq \varnothing$.

Problem 20.122 (IMO 1976 Longlist)**.** *Let $Q = [0, 1] \times [0, 1]$ and let $T : Q \to Q$ be defined as follows:*

$$T(x, y) = \begin{cases} \left(2x, \frac{y}{2}\right) & \text{if } 0 \leq x \leq 1/2, \\ \left(2x - 1, \frac{y+1}{2}\right), & \text{if } 1/2 < x \leq 1. \end{cases}$$

Show that, for every disc $D \subset Q$, there exists an integer $n > 0$ such that $T^n(D) \cap D \neq \varnothing$.

Problem 20.123 (IMO 1990 Shortlist). *Let a, b be natural numbers with $1 \leq a \leq b$ and $M = \left\lfloor \frac{a+b}{2} \right\rfloor$. Define the function $f : \mathbb{Z} \to \mathbb{Z}$ by*

$$f(n) = \begin{cases} n + a & \text{if } n < M, \\ n - b & \text{if } n \geq M. \end{cases}$$

Find the smallest natural number $k > 0$ such that $f^k(0) = 0$.

Problem 20.124 ([5], Problem 2003). *Let $(X, \langle\ \rangle)$ be a real inner product space, and let*

$$B = \{x \in X : \|x\| \leq 1\}$$

be the unit ball in X, where $\|x\| = \sqrt{\langle x, x \rangle}$. Let $f : B \to B$ be a function satisfying

$$\|f(x) - f(y)\| \leq \|x - y\| \quad \text{for all } x, y \in B.$$

Prove that the set of fixed points of f is convex.

Problem 20.125 ([1], Problem E3342, August/September 1989). *Let $S = \{1, 2, \ldots, n^2\}$ and define a permutation $f : S \to S$ as follows. Take n^2 cards numbered from 1 to n^2 and lay them in a square array, with the i-th row containing cards $(i-1)n + 1$, $(i-1)n + 2$, $\ldots$, in in order from left to right. Pick up the cards along rising diagonals, starting with the upper left-hand corner. If card j is the k-th card picked up, put $f(j) = k$. For example, if $n = 3$, then $f(1) = 1$, $f(4) = 2$, $f(2) = 3$, $f(7) = 4$, $f(5) = 5$, $f(3) = 6$, $f(8) = 7$, $f(6) = 8$, $f(9) = 9$.*

For each value of n, both 1 and n^2 are fixed points of f. If n is odd, then $(n^2 + 1)/2$ is also a fixed point of f. Characterize those n for which f has a fixed point not in $\{1,\ n^2,\ (n^2 + 1)/2\}$.

Problem 20.126 ([1], Problem E3428, March 1991). *Let I be a non-empty interval on the real line. Let $f : I \to I$ be a continuous function having the property that, for each $x \in I$, there exists a positive integer $n = n(x)$ with $f^n(x) = x$. For given I, characterize all such functions.*

Problem 20.127 (Sweden 1993). *Let a and b be real numbers and let*

$$f(x) = \frac{1}{a\,x + b}.$$

For which a and b are there three distinct real numbers x_1, x_2, x_3 such that

$$f(x_1) = x_2, \quad f(x_2) = x_3, \quad f(x_3) = x_1\ ?$$

20.7 Existence of Solutions

Problem 20.128. *Show that there is no function that is continuous at every rational and discontinuous at every irrational.*

Problem 20.129 (Bulgaria 1998). *Prove that there does not exist a function $f : \mathbb{R}_+^* \to \mathbb{R}_+^*$ such that*

$$f(x)^2 \geq f(x+y)\,(f(x)+y),$$

for any $x, y \in \mathbb{R}_+^*$.

Problem 20.130 (Iran 1998). *Let $f_1, f_2, f_3 : \mathbb{R} \to \mathbb{R}$ be functions such that $a_1 f_1 + a_2 f_2 + a_3 f_3$ is monotonic for all $a_1, a_2, a_3 \in \mathbb{R}$. Prove that there exist $c_1, c_2, c_3 \in \mathbb{R}$, not all zero, such that*

$$c_1 f_1(x) + c_2 f_2(x) + c_3 f_3(x) = 0,$$

for all $x \in \mathbb{R}$.

Problem 20.131 (Belarus 2000). *Does there exist a function $f : \mathbb{N}^* \to \mathbb{N}^*$ such that*

$$f(f(n-1)) = f(n+1) - f(n)$$

for all $n \geq 2$?

Problem 20.132 (LMO 1991). *Does there exist a function $F : \mathbb{N}^* \to \mathbb{N}^*$ such that, for any natural number x,*

$$F(F(F(\cdots F(x)\cdots))) = x + 1\ ?$$

Here F is applied F times.

Problem 20.133 (IMO 1989 Shortlist). *Let $g : \mathbb{C} \to \mathbb{C}$, $\omega \in \mathbb{C}$, $a \in \mathbb{C}$, $\omega^3 = 1$, and $\omega \neq 1$. Show that there is one and only one function $f : \mathbb{C} \to \mathbb{C}$ such that*

$$f(z) + f(\omega z + a) = g(z), \quad z \in \mathbb{C}.$$

Problem 20.134 (IMO 1967 Longlist). *The function $\phi(x, y, z)$ defined for all triples (x, y, z) of real numbers is such that there are two functions f and g defined for all pairs of real numbers such that*

$$\phi(x, y, z) = f(x + y, z) = g(x, y + z),$$

for all real numbers x, y, z. Show that there is a function h of one real variable such that

$$\phi(x, y, z) = h(x + y + z),$$

for all real numbers x, y, z.

Problem 20.135 (IMO 1992 Shortlist). *Let $\mathbb{R}_+$ be the set of all non-negative real numbers. Given two positive real numbers a and b, suppose that a mapping $f : \mathbb{R}_+ \to \mathbb{R}_+$ satisfies the functional equation*

$$f(f(x)) + a\,f(x) = b(a + b)x.$$

Prove that there exists a unique solution of this equation.

Problem 20.136 (IMO 1993). *Let $\mathbb{N}^* = \{1, 2, 3, \ldots\}$. Determine whether there exists a function $f : \mathbb{N}^* \to \mathbb{N}^*$ such that $f(1) = 2$ and*

$$\begin{aligned} f(f(n)) &= f(n) + n, \quad \text{for all } n \text{ in } \mathbb{N}^*, \\ f(n) &< f(n + 1), \quad \text{for all } n \text{ in } \mathbb{N}^*. \end{aligned}$$

Problem 20.137 (Russia 2000). *Is there a function $f : \mathbb{R} \to \mathbb{R}$ such that*

$$|f(x + y) + \sin x + \sin y| < 2$$

for all $x, y \in \mathbb{R}$?

Problem 20.138 (Turkey 2000). *Let $f : \mathbb{R} \to \mathbb{R}$ be a function such that*

$$|f(x + y) - f(x) - f(y)| \leq 1$$

for all $x, y \in \mathbb{R}$. Show that there exists a function $g : \mathbb{R} \to \mathbb{R}$ with

$$|f(x) - g(x)| \leq 1$$

for all $x \in \mathbb{R}$ and

$$g(x + y) = g(x) + g(y)$$

for all $x, y \in \mathbb{R}$.

Problem 20.139 (Romania 2000). *Prove that there is no function*

$$f : (0, +\infty) \to (0, +\infty)$$

such that

$$f(x + y) \geq f(x) + y\, f^2(x)$$

for all $x, y \in (0, +\infty)$.

Problem 20.140 (Romania 2005). *Let* $f : \mathbb{R} \to \mathbb{R}$ *be a convex function.*
(a) *Prove that* f *is continuous.*
(b) *Prove that there exists a unique function* $g : [0, +\infty) \to \mathbb{R}$ *such that, for all* $x \geq 0$,

$$f(x + g(x)) = f(g(x)) - g(x).$$

Problem 20.141 ([31], Problem 54). *Is there a non-trivial function* $f(x)$, *continuous on the whole line, that satisfies the functional equation*

$$f(x) + f(2x) + f(3x) = 0,$$

for all x*?*

Problem 20.142 (Iran 1995). *Does there exist a function* $f : \mathbb{R} \to \mathbb{R}$ *that fulfills all of the following conditions:*

(a) $f(1) = 1$,
(b) *there exists* $M > 0$ *such that* $-M < f(x) < M$,
(c) *if* $x \neq 0$, *then*

$$f\left(x + \frac{1}{x^2}\right) = f(x) + f\left(\frac{1}{x}\right)^2 \ ?$$

Problem 20.143 (IMO 1970 Longlist). *Let* E *be a finite set and let* $f : \mathcal{P}(E) \to \mathbb{R}_+$ *such that for any two disjoint sets* A, B *of* E,

$$f(A \cup B) = f(A) + f(B).$$

Prove that there exists a subset F *of* E *such that, if with each* $A \subset E$ *we associate a subset* A' *consisting of elements of* A *that are not in* F, *then* $f(A) = f(A')$, *and* $f(A)$ *is zero if and only if* A *is a subset of* F.

Problem 20.144 ([2], Problem 869, January 2008). *Prove that there does not exist a positive twice differentiable function* f *defined on* $[0, +\infty)$ *such that*

$$f(x)\, f''(x) \leq -1 \quad \textit{for all } x \geq 0.$$

20.8 Systems of Functional Equations

Problem 20.145. *Given a number $n \in \mathbb{N}$, find the continuous functions f_k, $k = 0, 1, \ldots, n$ such that*

$$f_n(x+y) = \sum_{k=0}^{n} f_k(x)\, f_{n-k}(y) \quad \text{for all } x, y \geq 0.$$

Problem 20.146 (APMC 1999). *Let $n \geq 2$ be a given integer. Determine all systems of n functions $(f_1, \ldots, f_n)$ where $f_i : \mathbb{R} \to \mathbb{R}$, $i = 1, 2, \ldots, n$, such that, for all $x, y \in \mathbb{R}$, the following equalities hold:*

$$\begin{aligned} f_1(x) - f_2(x)f_2(y) + f_1(y) &= 0, \\ f_2(x^2) - f_3(x)f_3(y) + f_2(y^2) &= 0, \\ \cdots \quad &\cdots \\ f_{n-1}(x^{n-1}) - f_n(x)f_n(y) + f_{n-1}(y^{n-1}) &= 0, \\ f_n(x^n) - f_1(x)f_1(y) + f_n(y^n) &= 0. \end{aligned}$$

Problem 20.147 (IMO 1977 Longlist). *Let $f : \mathbb{Q}^* \times \mathbb{Q}^* \to \mathbb{R}_+^*$ be a function satisfying the equations*

$$\begin{aligned} &f(x_1x_2, y) = f(x_1, y)\, f(x_2, y) \quad \text{for all } x_1, x_2, y, \\ &f(x, y_1y_2) = f(x, y_1)\, f(x, y_2) \quad \text{for all } x, y_1, y_2, \\ &f(x, 1-x) = 1 \quad \text{for all } x. \end{aligned}$$

Prove that

$$\begin{aligned} f(x,x) &= f(x,-x) = 1 \quad \text{for all } x, \\ f(x,y)\, f(y,x) &= 1 \quad \text{for all } x, y. \end{aligned}$$

Problem 20.148 (BMO 2003). *Find all functions $f : \mathbb{Q} \to \mathbb{R}$ that fulfill the following conditions:*

(a) $f(1) + 1 > 0$;

(b) $f(x+y) - xf(y) - yf(x) = f(x)f(y) - x - y + xy$, *for all* $x, y \in \mathbb{Q}$;

(c) $f(x) = 2f(x+1) + x + 2$, *for every* $x \in \mathbb{Q}$.

Problem 20.149 (IMO 1989 Longlist). *Let $f : \mathbb{R} \to \mathbb{R}$ be such that $f(1) = 1$ and*

$$f(x+y) = f(x) + f(y) \quad \text{for all } x, y,$$

$$f(x)\, f\left(\frac{1}{x}\right) = 1 \quad \text{for all } x \neq 0.$$

Find f.

Problem 20.150 (Putnam 1977). *Let $f, g, u : \mathbb{R} \to \mathbb{R}$ such that*

$$\begin{aligned} \frac{u(x+1)+u(x-1)}{2} &= f(x), \\ \frac{u(x+4)+u(x-4)}{2} &= g(x). \end{aligned}$$

Determine u in terms of f and g.

Problem 20.151 ([1], Problem 11053, December 2003). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that*

$$\begin{aligned} f\big(x + f(x)f(y)\big) &= f(x) + x\,f(y), \\ f\big(x + f(xy)\big) &= f(x) + x\,f(y), \end{aligned}$$

for all $x, y \in \mathbb{R}$.

20.9 Conditional Functional Equations

Problem 20.152 (IMO 2008). *Find all functions $f : (0, +\infty) \to (0, +\infty)$ such that*

$$\frac{f(w)^2 + f(x)^2}{f(y^2) + f(z^2)} = \frac{w^2 + x^2}{y^2 + z^2},$$

for all positive real numbers w, x, y, z satisfying $wx = yz$.

Problem 20.153 (Poland 1991). *On the Cartesian plane, consider the set V of all vectors with integer coordinates. Determine all functions $f : V \to \mathbb{R}$ satisfying the conditions:*

(a) *$f(\vec{u}) = 1$ for each of the four vectors $\vec{u} \in V$ of unit length.*

(b) *$f(\vec{x} + \vec{y}) = f(\vec{x}) + f(\vec{y})$ for every two perpendicular vectors $\vec{x}, \vec{y} \in V$. (The zero vector is considered to be perpendicular to every vector).*

Problem 20.154 (IMO 1989 Longlist). *Let $a \in (0, 1)$ and let f be a continuous function on $[0, 1]$ satisfying $f(0) = 0$, $f(1) = 1$ and*

$$f\left(\frac{x+y}{2}\right) = (1-a)\,f(x) + a\,f(y) \quad \text{for all } x \le y \text{ with } x, y \in [0, 1].$$

Determine $f(1/7)$.

Problem 20.155 (IMO 2004). *Find all polynomials $P(x)$ with real coefficients that satisfy the equality*

$$P(a-b)+P(b-c)+P(c-a) = 2\,P(a+b+c),$$

for all triples a, b, c of real numbers such that $ab+bc+ca=0$.

20.10 Polynomials

Problem 20.156 (IMO 1975). *Find all polynomials P, in two variables, with the following properties:*

(a) *For a positive integer n and all real t, x, y,*

$$P(tx, ty) = t^n\,P(x, y);$$

that is, P is homogeneous of degree n.

(b) *For all real a, b, c,*

$$P(b+c, a)+P(c+a, b)+P(a+b, c) = 0.$$

(c) $P(1,0) = 1$.

Problem 20.157 (Vietnam 1998). *Prove that for each positive odd integer n, there is exactly one polynomial $P(x)$ of degree n with real coefficients satisfying*

$$P\left(x-\frac{1}{x}\right) = x^n-\frac{1}{x^n} \quad \textit{for all } x \in \mathbb{R}^*.$$

Determine if the above assertion holds for positive even integers n.

Problem 20.158 (SPCMO 1998). *Find all polynomials $P(x, y)$ in two variables such that, for any x and y,*

$$P(x+y, y-x) = P(x-y).$$

Problem 20.159 (Greece 2005). *Find the polynomial $P(x)$ with real coefficients such that $P(2)=12$ and*

$$P\left(x^2\right) = x^2\left(x^2+1\right)\,P(x),$$

for each $x \in \mathbb{R}$.

Problem 20.160 (IMO 1992 Shortlist). *Let f, g and a be polynomials with real coefficients, f and g in one variable and a in two variables. Suppose*

$$f(x) - f(y) = a(x, y)\,\big(g(x) - g(y)\big),$$

for all $x, y \in \mathbb{R}$. Prove that there exists a polynomial h with

$$f(x) = h(g(x)) \quad \text{for all } x \in \mathbb{R}.$$

Problem 20.161 (China 1999). *Let a be a real number. Let $\{f_n(x)\}$ be a sequence of polynomials such that*
(a) $f_0(x) = 1$ *and*
(b) $f_{n+1}(x) = x f_n(x) + f_n(ax)$, *for* $n = 0, 1, 2, \ldots$.
Prove that

$$f_n(x) = x^n f_n\left(\frac{1}{x}\right), \quad n = 0, 1, 2, \ldots,$$

and find an explicit expression for $f_n(x)$.

Problem 20.162 (Georgia TST 2005). *Find all polynomials with real coefficients for which the equality*

$$P(2P(x)) = 2\,P^2(x) + 2\,P(x)^2$$

holds for any real number x.

Problem 20.163 (Japan 1995). *Find all non-constant rational functions $f(x)$ of x satisfying*

$$f(x)^2 - a = f(x^2),$$

where a is a real number.

*A **rational function** of x is one that can be expressed as the ratio of two polynomials of x.*

Problem 20.164 (CSM 1999). *Find all positive integers k for which the following assertion holds: if $F(x)$ is a polynomial with integer coefficients that satisfies*

$$F(c) \le k, \quad \text{for all } c \in \{0, 1, \ldots, k+1\},$$

then

$$F(0) = F(1) = \cdots = F(k+1).$$

Problem 20.165 (Bulgaria 1998). *The polynomials $P_n(x,y)$ for $n = 1, 2, \ldots$ are defined by*

$$\begin{aligned} P_1(x,y) &= 1, \\ P_{n+1}(x,y) &= (x+y-1)\,(y+1)\,P_n(x,y+2) + (y-y^2)\,P_n(x,y). \end{aligned}$$

Prove that

$$P_n(x,y) = P_n(y,x),$$

for all n and all x, y.

Problem 20.166 (CSM 1998). *A polynomial $P(x)$ of degree $n \geq 5$ with integer coefficients and n distinct integer roots is given. Find all integer roots of $P(P(x))$ given that 0 is a root of $P(x)$.*

Problem 20.167 (IMO 1976 Longlist). *Prove that if, for a polynomial $P(x,y)$, we have*

$$P(x-1, y-2x+1) = P(x,y),$$

then there exists a polynomial $Q(x)$ such that $P(x,y) = Q(y-x^2)$.

Problem 20.168 (IMO 1979 Shortlist). *Find all polynomials $P(x)$ with real coefficients for which*

$$P(x)\,P(2x^2) = P(2x^3+x).$$

Problem 20.169 (IMO 1978 Longlist). *Let the polynomials*

$$P(x) = x^n + \sum_{k=0}^{n-1} a_k\,x^k, \quad P(x) = x^m + \sum_{k=0}^{m-1} b_k\,x^k$$

be given, satisfying the identity

$$P(x)^2 = (x^2-1)\,Q(x)^2 + 1.$$

Prove that

$$P'(x) = n\,Q(x).$$

Problem 20.170 (IMO 1989 Longlist). *Let $P(x)$ be a polynomial such that the following inequalities are satisfied:*

$$\begin{aligned} P(1) &> P(0) \ > \ 0, \\ P(2) &> 2P(1) - P(0), \\ P(3) &> 3P(2) - 3P(1) + P(0), \end{aligned}$$

and also

$$P(n+4) > 4P(n+3) - 6P(n+2) + 4P(n+1) - P(n) \quad \textit{for all } n \in \mathbb{N}.$$

Prove that

$$P(n) > 0, \quad n \in \mathbb{N}.$$

Problem 20.171 (Putnam 1972). *Let n be an integer greater than 1. Show that there exists a polynomial $P(x, y, z)$ with integral coefficients such that*

$$P(x^n, x^{n+1}, x^{n+2} + x) = x.$$

Problem 20.172 (USA 1984). *$P(x)$ is a polynomial of degree $3n$ such that*

$$\begin{aligned} &P(0) = P(3) = \ldots = P(3n) = 2, \\ &P(1) = P(4) = \ldots = P(3n-2) = 1, \\ &P(2) = P(5) = \ldots = P(3n-1) = 0, \\ &P(3n+1) = 730. \end{aligned}$$

Determine n.

Problem 20.173 (IMO 1993). *Let $P(x) = x^n + 5x^{n-1} + 3$, where $n > 1$ is an integer. Prove that $P(x)$ cannot be expressed as the product of two polynomials, each of which has all of its coefficients, integers, and degree at least 1.*

Problem 20.174 ([5], Problem 1807, December 2008). *Let P be a polynomial with integer coefficients and let s be an integer such that, for some positive integer n, the number $s^{n+1}\, P(s)^n$ is a positive zero of P. Prove that $P(2) = 0$.*

Problem 20.175 ([2], Problem 879, May 2008). *Consider the polynomial*

$$f(x) = x^4 - 4ax^3 + 6b^2x^2 - 4c^3x + d^4,$$

where a, b, c, and d are positive real numbers. Prove that if f has four positive distinct roots, then $a > b > c > d$.

Problem 20.176 (Putnam 1976). *Let*

$$P(x,y) = x^2y + xy^2, \quad Q(x,y) = x^2 + xy + y^2.$$

For $n = 2, 3, \ldots,$ *let*

$$F_n(x,y) = (x+y)^n - (x^n + y^n),$$
$$G_n(x,y) = (x+y)^n + x^n + y^n.$$

Prove that, for each n*, either* F_n *or* G_n *is expressible as a polynomial in* P *and* Q *with integer coefficients.*

Problem 20.177 (Putnam 1981). *Let* $P(x)$ *be a polynomial with real coefficients and form the polynomial*

$$Q(x) = (x^2+1)P(x)\,P'(x) + x\left[P(x)^2 + P'(x)^2\right].$$

Given that $P(x)$ *has* n *distinct real roots exceeding 1, prove or disprove that* $Q(x)$ *has at least* $2n-1$ *distinct real roots.*

Problem 20.178 (HIMC 1999). *Let* $P(x)$ *be a polynomial of degree at least* 2*. Define the sequence* $\{Q_n(x)\}$ *by* $Q_1(x) = P(x)$ *and*

$$Q_{k+1}(x) = P(Q_k(x)), \quad k = 1, 2, \ldots.$$

Let r_n *be the average of the roots of* $Q_n(x)$*. It is given that* $r_{19} = 99$*. Find* r_{99}*.*

20.11 Functional Inequalities

Problem 20.179 (Popoviciu's Theorem [35]**).** *Let* f *be a convex real-valued function defined on an interval* I*. Then*

$$f(x)+f(y)+f(z)+3f\left(\frac{x+y+z}{3}\right) \geq 2\left[f\left(\frac{x+y}{2}\right)+f\left(\frac{y+z}{2}\right)+f\left(\frac{z+x}{2}\right)\right],$$

for all $x, y, z \in I$*.*

Problem 20.180 ([58]). *Let $f(x)$ be subadditive on $\mathbb{R}$ and let $f'(x)$ exist on $(0,+\infty)$. If $f(x)+f(-x) \leq 0$ for any $x \in (0,+\infty)$, then $f'(x)$ is decreasing on $(0,+\infty)$.*

Problem 20.181 (IMO 1977). *Let $f(n)$ be a function defined on the set of all positive integers and having all of its values in the same set. Prove that if*

$$f(n+1) > f(f(n))$$

for each positive integer n, then

$$f(n) = n, \quad \text{for each } n.$$

Problem 20.182 ([11], p. 35). *Determine $f : \mathbb{N}^* \to \mathbb{N}^*$ such that, for all k, m, n,*

$$f(km) + f(kn) - f(k)\, f(mn) \geq 1.$$

Problem 20.183 (APMO 1994). *Let $f : \mathbb{R} \to \mathbb{R}$ such that*
(a) *$f(x) + f(y) + 1 \geq f(x+y) \geq f(x) + f(y)$ for all x, y;*
(b) *$f(0) \geq f(x)$ for all $x \in [0, 1)$;*
(c) *$f(1) = f(-1) = 1$.*
Find all such functions f.

Problem 20.184 (HK 1999). *Let f be a function defined on the reals with the following properties:*
(A) $f(1) = 1$;
(B) $f(x+1) = x\, f(x)$;
(C) $f(x) = 10^{g(x)}$;
where $g(x)$ is a function defined on the reals satisfying

$$g(ty + (1-t)z) \leq t\, g(y) + (1-t)\, g(z)$$

for all real y, z and some $0 \leq t \leq 1$.

(a) *Prove that*

$$t\,[g(n) - g(n-1)] \leq g(n+t) - g(n) \leq t\,[g(n+1) - g(n)],$$

where n is an integer and $0 \leq t \leq 1$.

(b) *Prove that*

$$\frac{4}{3} \leq f\left(\frac{1}{2}\right) \leq \frac{4\sqrt{2}}{3}.$$

Problem 20.185 (IMSUC 2008). *Let $\gamma : [0,1] \to [0,1] \times [0,1]$ be a mapping such that, for each $s, t \in [0,1]$,*

$$|\gamma(s) - \gamma(t)| \leq M\,|s-t|^{\alpha}$$

in which α, M are fixed numbers. Prove that if γ is surjective, then $\alpha < 1/2$.

Problem 20.186 (IMO 2007 Shortlist; Ukraine TST 2008). *Consider those functions $f : \mathbb{N}^* \to \mathbb{N}^*$ that satisfy the condition*

$$f(m+n) \geq f(m) + f(f(n)) - 1$$

for all $m, n \in \mathbb{N}^*$. Find all possible values of $f(2007)$.

Problem 20.187 (Romania 1981). *Determine whether there exists an injective function $f : \mathbb{R} \to \mathbb{R}$ with the property that, for all x,*

$$f(x^2) - f(x)^2 \geq \frac{1}{4}.$$

Problem 20.188 ([11], p. 227). *If f is a surjective function and g is a bijective function, both defined on $\mathbb{N}$ and satisfying the property*

$$f(n) \geq g(n) \quad \text{for all } n,$$

then $f = g$.

Problem 20.189 (Russia 2000). *Find all functions $f : \mathbb{R} \to \mathbb{R}$ that satisfy the inequality*

$$f(x+y) + f(y+z) + f(z+x) \geq 3\,f(x+2y+3z)$$

for all $x, y, z \in \mathbb{R}$.

Problem 20.190 (Japan 2007). *Find all functions $f : \mathbb{R}_+^* \to \mathbb{R}$ such that*

$$f(x) + f(y) \leq \frac{f(x+y)}{2},$$
$$\frac{f(x)}{x} + \frac{f(y)}{y} \geq \frac{f(x+y)}{x+y},$$

for all $x, y > 0$.

Problem 20.191 (IMO 1979 Longlist). *If a function satisfies the conditions*

$$f(x) \leq x \quad \textit{for all } x \in \mathbb{R}$$

and

$$f(x+y) \leq f(x) + f(y) \quad \textit{for all } x, y \in \mathbb{R},$$

show that

$$f(x) = x \quad \textit{for all } x \in \mathbb{R}.$$

20.12 Functional Equations Containing Derivatives

Problem 20.192 (Romania 1999). *The function $f : \mathbb{R} \to \mathbb{R}$ is differentiable and*

$$f(x) = f\left(\frac{x}{2}\right) + \frac{x}{2} f'(x),$$

for every number x. Prove that f is a linear function.

Problem 20.193 (Putnam 1941). *Show that any solution $f(t)$ of the functional equation*

$$f(x+y)\, f(x-y) = f(x)^2 + f(y)^2 - 1 \quad \textit{for all } x, y \in \mathbb{R}$$

is such that

$$f''(t) = \pm m^2 f(t)$$

(m a non-negative constant) assuming the existence and continuity of the second derivative. Deduce that $f(t)$ is one of the functions

$$\pm\cos(mt), \qquad \pm\cosh(mt).$$

Problem 20.194 (Putnam 1997). *Let f be a twice-differentiable real-valued function satisfying*

$$f(x) + f''(x) = -x\, g(x)\, f'(x),$$

where $g(x) \geq 0$ for all real x. Prove that $|f(x)|$ is bounded.

Problem 20.195 (Poland 1977). *A function $h : \mathbb{R} \to \mathbb{R}$ is differentiable and satisfies $h(ax) = bh(x)$ for all x, where a and b are given positive numbers and $|a| \neq 0, 1$. Suppose that $h'(0) \neq 0$ and the function h' is continuous at $x = 0$. Prove that $a = b$ and that there is a real number c such that $h(x) = cx$ for all x.*

Problem 20.196 ([1], Problem E3338, August/September 1989). *Let s and t be given real numbers. Find all differentiable functions f on the real line that satisfy*

$$f'(sx+ty) = \frac{f(y)-f(x)}{y-x},$$

for all real x, y with $x \neq y$.

20.13 Functional Relations Containing Integrals

Problem 20.197 ([7], Problem 71-1). *Prove that there is only one non-negative function F for which*

$$F\left\{\int_0^x F(u)du\right\} = x, \quad x \geq 0,$$

namely, $F(x) = Ax^n$ for appropriate values of A and n.

Problem 20.198 (Romania 1999). *Let $\mathbb{R} \to \mathbb{R}$ be a monotonic function for which there exist $a, b, c, d \in \mathbb{R}$ with $a \neq 0$, $c \neq 0$ such that, for all x, the following equalities hold:*

$$\int_x^{x+\sqrt{3}} f(t)\,dt = a\,x + b,$$
$$\int_x^{x+\sqrt{2}} f(t)\,dt = c\,x + b.$$

Prove that f is a linear function.

Problem 20.199 (Putnam 1958). *Prove that if $f(x)$ is continuous for $a \leq x \leq b$ and*

$$\int_a^b x^n f(x)\,dx = 0, \quad n = 0, 1, 2, \ldots,$$

then $f(x)$ is identically zero on $a \leq x \leq b$.

Problem 20.200 (Romania 2004). *Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that, for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}^*$, we have*

$$n^2 \int_x^{x+\frac{1}{n}} f(t)\,dt = n\,f(x) + \frac{1}{2}.$$

Problem 20.201 (Romania 2004). *Let $f : [0,1] \to \mathbb{R}$ be an integrable function such that*

$$\int_0^1 f(x)\,dx = \int_0^1 x\,f(x)\,dx = 1.$$

Prove that

$$\int_0^1 f(x)^2\,dx \geq 4.$$

Problem 20.202 (Putnam 1972). *Let $f(x)$ be an integrable function in $0 \leq x \leq 1$ and suppose*

$$\int_0^1 f(x)\,x^k\,dx = 0, \quad k = 0, 1, \ldots, n-1,$$

and

$$\int_0^1 f(x)\,x^n\,dx = 1.$$

Show that

$$|f(x)| \geq 2^n\,(n+1)$$

in a set of positive measure.

Problem 20.203 (Putnam 1973). (a) *On [0,1], let f have a continuous derivative satisfying $0 < f'(x) \leq 1$. Also, suppose $f(0) = 0$. Prove that*

$$\left(\int_0^1 f(x)\,dx\right)^2 \geq \int_0^1 f(x)^3\,dx.$$

(b) *Show an example in which equality occurs.*

Problem 20.204 ([5], Problem 1824, June 2009). *Let f be a continuous real-valued function defined on [0, 1] and satisfying*

$$\int_0^1 f(x)\,dx = \int_0^1 x f(x)\,dx.$$

Prove that there exists a real number c, $0 < c < 1$, such that

$$cf(c) = \int_0^c x f(x)\,dx.$$

Problem 20.205 ([2], Problem 925, March 2010). *Let f be a twice differentiable function on $\mathbb{R}$ with f continuous on [0,1] such that*

$$\int_0^1 f(x)\,dx = 2\int_{1/4}^{3/4} f(x)\,dx.$$

Prove that there exists an $x_0 \in (0,1)$ such that $f''(x_0) = 0$.

Problem 20.206 ([2], Problem 898, March 2009). *Suppose f is a function defined on an open interval I such that $f''(x) \geq 0$ for all $x \in I$, and $[a,b] \subset I$. Prove that*

$$\int_0^1 f(a+(b-a)y)\,dy \geq \int_0^1 f\left(\frac{3a+b}{4} + \frac{b-a}{2}y\right)\,dy.$$

Part VII

AUXILIARY MATERIAL

ACRONYMS AND ABBREVIATIONS

APMO	Asian Pacific Mathematical Olympiad
APMC	Austrian-Polish Mathematics Competition
BH	Bosnia Herzegovina
BMO	Balkan Mathematical Olympiad
BWMC	Baltic Way Mathematical Contest
CSM	Czech and Slovak Match
HIMC	Hungary-Israel Mathematical Competition
HK	Hong Kong
IMSUC	Iranian Mathematical Society Undergraduate Competition
IMO	International Mathematical Olympiad
LMO	Leningrad Mathematical Olympiad
NMC	Nordic Mathematical Contest
Putnam	W. L. Putnam Mathematical Competition
RMM	Romanian Masters in Mathematics
ROM	Republic of Moldova
SPCMO	St. Petersburg City Mathematical Olympiad
TST	Team Selection Test
UK	United Kingdom

When the name of a country is given, it is implied that the problem was given in the national competition of that country. However, if the competition separates the exams for different grades, no effort has been made to include the corresponding grade.

SET CONVENTIONS

$\mathbb{N}$ the set of natural numbers including 0

$\mathbb{N}^*$ the set of positive natural numbers

In other books the symbol $\mathbb{N}$ may stand for the positive natural numbers, while $\mathbb{N}_0$ may be used to indicate the set of natural numbers including 0.

$\mathbb{Z}$ the set of integer numbers

$\mathbb{Z}^*$ the set of non-zero integer numbers

$\mathbb{Z}_+$ the set of non-negative integer numbers

$\mathbb{Z}_-$ the set of non-positive integer numbers

$\mathbb{Z}_+^*$ the set of positive natural numbers

$\mathbb{Z}_-^*$ the set of negative integer numbers

In other books the symbols $\mathbb{Z}_+$, $\mathbb{Z}_-$ may stand for the positive, negative integer numbers respectively.

$\mathbb{Q}$ the set of rational numbers

$\mathbb{Q}^*$ the set of non-zero rational numbers

$\mathbb{Q}_+$ the set of non-negative rational numbers

$\mathbb{Q}_-$ the set of non-positive rational numbers

$\mathbb{Q}_+^*$ the set of positive rational numbers

$\mathbb{Q}_-^*$ the set of negative rational numbers

In other books the symbols $\mathbb{Q}_+$, $\mathbb{Q}_-$ may stand for the positive, negative rational numbers respectively.

$\mathbb{R}$ the set of real numbers

$\mathbb{R}^*$ the set of non-zero real numbers

$\mathbb{R}_+$ the set of non-negative real numbers

$\mathbb{R}_-$ the set of non-positive real numbers

$\mathbb{R}_+^*$ the set of positive real numbers

$\mathbb{R}_-^*$ the set of negative real numbers

$\overline{\mathbb{R}}$ the set of extended real numbers, i.e., $\mathbb{R}$ with $\pm\infty$ included

In other books the symbols $\mathbb{R}_+$, $\mathbb{R}_-$ may stand for the positive, negative real numbers respectively, while the symbols $\overline{\mathbb{R}}_+$, $\overline{\mathbb{R}}_-$ may stand for the non-negative, non-positive real numbers.

$\mathbb{C}$ the set of complex numbers

$\mathbb{C}^*$ the set of non-zero complex numbers

$\overline{\mathbb{C}}$ the extended set of complex numbers $\mathbb{C} \cup \{\infty\}$ (Riemann sphere)

NAMED EQUATIONS

Abel	$f(g(x)) = f(x) + a$
Böttcher	$f(g(x)) = f(x)^p$
Cauchy I	$f(x+y) = f(x) + f(y)$
Cauchy II	$f(x+y) = f(x)\, f(y)$
Cauchy III	$f(x\,y) = f(x) + f(y)$
Cauchy IV	$f(x\,y) = f(x)\, f(y)$
conjugacy	$f(h(x)) = H(f(x))$
D'Alembert-Poisson I	$f(x+y) + f(x-y) = 2\, f(x)\, f(y)$
D'Alembert-Poisson II	$f(x+y)\, f(x-y) = f(x)^2 - f(y)^2$
Euler	$f(\lambda x, \lambda y) = \lambda\, f(x,y)$
Jensen	$f(\frac{x+y}{2}) = \frac{f(x)+f(y)}{2}$
Lobacevskii	$f(x)^2 = f(x+y)\, f(x-y)$
Pexider I	$f(x+y) = g(x) + h(y)$
Pexider II	$f(x+y) = g(x)\, h(y)$
Pexider III	$f(x\,y) = g(x) + h(y)$
Pexider IV	$f(x\,y) = g(x)\, f(y)$
Schröder	$f(g(x)) = \lambda\, f(x)$
Vincze I	$f(x+y) = g_1(x)\, g_2(y) + h(y)$
Vincze II	$f(x\,y) = g_1(x)\, g_2(y) + h(y)$
Wilson I	$f(x+y) + f(x-y) = 2\, f(x)\, g(y)$
Wilson II	$f(x+y) + g(x-y) = 2\, h(x)\, k(y)$

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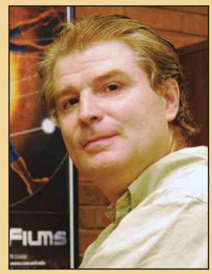
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